

PROMIN STANDARD

**promin**

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# Appendices

These tables are a complete generated projection of canonical owners. Every entry shows an owner path and an executable validator, schema reference, predicate, or mutation-probe identifier.

Owner: *core/promin.manifest.json* | Validator: *bundle\_digest* + *core\_components*

## Canonical owner trace

| Owner                      | Sole function                                                                                                                                                | Validator                                       | SHA256                                                                       |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| core/promin.manifest.json  | Own Core bundle identity and exact digest binding for the five other Core artifacts.                                                                         | urn:promin:contracts:1#/\$defs/StandardManifest | a04064a81507ba40c71a<br>aef453e70bf013cf31418<br>2ccb9ea1cd076acc7175<br>37a |
| core/semantic-model.json   | Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and provider capability contracts.                                     | urn:promin:contracts:1#/\$defs/SemanticModel    | 4bfb782f986bd65ec7f10<br>c16109587c6167af9c4ae<br>e468efcd8c6f8e3d72c8f9     |
| core/authority-model.json  | Own action capabilities, runtime Grant semantics, trust modes, configured signature-adapter verification, root ceilings, and separation-of-duty constraints. | urn:promin:contracts:1#/\$defs/AuthorityModel   | 6c2addec663f8f9796014<br>66fa947e9a4347a3053d<br>872ae89ced137fd174c31<br>5c |
| core/policy-set.json       | Own executable cross-record invariants through stable validator identifiers and no duplicated budget values.                                                 | urn:promin:contracts:1#/\$defs/PolicySet        | e9f8d0c51c624fbabf9b4<br>e973aed577480f17b878<br>5c043ac7701c25f239454<br>6a |
| core/conformance.json      | Own acceptance predicates, mutation families, structural hard ceilings, scale contracts, and production-claim rules.                                         | urn:promin:contracts:1#/\$defs/Conformance      | 3925dbac85b3cb99c719<br>2a915bdacffb8d6b10c19<br>e98b3c8d0711669cf3a0<br>65a |
| core/contracts.schema.json | Own all structural, format, kind-specific payload, and strict critical-field contracts in one bundle.                                                        | Draft 2020-12 oneOf                             | 6880be82afac157d4a24<br>8431c27da83d4963ad43<br>c78c1fdffd6c6476287e7c<br>d8 |

Core bundle digest: 47dfe980435788c3c7d12b97b6059251d5ee5d045f7597e937304a4ada507749

Selected preset outside Core identity: presets/semantic-standard.json | SHA256:  
094e95667950c1cd38f8d1b910f3e3ccda4dc9715f578c8b80eccb5c4674d1d2

Owner: *core/promin.manifest.json* | Validator: *canonical\_name\_only* + *core\_components* + *bundle\_digest*

## Schema catalog - 141 definitions

Schema ID: urn:promin:contracts:1 | Dialect: <https://json-schema.org/draft/2020-12/schema> |  
Definitions: 141

### 1. ActionCapabilityDef

Type summary: type="object"; additionalProperties=false

Required: id, purpose

| Property                                                                                                                                                                                                                                                                                                       | Required | Structural rule            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------|
| id                                                                                                                                                                                                                                                                                                             | yes      | enum=[16 items]            |
| purpose                                                                                                                                                                                                                                                                                                        | yes      | type="string"; minLength=1 |
| <pre>{   "additionalProperties": false,   "properties": {     "id": {       "enum": [         "standard.activate",         "authority.manage",         "task.plan",         "task.execute",         "lease.manage",         "evidence.publish",         "validation.evaluate",         "finding.record",</pre> |          |                            |



```

        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read"
    ]
},
"purpose": {
    "minLength": 1,
    "type": "string"
}
},
"required": [
    "id",
    "purpose"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ActionCapabilityDef | Validator: urn:promin:contracts:1#/\$defs/ActionCapabilityDef

## 2. Activation

Type summary: type="object"; additionalProperties=false

Required: record\_type, canonical\_name, core\_bundle\_digest, preset\_digest, init\_file\_digests, operating\_profile, trust\_mode, activation\_digest, proofs, implementation\_closure\_digest

| Property                      | Required | Structural rule                       |
|-------------------------------|----------|---------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"               |
| canonical_name                | yes      | const="promin"                        |
| core_bundle_digest            | yes      | \$ref="#/\$defs/Digest"               |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"               |
| init_file_digests             | yes      | \$ref="#/\$defs/InitFileDigests"      |
| operating_profile             | yes      | \$ref="#/\$defs/Id"                   |
| preset_digest                 | yes      | \$ref="#/\$defs/Digest"               |
| proofs                        | yes      | type="array"; minItems=1; maxItems=32 |
| record_type                   | yes      | const="Activation"                    |
| trust_mode                    | yes      | enum=["local-owner", "team-signed"]   |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_name": {
      "const": "promin"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "init_file_digests": {
      "$ref": "#/$defs/InitFileDigests"
    },
    "operating_profile": {
      "$ref": "#/$defs/Id"
    },
    "preset_digest": {
      "$ref": "#/$defs/Digest"
    },
    "proofs": {
      "items": {
        "$ref": "#/$defs/ActivationProof"
      },

```

```

    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  "record_type": {
    "const": "Activation"
  },
  "trust_mode": {
    "enum": [
      "local-owner",
      "team-signed"
    ]
  }
},
"required": [
  "record_type",
  "canonical_name",
  "core_bundle_digest",
  "preset_digest",
  "init_file_digests",
  "operating_profile",
  "trust_mode",
  "activation_digest",
  "proofs",
  "implementation_closure_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Activation | Validator: urn:promin:contracts:1#/\$defs/Activation

### 3. ActivationProof

Type summary: oneOf[2]=#/\$defs/LocalActivationProof | #/\$defs/SignatureActivationProof

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/LocalActivationProof"
    },
    {
      "$ref": "#/$defs/SignatureActivationProof"
    }
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/ActivationProof | Validator: urn:promin:contracts:1#/\$defs/ActivationProof

### 4. AffectedSemanticScope

Type summary: type="object"; additionalProperties=false

Required: invalidation\_class, input\_digest, truncated, pass\_credit

| Property           | Required | Structural rule                                                        |
|--------------------|----------|------------------------------------------------------------------------|
| input_digest       | yes      | \$ref="#/\$defs/Sha256"                                                |
| invalidation_class | yes      | enum=["BODY_ONLY", "IMPORT_SURFACE", "CMAKE_TOPOLOGY", "TOOLING_ONLY"] |
| pass_credit        | yes      | const=false                                                            |
| truncated          | yes      | type="boolean"                                                         |

```

{
  "additionalProperties": false,
  "properties": {
    "input_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "invalidation_class": {
      "enum": [
        "BODY_ONLY",
        "IMPORT_SURFACE",
        "CMAKE_TOPOLOGY",
        "TOOLING_ONLY"
      ]
    },
    "pass_credit": {

```

```

    "const": false
  },
  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "invalidation_class",
  "input_digest",
  "truncated",
  "pass_credit"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/AffectedSemanticScope | Validator: urn:promin:contracts:1#/\$defs/AffectedSemanticScope

## 5. Artifact

Type summary: type="object"; additionalProperties=false; allOf[3]=keys:else,if,then,title | keys:if,then,title | keys:else,if,then,title

Required: record\_type, artifact\_id, artifact\_kind, digest, media\_type, size\_bytes, retention\_class, created\_at

| Property                | Required | Structural rule                                                           |
|-------------------------|----------|---------------------------------------------------------------------------|
| artifact_id             | yes      | \$ref="#/\$defs/Id"                                                       |
| artifact_kind           | yes      | enum=["product", "evidence", "log", "report", "diff"]                     |
| candidate_delta         | no       | \$ref="#/\$defs/CandidateDelta"                                           |
| created_at              | yes      | \$ref="#/\$defs/Timestamp"                                                |
| digest                  | yes      | \$ref="#/\$defs/Digest"                                                   |
| evidence_binding        | no       | \$ref="#/\$defs/EvidenceBinding"                                          |
| evidence_class          | no       | enum=["product-execution", "validator", "harness-generated", "migration"] |
| evidence_purpose        | no       | enum=["diagnostic", "gate", "product"]                                    |
| media_type              | yes      | type="string"; minLength=1                                                |
| outcome                 | no       | enum=["pass", "fail", "blocked", "skipped", "error"]                      |
| product_credit_eligible | no       | type="boolean"                                                            |
| record_type             | yes      | const="Artifact"                                                          |
| retention_class         | yes      | enum=["ephemeral", "project", "release", "audit"]                         |
| size_bytes              | yes      | type="integer"; minimum=0                                                 |
| stale                   | no       | type="boolean"                                                            |
| unresolved              | no       | type="boolean"                                                            |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "evidence_binding"
              ]
            },
            {
              "required": [
                "evidence_purpose"
              ]
            },
            {
              "required": [
                "outcome"
              ]
            },
            {
              "required": [
                "stale"
              ]
            }
          ]
        }
      }
    }
  ]
}

```

```

    },
    {
      "required": [
        "unresolved"
      ]
    },
    {
      "required": [
        "evidence_class"
      ]
    },
    {
      "required": [
        "product_credit_eligible"
      ]
    }
  ]
}
},
"if": {
  "properties": {
    "artifact_kind": {
      "const": "evidence"
    }
  },
  "required": [
    "artifact_kind"
  ]
},
"then": {

```

```

    "required": [
      "evidence_binding",
      "evidence_purpose",
      "outcome",
      "stale",
      "unresolved",
      "evidence_class",
      "product_credit_eligible"
    ]
  },
  "title": "require replay-complete immutable evidence metadata"
},
{
  "if": {
    "properties": {
      "product_credit_eligible": {
        "const": true
      }
    },
    "required": [
      "product_credit_eligible"
    ]
  },
  "then": {
    "properties": {
      "artifact_kind": {
        "const": "evidence"
      }
    },

```

```

    "evidence_binding": {
      "required": [
        "provider_invocations"
      ]
    },
    "evidence_class": {
      "const": "product-execution"
    },
    "evidence_purpose": {
      "const": "product"
    },
    "outcome": {
      "const": "pass"
    },
    "stale": {
      "const": false
    },
    "unresolved": {
      "const": false
    }

```

```

    }
  },
  "required": [
    "evidence_binding",
    "evidence_purpose",
    "outcome",
    "stale",
    "unresolved",
    "evidence_class",
    "product_credit_eligible"
  ]
},
"title": "product credit requires fresh resolved passing product execution evidence"
},
{
  "else": {
    "not": {
      "required": [
        "candidate_delta"
      ]
    }
  },
  "if": {
    "properties": {
      "artifact_kind": {
        "const": "diff"
      }
    },
    "required": [
      "artifact_kind"
    ]
  },
  "then": {
    "required": [
      "candidate_delta"
    ]
  }
},

```

```

  "title": "diff Artifact is the canonical Candidate delta change manifest"
},
],
"properties": {
  "artifact_id": {
    "$ref": "#/$defs/Id"
  },
  "artifact_kind": {
    "enum": [
      "product",
      "evidence",
      "log",
      "report",
      "diff"
    ]
  },
  "candidate_delta": {
    "$ref": "#/$defs/CandidateDelta"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "digest": {
    "$ref": "#/$defs/Digest"
  },
  "evidence_binding": {
    "$ref": "#/$defs/EvidenceBinding"
  },

```

```

  "evidence_class": {
    "enum": [
      "product-execution",
      "validator",
      "harness-generated",
      "migration"
    ]
  },
  "evidence_purpose": {
    "enum": [
      "diagnostic",
      "gate",

```

```

        "product"
      ]
    },
    "media_type": {
      "minLength": 1,
      "type": "string"
    },
    "outcome": {
      "enum": [
        "pass",
        "fail",
        "blocked",
        "skipped",
        "error"
      ]
    }
  ],
  "product_credit_eligible": {
    "type": "boolean"
  },
  "record_type": {
    "const": "Artifact"
  },
  "retention_class": {
    "enum": [
      "ephemeral",
      "project",
      "release",
      "audit"
    ]
  },
  "size_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "stale": {
    "type": "boolean"
  },
  "unresolved": {
    "type": "boolean"
  }
},
"required": [
  "record_type",
  "artifact_id",
  "artifact_kind",
  "digest",
  "media_type",
  "size_bytes",
  "retention_class",
  "created_at"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Artifact | Validator: urn:promin:contracts:1#/\$defs/Artifact

## 6. AuthorityInit

Type summary: type="object"; additionalProperties=false; allOf[1]=keys;else;if,then

Required: record\_type, trust\_mode, subjects, roots

| Property    | Required | Structural rule                       |
|-------------|----------|---------------------------------------|
| keys        | no       | type="array"; minItems=1; maxItems=64 |
| record_type | yes      | const="AuthorityInit"                 |
| roots       | yes      | type="array"; minItems=1; maxItems=16 |
| subjects    | yes      | type="array"; minItems=1; maxItems=64 |
| team_policy | no       | \$ref="#/\$defs/TeamPolicy"           |
| trust_mode  | yes      | enum=["local-owner", "team-signed"]   |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {

```

```

    "allOf": [
      {
        "not": {
          "required": [
            "team_policy"
          ]
        }
      },
      {
        "not": {
          "required": [
            "keys"
          ]
        }
      }
    ],
    "if": {
      "properties": {
        "trust_mode": {
          "const": "team-signed"
        }
      }
    }
  },
  "then": {
    "required": [
      "team_policy",
      "keys"
    ]
  }
},
],
"properties": {
  "keys": {
    "items": {
      "$ref": "#/$defs/TrustKey"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  },
  "record_type": {
    "const": "AuthorityInit"
  },
  "roots": {
    "items": {
      "$ref": "#/$defs/RootAuthority"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array"
  },
  "subjects": {
    "items": {
      "$ref": "#/$defs/SubjectInit"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  },
  "team_policy": {
    "$ref": "#/$defs/TeamPolicy"
  },
  "trust_mode": {
    "enum": [
      "local-owner",
      "team-signed"
    ]
  }
},
"required": [
  "record_type",
  "trust_mode",
  "subjects",
  "roots"
],
"type": "object"
}

```

## 7. AuthorityModel

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, document, version, function, default, capabilities, command\_capability\_rules, command\_effect\_scope\_rules, bootstrap\_rule, trust\_modes, role\_presets\_non\_authoritative, separation\_of\_duties, model\_tier\_rule, command\_mutation\_claim\_rule, continuation\_access\_rule, standard\_release\_decision\_rule, release\_evidence\_authority\_rule, canonical\_timestamp\_contract, delegation\_depth\_max, derived\_event\_store\_policy\_contract, derived\_runtime\_policy\_contract, event\_contract, grant\_contract, scope\_contract, separation\_of\_duties\_contract

| Property                            | Required | Structural rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bootstrap_rule                      | yes      | const="authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json."                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| canonical_timestamp_contract        | yes      | const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| capabilities                        | yes      | const=[16 items]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| command_capability_rules            | yes      | const=[11 items]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| command_effect_scope_rules          | yes      | const=[11 items]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| command_mutation_claim_rule         | yes      | type="object"; additionalProperties=false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| continuation_access_rule            | yes      | const=schema(keys:authority_effect,authorization_rule,capability_id,default_budget_source,default_depth_source,default_ttl_seconds,depth_max_source,depth_min,hard_budget_source,max_continuation_state_bytes_source,max_query_bytes_source,max_resume_binding_bytes,max_resume_binding_fields,max_resume_binding_key_chars,max_resume_binding_value_bytes,min_result_bytes,persistent_entity_types_source,ranking_algorithm_id,required_claims,required_resume_binding_fields,revocation_rule,secret_lifecycle,selected_profile_digest_rule,selected_profile_id_source,state_digest_rule,state_persistence,token_envelope,token_max_bytes_source,token_version,traversal_algorithm_id,ttl_min_seconds,ttl_seconds_max)                                                                                                                  |
| default                             | yes      | const="deny"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| delegation_depth_max                | yes      | const=8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| derived_event_store_policy_contract | yes      | const=schema(keys:allowed_state_binding_leaf_types_source,command_conditional_fields_source,command_mutation_fields_source,event_contract_source,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,timestamp_contract_source)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| derived_runtime_policy_contract     | yes      | const=schema(keys:capability_ids_source,delegation_depth_source,grant_contract_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,separation_of_duties_capability_ids_source,separation_of_duties_contract_source,separation_of_duties_source,timestamp_contract_source)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| document                            | yes      | const="promin.authority-model"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| event_contract                      | yes      | const=schema(keys:auxiliary_event_kinds,command_bytes_max,command_conditional_fields,command_required_fields,command_to_primary_event,derived_tail_batch_threshold,derived_tail_byte_threshold,-envelope_bytes_max,event_batch_commitment_rule,event_batch_required_fields,event_required_fields,events_per_batch_max,genesis_event_semantic_digest,genesis_previous_authority_commitment,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,primary_event_count_per_command,requested_scope_items_max_source,runtime_overlay_compaction_depth,state_binding_algorithm,state_binding_algorithm_contract,state_binding_bytes_max,state_binding_identity_rule,state_binding_identity_rules,state_binding_leaf_type_source,state_binding_updates_per_batch_max,state_binding_value_rules,timestamp_contract_source) |
| function                            | yes      | const="Own action capabilities, runtime Grant semantics, trust modes, configured signature-adapter verification, root ceilings, and separation-of-duty constraints."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| grant_contract                      | yes      | const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| model_tier_rule                     | yes      | const="Model strength changes decomposition, context, parallelism, and analysis depth; it never changes authority."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| record_type                         | yes      | const="AuthorityModel"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



| Property                        | Required | Structural rule                                                                                                                                                                                                                                                                                 |
|---------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| release_evidence_authority_rule | yes      | const=schema(keys:attestation_algorithm,attestation_domain,decision_capability,evidence_capability,evidence_key_scope,required_claim_bindings,required_evidence_role_count,role_subject_key_uniqueness,separation_rule,unsigned_or_replayed)                                                    |
| role_presets_non_authoritative  | yes      | const=schema(keys:auditor,root-owner,validator,worker)                                                                                                                                                                                                                                          |
| schema_ref                      | yes      | const="urn:promin:contracts:1#/\$defs/AuthorityModel"                                                                                                                                                                                                                                           |
| scope_contract                  | yes      | const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)                               |
| separation_of_duties            | yes      | const=[5 items]                                                                                                                                                                                                                                                                                 |
| separation_of_duties_contract   | yes      | const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)                                                                                                                                                                 |
| standard_release_decision_rule  | yes      | const=schema(keys:chain,chronology,decision_outcomes,derived_truths,evidence_binding,history_and_current_state,independent_trust_binding,location,product_acceptance_pass,product_public_approval,publication_rule,rule,signature_algorithm,signature_provider_id,standard_name,version_source) |
| trust_modes                     | yes      | const=schema(keys:local-owner,team-signed)                                                                                                                                                                                                                                                      |
| version                         | yes      | const="1.0.0-alpha.4"                                                                                                                                                                                                                                                                           |

```

{
  "additionalProperties": false,
  "properties": {
    "bootstrap_rule": {
      "const": "authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json."
    },
    "canonical_timestamp_contract": {
      "const": {
        "calendar": "proleptic-gregorian",
        "format": "utc-second-z",
        "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str",
        "fractional_seconds": "reject",
        "normalization": "identity-round-trip",
        "offset": "Z-only",
        "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime",
        "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z$",
        "real_calendar_validation": true,
        "resolution_seconds": 1,
        "year_max": 9999,
        "year_min": 1
      }
    },
    "capabilities": {
      "const": [
        {
          "id": "standard.activate",
          "purpose": "Activate one digest-bound repository configuration."
        },
        {
          "id": "authority.manage",
          "purpose": "Issue or revoke runtime Grants within a configured root ceiling."
        },
        {
          "id": "task.plan",
          "purpose": "Create, split, order, or block Tasks."
        },
        {
          "id": "task.execute",
          "purpose": "Execute one Task inside its WorkCard and valid Lease."
        },
        {
          "id": "lease.manage",
          "purpose": "Acquire, heartbeat, close, expire, or revoke Leases."
        },
        {
          "id": "evidence.publish",
          "purpose": "Publish candidate-bound evidence Artifacts."
        },
        {
          "id": "validation.evaluate",

```

|                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                             | <pre>       "purpose": "Evaluate gates without promotion or release authority."     },     {       "id": "finding.record",       "purpose": "Record Findings."     },   ], </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| evidence, and separation of duties."                                                                                        | <pre>     {       "id": "finding.resolve",       "purpose": "Resolve one Finding only with evidence bound to that Finding's canonical digest."     },     {       "id": "finding.waive",       "purpose": "Waive one Finding only through an expiring exception-policy Decision, Finding-bound     },     {       "id": "candidate.promote",       "purpose": "Promote a validated Candidate."     },     {       "id": "release.decide",       "purpose": "Record an immutable historical release judgement bound to an exact acceptance-closure and provider-binding digest; current eligibility remains derived."     },     {       "id": "export.create",       "purpose": "Create a sanitized external package."     },     {       "id": "migration.manage",       "purpose": "Execute cutover, rollback, and deterministic recutover."     },     {       "id": "projection.rebuild",       "purpose": "Rebuild or verify non-authoritative projections."     },   ], </pre> |
| current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard distribution authority." | <pre>     {       "id": "projection.read",       "purpose": "Read one bounded current projection or resume its continuation under the exact current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard distribution authority."     }   ] }, "command_capability_rules": {   "const": [     {       "capability_id": "task.plan",       "command_kind": "task.record"     },     {       "capability_id": "task.plan",       "command_kind": "task.transition"     },     {       "capability_id": "task.execute",       "command_kind": "candidate.record"     },     {       "capability_by_value": {         "diff": "task.execute",         "evidence": "evidence.publish",         "log": "task.execute",         "product": "task.execute",         "report": "projection.rebuild"       }, </pre>                                                                                                                                     |
|                                                                                                                             | <pre>       "command_kind": "artifact.record",       "payload_field": "artifact_kind"     },     {       "capability_by_value": {         "execution": "task.execute",         "validation": "validation.evaluate"       },       "command_kind": "run.record",       "payload_field": "run_kind"     },   ], </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

```

    {
      "capability_id": "validation.evaluate",
      "command_kind": "gate.record"
    },
    {
      "capability_id": "finding.record",
      "command_kind": "finding.record"
    },
    {
      "capability_by_value": {
        "accept": "validation.evaluate",
        "promote": "candidate.promote",
        "reject": "validation.evaluate",
        "release": "release.decide",
        "resolve": "finding.resolve",
        "revoke": "authority.manage",
        "waive": "finding.waive"
      }
    },

```

```

    {
      "command_kind": "decision.record",
      "payload_field": "decision_kind"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.issue"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.revoke"
    },
    {
      "capability_id": "lease.manage",
      "command_kind": "lease.record"
    }
  ]
},

```

```

"command_effect_scope_rules": {
  "const": [
    {
      "command_kind": "task.record",
      "id_field": "task_id",
      "kind": "task",
      "mode": "payload-id"
    },
    {
      "command_kind": "task.transition",

```

```

      "id_field": "task_id",
      "kind": "task",
      "mode": "payload-id"
    },
    {
      "command_kind": "candidate.record",
      "id_field": "candidate_id",
      "kind": "candidate",
      "mode": "payload-id"
    },
    {
      "command_kind": "artifact.record",
      "id_field": "artifact_id",
      "kind": "artifact",
      "mode": "payload-id"
    },
    {
      "command_kind": "run.record",
      "mode": "payload-multi-id",
      "targets": [
        {
          "id_field": "task_id",
          "kind": "task"
        },
        {
          "id_field": "candidate_digest",
          "kind": "candidate"
        }
      ]
    }
  ]
},

```

```

    {
      "command_kind": "gate.record",

```

```

        "id_field": "candidate_digest",
        "kind": "candidate",
        "mode": "payload-id"
    },
    {
        "command_kind": "finding.record",
        "id_field": "finding_id",
        "kind": "finding",
        "mode": "payload-id"
    },
    {
        "command_kind": "decision.record",
        "grant_target_type": "Grant",
        "id_field": "target_id",
        "kind_field": "target_type",
        "kind_map": {
            "Candidate": "candidate",
            "Finding": "finding",
            "Task": "task"
        },
        "mode": "payload-dynamic-id-or-referenced-grant-scope"
    },
    {
        "command_kind": "grant.issue",
        "mode": "payload-scope",
        "scope_field": "scope"
    },
    {
        "command_kind": "grant.revoke",
        "grant_id_field": "grant_id",
        "mode": "referenced-grant-scope"
    },
    {
        "command_kind": "lease.record",
        "id_field": "task_id",
        "kind": "task",
        "mode": "payload-id"
    }
}
],
"command_mutation_claim_rule": {
    "additionalProperties": false,
    "properties": {
        "binding_equalities": {
            "items": {
                "minLength": 1,
                "type": "string"
            },
            "maxItems": 16,
            "minItems": 8,
            "type": "array",
            "uniqueItems": true
        },
        "claim_rule": {
            "minLength": 1,
            "type": "string"
        },
        "command_authorization_source": {
            "const": "CommandRequest.authorization.grant_id + grant_claim_digest"
        },
        "grant_separation_rule": {
            "minLength": 1,
            "type": "string"
        },
        "holder_grant_capability": {
            "const": "task.execute"
        },
        "holder_grant_field": {
            "const": "holder_authorization.grant_id"
        },
        "lease_bound_command_kinds": {
            "items": {
                "enum": [
                    "task.record",
                    "task.transition",
                    "candidate.record",
                    "artifact.record",
                    "run.record",

```

```

    "gate.record",
    "finding.record",

    "decision.record",
    "grant.issue",
    "grant.revoke",
    "lease.record"
  ]
},
"maxItems": 11,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"lease_bound_task_transition_states": {
  "items": {
    "enum": [
      "BLOCKED",
      "CANCELLED",
      "COMPLETED",
      "LEASED",
      "PLANNED",
      "READY",
      "RUNNING"
    ]
  },
  "maxItems": 7,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},

```

```

"lease_required_state": {
  "const": "ACTIVE"
},
"required_command_fields": {
  "const": [
    "workcard_task_id",
    "holder_authorization",
    "lease_id",
    "lease_generation",
    "fencing_token",
    "workcard_digest",
    "context_digest"
  ],
  "type": "array"
},
"required_workcard_fields": {
  "const": [
    "holder_grant_id",
    "lease_id",
    "lease_generation",
    "fencing_token",
    "context_digest"
  ],
  "type": "array"
},
"stale_or_unresolved": {
  "const": "reject"
}

```

```

},
"required": [
  "binding_equalities",
  "claim_rule",
  "command_authorization_source",
  "grant_separation_rule",
  "holder_grant_capability",
  "holder_grant_field",
  "lease_bound_command_kinds",
  "lease_bound_task_transition_states",
  "lease_required_state",
  "required_command_fields",
  "required_workcard_fields",
  "stale_or_unresolved"
],
"type": "object"
},
"continuation_access_rule": {
  "const": {

```

```

    "authority_effect": false,
    "authorization_rule": "Initial retrieval and every continuation resume require a current
projection.read Grant whose subject, grant_id, claim_digest, capability, Activation, conjunctive requested
scope, interval, and revocation state match the token claims exactly.",
    "capability_id": "projection.read",
    "default_budget_source": "selected Preset profile max_context_bytes, max_entities, max_relations,
max_fanout_per_entity, and top_k",
    "default_depth_source": "selected Preset profile default_dependency_depth",
    "default_ttl_seconds": 900,
    "depth_max_source": "Conformance.dependency_depth_hard_max",
    "depth_min": 1,
    "hard_budget_source": "Conformance.workcard_hard_ceiling",

    "max_continuation_state_bytes_source": "Conformance.scale_contracts.workcard.continuation_state_byt
es_max",
    "max_query_bytes_source": "PolicySet.derived_result_contracts.RetrievalPage.query_bytes_max",
    "max_resume_binding_bytes": 4096,
    "max_resume_binding_fields": 8,
    "max_resume_binding_key_chars": 64,
    "max_resume_binding_value_bytes": 512,
    "min_result_bytes": 1024,
    "persistent_entity_types_source": "SemanticModel.persistent_entities.kind-in-owner-order",
    "ranking_algorithm_id": "bm25-v1",
    "required_claims": [
        "activation_digest",
        "subject_id",
        "grant_id",
        "grant_claim_digest",
        "capability_id",
        "requested_scope",
        "normalized_query_digest",
        "cursor",
        "head_event_id",
        "head_sequence",
        "head_digest",
        "projection_digest",
        "ranking_id",
        "traversal_id",
        "dependency_depth",
        "budgets",
        "issued_at",
        "expires_at",

        "ttl_seconds",
        "revocation_epoch",
        "implementation_closure_digest",
        "state_id"
    ],
    "required_resume_binding_fields": [
        "activation_digest",
        "capability_id",
        "grant_claim_digest",
        "grant_id",
        "implementation_closure_digest",
        "requested_scope_digest",
        "revocation_epoch",
        "subject_id"
    ],
    "revocation_rule": "Expired, revoked, substituted, stale, mismatched, or unresolved Grant claims
reject before projection continuation.",
    "secret_lifecycle": "Generate one random 32-byte secret per Activation, store it only under .
promin/state/secrets with restrictive permissions, never derive it from public package or control digests,
never journal or export it, and rotate it with Activation.",
    "selected_profile_digest_rule": "sha256(canonical exact selected Preset profile object)",
    "selected_profile_id_source": "Activation.operating_profile",
    "state_digest_rule": "sha256(canonical full ContinuationTokenClaims); the compact token MAC binds
token version, opaque state_id, state digest, Activation, subject, Grant claim, and expiry",
    "state_persistence": "derived-live-only",
    "token_envelope": "promin-v2.<base64url opaque 128-bit state handle>.<lowercase state
digest>.<base64url HMAC-SHA256>",
    "token_max_bytes_source": "Conformance.scale_contracts.workcard.continuation_token_bytes_max",
    "token_version": 2,
    "traversal_algorithm_id": "typed-bfs-v2",
    "ttl_min_seconds": 1,
    "ttl_seconds_max": 3600
}

},
"default": {
    "const": "deny"
}

```

```

    },
    "delegation_depth_max": {
      "const": 8
    },
    "derived_event_store_policy_contract": {
      "const": {
        "allowed_state_binding_leaf_types_source": "Activation plus SemanticModel.persistent_entities.kind
plus Relation",
        "command_conditional_fields_source": "event_contract.command_conditional_fields",
        "command_mutation_fields_source": "command_mutation_claim_rule.required_command_fields",
        "event_contract_source": "event_contract",
        "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds",
        "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transiti
on_states",
        "persistence": "derived-live-only",
        "policy_digest_rule": "sha256(canonical EventStorePolicy without policy_digest)",
        "required_bindings": [
          "authority_model_digest",
          "max_command_bytes",
          "max_envelope_bytes",
          "max_state_binding_bytes",
          "max_state_binding_updates_per_batch",
          "max_events_per_batch",
          "max_requested_scope_items",
          "derived_tail_batch_threshold",
          "derived_tail_byte_threshold",
          "runtime_overlay_compaction_depth",
          "command_required_fields",
          "command_conditional_fields",
          "command_mutation_fields",
          "lease_bound_command_kinds",
          "lease_bound_task_transition_states",
          "command_to_primary_event",
          "allowed_state_binding_leaf_types",
          "state_binding_identity_rules",
          "state_binding_algorithm_contract",
          "state_binding_value_rules",
          "canonical_timestamp_contract",
          "genesis_previous_authority_commitment",
          "genesis_event_semantic_digest",
          "policy_digest"
        ],
        "scope_contract_source": "scope_contract",
        "timestamp_contract_source": "canonical_timestamp_contract"
      }
    },
    "derived_runtime_policy_contract": {
      "const": {
        "capability_ids_source": "capabilities[].id-in-owner-order",
        "delegation_depth_source": "delegation_depth_max",
        "grant_contract_source": "grant_contract",
        "persistence": "derived-live-only",
        "policy_digest_rule": "sha256(canonical AuthorityRuntimePolicy without policy_digest)",
        "required_bindings": [
          "authority_model_digest",
          "capability_ids",
          "separation_of_duties",
          "separation_of_duties_capability_ids",
          "separation_of_duties_contract",
          "delegation_depth_max",
          "scope_contract",
          "grant_contract",
          "canonical_timestamp_contract",
          "policy_digest"
        ],
        "scope_contract_source": "scope_contract",
        "separation_of_duties_capability_ids_source": "capabilities[].id plus
separation_of_duties_contract.external_capability_ids",
        "separation_of_duties_contract_source": "separation_of_duties_contract",
        "separation_of_duties_source": "separation_of_duties-in-owner-order",
        "timestamp_contract_source": "canonical_timestamp_contract"
      }
    },
    "document": {
      "const": "promin.authority-model"
    },
    "event_contract": {
      "const": {

```

```

"auxiliary_event_kinds": [
  "relation.recorded"
],
"command_bytes_max": 1048576,
"command_conditional_fields": [
  "definition_digest"

```

```

],
"command_required_fields": [
  "record_type",
  "command_id",
  "command_kind",
  "subject_id",
  "activation_digest",
  "idempotency_key",
  "requested_scope",
  "expected_head_digest",
  "issued_at",
  "payload",
  "intent_digest",
  "authorization"
],
"command_to_primary_event": {
  "artifact.record": "artifact.recorded",
  "candidate.record": "candidate.recorded",
  "decision.record": "decision.recorded",
  "finding.record": "finding.recorded",
  "gate.record": "gate.recorded",
  "grant.issue": "grant.issued",
  "grant.revoke": "grant.revoked",
  "lease.record": "lease.recorded",
  "run.record": "run.recorded",
  "task.record": "task.recorded",
  "task.transition": "task.transitioned"
},

```

```

"derived_tail_batch_threshold": 128,
"derived_tail_byte_threshold": 8388608,
"envelope_bytes_max": 2097152,
"event_batch_commitment_rule": {
  "authority_commitment": "sha256(canonical object with exactly previous_authority_commitment,
sequence, previous_batch_digest, event_count, event_semantic_digest, state_binding_update_count,
state_binding_digest)",
  "cumulative_event_count": "previous cumulative_event_count plus current events length",
  "cumulative_state_binding_update_count": "previous cumulative state-binding update count plus
current state_binding_delta length",
  "event_semantic_digest": "ordered append-only digest over every canonical Event",
  "previous_authority_commitment": "exact previous EventBatch authority_commitment; sequence one
uses genesis_previous_authority_commitment",
  "state_binding_delta": "nonempty canonical ordered unique set/delete updates over changed typed
authority leaves only",
  "state_binding_digest": "exact typed-sparse-merkle-v1 post-batch authority root using runtime
domain separators for leaf and internal nodes; never a full serialized state hash"
},

```

```

"event_batch_required_fields": [
  "record_type",
  "batch_id",
  "sequence",
  "previous_digest",
  "previous_authority_commitment",
  "authority_commitment",
  "cumulative_event_count",
  "event_semantic_digest",
  "state_binding_delta",
  "cumulative_state_binding_update_count",
  "state_binding_digest",
  "created_at",
  "command_id",
  "idempotency_key",
  "activation_record_digest",

```

```

"events",
"subject_id",
"command_intent_digest",
"command_digest",
"authorization_digest"
],
"event_required_fields": [
  "record_type",
  "event_id",

```



```

        "event_kind",
        "activation_digest",
        "payload"
    ],
    "events_per_batch_max": 20,
    "genesis_event_semantic_digest": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
    "genesis_previous_authority_commitment": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
    "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds",
    "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transition_states",
    "primary_event_count_per_command": 1,
    "requested_scope_items_max_source": "scope_contract.requested_scope_items_max",
    "runtime_overlay_compaction_depth": 64,
    "state_binding_algorithm": "typed-sparse-merkle-v1",
    "state_binding_algorithm_contract": {
        "delete_rule": "delete maps the keyed leaf to empty_leaf",
        "delta_order": [
            "leaf_type-ascending",
            "leaf_id-ascending"
        ],
    },
    "depth": 256,
    "depth_encoding": "unsigned-16-bit-big-endian",
    "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",
    "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
    "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
    "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
    "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
    "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
},
"state_binding_bytes_max": 1048576,
"state_binding_identity_rule": "direct uses the exact id field; typed_digest_id uses prefix plus first 48 lowercase hex characters of sha256(canonical ordered id_fields array)",
"state_binding_identity_rules": {
    "Activation": {
        "id_fields": [
            "activation_digest"
        ],
        "identity_mode": "direct"
    },
    "Artifact": {
        "id_fields": [
            "artifact_id"
        ],
        "identity_mode": "direct"
    },
    "Candidate": {
        "id_fields": [
            "candidate_id"
        ],
    },
    "identity_mode": "direct"
},
"Decision": {
    "id_fields": [
        "decision_id"
    ],
    "identity_mode": "direct"
},
"Finding": {
    "id_fields": [
        "finding_id"
    ],
    "identity_mode": "direct"
},
"GateResult": {
    "digest_algorithm": "sha256",
    "digest_hex_chars": 48,
    "digest_input": "canonical-ordered-id-field-values",
    "id_fields": [
        "gate_id",
        "run_id"
    ],
    "identity_mode": "typed_digest_id",
    "prefix": "gate-result:"
},
"Grant": {

```

```

    "id_fields": [
      "grant_id"
    ],
    "identity_mode": "direct"
  },
  "Lease": {
    "id_fields": [
      "lease_id"
    ],
    "identity_mode": "direct"
  },
  "Relation": {
    "id_fields": [
      "relation_id"
    ],
    "identity_mode": "direct"
  },
  "Run": {
    "id_fields": [
      "run_id"
    ],
    "identity_mode": "direct"
  },
  "Task": {
    "id_fields": [
      "task_id"
    ],
    "identity_mode": "direct"
  }
},

```

```

    "state_binding_leaf_type_source": "Activation plus SemanticModel.persistent_entities.kind plus
Relation",
    "state_binding_updates_per_batch_max": 20,
    "state_binding_value_rules": {
      "Activation": {
        "event_kinds": [],
        "value_definition": "Activation",
        "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence
one and immutable for the journal"
      },
      "Artifact": {
        "event_kinds": [
          "artifact.recorded"
        ],
        "value_definition": "Artifact",
        "value_digest_rule": "sha256(canonical exact Event.payload Artifact)"
      },
      "Candidate": {
        "event_kinds": [
          "candidate.recorded"
        ],
        "value_definition": "Candidate",
        "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
      },
      "Decision": {
        "event_kinds": [
          "decision.recorded"
        ],
        "value_definition": "Decision",
        "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
      }
    },

```

```

  },
  "Finding": {
    "event_kinds": [
      "finding.recorded"
    ],
    "value_definition": "Finding",
    "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
  },
  "GateResult": {
    "event_kinds": [
      "gate.recorded"
    ],
    "value_definition": "GateResult",
    "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
  },
  "Grant": {
    "event_kinds": [

```

```

        "grant.issued",
        "grant.revoked"
    ],
    "value_definition": "GrantAuthorityState",
    "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
},
"Lease": {
    "event_kinds": [
        "lease.recorded"
    ],
    "value_definition": "Lease",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
},
"Relation": {
    "event_kinds": [
        "relation.recorded"
    ],
    "value_definition": "Relation",
    "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
},
"Run": {
    "event_kinds": [
        "run.recorded"
    ],
    "value_definition": "Run",
    "value_digest_rule": "sha256(canonical exact Event.payload Run)"
},
"Task": {
    "event_kinds": [
        "task.recorded",
        "task.transitioned"
    ],
    "value_definition": "Task",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
}
},
"timestamp_contract_source": "canonical_timestamp_contract"
}
},
"function": {
    "const": "Own action capabilities, runtime Grant semantics, trust modes, configured
signature-adaptor verification, root ceilings, and separation-of-duty constraints."
},
"grant_contract": {
    "const": {
        "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)",
        "nonce": {
            "max_length": 128,
            "min_length": 16,
            "pattern": "^[A-Za-z0-9._-]{16,128}$"
        },
        "required_fields": [
            "record_type",
            "grant_id",
            "subject_id",
            "capability_id",
            "scope",
            "activation_digest",
            "issued_at",
            "expires_at",
            "nonce",
            "claim_digest",
            "trust_proofs"
        ],
        "scope_items_max_source": "scope_contract.grant_scope_items_max",
        "trust_proof_kinds": {
            "issuer-grant": {
                "required_fields": [
                    "kind",
                    "issuer_grant_id",
                    "issuer_signed_claim_digest",
                    "signed_claim_digest"
                ]
            }
        },
        "local-root": {
            "required_fields": [

```

```

        "kind",
        "root_subject_id",
        "authority_init_digest",
        "signed_claim_digest"
    ]
},
"signature": {
    "required_fields": [
        "kind",
        "key_id",
        "algorithm",
        "signed_digest",
        "signature"
    ],
    "signature_max_length": 8192,
    "signature_min_length": 1
}
},
"trust_proofs_max": 32,
"trust_proofs_min": 1,

    "trust_proofs_unique": true,
    "unknown_proof_kind": "reject"
}
},
"model_tier_rule": {
    "const": "Model strength changes decomposition, context, parallelism, and analysis depth; it never
changes authority."
},
"record_type": {
    "const": "AuthorityModel"
},
"release_evidence_authority_rule": {
    "const": {
        "attestation_algorithm": "Ed25519",
        "attestation_domain": "promin.release-evidence.producer-attestation.v1",
        "decision_capability": "standard.distribute",
        "evidence_capability": "evidence.produce",
        "evidence_key_scope": "Every evidence producer key is independently pinned to an exact subject,
evidence role, platform, public key, validity interval, and revocation state.",
        "required_claim_bindings": [
            "candidate_binding_digest",
            "record_type",
            "evidence_role",
            "platform",
            "invocation_id",
            "payload_digest",
            "raw_artifact_manifest_digest"
        ],
        "required_evidence_role_count": 7,
        "role_subject_key_uniqueness": "one-role-one-subject-one-key",

        "separation_rule": "For one candidate, an evidence-producing subject or key cannot hold
standard.distribute, and a distribution decider subject or key cannot hold evidence.produce.",
        "unsigned_or_replayed": "reject"
    }
},
"role_presets_non_authoritative": {
    "const": {
        "auditor": [
            "projection.read",
            "finding.record",
            "finding.resolve",
            "export.create"
        ],
        "root-owner": [
            "standard.activate",
            "authority.manage",
            "task.plan",
            "lease.manage",
            "finding.resolve",
            "finding.waive",
            "candidate.promote",
            "release.decide",
            "export.create",
            "migration.manage",
            "projection.rebuild",
            "projection.read"
        ]
    }
},

```

```

"validator": [
  "projection.read",
  "validation.evaluate",
  "evidence.publish",
  "finding.record"
],
"worker": [
  "projection.read",
  "task.execute",
  "evidence.publish"
]
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/AuthorityModel"
},
"scope_contract": {
  "const": {
    "all_selector": {
      "exclusive": true,
      "kind": "all",
      "value": "*"
    },
    "conjunctive_effect_scope": true,
    "duplicate_selector": "reject",
    "grant_scope_items_max": 32,
    "kinds": [
      "project",
      "path",
      "task",

```

```

      "candidate",
      "artifact",
      "finding",
      "all"
    ],
    "path_selector": {
      "absolute": "reject",
      "dot_segments": "reject",
      "kind": "path",
      "normalization": "NFC POSIX relative path"
    },
    "project_selector": {
      "items_max": 1,
      "kind": "project"
    },
    "requested_scope_items_max": 32,
    "selector_fields": [
      "kind",
      "value"
    ],
    "unique_items": true,
    "unknown_kind": "reject",
    "value_max_length": 4096,
    "value_min_length": 1,
    "wildcard_outside_all": "reject"
  }
},
"separation_of_duties": {

```

```

  "const": [
    {
      "forbid_lease_derived_capabilities": [
        "validation.evaluate",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide"
      ],
      "id": "SOD-001",
      "rule": "lease-is-not-acceptance",
      "scope_kind": "lease"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "task.execute",
        "release.decide"
      ],
      "id": "SOD-002",

```

```

    "rule": "executor-not-release-decider",
    "scope_kind": "candidate"
  },
  {
    "forbid_same_subject_for_same_candidate": [
      "validation.evaluate",
      "candidate.promote"
    ],
    "id": "SOD-003",
  },
  {
    "rule": "validator-not-self-promoter",
    "scope_kind": "candidate"
  },
  {
    "forbid_same_subject_for_same_finding": [
      "finding.record",
      "finding.waive"
    ],
    "id": "SOD-004",
    "rule": "finder-not-sole-waiver",
    "scope_kind": "finding"
  },
  {
    "forbid_same_key_capabilities": [
      "evidence.produce",
      "standard.distribute"
    ],
    "forbid_same_subject_for_same_candidate": [
      "evidence.produce",
      "standard.distribute"
    ],
    "id": "SOD-005",
    "rule": "evidence-producer-not-distribution-decider",
    "scope_kind": "candidate-and-key"
  }
]
},
"separation_of_duties_contract": {

```

```

  "const": {
    "base_fields": [
      "id",
      "rule",
      "scope_kind"
    ],
    "capability_values_max": 32,
    "external_capability_ids": [
      "evidence.produce",
      "standard.distribute"
    ],
    "rules_max": 64,
    "supported_rule_fields": {
      "forbid_lease_derived_capabilities": {
        "cardinality": "unique-list",
        "items_min": 1
      },
      "forbid_same_key_capabilities": {
        "cardinality": "unique-pair",
        "items_exact": 2
      },
      "forbid_same_subject_for_same_candidate": {
        "cardinality": "unique-pair",
        "items_exact": 2
      },
      "forbid_same_subject_for_same_finding": {
        "cardinality": "unique-pair",
        "items_exact": 2
      }
    }
  },
  "unknown_rule_field": "reject"
}
},
"standard_release_decision_rule": {
  "const": {
    "chain": "exact candidate bytes -> StandardReleaseCandidateBinding -> seven unique stable-read
authority-role records plus one non-authoritative exact eight-lane matrix and four CP314 supplemental
records -> physically resolved StandardReleaseEvidenceManifest -> signed approve or reject
StandardReleaseDecision -> independently pinned trust root -> derived distribution status",
    "chronology": "Every evidence invocation starts no later than it completes; its attestation time

```

```

equals completion and stays within bounded verifier skew. Nested physical runs complete no later than
their outer saturation audit, human review occurs inside its signed invocation, and
max_evidence_completed_at is derived from all seven authority records, all nested physical records, and
all four CP314 supplemental records. decided_at follows that maximum, is no more than 300 seconds beyond
verifier time, and every producer/decision key is valid at its event time.",
  "decision_outcomes": [
    "approve",
    "reject"
  ],
  "derived_truths": "ExactPackageVerification is recomputed by the verifier and multi-platform
implementation closure is derived only from fully authenticated Windows and Linux records; neither is an
independently supplied required role or public command.",
  "evidence_binding": "Every authority, supplemental, matrix, and nested physical record binds exact
path, SHA-256, and byte count and is read through a bounded handle-relative no-follow component walk. The
manifest rejects symlink/junction/reparse/root substitution and NFC/casefold collisions, validates exact
Draft 2020-12 schemas, verifies domain-separated producer attestations, binds producer/tool, role,
platform, invocation, raw artifacts, and exact candidate, and runs role-specific semantic recomputation.
The aggregate matrix remains non-authoritative with pass_credit=false and all acceptance/public flags
false.",
  "history_and_current_state": "The signed StandardReleaseDecision and its digest remain immutable
history. Current distribution eligibility is a derived view and becomes false when evidence, trust pin,
candidate identity, chronology, key validity, or any required gate no longer validates.",
  "independent_trust_binding": "Manifest construction and closure verification require an
independently operator-pinned SHA-256 for the exact trust configuration bytes. Exactly seven evidence
roles have one distinct key, subject, and public key each; CP313 and CP314 observations for one role reuse
only that role's configured producer. A mismatched, missing, duplicate, over-broad, or invalid scope
cannot yield evidence credit or approval.",
  "location": "external-to-canonical-standard-folder",
  "product_acceptance_pass": false,
  "product_public_approval": "not_approved",
  "publication_rule": "Only outcome approve signed by an active configured Ed25519 key with
standard.distribute capability and verified against an independently pinned trust-root SHA-256 yields
approved distribution status. Missing, rejected, stale, unsigned, mismatched, unpinned, fabricated, or
unresolved records fail closed.",
  "rule": "StandardReleaseDecision is external to the canonical standard bytes, binds candidate and
evidence manifest digests, includes explicit outcome and anti-replay nonce, proves decider authority
through a configured trust root and signature provider, and cannot elevate a trust root delivered by the
same untrusted source without an independent digest pin.",
  "signature_algorithm": "Ed25519",
  "signature_provider_id": "cryptography-ed25519-v1",
  "standard_name": "promin",
  "version_source": "StandardReleaseCandidateBinding.version"
}
},
{
  "trust_modes": {
    "const": {
      "local-owner": {
        "boundary": "single-machine owner-controlled root authority",
        "signer_required": false
      },
      "team-signed": {
        "boundary": "multiple actors or remote transport",
        "minimum_threshold": 1,
        "provider_capability": "signature",
        "provider_resolution": "Resolve the exact technologies.json signature binding for init and every
ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly
non-reconstructable and valid only while bound to the same project, Activation, provider, and identity;
cross-process CLI workflows require a reconstructable configured adapter.",
        "signer_required": true,
        "unsupported_binding": "reject-before-workflow"
      }
    }
  },
  "version": {
    "const": "1.0.0-alpha.4"
  },
  "required": [
    "record_type",
    "schema_ref",
    "document",
    "version",
    "function",
    "default",
    "capabilities",
    "command_capability_rules",
    "command_effect_scope_rules",
    "bootstrap_rule",

```

```

    "trust_modes",
    "role_presets_non_authoritative",
    "separation_of_duties",
    "model_tier_rule",
    "command_mutation_claim_rule",
    "continuation_access_rule",
    "standard_release_decision_rule",
    "release_evidence_authority_rule",
    "canonical_timestamp_contract",
    "delegation_depth_max",
    "derived_event_store_policy_contract",
    "derived_runtime_policy_contract",
    "event_contract",
    "grant_contract",
    "scope_contract",
    "separation_of_duties_contract"
  ],
  "type": "object"
}

```

Owner: `core/contracts.schema.json#/$defs/AuthorityModel` | Validator: `urn:promin:contracts:1#/$defs/AuthorityModel`

## 8. AuthorityRuntimePolicy

Type summary: `type="object"`; `additionalProperties=false`

Required: `record_type`, `authority_model_digest`, `capability_ids`, `separation_of_duties`, `separation_of_duties_capability_ids`, `separation_of_duties_contract`, `delegation_depth_max`, `scope_contract`, `grant_contract`, `canonical_timestamp_contract`, `policy_digest`

| Property                                         | Required | Structural rule                                                                                                                                                                                                                                                                |
|--------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>authority_model_digest</code>              | yes      | <code>\$ref="#/\$defs/Digest"</code>                                                                                                                                                                                                                                           |
| <code>canonical_timestamp_contract</code>        | yes      | <code>const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)</code>                                                                                         |
| <code>capability_ids</code>                      | yes      | <code>const=[16 items]</code>                                                                                                                                                                                                                                                  |
| <code>delegation_depth_max</code>                | yes      | <code>type="integer"; const=8</code>                                                                                                                                                                                                                                           |
| <code>grant_contract</code>                      | yes      | <code>const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)</code>                                                                                      |
| <code>policy_digest</code>                       | yes      | <code>\$ref="#/\$defs/Digest"</code>                                                                                                                                                                                                                                           |
| <code>record_type</code>                         | yes      | <code>const="AuthorityRuntimePolicy"</code>                                                                                                                                                                                                                                    |
| <code>scope_contract</code>                      | yes      | <code>const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)</code> |
| <code>separation_of_duties</code>                | yes      | <code>const=[5 items]</code>                                                                                                                                                                                                                                                   |
| <code>separation_of_duties_capability_ids</code> | yes      | <code>const=[18 items]</code>                                                                                                                                                                                                                                                  |
| <code>separation_of_duties_contract</code>       | yes      | <code>const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)</code>                                                                                                                                   |

```

{
  "additionalProperties": false,
  "properties": {
    "authority_model_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_timestamp_contract": {
      "const": {
        "calendar": "proleptic-gregorian",
        "format": "utc-second-z",
        "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str",
        "fractional_seconds": "reject",
        "normalization": "identity-round-trip",
        "offset": "Z-only",
        "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime",
        "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z$",
        "real_calendar_validation": true,
        "resolution_seconds": 1,
        "year_max": 9999,
        "year_min": 1
      }
    }
  }
},

```



```

"capability_ids": {
  "const": [
    "standard.activate",
    "authority.manage",
    "task.plan",
    "task.execute",
    "lease.manage",
    "evidence.publish",
    "validation.evaluate",
    "finding.record",
    "finding.resolve",
    "finding.waive",
    "candidate.promote",
    "release.decide",
    "export.create",
    "migration.manage",
    "projection.rebuild",
    "projection.read"
  ]
},
"delegation_depth_max": {
  "const": 8,
  "type": "integer"
},
"grant_contract": {
  "const": {
    "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)",
    "nonce": {
      "max_length": 128,
      "min_length": 16,
      "pattern": "^[A-Za-z0-9._-]{16,128}$"
    }
  },
  "required_fields": [
    "record_type",
    "grant_id",
    "subject_id",
    "capability_id",
    "scope",
    "activation_digest",
    "issued_at",
    "expires_at",
    "nonce",
    "claim_digest",
    "trust_proofs"
  ],
  "scope_items_max_source": "scope_contract.grant_scope_items_max",
  "trust_proof_kinds": {
    "issuer-grant": {
      "required_fields": [
        "kind",
        "issuer_grant_id",
        "issuer_signed_claim_digest",
        "signed_claim_digest"
      ]
    },
    "local-root": {
      "required_fields": [
        "kind",
        "root_subject_id",
        "authority_init_digest",
        "signed_claim_digest"
      ]
    }
  },
  "signature": {
    "required_fields": [
      "kind",
      "key_id",
      "algorithm",
      "signed_digest",
      "signature"
    ],
    "signature_max_length": 8192,
    "signature_min_length": 1
  },
  "trust_proofs_max": 32,
  "trust_proofs_min": 1,

```

```

    "trust_proofs_unique": true,
    "unknown_proof_kind": "reject"
  }
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "AuthorityRuntimePolicy"
},
"scope_contract": {
  "const": {
    "all_selector": {

```

```

      "exclusive": true,
      "kind": "all",
      "value": "*"
    },
    "conjunctive_effect_scope": true,
    "duplicate_selector": "reject",
    "grant_scope_items_max": 32,
    "kinds": [
      "project",
      "path",
      "task",
      "candidate",
      "artifact",
      "finding",
      "all"
    ],
    "path_selector": {
      "absolute": "reject",
      "dot_segments": "reject",
      "kind": "path",
      "normalization": "NFC POSIX relative path"
    },
    "project_selector": {
      "items_max": 1,
      "kind": "project"
    },
    "requested_scope_items_max": 32,
    "selector_fields": [

```

```

      "kind",
      "value"
    ],
    "unique_items": true,
    "unknown_kind": "reject",
    "value_max_length": 4096,
    "value_min_length": 1,
    "wildcard_outside_all": "reject"
  }
},
"separation_of_duties": {
  "const": [
    {
      "forbid_lease_derived_capabilities": [
        "validation.evaluate",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide"
      ],
      "id": "SOD-001",
      "rule": "lease-is-not-acceptance",
      "scope_kind": "lease"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "task.execute",
        "release.decide"

```

```

    ],
    "id": "SOD-002",
    "rule": "executor-not-release-decider",
    "scope_kind": "candidate"
  },
  {
    "forbid_same_subject_for_same_candidate": [
      "validation.evaluate",

```

```

        "candidate.promote"
    ],
    "id": "SOD-003",
    "rule": "validator-not-self-promoter",
    "scope_kind": "candidate"
  },
  {
    "forbid_same_subject_for_same_finding": [
      "finding.record",
      "finding.waive"
    ],
    "id": "SOD-004",
    "rule": "finder-not-sole-waiver",
    "scope_kind": "finding"
  },
  {
    "forbid_same_key_capabilities": [
      "evidence.produce",
      "standard.distribute"
    ],
    "forbid_same_subject_for_same_candidate": [
      "evidence.produce",
      "standard.distribute"
    ],
    "id": "SOD-005",
    "rule": "evidence-producer-not-distribution-decider",
    "scope_kind": "candidate-and-key"
  }
],
"separation_of_duties_capability_ids": {
  "const": [
    "standard.activate",
    "authority.manage",
    "task.plan",
    "task.execute",
    "lease.manage",
    "evidence.publish",
    "validation.evaluate",
    "finding.record",
    "finding.resolve",
    "finding.waive",
    "candidate.promote",
    "release.decide",
    "export.create",
    "migration.manage",
    "projection.rebuild",
    "projection.read",
    "evidence.produce",
    "standard.distribute"
  ]
},
"separation_of_duties_contract": {
  "const": {
    "base_fields": [
      "id",
      "rule",
      "scope_kind"
    ],
    "capability_values_max": 32,
    "external_capability_ids": [
      "evidence.produce",
      "standard.distribute"
    ],
    "rules_max": 64,
    "supported_rule_fields": {
      "forbid_lease_derived_capabilities": {
        "cardinality": "unique-list",
        "items_min": 1
      },
      "forbid_same_key_capabilities": {
        "cardinality": "unique-pair",
        "items_exact": 2
      },
      "forbid_same_subject_for_same_candidate": {
        "cardinality": "unique-pair",

```

```

        "items_exact": 2
      },
      "forbid_same_subject_for_same_finding": {
        "cardinality": "unique-pair",
        "items_exact": 2
      }
    },
    "unknown_rule_field": "reject"
  }
},
"required": [
  "record_type",
  "authority_model_digest",
  "capability_ids",
  "separation_of_duties",
  "separation_of_duties_capability_ids",
  "separation_of_duties_contract",
  "delegation_depth_max",
  "scope_contract",
  "grant_contract",
  "canonical_timestamp_contract",
  "policy_digest"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: *core/contracts.schema.json#/\$defs/AuthorityRuntimePolicy* | Validator: *urn:promin:contracts:1#/\$defs/AuthorityRuntimePolicy*

## 9. AuthorizationRef

Type summary: *oneOf[2]=#/\$defs/GrantAuthorization | #/\$defs/RootAuthorization*

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/GrantAuthorization"
    },
    {
      "$ref": "#/$defs/RootAuthorization"
    }
  ]
}

```

Owner: *core/contracts.schema.json#/\$defs/AuthorizationRef* | Validator: *urn:promin:contracts:1#/\$defs/AuthorizationRef*

## 10. Budget

Type summary: *type="object"; additionalProperties=false*

Required: *max\_bytes, max\_entities, max\_relations, max\_fanout\_per\_entity, top\_k*

| Property                     | Required | Structural rule                     |
|------------------------------|----------|-------------------------------------|
| <i>max_bytes</i>             | yes      | <i>type="integer"; minimum=1024</i> |
| <i>max_entities</i>          | yes      | <i>type="integer"; minimum=1</i>    |
| <i>max_fanout_per_entity</i> | yes      | <i>type="integer"; minimum=1</i>    |
| <i>max_relations</i>         | yes      | <i>type="integer"; minimum=0</i>    |
| <i>top_k</i>                 | yes      | <i>type="integer"; minimum=1</i>    |

```

{
  "additionalProperties": false,
  "properties": {
    "max_bytes": {
      "minimum": 1024,
      "type": "integer"
    },
    "max_entities": {
      "minimum": 1,
      "type": "integer"
    },
    "max_fanout_per_entity": {
      "minimum": 1,
      "type": "integer"
    }
  }
}

```

```

    "max_relations": {
      "minimum": 0,
      "type": "integer"
    },
    "top_k": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "max_bytes",
    "max_entities",
    "max_relations",
    "max_fanout_per_entity",
    "top_k"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Budget | Validator: urn:promin:contracts:1#/\$defs/Budget

## 11. Candidate

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then,title | keys:else,if,then,title

Required: record\_type, candidate\_id, candidate\_digest, inventory\_digest, product\_root\_digest, control\_excluded, candidate\_recipe\_digest, consistency\_mode, creditable

| Property                | Required | Structural rule                                          |
|-------------------------|----------|----------------------------------------------------------|
| baseline_kind           | no       | enum=["specification", "product", "hybrid"]              |
| candidate_digest        | yes      | \$ref="#/\$defs/Digest"                                  |
| candidate_id            | yes      | \$ref="#/\$defs/Id"                                      |
| candidate_recipe_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| consistency_mode        | yes      | enum=["observational-best-effort", "immutable-vcs-tree"] |
| control_excluded        | yes      | const=true                                               |
| creditable              | yes      | type="boolean"                                           |
| inventory_digest        | yes      | \$ref="#/\$defs/Digest"                                  |
| product_root_digest     | yes      | \$ref="#/\$defs/Digest"                                  |
| record_type             | yes      | const="Candidate"                                        |
| snapshot_digest         | no       | \$ref="#/\$defs/Digest"                                  |
| snapshot_provider_id    | no       | \$ref="#/\$defs/Id"                                      |
| specification_digest    | no       | \$ref="#/\$defs/Digest"                                  |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "specification_digest"
          ]
        }
      },
      "if": {
        "properties": {
          "baseline_kind": {
            "enum": [
              "specification",
              "hybrid"
            ]
          }
        }
      },
      "required": [
        "baseline_kind"
      ]
    },
    {
      "then": {
        "required": [
          "specification_digest"
        ]
      }
    }
  ]
}

```

```

    },
    "title": "Candidate specification baseline binding is exact"
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "snapshot_provider_id"
            ]
          },
          {
            "required": [
              "snapshot_digest"
            ]
          }
        ]
      },
      "properties": {
        "creditable": {
          "const": false
        }
      }
    },
    "if": {
      "properties": {
        "consistency_mode": {
          "enum": [
            "immutable-vcs-tree"
          ]
        }
      },
      "required": [
        "consistency_mode"
      ]
    },
    "then": {
      "properties": {
        "creditable": {
          "const": true
        }
      },
      "required": [
        "snapshot_provider_id",
        "snapshot_digest"
      ]
    },
    "title": "creditable Candidate requires provider-proven immutable snapshot"
  }
],
"properties": {
  "baseline_kind": {
    "enum": [
      "specification",
      "product",
      "hybrid"
    ]
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "candidate_id": {
    "$ref": "#/$defs/Id"
  },
  "candidate_recipe_digest": {
    "$ref": "#/$defs/Digest"
  },
  "consistency_mode": {
    "enum": [
      "observational-best-effort",
      "immutable-vcs-tree"
    ]
  },
  "control_excluded": {
    "const": true
  },
  "creditable": {

```

```

    "type": "boolean"
  },
  "inventory_digest": {
    "$ref": "#/$defs/Digest"
  },
  "product_root_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "Candidate"
  },
  "snapshot_digest": {
    "$ref": "#/$defs/Digest"
  },
  "snapshot_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "specification_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "record_type",
  "candidate_id",
  "candidate_digest",
  "inventory_digest",
  "product_root_digest",
  "control_excluded",
  "candidate_recipe_digest",
  "consistency_mode",
  "creditable"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/Candidate* | Validator: *urn:promin:contracts:1#/\$defs/Candidate*

## 12. CandidateDelta

Type summary: type="object"; additionalProperties=false

Required: base\_candidate\_digest, new\_candidate\_digest, workcard\_digest, changed\_paths

| Property              | Required | Structural rule                                             |
|-----------------------|----------|-------------------------------------------------------------|
| base_candidate_digest | yes      | \$ref="#/\$defs/Digest"                                     |
| changed_paths         | yes      | type="array"; minItems=1; maxItems=100000; uniqueItems=true |
| new_candidate_digest  | yes      | \$ref="#/\$defs/Digest"                                     |
| workcard_digest       | yes      | \$ref="#/\$defs/Digest"                                     |

```

{
  "additionalProperties": false,
  "properties": {
    "base_candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "changed_paths": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 100000,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "new_candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "workcard_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "base_candidate_digest",
    "new_candidate_digest",
    "workcard_digest",
    "changed_paths"
  ],
}

```

```

    "type": "object"
  }

```

Owner: [core/contracts.schema.json#/\\$defs/CandidateDelta](#) | Validator: [urn:promin:contracts:1#/\\$defs/CandidateDelta](#)

## 13. CandidateRecipe

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Required: inventory\_mode, include, exclude, symlink\_policy, path\_identity, collision\_policy, product\_identity\_excludes\_control\_state

| Property                                | Required | Structural rule                                          |
|-----------------------------------------|----------|----------------------------------------------------------|
| collision_policy                        | yes      | const="reject-nfc-and-casefold-collisions"               |
| exclude                                 | yes      | type="array"; minItems=1; maxItems=128; uniqueItems=true |
| include                                 | yes      | type="array"; minItems=1; maxItems=64; uniqueItems=true  |
| inventory_mode                          | yes      | const="explicit"                                         |
| path_identity                           | yes      | const="nfc-posix-relative"                               |
| product_identity_excludes_control_state | yes      | const=true                                               |
| snapshot_consistency                    | no       | enum=["observational-best-effort", "immutable-vcs-tree"] |
| snapshot_provider_id                    | no       | \$ref="#/\$defs/Id"                                      |
| symlink_policy                          | yes      | enum=["reject", "hash-link-metadata"]                    |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "snapshot_provider_id"
          ]
        }
      },
      "if": {
        "properties": {
          "snapshot_consistency": {
            "enum": [
              "immutable-vcs-tree"
            ]
          }
        },
        "required": [
          "snapshot_consistency"
        ]
      },
      "then": {
        "required": [
          "snapshot_provider_id"
        ]
      }
    },
    {
      "title": "immutable snapshot recipe requires exact provider"
    }
  ],
  "properties": {
    "collision_policy": {
      "const": "reject-nfc-and-casefold-collisions"
    },
    "exclude": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 128,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "include": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 64,
      "minItems": 1,

```



```

    "type": "array",
    "uniqueItems": true
  },
  "inventory_mode": {
    "const": "explicit"
  },
  "path_identity": {
    "const": "nfc-posix-relative"
  },
  "product_identity_excludes_control_state": {
    "const": true
  },
  "snapshot_consistency": {
    "enum": [
      "observational-best-effort",
      "immutable-vcs-tree"
    ]
  },
  "snapshot_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "symlink_policy": {
    "enum": [
      "reject",
      "hash-link-metadata"
    ]
  },
  "required": [
    "inventory_mode",
    "include",
    "exclude",
    "symlink_policy",
    "path_identity",
    "collision_policy",
    "product_identity_excludes_control_state"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CandidateRecipe | Validator: urn:promin:contracts:1#/\$defs/CandidateRecipe

## 14. CanonicalCompilationDatabase

Type summary: type="object"; additionalProperties=false

Required: status, digest, command\_count, pass\_credit

| Property      | Required | Structural rule                      |
|---------------|----------|--------------------------------------|
| command_count | yes      | type="integer"; minimum=0            |
| digest        | yes      | anyOf[2]=#/\$defs/Sha256   type:null |
| pass_credit   | yes      | const=false                          |
| status        | yes      | enum=["PASS", "FAIL", "UNAVAILABLE"] |

```

{
  "additionalProperties": false,
  "properties": {
    "command_count": {
      "minimum": 0,
      "type": "integer"
    },
    "digest": {
      "anyOf": [
        {
          "$ref": "#/$defs/Sha256"
        },
        {
          "type": "null"
        }
      ]
    },
    "pass_credit": {
      "const": false
    },
    "status": {
      "enum": [

```

```

    "PASS",
    "FAIL",
    "UNAVAILABLE"
  ]
},
},
},

```

```

"required": [
  "status",
  "digest",
  "command_count",
  "pass_credit"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CanonicalCompilationDatabase | Validator: urn:promin:contracts:1#/\$defs/CanonicalCompilationDatabase

## 15. CapacityReconciliation

Type summary: type="object"; additionalProperties=false

Required: reconciled\_by, grant\_id, grant\_claim\_digest, reconciled\_at, generation, fencing\_token

| Property           | Required | Structural rule            |
|--------------------|----------|----------------------------|
| fencing_token      | yes      | type="integer"; minimum=1  |
| generation         | yes      | type="integer"; minimum=1  |
| grant_claim_digest | yes      | \$ref="#/\$defs/Digest"    |
| grant_id           | yes      | \$ref="#/\$defs/Id"        |
| reconciled_at      | yes      | \$ref="#/\$defs/Timestamp" |
| reconciled_by      | yes      | \$ref="#/\$defs/Id"        |

```

{
  "additionalProperties": false,
  "properties": {
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "generation": {
      "minimum": 1,
      "type": "integer"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reconciled_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "reconciled_by": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "reconciled_by",
    "grant_id",
    "grant_claim_digest",
    "reconciled_at",
    "generation",
    "fencing_token"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CapacityReconciliation | Validator: urn:promin:contracts:1#/\$defs/CapacityReconciliation

## 16. CloseAck

Type summary: type="object"; additionalProperties=false

Required: acknowledged\_by, grant\_id, grant\_claim\_digest, acknowledged\_at

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required | Structural rule            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------|
| acknowledged_at                                                                                                                                                                                                                                                                                                                                                                                                                                                 | yes      | \$ref="#/\$defs/Timestamp" |
| acknowledged_by                                                                                                                                                                                                                                                                                                                                                                                                                                                 | yes      | \$ref="#/\$defs/Id"        |
| grant_claim_digest                                                                                                                                                                                                                                                                                                                                                                                                                                              | yes      | \$ref="#/\$defs/Digest"    |
| grant_id                                                                                                                                                                                                                                                                                                                                                                                                                                                        | yes      | \$ref="#/\$defs/Id"        |
| <pre>{   "additionalProperties": false,   "properties": {     "acknowledged_at": {       "\$ref": "#/\$defs/Timestamp"     },     "acknowledged_by": {       "\$ref": "#/\$defs/Id"     },     "grant_claim_digest": {       "\$ref": "#/\$defs/Digest"     },     "grant_id": {       "\$ref": "#/\$defs/Id"     }   },   "required": [     "acknowledged_by",     "grant_id",     "grant_claim_digest",     "acknowledged_at"   ],   "type": "object" }</pre> |          |                            |
| Owner: core/contracts.schema.json#/\$defs/CloseAck   Validator: urn:promin:contracts:1#/\$defs/CloseAck                                                                                                                                                                                                                                                                                                                                                         |          |                            |

## 17. CommandCapabilityRule

Type summary: type="object"; additionalProperties=false; oneOf[2]=keys:not,required | keys:not,required

Required: command\_kind

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required | Structural rule                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------|
| capability_by_value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | no       | type="object"; additionalProperties=schema(keys:\$ref) |
| capability_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | no       | \$ref="#/\$defs/Id"                                    |
| command_kind                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | yes      | \$ref="#/\$defs/Id"                                    |
| payload_field                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | no       | \$ref="#/\$defs/Id"                                    |
| <pre>{   "additionalProperties": false,   "oneOf": [     {       "not": {         "anyOf": [           {             "required": [               "payload_field"             ]           },           {             "required": [               "capability_by_value"             ]           }         ]       }     },     {       "required": [         "capability_id"       ]     }   ],   {     "not": {       "required": [         "capability_id"       ]     }   },   "required": [     "payload_field",</pre> |          |                                                        |

```

        "capability_by_value"
      ]
    },
    "properties": {
      "capability_by_value": {
        "additionalProperties": {
          "$ref": "#/$defs/Id"
        },
        "minProperties": 1,
        "type": "object"
      },
      "capability_id": {
        "$ref": "#/$defs/Id"
      },
      "command_kind": {
        "$ref": "#/$defs/Id"
      },
      "payload_field": {
        "$ref": "#/$defs/Id"
      }
    },
    "required": [
      "command_kind"
    ],
    "type": "object"
  }
}

```

Owner: core/contracts.schema.json#/\$defs/CommandCapabilityRule | Validator: urn:promin:contracts:1#/\$defs/CommandCapabilityRule

## 18. CommandEffectScopeRule

Type summary: type="object"; additionalProperties=false; allOf[6]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then

Required: command\_kind, mode

| Property          | Required | Structural rule                                                                                                                                          |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| command_kind      | yes      | \$ref="#/\$defs/Id"                                                                                                                                      |
| grant_id_field    | no       | \$ref="#/\$defs/Id"                                                                                                                                      |
| grant_target_type | no       | const="Grant"                                                                                                                                            |
| id_field          | no       | \$ref="#/\$defs/Id"                                                                                                                                      |
| kind              | no       | \$ref="#/\$defs/Id"                                                                                                                                      |
| kind_field        | no       | \$ref="#/\$defs/Id"                                                                                                                                      |
| kind_map          | no       | type="object"; additionalProperties=schema(keys:\$ref)                                                                                                   |
| mode              | yes      | enum=["payload-id", "payload-multi-id", "payload-dynamic-id", "payload-scope", "referenced-grant-scope", "payload-dynamic-id-or-referenced-grant-scope"] |
| scope_field       | no       | \$ref="#/\$defs/Id"                                                                                                                                      |
| targets           | no       | type="array"; minItems=1; maxItems=8; uniqueItems=true                                                                                                   |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-id"
          }
        }
      },
      "then": {
        "required": [
          "kind",
          "id_field"
        ]
      }
    },
    {
      "if": {
        "properties": {

```

```

        "mode": {
          "const": "payload-multi-id"
        }
      },
      "then": {
        "required": [

```

```

          "targets"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-dynamic-id"
          }
        }
      },
      "then": {
        "required": [
          "kind_field",
          "kind_map",
          "id_field"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-scope"
          }
        }
      }
    }
  ],

```

```

    "then": {
      "required": [
        "scope_field"
      ]
    }
  },
  {
    "if": {
      "properties": {
        "mode": {
          "const": "referenced-grant-scope"
        }
      }
    },
    "then": {
      "required": [
        "grant_id_field"
      ]
    }
  },
  {
    "if": {
      "properties": {
        "mode": {
          "const": "payload-dynamic-id-or-referenced-grant-scope"
        }
      }
    }
  }
],

```

```

    "then": {
      "required": [
        "kind_field",
        "kind_map",
        "id_field",
        "grant_target_type"
      ]
    }
  },
  {
    "properties": {
      "command_kind": {
        "$ref": "#/$defs/Id"
      }
    },

```

```

"grant_id_field": {
  "$ref": "#/$defs/Id"
},
"grant_target_type": {
  "const": "Grant"
},
"id_field": {
  "$ref": "#/$defs/Id"
},
"kind": {
  "$ref": "#/$defs/Id"
},
"kind_field": {
  "$ref": "#/$defs/Id"
},
},
"kind_map": {
  "additionalProperties": {
    "$ref": "#/$defs/Id"
  },
  "minProperties": 1,
  "type": "object"
},
"mode": {
  "enum": [
    "payload-id",
    "payload-multi-id",
    "payload-dynamic-id",
    "payload-scope",
    "referenced-grant-scope",
    "payload-dynamic-id-or-referenced-grant-scope"
  ]
},
"scope_field": {
  "$ref": "#/$defs/Id"
},
},
"targets": {
  "items": {
    "$ref": "#/$defs/EffectTarget"
  },
  "maxItems": 8,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "command_kind",
  "mode"
],
"type": "object"
}

```

```
Owner: core/contracts.schema.json#/$defs/CommandEffectScopeRule | Validator:
urn:promin:contracts:1#/$defs/CommandEffectScopeRule
```

## 19. CommandRequest

Type summary: type="object"; additionalProperties=false; allOf[16]=keys;if,then | keys;if,then | keys;if,then | keys;if,then | keys;if,then | keys;if,then | keys;if,then | keys;if,then | keys;if,then,title | keys;if,then,title | keys;if,then,title | keys;if,then,title | keys;if,then,title

Required: record\_type, command\_id, command\_kind, subject\_id, activation\_digest, idempotency\_key, requested\_scope, expected\_head\_digest, issued\_at, payload, intent\_digest, authorization

| Property          | Required | Structural rule                                                                                                                                                                                 |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                                         |
| authorization     | yes      | \$ref="#/\$defs/AuthorizationRef"                                                                                                                                                               |
| command_id        | yes      | \$ref="#/\$defs/Id"                                                                                                                                                                             |
| command_kind      | yes      | enum=["task.record", "task.transition", "candidate.record", "artifact.record", "run.record", "gate.record", "finding.record", "decision.record", "grant.issue", "grant.revoke", "lease.record"] |
| context_digest    | no       | \$ref="#/\$defs/Digest"                                                                                                                                                                         |
| definition_digest | no       | \$ref="#/\$defs/Digest"                                                                                                                                                                         |

| Property             | Required | Structural rule                                    |
|----------------------|----------|----------------------------------------------------|
| expected_head_digest | yes      | oneOf[2]=#/\$defs/Digest   type:null               |
| fencing_token        | no       | type="integer"; minimum=1                          |
| holder_authorization | no       | \$ref="#/\$defs/GrantAuthorization"                |
| idempotency_key      | yes      | type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$" |
| intent_digest        | yes      | \$ref="#/\$defs/Digest"                            |
| issued_at            | yes      | \$ref="#/\$defs/Timestamp"                         |
| lease_generation     | no       | type="integer"; minimum=1                          |
| lease_id             | no       | \$ref="#/\$defs/Id"                                |
| payload              | yes      | composed structural definition                     |
| record_type          | yes      | const="CommandRequest"                             |
| requested_scope      | yes      | type="array"; minItems=1; maxItems=32              |
| subject_id           | yes      | \$ref="#/\$defs/Id"                                |
| workcard_digest      | no       | \$ref="#/\$defs/Digest"                            |
| workcard_task_id     | no       | \$ref="#/\$defs/Id"                                |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "task.record"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Task"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "task.transition"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/TaskTransition"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "candidate.record"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Candidate"
          }
        }
      }
    },
    {
      "if": {
        "properties": {

```

```

        "command_kind": {
          "const": "artifact.record"
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Artifact"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "run.record"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Run"
          }
        }
      }
    }
  ],
  {

```

```

    "if": {
      "properties": {
        "command_kind": {
          "const": "gate.record"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GateResult"
        }
      }
    }
  ],
  {
    "if": {
      "properties": {
        "command_kind": {
          "const": "finding.record"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Finding"
        }
      }
    }
  }

```

```

  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "decision.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
},

```



```

{
  "if": {
    "properties": {
      "command_kind": {
        "const": "grant.issue"
      }
    }
  },
  "then": {

```

```

    "properties": {
      "payload": {
        "$ref": "#/$defs/Grant"
      }
    }
  },
  {
    "if": {
      "properties": {
        "command_kind": {
          "const": "grant.revoke"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GrantRevocation"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "command_kind": {
          "const": "lease.record"

```

```

        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Lease"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "command_kind": {
          "enum": [
            "artifact.record",
            "run.record",
            "gate.record",
            "finding.record"
          ]
        }
      }
    },
    "required": [
      "command_kind"
    ]
  },
  "then": {

```

```

    "required": [
      "workcard_task_id",
      "holder_authorization",
      "lease_id",
      "lease_generation",
      "fencing_token",
      "workcard_digest",
      "context_digest"
    ]
  },
  "title": "require current Lease and bounded WorkCard for direct mutation"
},

```

```

{
  "if": {
    "properties": {
      "command_kind": {
        "const": "task.transition"
      },
      "payload": {
        "properties": {
          "to_state": {
            "enum": [
              "LEASED",
              "RUNNING",
              "COMPLETED"
            ]
          }
        }
      }
    },
    "required": [
      "to_state"
    ]
  },
  "required": [
    "command_kind",
    "payload"
  ]
},
"then": {
  "required": [
    "workcard_task_id",
    "holder_authorization",
    "lease_id",
    "lease_generation",
    "fencing_token",
    "workcard_digest",
    "context_digest"
  ]
},
"title": "require current Lease and bounded WorkCard for execution transition"
},
{
  "if": {
    "not": {
      "anyOf": [
        {
          "properties": {
            "command_kind": {
              "enum": [
                "artifact.record",
                "run.record",
                "gate.record",
                "finding.record"
              ]
            }
          },
          "required": [
            "command_kind"
          ]
        },
        {
          "properties": {
            "command_kind": {
              "const": "task.transition"
            },
            "payload": {
              "properties": {
                "to_state": {
                  "enum": [
                    "LEASED",
                    "RUNNING",
                    "COMPLETED"
                  ]
                }
              }
            },
            "required": [
              "to_state"
            ]
          }
        }
      ]
    }
  }
}

```

```

        },
        "required": [
            "command_kind",
            "payload"
        ]
    }
}
},
"then": {
    "not": {
        "anyOf": [
            {
                "required": [
                    "workcard_task_id"
                ]
            },
            {
                "required": [
                    "holder_authorization"
                ]
            },
        ],
    }
}

```

```

        "required": [
            "lease_id"
        ]
    },
    {
        "required": [
            "lease_generation"
        ]
    },
    {
        "required": [
            "fencing_token"
        ]
    },
    {
        "required": [
            "workcard_digest"
        ]
    },
    {
        "required": [
            "context_digest"
        ]
    }
]
}
},
"title": "forbid mutation binding on non-lease command"

```

```

},
{
    "if": {
        "properties": {
            "command_kind": {
                "enum": [
                    "run.record",
                    "gate.record"
                ]
            }
        }
    },
    "then": {
        "required": [
            "definition_digest"
        ]
    },
    "title": "run and gate commands bind an immutable definition digest"
},
{
    "if": {
        "properties": {
            "command_kind": {
                "enum": [
                    "task.record",
                    "task.transition",

```

```

        "candidate.record",
        "artifact.record",

        "finding.record",
        "decision.record",
        "grant.issue",
        "grant.revoke",
        "lease.record"
    ]
}
},
"then": {
    "not": {
        "required": [
            "definition_digest"
        ]
    }
},
"title": "other commands cannot claim a gate run definition"
}
],
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "authorization": {
        "$ref": "#/$defs/AuthorizationRef"
    },
    "command_id": {
        "$ref": "#/$defs/Id"
    }
},
"command_kind": {
    "enum": [
        "task.record",
        "task.transition",
        "candidate.record",
        "artifact.record",
        "run.record",
        "gate.record",
        "finding.record",
        "decision.record",
        "grant.issue",
        "grant.revoke",
        "lease.record"
    ]
},
"context_digest": {
    "$ref": "#/$defs/Digest"
},
"definition_digest": {
    "$ref": "#/$defs/Digest"
},
"expected_head_digest": {
    "oneOf": [
        {
            "$ref": "#/$defs/Digest"
        },
        {
            "type": "null"
        }
    ]
},
"fencing_token": {
    "minimum": 1,
    "type": "integer"
},
"holder_authorization": {
    "$ref": "#/$defs/GrantAuthorization"
},
"idempotency_key": {
    "pattern": "^[A-Za-z0-9._-]{16,128}$",
    "type": "string"
},
"intent_digest": {
    "$ref": "#/$defs/Digest"
},
"issued_at": {

```

```

    "$ref": "#/$defs/Timestamp"
  },
  "lease_generation": {
    "minimum": 1,
    "type": "integer"
  },
  "lease_id": {
    "$ref": "#/$defs/Id"
  },
  "payload": {},
  "record_type": {
    "const": "CommandRequest"
  },
  "requested_scope": {
    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "workcard_digest": {
    "$ref": "#/$defs/Digest"
  },
  "workcard_task_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "record_type",
  "command_id",
  "command_kind",
  "subject_id",
  "activation_digest",
  "idempotency_key",
  "requested_scope",
  "expected_head_digest",
  "issued_at",
  "payload",
  "intent_digest",
  "authorization"
],
"type": "object",
"x-promin-max-canonical-bytes": 1048576
}

```

Owner: core/contracts.schema.json#/\$defs/CommandRequest | Validator: urn:promin:contracts:1#/\$defs/CommandRequest

## 20. CommandResult

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Required: record\_type, command\_id, outcome, batch\_digest, reason\_codes, facts, created\_at

| Property          | Required | Structural rule                                                 |
|-------------------|----------|-----------------------------------------------------------------|
| batch_digest      | yes      | oneOf[2]=#/\$defs/Digest   type:null                            |
| command_id        | yes      | \$ref="#/\$defs/Id"                                             |
| created_at        | yes      | \$ref="#/\$defs/Timestamp"                                      |
| facts             | yes      | type="object"                                                   |
| operation_metrics | no       | \$ref="#/\$defs/OperationMetrics"                               |
| outcome           | yes      | enum=["committed", "rejected", "idempotent-replay", "conflict"] |
| reason_codes      | yes      | type="array"; minItems=0; maxItems=32; uniqueItems=true         |
| record_type       | yes      | const="CommandResult"                                           |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [

```

```

        "operation_metrics"
      ]
    },
    "if": {
      "properties": {
        "outcome": {
          "enum": [
            "committed",
            "idempotent-replay"
          ]
        }
      },
      "required": [
        "outcome"
      ]
    },
    "then": {
      "required": [
        "operation_metrics"
      ]
    }
  },
  "title": "committed command result carries exact operation metrics"
},
"properties": {
  "batch_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "command_id": {
    "$ref": "#/$defs/Id"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "facts": {
    "type": "object"
  },
  "operation_metrics": {
    "$ref": "#/$defs/OperationMetrics"
  },
  "outcome": {
    "enum": [
      "committed",
      "rejected",
      "idempotent-replay",
      "conflict"
    ]
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "record_type": {
    "const": "CommandResult"
  }
},
"required": [
  "record_type",
  "command_id",
  "outcome",
  "batch_digest",
  "reason_codes",
  "facts",
  "created_at"
]

```

```

],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CommandResult | Validator: urn:promin:contracts:1#/\$defs/CommandResult

## 21. Conformance

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, document, version, function, structural\_budgets, workcard\_hard\_ceiling, scale\_contracts, reference\_benchmarks, required\_acceptance, mutation\_families, production\_claim\_rule, dependency\_depth\_hard\_max, alpha4\_no\_credit\_rules, acceptance\_proof\_class\_counts, acceptance\_proof\_classes, command\_latency\_contracts

| Property                      | Required | Structural rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acceptance_proof_class_counts | yes      | const=schema(keys:direct-semantic,human-decision,local-runtime-reexecute,migration-external-no-credit,package-static,saturation-signed,signed-role-platfor m-evidence,standard-release-signed-manifest)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| acceptance_proof_classes      | yes      | const=schema(keys:activation-digest-binding,artifact-kind-resolves-correct-cap ability,authenticated-role-platform-evidence-attestations,authorization-binds-ex act-grant-claim,bounded-incremental-commit-and-compaction,bounded-workcar d-through-depth-twelve,broad-query-bounded-seed-refinement,candidate-delta-within-task-and-grant-scope,candidate-evidence-decision-acyclic-chain,candidat e-evidence-provider-binding,candidate-recipe-applied-once,canonical-exact-zip-distribution,canonical-name-only,command-capability-and-scope-resolution,com mand-effect-target-in-requested-scope,configuration-exact-idempotency,conjunctive-scope-containment,continuation-secret-subject-grant-binding,continuation -token-bytes-at-most-256,continuation-v2-complete-page-union,creditable-cand idate-immutable-vcs-tree,current-interpreter-and-clean-install-modes,current-re lease-closure-revalidation,decision-after-evidence-with-bounded-skew,decision-grant-matches-command-authorization,decision-kind-target-scope-binding,depe ndency-graph-acyclic-and-current-ready-frontier,depth-aware-seed-budget,dete rministic-activation-identity,deterministic-replay,evidence-manifest-exact-eight-lane-matrix-resolution,evidence-manifest-four-cp314-supplemental-records,evid ence-manifest-nested-and-supplemental-chronology,evidence-manifest-path-di gest-size-closure,evidence-manifest-physical-resolution,exact-command-primar y-event-correspondence,exact-five-init-records,exact-one-selected-preset,exact -six-base-command-surface,exact-six-core-artifacts,explicit-scale-marker-selecti on,external-standard-release-decision-exact-binding,finding-bound-resolution-a nd-waiver-evidence,fresh-semantic-control-state-per-saturation-iteration,grant-i ssuance-chain-closure,grant-scope-expiry-nonce-proof,handle-relative-evidence -root-safety,historical-decision-current-eligibility-separation,idempotent-state-co mmands,implementation-closure-bound-and-current,installed-platform-observat ion-closure,inventory-stream-memory-amplification-at-most-32,lease-generatio n-fence-and-close-ack,lease-holder-has-task-execute-grant,lossless-continuation -in-reference-100k-scenario,mutation-workcard-requires-current-lease,no-false-success,non-skipped-credit-result-has-evidence,normalization-collision-rejection, offline-installed-no-degradation,offline-release-online-compatibility,one-comm and-one-atomic-batch,physical-100k-one-artifact-proxy-per-file,policy-owned-tas k-and-lease-state-machines,preset-outside-core-digest,product-public-approval-separation,profile-bound-physical-100k-performance,profile-default-with-core-d epth-twelve,projection-disposable-and-current,provider-substitution-invalidates-derived-state,raw-scale-summary-recomputation,read-workcard-without-lease,r equired-no-degradation-tests-fail-closed,required-provider-preflight-before-com mit,rollback-and-recutover-equivalence,root-bootstrap-command-limited,root-ce iling-without-bootstrap-grant-cycle,safe-recursive-export,selected-provider-adap ter-evidence,semver-from-candidate-binding,seven-distinct-role-producer-scope s,signed-standard-decision-authority-proof,single-derived-platform-closure-own er,single-pass-inventory,standard-distribution-separate-from-product-acceptanc e,strict-critical-fields,team-signatures-cryptographically-verified,team-signed-cli-end-to-end,three-zero-new-saturation-iterations,truthful-operation-reporting,trut hful-process-invocation-status,typed-human-document-verification,typed-refere nce-domain-range-closure,verified-head-checkpoint-delta-replay,verified-invent ory-result-provenance,windows-linux-exact-zip-matrix,zero-compiled-cache-files, zero-core-provider-lock-in,zero-dead-evidence-budget-fields,zero-forbidden-nam e-occurrences,zero-product-scan-during-init,zero-unowned-normative-facts) |
| alpha4_no_credit_rules        | yes      | const=schema(keys:acceptance_pass,diagnostic_or_static_only_pass_credit,dyn amic_handoff_pass_credit,product_acceptance_pass,release_eligible,unavailable _is_pass)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| command_latency_contracts     | yes      | const=[9 items]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| dependency_depth_hard_max     | yes      | type="integer"; const=12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| document                      | yes      | const="promin.conformance"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| function                      | yes      | const="Own acceptance predicates, mutation families, structural hard ceilings, scale contracts, and production-claim rules."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| mutation_families             | yes      | const=[77 items]; minItems=77; maxItems=77                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Property              | Required | Structural rule                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| production_claim_rule | yes      | const="Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recutover equivalence, and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution evidence." |
| record_type           | yes      | const="Conformance"                                                                                                                                                                                                                                                                                                                                                                                                         |
| reference_benchmarks  | yes      | const=schema(keys:core_valid_relation_count,depths_tested,file_count,forced_continuation,inventory_passes,performance_profiles,physical_relation_corpus,query_count,raw_file_proxy_ratio,rebuild_product_passes,reference_targets_are_environment_specific,semantic_corpus,silent_truncations_max,synthetic_task_ratio)                                                                                                     |
| required_acceptance   | yes      | const=[102 items]; minItems=102; maxItems=102                                                                                                                                                                                                                                                                                                                                                                               |
| scale_contracts       | yes      | const=schema(keys:event_commit,init,inventory,normal_one_shot_replay,projection,replay,saturation,workcard)                                                                                                                                                                                                                                                                                                                 |
| schema_ref            | yes      | const="urn:promin:contracts:1#/\$defs/Conformance"                                                                                                                                                                                                                                                                                                                                                                          |
| structural_budgets    | yes      | const=schema(keys:authority_bearing_init_records,base_user_commands_max,core_normative_artifacts,persistent_entity_kinds_max,relation_kinds_max,resolved_plan_bytes_max,resolved_plan_source_samples_max,selected_presets)                                                                                                                                                                                                  |
| version               | yes      | const="1.0.0-alpha.4"                                                                                                                                                                                                                                                                                                                                                                                                       |
| workcard_hard_ceiling | yes      | const=schema(keys:max_bytes,max_entities,max_fanout_per_entity,max_relations,top_k)                                                                                                                                                                                                                                                                                                                                         |

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  "properties": {
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        "human-decision": 9,
        "local-runtime-reexecute": 30,
        "migration-external-no-credit": 1,
        "package-static": 4,
        "saturation-signed": 1,
        "signed-role-platform-evidence": 17,
        "standard-release-signed-manifest": 5
      }
    },
    "acceptance_proof_classes": {
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        "artifact-kind-resolves-correct-capability": "local-runtime-reexecute",
        "authenticated-role-platform-evidence-attestations": "signed-role-platform-evidence",
        "authorization-binds-exact-grant-claim": "local-runtime-reexecute",
        "bounded-incremental-commit-and-compaction": "signed-role-platform-evidence",
        "bounded-workcard-through-depth-twelve": "direct-semantic",
        "broad-query-bounded-seed-refinement": "direct-semantic",
        "candidate-delta-within-task-and-grant-scope": "direct-semantic",
        "candidate-evidence-decision-acyclic-chain": "human-decision",
        "candidate-evidence-provider-binding": "local-runtime-reexecute",
        "candidate-recipe-applied-once": "direct-semantic",
        "canonical-exact-zip-distribution": "direct-semantic",
        "canonical-name-only": "direct-semantic",
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        "command-effect-target-in-requested-scope": "local-runtime-reexecute",
        "configuration-exact-idempotency": "local-runtime-reexecute",
        "conjunctive-scope-containment": "local-runtime-reexecute",
        "continuation-secret-subject-grant-binding": "direct-semantic",
        "continuation-token-bytes-at-most-256": "direct-semantic",
        "continuation-v2-complete-page-union": "direct-semantic",
        "creditable-candidate-immutable-vcs-tree": "direct-semantic",
        "current-interpreter-and-clean-install-modes": "signed-role-platform-evidence",
        "current-release-closure-revalidation": "human-decision",
        "decision-after-evidence-with-bounded-skew": "human-decision",
        "decision-grant-matches-command-authorization": "local-runtime-reexecute",
        "decision-kind-target-scope-binding": "local-runtime-reexecute",
        "dependency-graph-acyclic-and-current-ready-frontier": "direct-semantic",
        "depth-aware-seed-budget": "local-runtime-reexecute",
        "deterministic-activation-identity": "local-runtime-reexecute",
        "deterministic-replay": "local-runtime-reexecute",
        "evidence-manifest-exact-eight-lane-matrix-resolution": "standard-release-signed-manifest",
        "evidence-manifest-four-cp314-supplemental-records": "standard-release-signed-manifest",
        "evidence-manifest-nested-and-supplemental-chronology": "standard-release-signed-manifest",

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"evidence-manifest-path-digest-size-closure": "standard-release-signed-manifest",
"evidence-manifest-physical-resolution": "signed-role-platform-evidence",
"exact-command-primary-event-correspondence": "local-runtime-reexecute",
"exact-five-init-records": "direct-semantic",
"exact-one-selected-preset": "direct-semantic",
"exact-six-base-command-surface": "direct-semantic",

"exact-six-core-artifacts": "direct-semantic",
"explicit-scale-marker-selection": "direct-semantic",
"external-standard-release-decision-exact-binding": "human-decision",
"finding-bound-resolution-and-waiver-evidence": "direct-semantic",
"fresh-semantic-control-state-per-saturation-iteration": "saturation-signed",
"grant-issuance-chain-closure": "local-runtime-reexecute",
"grant-scope-expiry-nonce-proof": "local-runtime-reexecute",
"handle-relative-evidence-root-safety": "signed-role-platform-evidence",
"historical-decision-current-eligibility-separation": "human-decision",
"idempotent-state-commands": "local-runtime-reexecute",
"implementation-closure-bound-and-current": "direct-semantic",
"installed-platform-observation-closure": "signed-role-platform-evidence",
"inventory-stream-memory-amplification-at-most-32": "direct-semantic",
"lease-generation-fence-and-close-ack": "local-runtime-reexecute",
"lease-holder-has-task-execute-grant": "local-runtime-reexecute",
"lossless-continuation-in-reference-100k-scenario": "direct-semantic",
"mutation-workcard-requires-current-lease": "direct-semantic",
"no-false-success": "direct-semantic",
"non-skipped-credit-result-has-evidence": "local-runtime-reexecute",
"normalization-collision-rejection": "direct-semantic",
"offline-installed-no-degradation": "signed-role-platform-evidence",
"offline-release-online-compatibility": "signed-role-platform-evidence",
"one-command-one-atomic-batch": "local-runtime-reexecute",
"physical-100k-one-artifact-proxy-per-file": "signed-role-platform-evidence",
"policy-owned-task-and-lease-state-machines": "local-runtime-reexecute",
"preset-outside-core-digest": "direct-semantic",
"product-public-approval-separation": "human-decision",
"profile-bound-physical-100k-performance": "signed-role-platform-evidence",

"profile-default-with-core-depth-twelve": "direct-semantic",
"projection-disposable-and-current": "local-runtime-reexecute",
"provider-substitution-invalidates-derived-state": "local-runtime-reexecute",
"raw-scale-summary-recomputation": "signed-role-platform-evidence",
"read-workcard-without-lease": "direct-semantic",
"required-no-degradation-tests-fail-closed": "signed-role-platform-evidence",
"required-provider-preflight-before-commit": "local-runtime-reexecute",
"rollback-and-recutover-equivalence": "migration-external-no-credit",
"root-bootstrap-command-limited": "local-runtime-reexecute",
"root-ceiling-without-bootstrap-grant-cycle": "local-runtime-reexecute",
"safe-recursive-export": "direct-semantic",
"selected-provider-adapter-evidence": "direct-semantic",
"semver-from-candidate-binding": "human-decision",
"seven-distinct-role-producer-scopes": "standard-release-signed-manifest",
"signed-standard-decision-authority-proof": "human-decision",
"single-derived-platform-closure-owner": "signed-role-platform-evidence",
"single-pass-inventory": "local-runtime-reexecute",
"standard-distribution-separate-from-product-acceptance": "human-decision",
"strict-critical-fields": "direct-semantic",
"team-signatures-cryptographically-verified": "local-runtime-reexecute",
"team-signed-cli-end-to-end": "direct-semantic",
"three-zero-new-saturation-iterations": "signed-role-platform-evidence",
"truthful-operation-reporting": "local-runtime-reexecute",
"truthful-process-invocation-status": "signed-role-platform-evidence",
"typed-human-document-verification": "signed-role-platform-evidence",
"typed-reference-domain-range-closure": "direct-semantic",
"verified-head-checkpoint-delta-replay": "direct-semantic",
"verified-inventory-result-provenance": "direct-semantic",

"windows-linux-exact-zip-matrix": "signed-role-platform-evidence",
"zero-compiled-cache-files": "package-static",
"zero-core-provider-lock-in": "package-static",
"zero-dead-evidence-budget-fields": "direct-semantic",
"zero-forbidden-name-occurrences": "package-static",
"zero-product-scan-during-init": "local-runtime-reexecute",
"zero-unowned-normative-facts": "package-static"
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},
"alpha4_no_credit_rules": {
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    "dynamic_handoff_pass_credit": false,
    "product_acceptance_pass": false,
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            "percentile": "p95",
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            "provider_state": "installed-current",
            "status": "provisional",
            "warmup_runs": 1,
            "workload_id": "current-projection-32-v1"
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            "max_total_bytes": 65536,
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            "measurement_mode": "cli_cold",
            "percentile": "p95",
            "projection_state": "empty",
            "provider_state": "no-install",
            "status": "provisional",
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            "status": "provisional",
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            "provider_state": "installed-current",
            "status": "provisional",
            "warmup_runs": 1,
            "workload_id": "current-projection-32-v1"
        },
        {

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    "max_total_bytes": 65536,
    "measured_runs": 3,
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    "percentile": "p95",
    "projection_state": "current",
    "provider_state": "installed-current",
    "status": "provisional",
    "warmup_runs": 1,
    "workload_id": "current-project-no-replay-v1"
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]
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  "type": "integer"
},
"document": {
  "const": "promin.conformance"
},
"function": {
  "const": "Own acceptance predicates, mutation families, structural hard ceilings, scale contracts,
and production-claim rules."
},
"mutation_families": {
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    "status-credit-contradiction",
    "missing-critical-field",
    "unknown-critical-field",
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    "grant-replay",
    "scope-escalation",
    "expired-grant",

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"revoked-grant",
"stale-fence",
"candidate-drift",
"provider-substitution",
"event-fork",
"batch-command-mismatch",
"duplicate-idempotency-key",
"projection-corruption",
"projection-staleness",
"reference-break",
"relation-domain-break",
"context-flood",
"symlink-core-input",
"shadow-core-file",
"symlink-product-entry",
"secret-export",
"archive-expansion",
"preset-authority-escalation",
"normalization-key-collision",
"path-casefold-collision",
"forbidden-name-reintroduction",
"reinit-configuration-conflict",
"clock-order-violation",
"capability-confusion",
"command-scope-escalation",
"grant-issuer-chain-break",
"unverified-signature-acceptance",
"provider-declared-but-unavailable",

"illegal-task-transition",
"illegal-lease-transition",
"decision-target-confusion",
"credit-without-evidence",
"command-primary-event-mismatch",
"duplicate-event-id",
"scope-selector-drop-broadening",
"authorization-grant-claim-substitution",
"command-effect-outside-requested-scope",
"lease-holder-grant-mismatch",
"decision-grant-provenance-mismatch",
"artifact-kind-capability-confusion",
"continuation-frontier-loss",
"continuation-duplicate-emission",
"candidate-recipe-bypass",
"observational-candidate-pass-credit",
"snapshot-provider-substitution",
"unsupported-provider-adapter",
"stale-release-closure-credit",
"finding-evidence-target-mismatch",
"candidate-delta-scope-escape",
"checkpoint-head-mismatch",
"checkpoint-corruption-credit",
"normal-command-full-replay",
"profile-default-depth-bypass",
"continuation-public-material-mismatch",
"inventory-provenance-override",
"implementation-closure-drift",

"noncanonical-archive-layout",
"inventory-graph-inflation",
"workcard-budget-bypass",
"fabricated-evidence-digest",
"unsigned-standard-decision",
"unconfigured-standard-decision-key",
"standard-decision-version-replay",
"candidate-decision-cycle",
"broad-query-corpus-pagination",
"continuation-token-context-inflation",
"inventory-stream-materialization",
"pdf-header-only-credit",
"scale-environment-skip-credit",
"content-search-omission"
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"items": {
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    "unknown-critical-field",
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```

```
"scope-escalation",
"expired-grant",
"revoked-grant",
"stale-fence",
"candidate-drift",
"provider-substitution",
```

```
"event-fork",
"batch-command-mismatch",
"duplicate-idempotency-key",
"projection-corruption",
"projection-staleness",
"reference-break",
"relation-domain-break",
"context-flood",
"symlink-core-input",
"shadow-core-file",
"symlink-product-entry",
"secret-export",
"archive-expansion",
"preset-authority-escalation",
"normalization-key-collision",
"path-casefold-collision",
"forbidden-name-reintroduction",
"reinit-configuration-conflict",
"clock-order-violation",
"capability-confusion",
"command-scope-escalation",
"grant-issuer-chain-break",
"unverified-signature-acceptance",
"provider-declared-but-unavailable",
"illegal-task-transition",
"illegal-lease-transition",
"decision-target-confusion",
"credit-without-evidence",
```

```
"command-primary-event-mismatch",
"duplicate-event-id",
"scope-selector-drop-broadening",
"authorization-grant-claim-substitution",
"command-effect-outside-requested-scope",
"lease-holder-grant-mismatch",
"decision-grant-provenance-mismatch",
"artifact-kind-capability-confusion",
"continuation-frontier-loss",
"continuation-duplicate-emission",
"candidate-recipe-bypass",
"observational-candidate-pass-credit",
"snapshot-provider-substitution",
"unsupported-provider-adapter",
"stale-release-closure-credit",
"finding-evidence-target-mismatch",
"candidate-delta-scope-escape",
"checkpoint-head-mismatch",
"checkpoint-corruption-credit",
"normal-command-full-replay",
"profile-default-depth-bypass",
"continuation-public-material-mismatch",
"inventory-provenance-override",
"implementation-closure-drift",
"noncanonical-archive-layout",
"inventory-graph-inflation",
"workcard-budget-bypass",
"fabricated-evidence-digest",
```

```
"unsigned-standard-decision",
"unconfigured-standard-decision-key",
"standard-decision-version-replay",
"candidate-decision-cycle",
"broad-query-corpus-pagination",
"continuation-token-context-inflation",
"inventory-stream-materialization",
"pdf-header-only-credit",
"scale-environment-skip-credit",
"content-search-omission"
```

```
]
},
"maxItems": 77,
"minItems": 77
},
```

```

"production_claim_rule": {
  "const": "Standard distribution success never grants product acceptance. Live product credit
separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact
target-bound evidence, current project release closure eligibility, rollback and recutover equivalence,
and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution
evidence."
},
"record_type": {
  "const": "Conformance"
},
"reference_benchmarks": {
  "const": {
    "core_valid_relation_count": 198999,
    "depths_tested": [
      1,
      2,
      3,
      4,
      5,
      6,
      7,
      8,
      9,
      10,
      11,
      12
    ],
    "file_count": 100000,
    "forced_continuation": true,
    "inventory_passes": 1,
    "performance_profiles": {
      "portable-local-v1": {
        "commit_bytes_per_changed_record_max": 32768,
        "commit_p95_ms_max": 500,
        "commit_p99_ms_max": 1000,
        "compatible_platforms": [
          "linux",
          "windows"
        ],
        "database_bytes_max": 402653184,
        "p50_ms_max": 100,
        "p95_ms_max": 150,
        "p99_ms_max": 500,
        "peak_rss_bytes_max": 805306368,
        "projection_amplification_max": 20,
        "requires_same_runner_no_degradation": true,
        "runtime_checkpoint_count_max": 24,
        "semantic_inflation_max": 3.25,
        "semantic_ingestion_seconds_max": 600
      }
    },
    "physical_relation_corpus": {
      "artifact_target_coverage": 1,
      "expected_counts_source": "fixture-manifest",
      "generation": "explicit-authorized-command-events",
      "relation_count": 198971,
      "relation_kind": "READS",
      "separate_from_raw_inventory": true,
      "task_count": 1567,
      "total_core_valid_relations": 198999
    },
    "query_count": 600,
    "raw_file_proxy_ratio": 1,
    "rebuild_product_passes": 0,
    "reference_targets_are_environment_specific": true,
    "semantic_corpus": {
      "depths": [
        1,
        2,
        3,
        4,
        5,
        6,
        7,
        8,
        9,
        10,

```

```

    11,
    12
  ],
  "expected_counts_source": "fixture-manifest",
  "relation_count": 28,
  "relation_kind": "DEPENDS_ON",
  "separate_from_raw_inventory": true,
  "task_count": 32
},
"silent_truncations_max": 0,
"synthetic_task_ratio": 0
}
},
"required_acceptance": {
  "const": [
    "canonical-name-only",
    "exact-six-core-artifacts",
    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",
    "candidate-evidence-provider-binding",
    "one-command-one-atomic-batch",
    "deterministic-replay",
    "projection-disposable-and-current",
    "bounded-workcard-through-depth-twelve",
    "typed-reference-domain-range-closure",
    "dependency-graph-acyclic-and-current-ready-frontier",
    "safe-recursive-export",
    "idempotent-state-commands",
    "normalization-collision-rejection",
    "provider-substitution-invalidates-derived-state",
    "rollback-and-recutover-equivalence",
    "zero-forbidden-name-occurrences",
    "zero-unowned-normative-facts",
    "zero-core-provider-lock-in",
    "zero-compiled-cache-files",
    "physical-100k-one-artifact-proxy-per-file",
    "external-standard-release-decision-exact-binding",
    "root-bootstrap-command-limited",
    "deterministic-activation-identity",
    "configuration-exact-idempotency",
    "truthful-operation-reporting",
    "depth-aware-seed-budget",
    "lossless-continuation-in-reference-100k-scenario",
    "read-workcard-without-lease",
    "mutation-workcard-requires-current-lease",
    "command-capability-and-scope-resolution",
    "grant-issuance-chain-closure",
    "team-signatures-cryptographically-verified",
    "required-provider-preflight-before-commit",
    "policy-owned-task-and-lease-state-machines",
    "decision-kind-target-scope-binding",
    "non-skipped-credit-result-has-evidence",
    "exact-command-primary-event-correspondence",
    "conjunctive-scope-containment",
    "authorization-binds-exact-grant-claim",
    "command-effect-target-in-requested-scope",
    "lease-holder-has-task-execute-grant",
    "decision-grant-matches-command-authorization",
    "artifact-kind-resolves-correct-capability",
    "continuation-v2-complete-page-union",
    "candidate-recipe-applied-once",
    "creditable-candidate-immutable-vcs-tree",
    "selected-provider-adapter-evidence",
    "team-signed-cli-end-to-end",
    "current-release-closure-revalidation",
    "finding-bound-resolution-and-waiver-evidence",
    "candidate-delta-within-task-and-grant-scope",
    "verified-head-checkpoint-delta-replay",
    "profile-default-with-core-depth-twelve",

```

```

"canonical-exact-zip-distribution",
"current-interpretter-and-clean-install-modes",

"required-no-degradation-tests-fail-closed",
"continuation-secret-subject-grant-binding",
"implementation-closure-bound-and-current",
"verified-inventory-result-provenance",
"exact-six-base-command-surface",
"zero-dead-evidence-budget-fields",
"profile-bound-physical-100k-performance",
"windows-linux-exact-zip-matrix",
"fresh-semantic-control-state-per-saturation-iteration",
"three-zero-new-saturation-iterations",
"standard-distribution-separate-from-product-acceptance",
"candidate-evidence-decision-acyclic-chain",
"evidence-manifest-physical-resolution",
"evidence-manifest-exact-eight-lane-matrix-resolution",
"evidence-manifest-four-cp314-supplemental-records",
"evidence-manifest-path-digest-size-closure",
"evidence-manifest-nested-and-supplemental-chronology",
"seven-distinct-role-producer-scopes",
"signed-standard-decision-authority-proof",
"semver-from-candidate-binding",
"typed-human-document-verification",
"broad-query-bounded-seed-refinement",
"continuation-token-bytes-at-most-256",
"inventory-stream-memory-amplification-at-most-32",
"explicit-scale-marker-selection",
"authenticated-role-platform-evidence-attestations",
"installed-platform-observation-closure",
"handle-relative-evidence-root-safety",

"raw-scale-summary-recomputation",
"bounded-incremental-commit-and-compaction",
"offline-installed-no-degradation",
"offline-release-online-compatibility",
"decision-after-evidence-with-bounded-skew",
"single-derived-platform-closure-owner",
"truthful-process-invocation-status",
"product-public-approval-separation",
"historical-decision-current-eligibility-separation"
],
"items": {
  "enum": [
    "canonical-name-only",
    "exact-six-core-artifacts",
    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",
    "candidate-evidence-provider-binding",
    "one-command-one-atomic-batch",
    "deterministic-replay",

    "projection-disposable-and-current",
    "bounded-workcard-through-depth-twelve",
    "typed-reference-domain-range-closure",
    "dependency-graph-acyclic-and-current-ready-frontier",
    "safe-recursive-export",
    "idempotent-state-commands",
    "normalization-collision-rejection",
    "provider-substitution-invalidates-derived-state",
    "rollback-and-recutover-equivalence",
    "zero-forbidden-name-occurrences",
    "zero-unowned-normative-facts",
    "zero-core-provider-lock-in",
    "zero-compiled-cache-files",
    "physical-100k-one-artifact-proxy-per-file",
    "external-standard-release-decision-exact-binding",
    "root-bootstrap-command-limited",
    "deterministic-activation-identity",
    "configuration-exact-idempotency",
    "truthful-operation-reporting",

```



```

"depth-aware-seed-budget",
"lossless-continuation-in-reference-100k-scenario",
"read-workcard-without-lease",
"mutation-workcard-requires-current-lease",
"command-capability-and-scope-resolution",
"grant-issuance-chain-closure",
"team-signatures-cryptographically-verified",
"required-provider-preflight-before-commit",
"policy-owned-task-and-lease-state-machines",

```

```

"decision-kind-target-scope-binding",
"non-skipped-credit-result-has-evidence",
"exact-command-primary-event-correspondence",
"conjunctive-scope-containment",
"authorization-binds-exact-grant-claim",
"command-effect-target-in-requested-scope",
"lease-holder-has-task-execute-grant",
"decision-grant-matches-command-authorization",
"artifact-kind-resolves-correct-capability",
"continuation-v2-complete-page-union",
"candidate-recipe-applied-once",
"creditable-candidate-immutable-vcs-tree",
"selected-provider-adapter-evidence",
"team-signed-cli-end-to-end",
"current-release-closure-revalidation",
"finding-bound-resolution-and-waiver-evidence",
"candidate-delta-within-task-and-grant-scope",
"verified-head-checkpoint-delta-replay",
"profile-default-with-core-depth-twelve",
"canonical-exact-zip-distribution",
"current-interpreter-and-clean-install-modes",
"required-no-degradation-tests-fail-closed",
"continuation-secret-subject-grant-binding",
"implementation-closure-bound-and-current",
"verified-inventory-result-provenance",
"exact-six-base-command-surface",
"zero-dead-evidence-budget-fields",
"profile-bound-physical-100k-performance",

```

```

"windows-linux-exact-zip-matrix",
"fresh-semantic-control-state-per-saturation-iteration",
"three-zero-new-saturation-iterations",
"standard-distribution-separate-from-product-acceptance",
"candidate-evidence-decision-acyclic-chain",
"evidence-manifest-physical-resolution",
"evidence-manifest-exact-eight-lane-matrix-resolution",
"evidence-manifest-four-cp314-supplemental-records",
"evidence-manifest-path-digest-size-closure",
"evidence-manifest-nested-and-supplemental-chronology",
"seven-distinct-role-producer-scopes",
"signed-standard-decision-authority-proof",
"semver-from-candidate-binding",
"typed-human-document-verification",
"broad-query-bounded-seed-refinement",
"continuation-token-bytes-at-most-256",
"inventory-stream-memory-amplification-at-most-32",
"explicit-scale-marker-selection",
"authenticated-role-platform-evidence-attestations",
"installed-platform-observation-closure",
"handle-relative-evidence-root-safety",
"raw-scale-summary-recomputation",
"bounded-incremental-commit-and-compaction",
"offline-installed-no-degradation",
"offline-release-online-compatibility",
"decision-after-evidence-with-bounded-skew",
"single-derived-platform-closure-owner",
"truthful-process-invocation-status",

```

```

"product-public-approval-separation",
"historical-decision-current-eligibility-separation"
]
},
"maxItems": 102,
"minItems": 102
},
"scale_contracts": {
  "const": {
    "event_commit": {
      "complexity": "O(batch)",
      "head_lookup": "O(1)"
    }
  }
}

```

```

    },
    "init": {
      "complexity": "0(core+plan)",
      "project_tree_passes": 0
    },
    "inventory": {
      "complexity": "0(files+bytes)",
      "creditable_consistency_modes": [
        "immutable-vcs-tree"
      ],
      "derived_artifact_proxies_per_raw_file": 1,
      "manifest_verified_jsonl": true,
      "memory_amplification_max": 32,
      "observational_mode_pass_credit": false,
      "project_tree_passes_max": 1,
      "search_text_bytes_max": 4096,

      "synthetic_relations_per_raw_file": 0,
      "synthetic_tasks_per_raw_file": 0
    },
    "normal_one_shot_replay": {
      "checkpoint_authoritative": false,
      "complexity": "0(delta-since-verified-checkpoint)",
      "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"
    },
    "projection": {
      "public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only",
      "rebuild_product_tree_passes": 0,
      "reserved_provenance_rejected": true
    },
    "replay": {
      "complexity": "0(events)"
    },
    "saturation": {
      "fresh_control_state_per_iteration": true,
      "init_product_tree_scans": 0,
      "semantic_state_reused": false
    },
    "workcard": {
      "broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus
pagination",
      "complexity": "0(top_k+capped_typed_closure)",
      "continuation_state_bytes_max": 16384,
      "continuation_token_bytes_max": 256,
      "corpus_independent_output_bound": true,
      "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*
requested_depth))))",

      "selected_closure_union_completeness": 1
    }
  },
  "schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/Conformance"
  },
  "structural_budgets": {
    "const": {
      "authority_bearing_init_records": 5,
      "base_user_commands_max": 11,
      "core_normative_artifacts": 6,
      "persistent_entity_kinds_max": 9,
      "relation_kinds_max": 6,
      "resolved_plan_bytes_max": 8192,
      "resolved_plan_source_samples_max": 64,
      "selected_presets": 1
    }
  },
  "version": {
    "const": "1.0.0-alpha.4"
  },
  "workcard_hard_ceiling": {
    "const": {
      "max_bytes": 16384,
      "max_entities": 32,
      "max_fanout_per_entity": 8,
      "max_relations": 48,

      "top_k": 12
    }
  }
}

```

```

},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "structural_budgets",
  "workcard_hard_ceiling",
  "scale_contracts",
  "reference_benchmarks",
  "required_acceptance",
  "mutation_families",
  "production_claim_rule",
  "dependency_depth_hard_max",
  "alpha4_no_credit_rules",
  "acceptance_proof_class_counts",
  "acceptance_proof_classes",
  "command_latency_contracts"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Conformance | Validator: urn:promin:contracts:1#/\$defs/Conformance

## 22. ContinuationTokenClaims

Type summary: type="object"; additionalProperties=false

Required: record\_type, token\_version, activation\_digest, subject\_id, grant\_id, grant\_claim\_digest, capability\_id, requested\_scope, normalized\_query\_digest, cursor, head\_event\_id, head\_sequence, head\_digest, projection\_digest, ranking\_id, traversal\_id, dependency\_depth, budgets, issued\_at, expires\_at, ttl\_seconds, revocation\_epoch, implementation\_closure\_digest, state\_id

| Property                      | Required | Structural rule                         |
|-------------------------------|----------|-----------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"                 |
| budgets                       | yes      | \$ref="#/\$defs/Budget"                 |
| capability_id                 | yes      | const="projection.read"                 |
| cursor                        | yes      | type="integer"; minimum=0               |
| dependency_depth              | yes      | type="integer"; minimum=1; maximum=12   |
| expires_at                    | yes      | \$ref="#/\$defs/Timestamp"              |
| grant_claim_digest            | yes      | \$ref="#/\$defs/Digest"                 |
| grant_id                      | yes      | \$ref="#/\$defs/Id"                     |
| head_digest                   | yes      | oneOf[2]=#/\$defs/Digest   type:null    |
| head_event_id                 | yes      | oneOf[2]=#/\$defs/Id   type:null        |
| head_sequence                 | yes      | type="integer"; minimum=0               |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                 |
| issued_at                     | yes      | \$ref="#/\$defs/Timestamp"              |
| normalized_query_digest       | yes      | \$ref="#/\$defs/Digest"                 |
| projection_digest             | yes      | \$ref="#/\$defs/Digest"                 |
| ranking_id                    | yes      | const="bm25-v1"                         |
| record_type                   | yes      | const="ContinuationTokenClaims"         |
| requested_scope               | yes      | type="array"; minItems=1; maxItems=32   |
| revocation_epoch              | yes      | \$ref="#/\$defs/Digest"                 |
| state_id                      | yes      | \$ref="#/\$defs/Id"                     |
| subject_id                    | yes      | \$ref="#/\$defs/Id"                     |
| token_version                 | yes      | type="integer"; const=2                 |
| traversal_id                  | yes      | const="typed-bfs-v2"                    |
| ttl_seconds                   | yes      | type="integer"; minimum=1; maximum=3600 |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    }
  }
}

```

```

    },
    "budgets": {
      "$ref": "#/$defs/Budget"
    },
    "capability_id": {
      "const": "projection.read"
    },
    "cursor": {
      "minimum": 0,
      "type": "integer"
    },
    "dependency_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {

```

```

      "$ref": "#/$defs/Id"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_event_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_sequence": {
      "minimum": 0,
      "type": "integer"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"

```

```

    },
    "issued_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "normalized_query_digest": {
      "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking_id": {
      "const": "bm25-v1"
    },
    "record_type": {
      "const": "ContinuationTokenClaims"
    },
    "requested_scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "revocation_epoch": {
      "$ref": "#/$defs/Digest"

```

```

    },
    "state_id": {
      "$ref": "#/$defs/Id"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    },
    "token_version": {
      "const": 2,
      "type": "integer"
    },
    "traversal_id": {
      "const": "typed-bfs-v2"
    },
    "ttl_seconds": {
      "maximum": 3600,
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "record_type",
    "token_version",
    "activation_digest",
    "subject_id",
    "grant_id",
    "grant_claim_digest",
    "capability_id",
    "requested_scope",
    "normalized_query_digest",
    "cursor",
    "head_event_id",
    "head_sequence",
    "head_digest",
    "projection_digest",
    "ranking_id",
    "traversal_id",
    "dependency_depth",
    "budgets",
    "issued_at",
    "expires_at",
    "ttl_seconds",
    "revocation_epoch",
    "implementation_closure_digest",
    "state_id"
  ],
  "type": "object",
  "x-promin-max-state-bytes": 16384,
  "x-promin-max-token-bytes": 256,
  "x-promin-persistence": "derived-live-only"
}

```

Owner: *core/contracts.schema.json#/\$defs/ContinuationTokenClaims* | Validator: *urn:promin:contracts:1#/\$defs/ContinuationTokenClaims*

## 23. ContinueResult

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:if,then

Required: record\_type, status, subject\_id, activation\_digest, work\_card, continuation

| Property          | Required | Structural rule                        |
|-------------------|----------|----------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                |
| continuation      | yes      | oneOf[2]=type:null   type:string       |
| record_type       | yes      | const="ContinueResult"                 |
| status            | yes      | enum=["ready", "empty"]                |
| subject_id        | yes      | \$ref="#/\$defs/Id"                    |
| work_card         | yes      | oneOf[2]=#/\$defs/WorkCard   type:null |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {

```

```

        "continuation": {
            "type": "null"
        },
        "work_card": {
            "type": "null"
        }
    },
    "if": {
        "properties": {
            "status": {
                "const": "ready"
            }
        },
        "required": [
            "status"
        ]
    },
    "then": {
        "properties": {
            "work_card": {
                "$ref": "#/$defs/WorkCard"
            }
        }
    }
}

```

```

    }
}
},
{
    "if": {
        "properties": {
            "continuation": {
                "type": "string"
            }
        },
        "required": [
            "continuation"
        ]
    },
    "then": {
        "properties": {
            "work_card": {
                "properties": {
                    "truncated": {
                        "const": true
                    }
                },
                "required": [
                    "truncated"
                ]
            }
        }
    }
}
}

```

```

    }
},
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            },
            {
                "maxLength": 256,
                "minLength": 1,
                "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\. [0-9a-f]{64}\\. [A-Za-z0-9_-]{43}$",
                "type": "string"
            }
        ]
    },
    "record_type": {
        "const": "ContinueResult"
    },
    "status": {
        "enum": [
            "ready",
            "empty"
        ]
    }
}

```

```

    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    },
    "work_card": {
      "oneOf": [
        {
          "$ref": "#/$defs/WorkCard"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "required": [
    "record_type",
    "status",
    "subject_id",
    "activation_digest",
    "work_card",
    "continuation"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/ContinueResult | Validator: urn:promin:contracts:1#/\$defs/ContinueResult

## 24. CoreComponent

Type summary: type="object"; additionalProperties=false

Required: path, sha256

| Property | Required | Structural rule                                                                                                      |
|----------|----------|----------------------------------------------------------------------------------------------------------------------|
| path     | yes      | enum=["semantic-model.json", "authority-model.json", "policy-set.json", "contracts.schema.json", "conformance.json"] |
| sha256   | yes      | \$ref="#/\$defs/Digest"                                                                                              |

```

{
  "additionalProperties": false,
  "properties": {
    "path": {
      "enum": [
        "semantic-model.json",
        "authority-model.json",
        "policy-set.json",
        "contracts.schema.json",
        "conformance.json"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "path",
    "sha256"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/CoreComponent | Validator: urn:promin:contracts:1#/\$defs/CoreComponent

## 25. CurrentInventory

Type summary: type="object"; additionalProperties=false

Required: record\_type, candidate\_digest, manifest\_digest, inventory\_digest, stream\_digest, stream\_bytes, entry\_count, projection\_row\_count, project\_tree\_passes, inventory\_passes, rebuild\_product\_passes, search\_text\_rule, projection\_authoritative

| Property         | Required | Structural rule           |
|------------------|----------|---------------------------|
| candidate_digest | yes      | \$ref="#/\$defs/Digest"   |
| entry_count      | yes      | type="integer"; minimum=0 |
| inventory_digest | yes      | \$ref="#/\$defs/Digest"   |

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Required | Structural rule                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------|
| inventory_passes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | yes      | type="integer"; const=1                                                                                                   |
| manifest_digest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | yes      | \$ref="#/\$defs/Digest"                                                                                                   |
| project_tree_passes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | type="integer"; minimum=0; maximum=1                                                                                      |
| projection_authoritative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | yes      | const=false                                                                                                               |
| projection_row_count                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | yes      | type="integer"; minimum=0                                                                                                 |
| rebuild_product_passes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | yes      | type="integer"; const=0                                                                                                   |
| record_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | yes      | const="CurrentInventory"                                                                                                  |
| search_text_rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | yes      | const="exact NFC-normalized canonical path, UTF-8 bounded by Conformance scale_contracts.inventory.search_text_bytes_max" |
| stream_bytes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | yes      | type="integer"; minimum=1                                                                                                 |
| stream_digest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | yes      | \$ref="#/\$defs/Digest"                                                                                                   |
| <pre> {   "additionalProperties": false,   "properties": {     "candidate_digest": {       "\$ref": "#/\$defs/Digest"     },     "entry_count": {       "minimum": 0,       "type": "integer"     },     "inventory_digest": {       "\$ref": "#/\$defs/Digest"     },     "inventory_passes": {       "const": 1,       "type": "integer"     },     "manifest_digest": {       "\$ref": "#/\$defs/Digest"     },     "project_tree_passes": {       "maximum": 1,       "minimum": 0,       "type": "integer"     },     "projection_authoritative": {       "const": false     },     "projection_row_count": {       "minimum": 0,       "type": "integer"     },     "rebuild_product_passes": {       "const": 0,       "type": "integer"     },     "record_type": {       "const": "CurrentInventory"     },     "search_text_rule": {       "const": "exact NFC-normalized canonical path, UTF-8 bounded by Conformance scale_contracts.inventory.search_text_bytes_max"     },     "stream_bytes": {       "minimum": 1,       "type": "integer"     },     "stream_digest": {       "\$ref": "#/\$defs/Digest"     }   },   "required": [     "record_type",     "candidate_digest",     "manifest_digest",     "inventory_digest",     "stream_digest",     "stream_bytes",     "entry_count"   ] } </pre> |          |                                                                                                                           |



```

    "projection_row_count",
    "project_tree_passes",
    "inventory_passes",
    "rebuild_product_passes",
    "search_text_rule",
    "projection_authoritative"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "derived-rebuild-only"
}

```

Owner: core/contracts.schema.json#/\$defs/CurrentInventory | Validator: urn:promin:contracts:1#/\$defs/CurrentInventory

## 26. Decision

Type summary: type="object"; additionalProperties=false; allOf[7]=keys:if,then | keys:if,then | keys:if,then | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title

Required: record\_type, decision\_id, decision\_kind, subject\_id, grant\_id, grant\_claim\_digest, activation\_digest, target\_type, target\_id, target\_digest, rationale, evidence\_artifacts, created\_at

| Property                | Required | Structural rule                                                               |
|-------------------------|----------|-------------------------------------------------------------------------------|
| activation_digest       | yes      | \$ref="#/\$defs/Digest"                                                       |
| candidate_digest        | no       | \$ref="#/\$defs/Digest"                                                       |
| created_at              | yes      | \$ref="#/\$defs/Timestamp"                                                    |
| decision_id             | yes      | \$ref="#/\$defs/Id"                                                           |
| decision_kind           | yes      | enum=["accept", "reject", "resolve", "waive", "promote", "release", "revoke"] |
| evidence_artifacts      | yes      | type="array"; minItems=1; maxItems=64; uniqueItems=true                       |
| exception_policy_digest | no       | \$ref="#/\$defs/Digest"                                                       |
| expires_at              | no       | \$ref="#/\$defs/Timestamp"                                                    |
| finding_digest          | no       | \$ref="#/\$defs/Digest"                                                       |
| grant_claim_digest      | yes      | \$ref="#/\$defs/Digest"                                                       |
| grant_id                | yes      | \$ref="#/\$defs/Id"                                                           |
| provider_binding_digest | no       | \$ref="#/\$defs/Digest"                                                       |
| rationale               | yes      | type="string"; minLength=1                                                    |
| record_type             | yes      | const="Decision"                                                              |
| release_closure_digest  | no       | \$ref="#/\$defs/Digest"                                                       |
| subject_id              | yes      | \$ref="#/\$defs/Id"                                                           |
| target_digest           | yes      | \$ref="#/\$defs/Digest"                                                       |
| target_id               | yes      | \$ref="#/\$defs/Id"                                                           |
| target_type             | yes      | enum=["Task", "Candidate", "Finding", "Grant"]                                |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "decision_kind": {
            "enum": [
              "resolve",
              "waive"
            ]
          }
        }
      },
      "then": {
        "properties": {
          "target_type": {
            "const": "Finding"
          }
        }
      }
    }
  ],
  "type": "object"
}

```

```

    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "promote",
            "release"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "target_type": {
          "const": "Candidate"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "accept",
            "reject"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "target_type": {
          "enum": [
            "Task",
            "Candidate"
          ]
        }
      }
    }
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "release_closure_digest"
            ]
          },
          {
            "required": [
              "provider_binding_digest"
            ]
          }
        ]
      }
    },
    "if": {
      "properties": {
        "decision_kind": {
          "const": "release"
        }
      }
    },
    "required": [
      "decision_kind"
    ]
  },
  {
    "then": {
      "required": [
        "release_closure_digest",
        "provider_binding_digest"
      ]
    },
    "title": "release Decision binds exact historical closure and providers"
  },
  {
    "else": {

```

```

    "not": {
      "required": [
        "finding_digest"
      ]
    },
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "resolve",
            "waive"
          ]
        }
      },
      "required": [
        "decision_kind"
      ]
    },
    "then": {
      "required": [
        "finding_digest"
      ]
    },
    "title": "Finding disposition binds exact canonical Finding digest"
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "exception_policy_digest"
            ]
          },
          {
            "required": [
              "expires_at"
            ]
          }
        ]
      }
    }
  },
  {
    "if": {
      "properties": {
        "decision_kind": {
          "const": "waive"
        }
      },
      "required": [
        "decision_kind"
      ]
    },
    "then": {
      "required": [
        "exception_policy_digest",
        "expires_at"
      ]
    },
    "title": "waiver binds exception policy and expiry"
  },
  {
    "else": {
      "required": [
        "candidate_digest"
      ]
    },
    "if": {
      "properties": {
        "decision_kind": {
          "const": "revoke"
        }
      },
      "required": [
        "decision_kind"
      ]
    },
  },

```

```

    "then": {
      "not": {
        "required": [
          "candidate_digest"
        ]
      },
      "properties": {
        "target_type": {
          "const": "Grant"
        }
      }
    },
    "title": "Grant revocation has no Candidate binding"
  }
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "decision_kind": {
      "enum": [
        "accept",
        "reject",
        "resolve",
        "waive",
        "promote",
        "release",
        "revoke"
      ]
    },
    "evidence_artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "artifact_id": {
            "$ref": "#/$defs/Id"
          },
          "artifact_record_digest": {

```

```

            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "artifact_id",
          "artifact_record_digest"
        ],
        "type": "object"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "exception_policy_digest": {
      "$ref": "#/$defs/Digest"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "finding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"

```

```

  },
  "provider_binding_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "rationale": {
    "minLength": 1,
    "type": "string"
  },
  "record_type": {
    "const": "Decision"
  },
  "release_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "target_digest": {
    "$ref": "#/$defs/Digest"
  },
  "target_id": {
    "$ref": "#/$defs/Id"
  },
  "target_type": {
    "enum": [
      "Task",
      "Candidate",
      "Finding",
      "Grant"
    ]
  },
  "required": [
    "record_type",
    "decision_id",
    "decision_kind",
    "subject_id",
    "grant_id",
    "grant_claim_digest",
    "activation_digest",
    "target_type",
    "target_id",
    "target_digest",
    "rationale",
    "evidence_artifacts",
    "created_at"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Decision | Validator: urn:promin:contracts:1#/\$defs/Decision

## 27. Digest

Type summary: type="string"; pattern="^[0-9a-f]{64}\$"

Required: -

```

{
  "pattern": "^[0-9a-f]{64}$",
  "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/Digest | Validator: urn:promin:contracts:1#/\$defs/Digest

## 28. DoctorResult

Type summary: type="object"; additionalProperties=false

Required: record\_type, status, claim\_scope, source\_scope, report\_authoritative, pass\_credit, product\_acceptance\_pass, product\_public\_approval, activation\_digest, implementation\_closure\_digest, head, recovery, replay, projection, provider\_adapters, metrics, components, archival\_diagnostics, operation\_metrics, product\_tree\_scans

| Property             | Required | Structural rule                           |
|----------------------|----------|-------------------------------------------|
| activation_digest    | yes      | \$ref="#/\$defs/Digest"                   |
| archival_diagnostics | yes      | type="object"; additionalProperties=false |
| claim_scope          | yes      | const="runtime-health-only"               |
| components           | yes      | type="object"; additionalProperties=false |

| Property                      | Required | Structural rule                                        |
|-------------------------------|----------|--------------------------------------------------------|
| head                          | yes      | type="object"; additionalProperties=false              |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                                |
| metrics                       | yes      | type="object"; additionalProperties=false              |
| operation_metrics             | yes      | oneOf[2]=#/\$defs/OperationMetrics   type:null         |
| pass_credit                   | yes      | const=false                                            |
| product_acceptance_pass       | yes      | const=false                                            |
| product_public_approval       | yes      | const="not_approved"                                   |
| product_tree_scans            | yes      | const=0                                                |
| projection                    | yes      | oneOf[2]=type:object   type:object                     |
| provider_adapters             | yes      | type="array"; minItems=1; maxItems=32                  |
| record_type                   | yes      | const="DoctorResult"                                   |
| recovery                      | yes      | type="object"; additionalProperties=false              |
| replay                        | yes      | oneOf[2]=type:null   type:object                       |
| report_authoritative          | yes      | const=false                                            |
| source_scope                  | yes      | const="installed-core-and-live-operational-state-only" |
| status                        | yes      | enum=["healthy", "degraded", "failed", "incomplete"]   |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "archival_diagnostics": {
      "additionalProperties": false,
      "properties": {
        "affects_current_status": {
          "const": false
        },
        "entries": {
          "items": {
            "additionalProperties": false,
            "properties": {
              "detail_digest": {
                "$ref": "#/$defs/Digest"
              },
              "diagnostic_id": {
                "$ref": "#/$defs/Id"
              },
              "recorded_at": {
                "$ref": "#/$defs/Timestamp"
              },
              "status": {
                "enum": [
                  "healthy",
                  "degraded",
                  "failed",
                  "incomplete"
                ]
              }
            }
          },
          "required": [
            "diagnostic_id",
            "status",
            "recorded_at",
            "detail_digest"
          ],
          "type": "object"
        },
        "maxItems": 256,
        "minItems": 0,
        "type": "array"
      },
      "pass_credit": {
        "const": false
      }
    },
    "required": [

```

```

    "affects_current_status",
    "pass_credit",
    "entries"
  ],
  "type": "object"

```

```

  },
  "claim_scope": {
    "const": "runtime-health-only"
  },
  "components": {
    "additionalProperties": false,
    "properties": {
      "core": {
        "additionalProperties": false,
        "properties": {
          "observed_digest": {
            "oneOf": [
              {
                "$ref": "#/$defs/Digest"
              },
              {
                "type": "null"
              }
            ]
          },
          "reason_codes": {
            "items": {
              "$ref": "#/$defs/Id"
            },
            "maxItems": 32,
            "minItems": 0,
            "type": "array",
            "uniqueItems": true
          }
        }
      }
    }
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",
      "incomplete"
    ]
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
},
"init": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
}

```

```

  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",

```

```

        "incomplete"
      ]
    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
},
"projection": {

```

```

  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  }

```

```

}
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"providers": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,

```

```

    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",
      "incomplete"
    ]
  }
}

```



```

    ]
  }
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"recovery": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {

```

```

          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"

```

```

},
"replay": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  }
}

```

```

    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
}
},
"required": [
  "core",
  "init",
  "providers",
  "recovery",
  "replay",
  "projection"
],
"type": "object"
},
"head": {
  "additionalProperties": false,
  "properties": {
    "batch_digest": {
      "oneOf": [

```

```

        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "batch_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "sequence": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "sequence",
    "batch_id",
    "batch_digest"
  ],
  "type": "object"

```

```

},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"metrics": {
  "additionalProperties": false,
  "properties": {
    "core_artifacts_verified": {
      "minimum": 0,
      "type": "integer"
    },
    "event_batches": {
      "minimum": 0,
      "type": "integer"
    },
    "init_records_verified": {
      "minimum": 0,
      "type": "integer"
    },
    "projection_entities": {
      "oneOf": [
        {
          "minimum": 0,
          "type": "integer"

```

```

    },
    {
      "type": "null"
    }
  ],
  },
  "projection_relations": {
    "oneOf": [
      {
        "minimum": 0,
        "type": "integer"
      },
      {
        "type": "null"
      }
    ]
  },
  "provider_healthchecks_executed": {
    "minimum": 0,
    "type": "integer"
  }
},
"required": [
  "core_artifacts_verified",
  "event_batches",
  "init_records_verified",
  "projection_entities",
  "projection_relations",
  "provider_healthchecks_executed"
],
"type": "object"
},

```

```

"operation_metrics": {
  "oneOf": [
    {
      "$ref": "#/$defs/OperationMetrics"
    },
    {
      "type": "null"
    }
  ]
},
"pass_credit": {
  "const": false
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"product_tree_scans": {
  "const": 0
},
"projection": {
  "oneOf": [
    {
      "additionalProperties": false,
      "properties": {
        "projection_authoritative": {

```

```

          "const": false
        },
        "status": {
          "const": "missing"
        }
      }
    },
    "required": [
      "status",
      "projection_authoritative"
    ],
    "type": "object"
  },
  {
    "additionalProperties": false,
    "properties": {
      "activation_digest": {
        "$ref": "#/$defs/Digest"

```

```

    },
    "built_at": {
      "$ref": "#/$defs/Timestamp"
    },
    },
    "entity_count": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "event_count": {
      "minimum": 0,
      "type": "integer"
    }
  }

```

```

    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    },
    "head_sequence": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    },
    "incremental_changed_records": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "incremental_commit_count": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "inventory_entries": {
      "minimum": 0,

```

```

      "type": "integer"
    },
    },
    "inventory_passes": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "inventory_proxies": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "inventory_relations": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "product_passes": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "projection_authoritative": {
      "const": false
    },
    },
    "projection_compaction_count": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "projection_db_bytes": {
      "minimum": 0,
      "type": "integer"
    }
  }

```

```

    },
    "raw_file_proxy_ratio": {
      "minimum": 0,
      "type": "number"
    },
    },
    "relation_count": {
      "minimum": 0,
      "type": "integer"
    },
    },
    "semantic_digest": {

```

```

        "$ref": "#/$defs/Digest"
    },
    "semantic_digest_algorithm": {
        "maxLength": 128,
        "minLength": 1,
        "type": "string"
    },
    "storage_layout": {
        "maxLength": 128,
        "minLength": 1,
        "type": "string"
    },
    "synthetic_task_count": {
        "const": 0,
        "type": "integer"
    },
    "synthetic_task_ratio": {
        "const": 0.0,

```

```

        "type": "number"
    }
},
"required": [
    "activation_digest",
    "head_digest",
    "head_sequence",
    "semantic_digest",
    "entity_count",
    "relation_count",
    "inventory_entries",
    "inventory_proxies",
    "inventory_relations",
    "inventory_passes",
    "product_passes",
    "event_count",
    "built_at",
    "projection_authoritative",
    "implementation_closure_digest",
    "semantic_digest_algorithm",
    "storage_layout",
    "incremental_commit_count",
    "incremental_changed_records",
    "projection_compaction_count",
    "raw_file_proxy_ratio",
    "synthetic_task_count",
    "synthetic_task_ratio",
    "projection_db_bytes"

```

```

    ],
    "type": "object"
}
]
},
"provider_adapters": {
    "items": {
        "additionalProperties": false,
        "oneOf": [
            {
                "properties": {
                    "adapter_id": {
                        "const": "python-runtime-v1"
                    },
                    "capability_id": {
                        "const": "control-runtime"
                    },
                    "identity_kind": {
                        "const": "file-digest"
                    },
                    "invocation_kind": {
                        "const": "python-runtime"
                    },
                    "operation": {
                        "const": "healthcheck"
                    },
                    "operation_contract_digest": {
                        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
                    }
                }
            },
            {
                "protocol_id": {
                    "const": "promin.control-runtime.v1"
                }
            }
        ]
    }
}

```

```

    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "shape-validation"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.shape-validation.v1"
      }
    }
  }
},

```

```

    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "content-identity"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.content-identity.v1"
      }
    }
  }
},

```

```

  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "local-serialization"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.local-serialization.v1"
      }
    }
  }
},

```

```

{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "query-projection"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.query-projection.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "git-executable-v1"
    },
    "capability_id": {
      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "053c4181d6d35bd4ca5a13b8a1f18c19571c21b3426644bfa6bda99066dce5b3"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "0e2de1aec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-module-signature-v1"
      },
      "capability_id": {
        "const": "signature"
      },
      "identity_kind": {
        "const": "module-file-digest"
      },
      "invocation_kind": {
        "const": "python-module"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "0e2de1aec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
      },
      "protocol_id": {
        "const": "promin.signature.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "executable-build-dependency-v1"
      },
      "capability_id": {
        "const": "build-dependency"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
      },
      "protocol_id": {
        "const": "promin.build-dependency.v1"
      }
    }
  }
],
"properties": {
  "adapter_id": {
    "$ref": "#/defs/Id"
  }
}

```



```

    },
    "authoritative": {
      "const": false
    },
    "capability_id": {
      "enum": [
        "control-runtime",
        "shape-validation",
        "content-identity",
        "local-serialization",
        "query-projection",
        "filesystem-inventory",
        "export-scan",
        "signature",
        "build-dependency"
      ]
    },
    "completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "dependency_receipt_digest": {
      "$ref": "#/$defs/Digest"
    },
    "exit_code": {
      "oneOf": [
        {
          "type": "integer"
        },
        {
          "type": "null"
        }
      ]
    },
    "identity_digest": {
      "$ref": "#/$defs/Digest"
    },
    "identity_kind": {
      "enum": [
        "file-digest",
        "module-file-digest",
        "distribution-tree-digest",
        "module-closure-digest",
        "platform-attestation"
      ]
    },
    "invocation_kind": {
      "enum": [
        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
      ]
    },
    "invocation_receipt_digest": {
      "$ref": "#/$defs/Digest"
    },
    "invoked": {
      "const": true
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "$ref": "#/$defs/Digest"
    },
    "outcome": {
      "enum": [
        "healthy",
        "degraded",
        "failed"
      ]
    },
    "pass_credit": {
      "const": false
    },
    "persistence_scope": {
      "const": "content-addressed-receipt"
    }
  }

```

```

    },
    "protocol_id": {
      "$ref": "#/$defs/Id"
    },
    "provider_id": {
      "$ref": "#/$defs/Id"
    },
    "reconstructable": {
      "const": true
    },
    "started_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "stderr_digest": {
      "$ref": "#/$defs/Digest"
    },
    "stderr_size_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stdout_digest": {
      "$ref": "#/$defs/Digest"
    },
  },

```

```

    "stdout_size_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "capability_id",
    "provider_id",
    "invocation_kind",
    "identity_kind",
    "identity_digest",
    "adapter_id",
    "protocol_id",
    "operation",
    "operation_contract_digest",
    "invocation_receipt_digest",
    "dependency_receipt_digest",
    "persistence_scope",
    "reconstructable",
    "invoked",
    "started_at",
    "completed_at",
    "outcome",
    "exit_code",
    "stdout_digest",
    "stdout_size_bytes",
    "stderr_digest",

```

```

    "stderr_size_bytes",
    "authoritative",
    "pass_credit"
  ],
  "type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array"
},
"record_type": {
  "const": "DoctorResult"
},
"recovery": {
  "additionalProperties": false,
  "properties": {
    "batch_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  }
}
},

```

```

"batch_id": {
  "oneOf": [
    {
      "$ref": "#/$defs/Id"
    },
    {
      "type": "null"
    }
  ]
},
"sequence": {
  "minimum": 0,
  "type": "integer"
}
},
"required": [
  "sequence",
  "batch_id",
  "batch_digest"
],
"type": "object"
},
"replay": {
  "oneOf": [
    {
      "type": "null"
    },
    {
      "additionalProperties": false,
      "properties": {

```

```

"event_batches": {
  "minimum": 0,
  "type": "integer"
},
"head": {
  "additionalProperties": false,
  "properties": {
    "batch_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "batch_id": {
    "oneOf": [
      {
        "$ref": "#/$defs/Id"
      },
      {
        "type": "null"
      }
    ]
  },
  "sequence": {

```

```

    "minimum": 0,
    "type": "integer"
  }
},
"required": [
  "sequence",
  "batch_id",
  "batch_digest"
],
"type": "object"
},
"record_type": {
  "const": "ReplayResult"
},
"runtime_checkpoint": {
  "oneOf": [
    {
      "additionalProperties": false,
      "properties": {

```

```

    "authoritative": {
      "const": false
    },
    "batch_threshold": {
      "minimum": 1,
      "type": "integer"
    },
    "byte_threshold": {
      "minimum": 1,

```

```

      "type": "integer"
    },
    "checkpoint_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "checkpoint_count": {
      "minimum": 0,
      "type": "integer"
    },
    "tail_batches": {
      "minimum": 0,
      "type": "integer"
    },
    "tail_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "written": {
      "const": false
    }
  },
  "required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",

```

```

    "tail_batches",
    "tail_bytes",
    "written"
  ],
  "type": "object"
},
{
  "additionalProperties": false,
  "properties": {
    "authoritative": {
      "const": false
    },
    "batch_threshold": {
      "minimum": 1,
      "type": "integer"
    },
    "byte_threshold": {
      "minimum": 1,
      "type": "integer"
    },
    "checkpoint_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "checkpoint_count": {
      "minimum": 0,
      "type": "integer"
    },
  },

```

```

  "compacted_tail_batches": {
    "minimum": 0,
    "type": "integer"
  },
  "compacted_tail_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "tail_batches": {
    "minimum": 0,
    "type": "integer"
  },

```

```

        "tail_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "written": {
            "const": true
        }
    },
    "required": [
        "authoritative",
        "batch_threshold",
        "byte_threshold",
        "checkpoint_bytes",
        "checkpoint_count",
        "tail_batches",
        "tail_bytes",
        "written",
        "compacted_tail_batches",
        "compacted_tail_bytes"
    ],
    "type": "object"
}
]
},
"runtime_checkpoint_authoritative": {
    "const": false
},
"runtime_checkpoint_written": {
    "type": "boolean"
},
"status": {
    "const": "pass"
}
},
"required": [
    "record_type",
    "status",
    "event_batches",
    "head",
    "runtime_checkpoint_written",
    "runtime_checkpoint",
    "runtime_checkpoint_authoritative"
],
"type": "object"

```

```

    }
}
},
"report_authoritative": {
    "const": false
},
"source_scope": {
    "const": "installed-core-and-live-operational-state-only"
},
"status": {
    "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
    ]
}
},
"required": [
    "record_type",
    "status",
    "claim_scope",
    "source_scope",
    "report_authoritative",
    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "activation_digest",
    "implementation_closure_digest",
    "head",
    "recovery",
    "replay",
    "projection",

```

```

    "provider_adapters",
    "metrics",
    "components",
    "archival_diagnostics",
    "operation_metrics",
    "product_tree_scans"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: *core/contracts.schema.json#/\$defs/DoctorResult* | Validator: *urn:promin:contracts:1#/\$defs/DoctorResult*

## 29. DynamicValidationHandoff

Type summary: type="object"; additionalProperties=false

Required: status, allowed\_write\_scope, forbidden, pass\_credit

| Property            | Required | Structural rule          |
|---------------------|----------|--------------------------|
| allowed_write_scope | yes      | type="array"; minItems=1 |
| forbidden           | yes      | type="array"; minItems=1 |
| pass_credit         | yes      | const=false              |
| status              | yes      | const="PENDING_DYNAMIC"  |

```

{
  "additionalProperties": false,
  "properties": {
    "allowed_write_scope": {
      "minItems": 1,
      "type": "array"
    },
    "forbidden": {
      "minItems": 1,
      "type": "array"
    },
    "pass_credit": {
      "const": false
    },
    "status": {
      "const": "PENDING_DYNAMIC"
    }
  },
  "required": [
    "status",
    "allowed_write_scope",
    "forbidden",
    "pass_credit"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/DynamicValidationHandoff* | Validator: *urn:promin:contracts:1#/\$defs/DynamicValidationHandoff*

## 30. EffectTarget

Type summary: type="object"; additionalProperties=false

Required: kind, id\_field

| Property | Required | Structural rule     |
|----------|----------|---------------------|
| id_field | yes      | \$ref="#/\$defs/Id" |
| kind     | yes      | \$ref="#/\$defs/Id" |

```

{
  "additionalProperties": false,
  "properties": {
    "id_field": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",

```

```

    "id_field"
  },
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/EffectTarget | Validator: urn:promin:contracts:1#/\$defs/EffectTarget

## 31. Event

Type summary: type="object"; additionalProperties=false; allOf[12]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then

Required: record\_type, event\_id, event\_kind, activation\_digest, payload

| Property          | Required | Structural rule                                                                                                                                                                                                                          |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                                                                                  |
| event_id          | yes      | \$ref="#/\$defs/Id"                                                                                                                                                                                                                      |
| event_kind        | yes      | enum=["artifact.recorded", "candidate.recorded", "decision.recorded", "finding.recorded", "gate.recorded", "grant.issued", "grant.revoked", "lease.recorded", "run.recorded", "task.recorded", "task.transitioned", "relation.recorded"] |
| payload           | yes      | composed structural definition                                                                                                                                                                                                           |
| record_type       | yes      | const="Event"                                                                                                                                                                                                                            |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "event_kind": {
            "const": "task.recorded"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Task"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "event_kind": {
            "const": "task.transitioned"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/TaskTransition"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "event_kind": {
            "const": "candidate.recorded"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Candidate"
          }
        }
      }
    }
  ],
}

```

```

{
  "if": {
    "properties": {
      "event_kind": {
        "const": "artifact.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Artifact"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "run.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Run"
      }
    }
  }
},
{

```

```

  "if": {
    "properties": {
      "event_kind": {
        "const": "gate.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GateResult"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "finding.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Finding"
      }
    }
  }
}

```

```

}
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "decision.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
}

```



```

    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "grant.issued"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Grant"
        }
      }
    },
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "grant.revoked"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GrantRevocation"
        }
      }
    },
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "lease.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Lease"
        }
      }
    },
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "relation.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Relation"
        }
      }
    },
  },
},
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "event_id": {
    "$ref": "#/$defs/Id"
  },
  "event_kind": {
    "enum": [

```

```

        "artifact.recorded",
        "candidate.recorded",
        "decision.recorded",
        "finding.recorded",
        "gate.recorded",
        "grant.issued",
        "grant.revoked",
        "lease.recorded",
        "run.recorded",
        "task.recorded",
        "task.transitioned",
        "relation.recorded"
    ]
},
"payload": {},
"record_type": {
    "const": "Event"
}
},
{
    "required": [
        "record_type",
        "event_id",
        "event_kind",
        "activation_digest",
        "payload"
    ],
    "type": "object"
}
}

```

Owner: core/contracts.schema.json#/\$defs/Event | Validator: urn:promin:contracts:1#/\$defs/Event

## 32. EventBatch

Type summary: type="object"; additionalProperties=false

Required: record\_type, batch\_id, sequence, previous\_digest, previous\_authority\_commitment, authority\_commitment, cumulative\_event\_count, event\_semantic\_digest, state\_binding\_delta, cumulative\_state\_binding\_update\_count, state\_binding\_digest, created\_at, command\_id, idempotency\_key, activation\_record\_digest, events, subject\_id, command\_intent\_digest, command\_digest, authorization\_digest

| Property                                                                                                    | Required | Structural rule                                         |
|-------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------|
| activation_record_digest                                                                                    | yes      | \$ref="#/\$defs/Digest"                                 |
| authority_commitment                                                                                        | yes      | \$ref="#/\$defs/Digest"                                 |
| authorization_digest                                                                                        | yes      | \$ref="#/\$defs/Digest"                                 |
| batch_id                                                                                                    | yes      | \$ref="#/\$defs/Id"                                     |
| command_digest                                                                                              | yes      | \$ref="#/\$defs/Digest"                                 |
| command_id                                                                                                  | yes      | \$ref="#/\$defs/Id"                                     |
| command_intent_digest                                                                                       | yes      | \$ref="#/\$defs/Digest"                                 |
| created_at                                                                                                  | yes      | \$ref="#/\$defs/Timestamp"                              |
| cumulative_event_count                                                                                      | yes      | type="integer"; minimum=1                               |
| cumulative_state_binding_update_count                                                                       | yes      | type="integer"; minimum=1                               |
| event_semantic_digest                                                                                       | yes      | \$ref="#/\$defs/Digest"                                 |
| events                                                                                                      | yes      | type="array"; minItems=1; maxItems=20                   |
| idempotency_key                                                                                             | yes      | type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"      |
| previous_authority_commitment                                                                               | yes      | \$ref="#/\$defs/Digest"                                 |
| previous_digest                                                                                             | yes      | oneOf[2]=#/\$defs/Digest   type:null                    |
| record_type                                                                                                 | yes      | const="EventBatch"                                      |
| sequence                                                                                                    | yes      | type="integer"; minimum=1                               |
| state_binding_delta                                                                                         | yes      | type="array"; minItems=1; maxItems=20; uniqueItems=true |
| state_binding_digest                                                                                        | yes      | \$ref="#/\$defs/Digest"                                 |
| subject_id                                                                                                  | yes      | \$ref="#/\$defs/Id"                                     |
| <pre> {     "additionalProperties": false,     "properties": {         "activation_record_digest": { </pre> |          |                                                         |

```

    "$ref": "#/$defs/Digest"
  },
  "authority_commitment": {
    "$ref": "#/$defs/Digest"
  },
  "authorization_digest": {
    "$ref": "#/$defs/Digest"
  },
  "batch_id": {
    "$ref": "#/$defs/Id"
  },
  "command_digest": {
    "$ref": "#/$defs/Digest"
  },
  "command_id": {
    "$ref": "#/$defs/Id"
  },
  "command_intent_digest": {
    "$ref": "#/$defs/Digest"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "cumulative_event_count": {

```

```

    "minimum": 1,
    "type": "integer"
  },
  "cumulative_state_binding_update_count": {
    "minimum": 1,
    "type": "integer"
  },
  "event_semantic_digest": {
    "$ref": "#/$defs/Digest"
  },
  "events": {
    "items": {
      "$ref": "#/$defs/Event"
    },
    "maxItems": 20,
    "minItems": 1,
    "type": "array"
  },
  "idempotency_key": {
    "pattern": "^[A-Za-z0-9._-]{16,128}$",
    "type": "string"
  },
  "previous_authority_commitment": {
    "$ref": "#/$defs/Digest"
  },
  "previous_digest": {
    "oneOf": [
      {

```

```

        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "record_type": {
    "const": "EventBatch"
  },
  "sequence": {
    "minimum": 1,
    "type": "integer"
  },
  "state_binding_delta": {
    "items": {
      "$ref": "#/$defs/StateBindingUpdate"
    },
    "maxItems": 20,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "state_binding_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

"subject_id": {
  "$ref": "#/$defs/Id"
},
},
"required": [
  "record_type",
  "batch_id",
  "sequence",
  "previous_digest",
  "previous_authority_commitment",
  "authority_commitment",
  "cumulative_event_count",
  "event_semantic_digest",
  "state_binding_delta",
  "cumulative_state_binding_update_count",
  "state_binding_digest",
  "created_at",
  "command_id",
  "idempotency_key",
  "activation_record_digest",
  "events",
  "subject_id",
  "command_intent_digest",
  "command_digest",
  "authorization_digest"
],
"type": "object",
"x-promin-max-canonical-bytes": 2097152,
"x-promin-max-state-binding-bytes": 1048576,
"x-promin-primary-event-count-per-command": 1
}

```

Owner: core/contracts.schema.json#/\$defs/EventBatch | Validator: urn:promin:contracts:1#/\$defs/EventBatch

### 33. EventStorePolicy

Type summary: type="object"; additionalProperties=false

Required: record\_type, authority\_model\_digest, max\_command\_bytes, max\_envelope\_bytes, max\_state\_binding\_bytes, max\_state\_binding\_updates\_per\_batch, max\_events\_per\_batch, max\_requested\_scope\_items, derived\_tail\_batch\_threshold, derived\_tail\_byte\_threshold, runtime\_overlay\_compaction\_depth, command\_required\_fields, command\_conditional\_fields, command\_mutation\_fields, lease\_bound\_command\_kinds, lease\_bound\_task\_transition\_states, command\_to\_primary\_event, allowed\_state\_binding\_leaf\_types, state\_binding\_identity\_rules, state\_binding\_algorithm\_contract, state\_binding\_value\_rules, canonical\_timestamp\_contract, genesis\_previous\_authority\_commitment, genesis\_event\_semantic\_digest, policy\_digest

| Property                              | Required | Structural rule                                                                                                                                                                                                |
|---------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allowed_state_binding_leaf_types      | yes      | const=["Activation", "Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease", "Relation"]                                                                                |
| authority_model_digest                | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                                                        |
| canonical_timestamp_contract          | yes      | const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)                                      |
| command_conditional_fields            | yes      | const=["definition_digest"]                                                                                                                                                                                    |
| command_mutation_fields               | yes      | const=["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]                                                                       |
| command_required_fields               | yes      | const=["record_type", "command_id", "command_kind", "subject_id", "activation_digest", "idempotency_key", "requested_scope", "expected_head_digest", "issued_at", "payload", "intent_digest", "authorization"] |
| command_to_primary_event              | yes      | const=[11 items]                                                                                                                                                                                               |
| derived_tail_batch_threshold          | yes      | type="integer"; const=128                                                                                                                                                                                      |
| derived_tail_byte_threshold           | yes      | type="integer"; const=8388608                                                                                                                                                                                  |
| genesis_event_semantic_digest         | yes      | const="e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"                                                                                                                                       |
| genesis_previous_authority_commitment | yes      | const="e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"                                                                                                                                       |
| lease_bound_command_kinds             | yes      | const=["artifact.record", "run.record", "gate.record", "finding.record"]                                                                                                                                       |
| lease_bound_task_transition_states    | yes      | const=["LEASED", "RUNNING", "COMPLETED"]                                                                                                                                                                       |

| Property                            | Required | Structural rule                                                                                                                                                                  |
|-------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| max_command_bytes                   | yes      | type="integer"; const=1048576                                                                                                                                                    |
| max_envelope_bytes                  | yes      | type="integer"; const=2097152                                                                                                                                                    |
| max_events_per_batch                | yes      | type="integer"; const=20                                                                                                                                                         |
| max_requested_scope_items           | yes      | type="integer"; const=32                                                                                                                                                         |
| max_state_binding_bytes             | yes      | type="integer"; const=1048576                                                                                                                                                    |
| max_state_binding_updates_per_batch | yes      | type="integer"; const=20                                                                                                                                                         |
| policy_digest                       | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                          |
| record_type                         | yes      | const="EventStorePolicy"                                                                                                                                                         |
| runtime_overlay_compaction_depth    | yes      | type="integer"; const=64                                                                                                                                                         |
| state_binding_algorithm_contract    | yes      | const=schema(keys:delete_rule,delta_order,depth,depth_encoding,empty_leaf_domain_utf8,internal_node_domain_utf8,internal_node_rule,leaf_domain_utf8,leaf_key_rule,set_leaf_rule) |
| state_binding_identity_rules        | yes      | const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult,Grant,Lease,Relation,Run,Task)                                                                       |
| state_binding_value_rules           | yes      | const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult,Grant,Lease,Relation,Run,Task)                                                                       |

```
{
  "additionalProperties": false,
  "properties": {
    "allowed_state_binding_leaf_types": {
      "const": [
        "Activation",
        "Task",
        "Candidate",
        "Artifact",
        "Run",
        "Finding",
        "Decision",
        "GateResult",
        "Grant",
        "Lease",
        "Relation"
      ]
    },
    "authority_model_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_timestamp_contract": {
      "const": {
        "calendar": "proleptic-gregorian",
        "format": "utc-second-z",
        "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str",
        "fractional_seconds": "reject",
        "normalization": "identity-round-trip",
        "offset": "Z-only",
        "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime",
        "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}::[0-9]{2}::[0-9]{2}Z$",
        "real_calendar_validation": true,
        "resolution_seconds": 1,
        "year_max": 9999,
        "year_min": 1
      }
    },
    "command_conditional_fields": {
      "const": [
        "definition_digest"
      ]
    },
    "command_mutation_fields": {
      "const": [
        "workcard_task_id",
        "holder_authorization",
        "lease_id",
        "lease_generation",
        "fencing_token",
        "workcard_digest",
        "context_digest"
      ]
    }
  }
}
```

```

},
"command_required_fields": {
  "const": [
    "record_type",
    "command_id",
    "command_kind",
    "subject_id",
    "activation_digest",
    "idempotency_key",
    "requested_scope",
    "expected_head_digest",
    "issued_at",
    "payload",
    "intent_digest",
    "authorization"
  ]
},
"command_to_primary_event": {
  "const": [
    [
      "artifact.record",
      "artifact.recorded"
    ],
    [
      "candidate.record",
      "candidate.recorded"
    ],
    [
      "decision.record",
      "decision.recorded"
    ],
    [
      "finding.record",
      "finding.recorded"
    ],
    [
      "gate.record",
      "gate.recorded"
    ],
    [
      "grant.issue",
      "grant.issued"
    ],
    [
      "grant.revoke",
      "grant.revoked"
    ],
    [
      "lease.record",
      "lease.recorded"
    ],
    [
      "run.record",
      "run.recorded"
    ],
    [
      "task.record",
      "task.recorded"
    ],
    [
      "task.transition",
      "task.transitioned"
    ]
  ]
},
"derived_tail_batch_threshold": {
  "const": 128,
  "type": "integer"
},
"derived_tail_byte_threshold": {
  "const": 8388608,
  "type": "integer"
},
"genesis_event_semantic_digest": {
  "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"genesis_previous_authority_commitment": {

```

```

    "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
  },
  "lease_bound_command_kinds": {
    "const": [
      "artifact.record",
      "run.record",
      "gate.record",
      "finding.record"
    ]
  },
  "lease_bound_task_transition_states": {

```

```

    "const": [
      "LEASED",
      "RUNNING",
      "COMPLETED"
    ]
  },
  "max_command_bytes": {
    "const": 1048576,
    "type": "integer"
  },
  "max_envelope_bytes": {
    "const": 2097152,
    "type": "integer"
  },
  "max_events_per_batch": {
    "const": 20,
    "type": "integer"
  },
  "max_requested_scope_items": {
    "const": 32,
    "type": "integer"
  },
  "max_state_binding_bytes": {
    "const": 1048576,
    "type": "integer"
  },
  "max_state_binding_updates_per_batch": {
    "const": 20,

```

```

    "type": "integer"
  },
  "policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "EventStorePolicy"
  },
  "runtime_overlay_compaction_depth": {
    "const": 64,
    "type": "integer"
  },
  "state_binding_algorithm_contract": {
    "const": {
      "delete_rule": "delete maps the keyed leaf to empty_leaf",
      "delta_order": [
        "leaf_type-ascending",
        "leaf_id-ascending"
      ],
      "depth": 256,
      "depth_encoding": "unsigned-16-bit-big-endian",
      "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",
      "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
      "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
      "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
      "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
      "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
    }
  },

```

```

  },
  "state_binding_identity_rules": {
    "const": {
      "Activation": {
        "id_fields": [
          "activation_digest"
        ],
        "identity_mode": "direct"
      },
      "Artifact": {

```

```

    "id_fields": [
      "artifact_id"
    ],
    "identity_mode": "direct"
  },
  "Candidate": {
    "id_fields": [
      "candidate_id"
    ],
    "identity_mode": "direct"
  },
  "Decision": {
    "id_fields": [
      "decision_id"
    ],
    "identity_mode": "direct"
  },
  "Finding": {
    "id_fields": [
      "finding_id"
    ],
    "identity_mode": "direct"
  },
  "GateResult": {
    "digest_algorithm": "sha256",
    "digest_hex_chars": 48,
    "digest_input": "canonical-ordered-id-field-values",
    "id_fields": [
      "gate_id",
      "run_id"
    ],
    "identity_mode": "typed_digest_id",
    "prefix": "gate-result:"
  },
  "Grant": {
    "id_fields": [
      "grant_id"
    ],
    "identity_mode": "direct"
  },
  "Lease": {
    "id_fields": [
      "lease_id"
    ],
    "identity_mode": "direct"
  },
  "Relation": {
    "id_fields": [
      "relation_id"
    ],
    "identity_mode": "direct"
  },
  "Run": {
    "id_fields": [
      "run_id"
    ],
    "identity_mode": "direct"
  },
  "Task": {
    "id_fields": [
      "task_id"
    ],
    "identity_mode": "direct"
  }
},
"state_binding_value_rules": {
  "const": {
    "Activation": {
      "event_kinds": [],
      "value_definition": "Activation",
      "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence one and immutable for the journal"
    },
    "Artifact": {
      "event_kinds": [
        "artifact.recorded"
      ]
    }
  }
}

```



```

    ],
    "value_definition": "Artifact",
    "value_digest_rule": "sha256(canonical exact Event.payload Artifact)"
  },
  "Candidate": {
    "event_kinds": [
      "candidate.recorded"
    ],
    "value_definition": "Candidate",
    "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
  },
  "Decision": {
    "event_kinds": [
      "decision.recorded"
    ],
    "value_definition": "Decision",
    "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
  },
  "Finding": {
    "event_kinds": [
      "finding.recorded"
    ],
    "value_definition": "Finding",
    "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
  },
  "GateResult": {
    "event_kinds": [
      "gate.recorded"
    ],
    "value_definition": "GateResult",
    "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
  },
  "Grant": {
    "event_kinds": [
      "grant.issued",
      "grant.revoked"
    ],
    "value_definition": "GrantAuthorityState",
    "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
  },
  "Lease": {
    "event_kinds": [
      "lease.recorded"
    ],
    "value_definition": "Lease",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
  },
  "Relation": {
    "event_kinds": [
      "relation.recorded"
    ],
    "value_definition": "Relation",
    "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
  },
  "Run": {
    "event_kinds": [
      "run.recorded"
    ],
    "value_definition": "Run",
    "value_digest_rule": "sha256(canonical exact Event.payload Run)"
  },
  "Task": {
    "event_kinds": [
      "task.recorded",
      "task.transitioned"
    ],
    "value_definition": "Task",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
  }
}
},
"required": [
  "record_type",
  "authority_model_digest",
  "max_command_bytes",
  "max_envelope_bytes",

```

```

    "max_state_binding_bytes",
    "max_state_binding_updates_per_batch",
    "max_events_per_batch",
    "max_requested_scope_items",
    "derived_tail_batch_threshold",

    "derived_tail_byte_threshold",
    "runtime_overlay_compaction_depth",
    "command_required_fields",
    "command_conditional_fields",
    "command_mutation_fields",
    "lease_bound_command_kinds",
    "lease_bound_task_transition_states",
    "command_to_primary_event",
    "allowed_state_binding_leaf_types",
    "state_binding_identity_rules",
    "state_binding_algorithm_contract",
    "state_binding_value_rules",
    "canonical_timestamp_contract",
    "genesis_previous_authority_commitment",
    "genesis_event_semantic_digest",
    "policy_digest"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/EventStorePolicy | Validator: urn:promin:contracts:1#/\$defs/EventStorePolicy

## 34. EvidenceBinding

Type summary: type="object"; additionalProperties=false

Required: activation\_digest, implementation\_closure\_digest, candidate\_digest, policy\_digest, tool\_digest, input\_digests

| Property                      | Required | Structural rule                                          |
|-------------------------------|----------|----------------------------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"                                  |
| candidate_digest              | yes      | \$ref="#/\$defs/Digest"                                  |
| finding_digest                | no       | \$ref="#/\$defs/Digest"                                  |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| input_digests                 | yes      | type="array"; minItems=0; maxItems=128; uniqueItems=true |
| policy_digest                 | yes      | \$ref="#/\$defs/Digest"                                  |
| provider_invocations          | no       | type="array"; minItems=1; maxItems=32; uniqueItems=true  |
| tool_digest                   | yes      | \$ref="#/\$defs/Digest"                                  |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "finding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "input_digests": {
      "items": {
        "$ref": "#/$defs/Digest"
      },
      "maxItems": 128,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "policy_digest": {
      "$ref": "#/$defs/Digest"
    },
    "provider_invocations": {

```

```

    "items": {
      "$ref": "#/$defs/ProviderInvocationEvidence"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "tool_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "activation_digest",
  "implementation_closure_digest",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "input_digests"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/EvidenceBinding | Validator: urn:promin:contracts:1#/\$defs/EvidenceBinding

## 35. EvidenceProducerAttestation

Type summary: type="object"; additionalProperties=false

Required: record\_type, claim\_domain, candidate\_binding\_digest, evidence\_role, evidence\_record\_type, platform, invocation\_id, payload\_digest, raw\_artifact\_manifest\_digest, producer\_id, trust\_root\_id, signature\_provider\_id, key\_id, public\_key, nonce, signed\_at, signed\_claim\_digest, signature

| Property                     | Required | Structural rule                                                                                                                      |
|------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| candidate_binding_digest     | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| claim_domain                 | yes      | const="promin.release-evidence.producer-attestation.v1"                                                                              |
| evidence_record_type         | yes      | enum=["HumanDocumentVerification", "NoDegradationResult", "PlatformVerificationResult", "SaturationAudit", "SaturationEvidence"]     |
| evidence_role                | yes      | enum=["human-documents", "linux", "linux-no-degradation", "physical-scale", "saturation-audit", "windows", "windows-no-degradation"] |
| invocation_id                | yes      | \$ref="#/\$defs/Id"                                                                                                                  |
| key_id                       | yes      | \$ref="#/\$defs/Id"                                                                                                                  |
| nonce                        | yes      | type="string"; minLength=24; maxLength=128                                                                                           |
| payload_digest               | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| platform                     | yes      | enum=["cross-platform", "darwin", "linux", "windows"]                                                                                |
| producer_id                  | yes      | \$ref="#/\$defs/Id"                                                                                                                  |
| public_key                   | yes      | type="string"; pattern="^[A-Za-z0-9+/]{43}=\$"; minLength=44; maxLength=44                                                           |
| raw_artifact_manifest_digest | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| record_type                  | yes      | const="EvidenceProducerAttestation"                                                                                                  |
| signature                    | yes      | type="string"; minLength=80; maxLength=128                                                                                           |
| signature_provider_id        | yes      | const="cryptography-ed25519-v1"                                                                                                      |
| signed_at                    | yes      | \$ref="#/\$defs/Timestamp"                                                                                                           |
| signed_claim_digest          | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| trust_root_id                | yes      | \$ref="#/\$defs/Id"                                                                                                                  |

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "claim_domain": {
      "const": "promin.release-evidence.producer-attestation.v1"
    },
    "evidence_record_type": {
      "enum": [
        "HumanDocumentVerification",
        "NoDegradationResult",

```

```

        "PlatformVerificationResult",
        "SaturationAudit",
        "SaturationEvidence"
    ]
},
"evidence_role": {
    "enum": [
        "human-documents",
        "linux",
        "linux-no-degradation",
        "physical-scale",
        "saturation-audit",
        "windows",
        "windows-no-degradation"
    ]
}

```

```

},
"invocation_id": {
    "$ref": "#/$defs/Id"
},
"key_id": {
    "$ref": "#/$defs/Id"
},
"nonce": {
    "maxLength": 128,
    "minLength": 24,
    "type": "string"
},
"payload_digest": {
    "$ref": "#/$defs/Digest"
},
"platform": {
    "enum": [
        "cross-platform",
        "darwin",
        "linux",
        "windows"
    ]
},
"producer_id": {
    "$ref": "#/$defs/Id"
},
"public_key": {
    "maxLength": 44,

```

```

    "minLength": 44,
    "pattern": "^[A-Za-z0-9+/{43}]=$",
    "type": "string"
},
"raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"record_type": {
    "const": "EvidenceProducerAttestation"
},
"signature": {
    "maxLength": 128,
    "minLength": 80,
    "type": "string"
},
"signature_provider_id": {
    "const": "cryptography-ed25519-v1"
},
"signed_at": {
    "$ref": "#/$defs/Timestamp"
},
"signed_claim_digest": {
    "$ref": "#/$defs/Digest"
},
"trust_root_id": {
    "$ref": "#/$defs/Id"
}
},

```

```

"required": [
    "record_type",
    "claim_domain",
    "candidate_binding_digest",
    "evidence_role",
    "evidence_record_type",

```

```

    "platform",
    "invocation_id",
    "payload_digest",
    "raw_artifact_manifest_digest",
    "producer_id",
    "trust_root_id",
    "signature_provider_id",
    "key_id",
    "public_key",
    "nonce",
    "signed_at",
    "signed_claim_digest",
    "signature"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/EvidenceProducerAttestation | Validator: urn:promin:contracts:1#/\$defs/EvidenceProducerAttestation

## 36. FinalStaticAdmission

Type summary: type="object"; additionalProperties=false

Required: schema, checks, acceptance\_pass, pass\_credit, runtime\_validated

| Property          | Required | Structural rule                          |
|-------------------|----------|------------------------------------------|
| acceptance_pass   | yes      | const=false                              |
| checks            | yes      | type="array"; minItems=1                 |
| pass_credit       | yes      | const=false                              |
| runtime_validated | yes      | const=false                              |
| schema            | yes      | const="promin.final-static-admission.v1" |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "checks": {
      "minItems": 1,
      "type": "array"
    },
    "pass_credit": {
      "const": false
    },
    "runtime_validated": {
      "const": false
    },
    "schema": {
      "const": "promin.final-static-admission.v1"
    }
  },
  "required": [
    "schema",
    "checks",
    "acceptance_pass",
    "pass_credit",
    "runtime_validated"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/FinalStaticAdmission | Validator: urn:promin:contracts:1#/\$defs/FinalStaticAdmission

## 37. Finding

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: record\_type, finding\_id, status, severity, blocking, statement, activation\_digest, candidate\_digest, evidence\_artifacts, created\_at

| Property          | Required | Structural rule         |
|-------------------|----------|-------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest" |
| blocking          | yes      | type="boolean"          |

| Property                | Required | Structural rule                                         |
|-------------------------|----------|---------------------------------------------------------|
| candidate_digest        | yes      | \$ref="#/\$defs/Digest"                                 |
| created_at              | yes      | \$ref="#/\$defs/Timestamp"                              |
| disposition_decision_id | no       | \$ref="#/\$defs/Id"                                     |
| evidence_artifacts      | yes      | type="array"; minItems=1; maxItems=64; uniqueItems=true |
| finding_id              | yes      | \$ref="#/\$defs/Id"                                     |
| record_type             | yes      | const="Finding"                                         |
| severity                | yes      | enum=["P0", "P1", "P2", "P3"]                           |
| statement               | yes      | type="string"; minLength=1                              |
| status                  | yes      | enum=["OPEN", "RESOLVED", "WAIVED"]                     |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "disposition_decision_id"
          ]
        }
      },
      "if": {
        "properties": {
          "status": {
            "enum": [
              "RESOLVED",
              "WAIVED"
            ]
          }
        }
      },
      "then": {
        "properties": {
          "blocking": {
            "const": false
          }
        }
      },
      "required": [
        "disposition_decision_id"
      ]
    }
  ],
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "blocking": {
      "type": "boolean"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "disposition_decision_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "artifact_id": {
            "$ref": "#/$defs/Id"
          }
        }
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [

```

```

        "artifact_id",
        "artifact_record_digest"
    ],
    "type": "object"
  },
  "maxItems": 64,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"finding_id": {
  "$ref": "#/$defs/Id"
},
"record_type": {
  "const": "Finding"
},
"severity": {
  "enum": [
    "P0",
    "P1",
    "P2",
    "P3"
  ]
}
},
"statement": {
  "minLength": 1,
  "type": "string"
},
"status": {
  "enum": [
    "OPEN",
    "RESOLVED",
    "WAIVED"
  ]
}
},
"required": [
  "record_type",
  "finding_id",
  "status",
  "severity",
  "blocking",
  "statement",
  "activation_digest",
  "candidate_digest",
  "evidence_artifacts",
  "created_at"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Finding | Validator: urn:promin:contracts:1#/\$defs/Finding

## 38. FindingClassification

Type summary: enum=["PROVEN", "REVIEW", "SAFE"]

Required: -

```

{
  "enum": [
    "PROVEN",
    "REVIEW",
    "SAFE"
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/FindingClassification | Validator: urn:promin:contracts:1#/\$defs/FindingClassification

## 39. GateEvidenceArtifactBinding

Type summary: type="object"; additionalProperties=false

Required: artifact\_id, artifact\_record\_digest, run\_id, run\_digest

| Property               | Required | Structural rule         |
|------------------------|----------|-------------------------|
| artifact_id            | yes      | \$ref="#/\$defs/Id"     |
| artifact_record_digest | yes      | \$ref="#/\$defs/Digest" |

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required | Structural rule         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------|
| run_digest                                                                                                                                                                                                                                                                                                                                                                                                                                     | yes      | \$ref="#/\$defs/Digest" |
| run_id                                                                                                                                                                                                                                                                                                                                                                                                                                         | yes      | \$ref="#/\$defs/Id"     |
| <pre>{   "additionalProperties": false,   "properties": {     "artifact_id": {       "\$ref": "#/\$defs/Id"     },     "artifact_record_digest": {       "\$ref": "#/\$defs/Digest"     },     "run_digest": {       "\$ref": "#/\$defs/Digest"     },     "run_id": {       "\$ref": "#/\$defs/Id"     }   },   "required": [     "artifact_id",     "artifact_record_digest",     "run_id",     "run_digest"   ],   "type": "object" }</pre> |          |                         |
| Owner: core/contracts.schema.json#/\$defs/GateEvidenceArtifactBinding   Validator: urn:promin:contracts:1#/\$defs/GateEvidenceArtifactBinding                                                                                                                                                                                                                                                                                                  |          |                         |

## 40. GateResult

Type summary: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required

Required: record\_type, task\_id, gate\_id, run\_id, run\_digest, definition\_digest, status, outcome, pass\_credit, activation\_digest, candidate\_digest, policy\_digest, tool\_digest, evidence\_class, evidence\_artifacts

| Property                                                                                                                                                                                                                                | Required | Structural rule                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------|
| activation_digest                                                                                                                                                                                                                       | yes      | \$ref="#/\$defs/Digest"                                                   |
| candidate_digest                                                                                                                                                                                                                        | yes      | \$ref="#/\$defs/Digest"                                                   |
| definition_digest                                                                                                                                                                                                                       | yes      | \$ref="#/\$defs/Digest"                                                   |
| evidence_artifacts                                                                                                                                                                                                                      | yes      | type="array"; minItems=0; maxItems=64; uniqueItems=true                   |
| evidence_class                                                                                                                                                                                                                          | yes      | enum=["product-execution", "validator", "harness-generated", "migration"] |
| gate_id                                                                                                                                                                                                                                 | yes      | \$ref="#/\$defs/Id"                                                       |
| outcome                                                                                                                                                                                                                                 | yes      | enum=["pass", "fail", "blocked", "skipped"]                               |
| pass_credit                                                                                                                                                                                                                             | yes      | type="boolean"                                                            |
| policy_digest                                                                                                                                                                                                                           | yes      | \$ref="#/\$defs/Digest"                                                   |
| reason                                                                                                                                                                                                                                  | no       | type="string"; minLength=1; maxLength=2048                                |
| record_type                                                                                                                                                                                                                             | yes      | const="GateResult"                                                        |
| run_digest                                                                                                                                                                                                                              | yes      | \$ref="#/\$defs/Digest"                                                   |
| run_id                                                                                                                                                                                                                                  | yes      | \$ref="#/\$defs/Id"                                                       |
| status                                                                                                                                                                                                                                  | yes      | enum=["pass", "fail", "blocked", "skipped"]                               |
| task_id                                                                                                                                                                                                                                 | yes      | \$ref="#/\$defs/Id"                                                       |
| tool_digest                                                                                                                                                                                                                             | yes      | \$ref="#/\$defs/Digest"                                                   |
| <pre>{   "additionalProperties": false,   "oneOf": [     {       "properties": {         "evidence_artifacts": {           "minItems": 1         },         "outcome": {           "const": "pass"         }       },     }   ] }</pre> |          |                                                                           |



```

        "status": {
          "const": "pass"
        }
      },
      "required": [
        "status",
        "outcome",
        "pass_credit"
      ]
    },
    {
      "properties": {
        "evidence_artifacts": {
          "minItems": 1
        },
        "outcome": {
          "const": "fail"

```

```

      },
      "pass_credit": {
        "const": false
      },
      "status": {
        "const": "fail"
      }
    },
    "required": [
      "status",
      "outcome",
      "pass_credit"
    ]
  },
  {
    "properties": {
      "evidence_artifacts": {
        "minItems": 1
      },
      "outcome": {
        "const": "blocked"
      },
      "pass_credit": {
        "const": false
      },
      "status": {
        "const": "blocked"
      }

```

```

    },
    "required": [
      "status",
      "outcome",
      "pass_credit",
      "reason"
    ]
  },
  {
    "properties": {
      "evidence_artifacts": {
        "maxItems": 0
      },
      "outcome": {
        "const": "skipped"
      },
      "pass_credit": {
        "const": false
      },
      "status": {
        "const": "skipped"
      }
    },
    "required": [
      "status",
      "outcome",
      "pass_credit",
      "reason"

```

```

  ]
},
"properties": {

```

```

"activation_digest": {
  "$ref": "#/$defs/Digest"
},
"candidate_digest": {
  "$ref": "#/$defs/Digest"
},
"definition_digest": {
  "$ref": "#/$defs/Digest"
},
"evidence_artifacts": {
  "items": {
    "$ref": "#/$defs/GateEvidenceArtifactBinding"
  },
  "maxItems": 64,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"evidence_class": {
  "enum": [
    "product-execution",
    "validator",
    "harness-generated",
    "migration"

```

```

]
},
"gate_id": {
  "$ref": "#/$defs/Id"
},
"outcome": {
  "enum": [
    "pass",
    "fail",
    "blocked",
    "skipped"
  ]
},
"pass_credit": {
  "type": "boolean"
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"reason": {
  "maxLength": 2048,
  "minLength": 1,
  "type": "string"
},
"record_type": {
  "const": "GateResult"
},
"run_digest": {

```

```

  "$ref": "#/$defs/Digest"
},
"run_id": {
  "$ref": "#/$defs/Id"
},
"status": {
  "enum": [
    "pass",
    "fail",
    "blocked",
    "skipped"
  ]
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"tool_digest": {
  "$ref": "#/$defs/Digest"
}
},
"required": [
  "record_type",
  "task_id",
  "gate_id",
  "run_id",
  "run_digest",

```

```

    "definition_digest",
    "status",

    "outcome",
    "pass_credit",
    "activation_digest",
    "candidate_digest",
    "policy_digest",
    "tool_digest",
    "evidence_class",
    "evidence_artifacts"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GateResult | Validator: urn:promin:contracts:1#/\$defs/GateResult

## 41. GateRunDefinition

Type summary: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required; allOf[1]=keys:if,then

Required: definition\_kind, definition\_id, owner\_kind, owner\_digest, defined\_at\_head\_digest, gate\_id, run\_kind, expected\_evidence\_class, expected\_evidence\_purpose, product\_credit\_required, target\_kind, target\_digest, target\_scope, candidate\_digest, policy\_digest, tool\_digest, implementation\_closure\_digest, provider\_binding\_digest, input\_digests, activation\_digest

| Property                      | Required | Structural rule                                                           |
|-------------------------------|----------|---------------------------------------------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"                                                   |
| candidate_digest              | yes      | \$ref="#/\$defs/Digest"                                                   |
| defined_at_head_digest        | yes      | oneOf[2]=#/\$defs/Digest   type:null                                      |
| definition_id                 | yes      | \$ref="#/\$defs/Id"                                                       |
| definition_kind               | yes      | const="GateRunDefinition"                                                 |
| expected_evidence_class       | yes      | enum=["product-execution", "validator", "harness-generated", "migration"] |
| expected_evidence_purpose     | yes      | enum=["diagnostic", "gate", "product"]                                    |
| gate_id                       | yes      | \$ref="#/\$defs/Id"                                                       |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                                                   |
| input_digests                 | yes      | type="array"; minItems=0; maxItems=128; uniqueItems=true                  |
| owner_digest                  | yes      | \$ref="#/\$defs/Digest"                                                   |
| owner_kind                    | yes      | const="Task"                                                              |
| policy_digest                 | yes      | \$ref="#/\$defs/Digest"                                                   |
| product_credit_required       | yes      | type="boolean"                                                            |
| provider_binding_digest       | yes      | \$ref="#/\$defs/Digest"                                                   |
| run_kind                      | yes      | enum=["execution", "validation"]                                          |
| target_digest                 | yes      | \$ref="#/\$defs/Digest"                                                   |
| target_kind                   | yes      | enum=["candidate", "finding"]                                             |
| target_scope                  | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true                   |
| tool_digest                   | yes      | \$ref="#/\$defs/Digest"                                                   |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "product_credit_required": {
            "const": true
          }
        },
        "required": [
          "product_credit_required"
        ]
      },
      "then": {
        "properties": {
          "expected_evidence_class": {
            "const": "product-execution"
          },

```

```

        "expected_evidence_purpose": {
            "const": "product"
        }
    }
}
],
"oneOf": [
{

```

```

    "properties": {
        "expected_evidence_class": {
            "const": "validator"
        },
        "expected_evidence_purpose": {
            "const": "diagnostic"
        }
    },
    "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
},
{
    "properties": {
        "expected_evidence_class": {
            "const": "validator"
        },
        "expected_evidence_purpose": {
            "const": "gate"
        }
    },
    "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
},
{

```

```

    "properties": {
        "expected_evidence_class": {
            "const": "product-execution"
        },
        "expected_evidence_purpose": {
            "const": "gate"
        }
    },
    "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
},
{
    "properties": {
        "expected_evidence_class": {
            "const": "product-execution"
        },
        "expected_evidence_purpose": {
            "const": "product"
        }
    },
    "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
    ]
}
],

```

```

"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
        "$ref": "#/$defs/Digest"
    },
    "defined_at_head_digest": {
        "oneOf": [
            {
                "$ref": "#/$defs/Digest"
            },

```

```

    {
      "type": "null"
    }
  ],
  "definition_id": {
    "$ref": "#/$defs/Id"
  },
  "definition_kind": {
    "const": "GateRunDefinition"
  },
  "expected_evidence_class": {
    "enum": [
      "product-execution",
      "validator",
      "harness-generated",

```

```

      "migration"
    ]
  },
  "expected_evidence_purpose": {
    "enum": [
      "diagnostic",
      "gate",
      "product"
    ]
  },
  "gate_id": {
    "$ref": "#/$defs/Id"
  },
  "implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "input_digests": {
    "items": {
      "$ref": "#/$defs/Digest"
    },
    "maxItems": 128,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "owner_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

  "owner_kind": {
    "const": "Task"
  },
  "policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "product_credit_required": {
    "type": "boolean"
  },
  "provider_binding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "run_kind": {
    "enum": [
      "execution",
      "validation"
    ]
  },
  "target_digest": {
    "$ref": "#/$defs/Digest"
  },
  "target_kind": {
    "enum": [
      "candidate",
      "finding"
    ]
  },
  "target_scope": {

```

```

    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,

```

```

    "type": "array",
    "uniqueItems": true
  },
  "tool_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "definition_kind",
  "definition_id",
  "owner_kind",
  "owner_digest",
  "defined_at_head_digest",
  "gate_id",
  "run_kind",
  "expected_evidence_class",
  "expected_evidence_purpose",
  "product_credit_required",
  "target_kind",
  "target_digest",
  "target_scope",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests",
  "activation_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GateRunDefinition | Validator: urn:promin:contracts:1#/\$defs/GateRunDefinition

## 42. Grant

Type summary: type="object"; additionalProperties=false

Required: record\_type, grant\_id, subject\_id, capability\_id, scope, activation\_digest, issued\_at, expires\_at, nonce, claim\_digest, trust\_proofs

| Property          | Required | Structural rule                                                                 |
|-------------------|----------|---------------------------------------------------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                                                         |
| capability_id     | yes      | enum=[16 items]                                                                 |
| claim_digest      | yes      | \$ref="#/\$defs/Digest"                                                         |
| expires_at        | yes      | \$ref="#/\$defs/Timestamp"                                                      |
| grant_id          | yes      | \$ref="#/\$defs/Id"                                                             |
| issued_at         | yes      | \$ref="#/\$defs/Timestamp"                                                      |
| nonce             | yes      | type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"; minLength=16; maxLength=128 |
| record_type       | yes      | const="Grant"                                                                   |
| scope             | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true                         |
| subject_id        | yes      | \$ref="#/\$defs/Id"                                                             |
| trust_proofs      | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true                         |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capability_id": {
      "enum": [
        "standard.activate",
        "authority.manage",
        "task.plan",
        "task.execute",
        "lease.manage",
        "evidence.publish",
        "validation.evaluate",
        "finding.record",
        "finding.resolve",
        "finding.waive",

```

```

        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read"
    ]
},
"claim_digest": {
    "$ref": "#/$defs/Digest"
},
"expires_at": {
    "$ref": "#/$defs/Timestamp"
},
"grant_id": {
    "$ref": "#/$defs/Id"
},
"issued_at": {
    "$ref": "#/$defs/Timestamp"
},
"nonce": {
    "maxLength": 128,
    "minLength": 16,
    "pattern": "^[A-Za-z0-9._:-]{16,128}$",
    "type": "string"
},
"record_type": {
    "const": "Grant"
},
"scope": {
    "items": {
        "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"subject_id": {
    "$ref": "#/$defs/Id"
},
"trust_proofs": {
    "items": {
        "$ref": "#/$defs/GrantTrustProof"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
}
},
"required": [
    "record_type",
    "grant_id",
    "subject_id",
    "capability_id",
    "scope",
    "activation_digest",
    "issued_at",
    "expires_at",
    "nonce",
    "claim_digest",
    "trust_proofs"
],
"type": "object"
}

```

Owner: `core/contracts.schema.json#/$defs/Grant` | Validator: `urn:promin:contracts:1#/$defs/Grant`

## 43. GrantAuthorityState

Type summary: type="object"; additionalProperties=false

Required: record\_type, grant, revocation

| Property | Required | Structural rule                     |
|----------|----------|-------------------------------------|
| grant    | yes      | <code>\$ref="#/\$defs/Grant"</code> |

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Required | Structural rule                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------|
| record_type                                                                                                                                                                                                                                                                                                                                                                                                                                                 | yes      | const="GrantAuthorityState"                   |
| revocation                                                                                                                                                                                                                                                                                                                                                                                                                                                  | yes      | oneOf[2]=#/\$defs/GrantRevocation   type:null |
| <pre> {   "additionalProperties": false,   "properties": {     "grant": {       "\$ref": "#/\$defs/Grant"     },     "record_type": {       "const": "GrantAuthorityState"     },     "revocation": {       "oneOf": [         {           "\$ref": "#/\$defs/GrantRevocation"         },         {           "type": "null"         }       ]     }   },   "required": [     "record_type",     "grant",     "revocation"   ],   "type": "object" } </pre> |          |                                               |
| Owner: core/contracts.schema.json#/\$defs/GrantAuthorityState   Validator: urn:promin:contracts:1#/\$defs/GrantAuthorityState                                                                                                                                                                                                                                                                                                                               |          |                                               |

## 44. GrantAuthorization

Type summary: type="object"; additionalProperties=false

Required: kind, grant\_id, grant\_claim\_digest

| Property                                                                                                                                                                                                                                                                                                                                   | Required | Structural rule         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------|
| grant_claim_digest                                                                                                                                                                                                                                                                                                                         | yes      | \$ref="#/\$defs/Digest" |
| grant_id                                                                                                                                                                                                                                                                                                                                   | yes      | \$ref="#/\$defs/Id"     |
| kind                                                                                                                                                                                                                                                                                                                                       | yes      | const="grant"           |
| <pre> {   "additionalProperties": false,   "properties": {     "grant_claim_digest": {       "\$ref": "#/\$defs/Digest"     },     "grant_id": {       "\$ref": "#/\$defs/Id"     },     "kind": {       "const": "grant"     }   },   "required": [     "kind",     "grant_id",     "grant_claim_digest"   ],   "type": "object" } </pre> |          |                         |
| Owner: core/contracts.schema.json#/\$defs/GrantAuthorization   Validator: urn:promin:contracts:1#/\$defs/GrantAuthorization                                                                                                                                                                                                                |          |                         |

## 45. GrantRevocation

Type summary: type="object"; additionalProperties=false

Required: grant\_id, decision\_id, reason, revoked\_at

| Property    | Required | Structural rule     |
|-------------|----------|---------------------|
| decision_id | yes      | \$ref="#/\$defs/Id" |



| Property   | Required | Structural rule            |
|------------|----------|----------------------------|
| grant_id   | yes      | \$ref="#/\$defs/Id"        |
| reason     | yes      | type="string"; minLength=1 |
| revoked_at | yes      | \$ref="#/\$defs/Timestamp" |

```

{
  "additionalProperties": false,
  "properties": {
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "revoked_at": {
      "$ref": "#/$defs/Timestamp"
    }
  },
  "required": [
    "grant_id",
    "decision_id",
    "reason",
    "revoked_at"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/GrantRevocation | Validator: urn:promin:contracts:1#/\$defs/GrantRevocation

## 46. GrantTrustProof

Type summary: oneOf[3]=#/\$defs/IssuerGrantTrustProof | #/\$defs/LocalTrustProof | #/\$defs/SignatureTrustProof

Required: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/IssuerGrantTrustProof"
    },
    {
      "$ref": "#/$defs/LocalTrustProof"
    },
    {
      "$ref": "#/$defs/SignatureTrustProof"
    }
  ]
}

```

Owner: core/contracts.schema.json#/\$defs/GrantTrustProof | Validator: urn:promin:contracts:1#/\$defs/GrantTrustProof

## 47. Healthcheck

Type summary: type="object"; additionalProperties=false

Required: argv, timeout\_ms, expected\_exit

| Property      | Required | Structural rule                            |
|---------------|----------|--------------------------------------------|
| argv          | yes      | type="array"; minItems=1; maxItems=16      |
| expected_exit | yes      | type="integer"; minimum=0; maximum=255     |
| timeout_ms    | yes      | type="integer"; minimum=100; maximum=60000 |

```

{
  "additionalProperties": false,
  "properties": {
    "argv": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 1,
      "type": "array"
    }
  }
}

```

```

    },
    "expected_exit": {
      "maximum": 255,
      "minimum": 0,
      "type": "integer"
    },
    "timeout_ms": {
      "maximum": 60000,
      "minimum": 100,
      "type": "integer"
    }
  },
  "required": [
    "argv",
    "timeout_ms",
    "expected_exit"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Healthcheck | Validator: urn:promin:contracts:1#/\$defs/Healthcheck

## 48. HumanDocumentVerification

Type summary: type="object"; additionalProperties=false

Required: record\_type, status, candidate\_binding\_digest, documents, deterministic\_rebuild, rebuild\_digest, page\_count, extraction\_diagnostics, visual\_review\_scope, font\_bindings, generator\_result\_digest, render\_environment, render\_manifest, render\_manifest\_digest, product\_acceptance\_pass, producer, invocation, raw\_artifact\_manifest\_digest, producer\_attestation, result\_digest

| Property                     | Required | Structural rule                                           |
|------------------------------|----------|-----------------------------------------------------------|
| candidate_binding_digest     | yes      | \$ref="#/\$defs/Digest"                                   |
| deterministic_rebuild        | yes      | const=true                                                |
| documents                    | yes      | type="array"; minItems=4; maxItems=4; uniqueItems=true    |
| extraction_diagnostics       | yes      | type="object"; additionalProperties=false                 |
| font_bindings                | yes      | type="object"; additionalProperties=false                 |
| generator_result_digest      | yes      | \$ref="#/\$defs/Digest"                                   |
| invocation                   | yes      | \$ref="#/\$defs/ReleaseEvidenceInvocation"                |
| page_count                   | yes      | type="integer"; minimum=1                                 |
| producer                     | yes      | \$ref="#/\$defs/ReleaseEvidenceProducer"                  |
| producer_attestation         | yes      | \$ref="#/\$defs/EvidenceProducerAttestation"              |
| product_acceptance_pass      | yes      | const=false                                               |
| raw_artifact_manifest_digest | yes      | \$ref="#/\$defs/Digest"                                   |
| rebuild_digest               | yes      | \$ref="#/\$defs/Digest"                                   |
| record_type                  | yes      | const="HumanDocumentVerification"                         |
| render_environment           | yes      | type="object"; additionalProperties=false                 |
| render_manifest              | yes      | type="array"; minItems=4; maxItems=4096; uniqueItems=true |
| render_manifest_digest       | yes      | \$ref="#/\$defs/Digest"                                   |
| result_digest                | yes      | \$ref="#/\$defs/Digest"                                   |
| status                       | yes      | const="pass"                                              |
| visual_review_scope          | yes      | type="object"; additionalProperties=false                 |

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "deterministic_rebuild": {
      "const": true
    },
    "documents": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "blank_pages": {

```

```

    "maxItems": 0,
    "type": "array"
  },
  "extracted_characters": {
    "minimum": 1,
    "type": "integer"
  },
  "extraction_errors": {
    "maxItems": 0,
    "type": "array"
  },
  "page_count": {
    "minimum": 1,
    "type": "integer"
  }

```

```

  },
  "path": {
    "enum": [
      "human/promin_appendices_en.pdf",
      "human/promin_appendices_ua.pdf",
      "human/promin_main_en.pdf",
      "human/promin_main_ua.pdf"
    ]
  },
  "sha256": {
    "$ref": "#/$defs/Digest"
  },
  "size_bytes": {
    "minimum": 1,
    "type": "integer"
  }
},
"required": [
  "path",
  "sha256",
  "size_bytes",
  "page_count",
  "extracted_characters",
  "blank_pages",
  "extraction_errors"
],
"type": "object"
},

```

```

    "maxItems": 4,
    "minItems": 4,
    "type": "array",
    "uniqueItems": true
  },
  "extraction_diagnostics": {
    "additionalProperties": false,
    "properties": {
      "blank_pages": {
        "maxItems": 0,
        "type": "array"
      },
      "documents_parsed": {
        "const": 4
      },
      "errors": {
        "maxItems": 0,
        "type": "array"
      },
      "total_extracted_characters": {
        "minimum": 1,
        "type": "integer"
      }
    }
  },
  "required": [
    "documents_parsed",
    "total_extracted_characters",
    "blank_pages",

```

```

    "errors"
  ],
  "type": "object"
},
"font_bindings": {
  "additionalProperties": false,
  "properties": {

```

```

"bold": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"

```

```

},
"italic": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
},
"mono": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,

```

```

      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
},
"regular": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "pdf_font_name": {
      "minLength": 1,
      "type": "string"
    },
  },

```

```

        "sha256": {
            "$ref": "#/$defs/Digest"
        },
    },
    "required": [
        "bytes",
        "pdf_font_name",
        "sha256"
    ],
    "type": "object"
},
{
    "required": [
        "regular",
        "bold",
        "italic",
        "mono"
    ],
    "type": "object"
},
"generator_result_digest": {
    "$ref": "#/$defs/Digest"
},
"invocation": {
    "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"page_count": {
    "minimum": 1,
    "type": "integer"
},
"producer": {
    "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
    "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
    "const": false
},
"raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"rebuild_digest": {
    "$ref": "#/$defs/Digest"
},
"record_type": {
    "const": "HumanDocumentVerification"
},
"render_environment": {
    "additionalProperties": false,
    "properties": {
        "font_binding_digest": {
            "$ref": "#/$defs/Digest"
        },
        "generator_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "platform": {
            "minLength": 1,
            "type": "string"
        },
        "python_version": {
            "minLength": 1,
            "type": "string"
        },
        "reportlab_version": {
            "minLength": 1,
            "type": "string"
        }
    }
},
"required": [
    "generator_sha256",
    "reportlab_version",
    "python_version",
    "platform",
    "font_binding_digest"
],
"type": "object"

```

```

},
"render_manifest": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "document_path": {

```

```

        "enum": [
          "human/promin_appendices_en.pdf",
          "human/promin_appendices_ua.pdf",
          "human/promin_main_en.pdf",
          "human/promin_main_ua.pdf"
        ]
      },
      "image_bytes": {
        "maximum": 33554432,
        "minimum": 1,
        "type": "integer"
      },
      "image_path": {
        "$ref": "#/$defs/RelativePath"
      },
      "image_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "page": {
        "minimum": 1,
        "type": "integer"
      }
    }
  },
  "required": [
    "document_path",
    "page",
    "image_path",
    "image_sha256",

```

```

    "image_bytes"
  ],
  "type": "object"
},
"maxItems": 4096,
"minItems": 4,
"type": "array",
"uniqueItems": true
},
"render_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
},
"visual_review_scope": {
  "additionalProperties": false,
  "properties": {
    "clipping_detected": {
      "const": false
    },
    "completed": {
      "const": true
    }
  },
  "pages_reviewed": {

```

```

    "minimum": 1,
    "type": "integer"
  },
  "render_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rendered_page_count": {
    "minimum": 1,
    "type": "integer"
  },
  "reviewed_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "reviewer_id": {
    "$ref": "#/$defs/Id"
  }
}

```

```

    },
    "unreadable_text_detected": {
      "const": false
    }
  },
  "required": [
    "completed",
    "reviewer_id",
    "reviewed_at",
    "rendered_page_count",
    "pages_reviewed",
    "clipping_detected",
    "unreadable_text_detected",
    "render_manifest_digest"
  ],
  "type": "object"
}
},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "documents",
  "deterministic_rebuild",
  "rebuild_digest",
  "page_count",
  "extraction_diagnostics",
  "visual_review_scope",
  "font_bindings",
  "generator_result_digest",
  "render_environment",
  "render_manifest",
  "render_manifest_digest",
  "product_acceptance_pass",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/HumanDocumentVerification | Validator: urn:promin:contracts:1#/\$defs/HumanDocumentVerification

## 49. Id

Type summary: type="string"; pattern="^[A-Za-z0-9][A-Za-z0-9.\_-]{0,127}\$"

Required: -

```

{
  "pattern": "^[A-Za-z0-9][A-Za-z0-9._-]{0,127}$",
  "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/Id | Validator: urn:promin:contracts:1#/\$defs/Id

## 50. ImplementationClosure

Type summary: type="object"; additionalProperties=false

Required: runtime, interpreter, components, adapters, closure\_digest

| Property       | Required | Structural rule                                           |
|----------------|----------|-----------------------------------------------------------|
| adapters       | yes      | type="array"; minItems=1; maxItems=1; uniqueItems=true    |
| closure_digest | yes      | \$ref="#/\$defs/Digest"                                   |
| components     | yes      | type="array"; minItems=1; maxItems=4096; uniqueItems=true |
| interpreter    | yes      | type="object"; additionalProperties=false                 |
| runtime        | yes      | type="object"; additionalProperties=false                 |

```

{
  "additionalProperties": false,
  "properties": {
    "adapters": {

```

```

"items": {
  "additionalProperties": false,
  "properties": {
    "adapter_id": {
      "$ref": "#/$defs/Id"
    },
    "capability_id": {
      "enum": [
        "control-runtime",
        "shape-validation",
        "content-identity",
        "local-serialization",
        "query-projection",
        "filesystem-inventory",
        "export-scan",
        "signature",
        "build-dependency"
      ]
    },
    "dependency_receipt_digest": {
      "$ref": "#/$defs/Digest"
    },
    "digest": {
      "$ref": "#/$defs/Digest"
    }
  }
},

```

```

    "operation_contract_digest": {
      "$ref": "#/$defs/Digest"
    },
    "protocol_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "adapter_id",
    "capability_id",
    "version",
    "digest",
    "protocol_id",
    "operation_contract_digest",
    "dependency_receipt_digest"
  ],
  "type": "object"
},
"maxItems": 1,
"minItems": 1,
"type": "array",
"uniqueItems": true
},

```

```

"closure_digest": {
  "$ref": "#/$defs/Digest"
},
"components": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "component_id": {
        "$ref": "#/$defs/Id"
      },
      "component_kind": {
        "enum": [
          "native-runtime",
          "provider-file",
          "provider-tree",
          "python-distribution"
        ]
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "size_bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "source": {

```



```

        "maxLength": 4096,
        "minLength": 1,

        "type": "string"
    },
    "version": {
        "maxLength": 256,
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "component_id",
    "component_kind",
    "source",
    "version",
    "digest",
    "size_bytes"
],
"type": "object"
},
"maxItems": 4096,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"interpreter": {
    "additionalProperties": false,
    "properties": {
        "executable_digest": {
            "$ref": "#/$defs/Digest"

```

```

    },
    "name": {
        "minLength": 1,
        "type": "string"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "name",
    "version",
    "executable_digest"
],
"type": "object"
},
"runtime": {
    "additionalProperties": false,
    "properties": {
        "digest": {
            "$ref": "#/$defs/Digest"
        },
        "name": {
            "const": "promin"
        },
        "version": {
            "minLength": 1,

```

```

        "type": "string"
    }
},
"required": [
    "name",
    "version",
    "digest"
],
"type": "object"
}
},
"required": [
    "runtime",
    "interpreter",
    "components",
    "adapters",
    "closure_digest"
],

```

```
"type": "object"
}
```

Owner: [core/contracts.schema.json#/\\$defs/ImplementationClosure](#) | Validator: [urn:promin:contracts:1#/\\$defs/ImplementationClosure](#)

## 51. InitFileDigests

Type summary: type="object"; additionalProperties=false

Required: project.json, standards.json, technologies.json, authority.json

| Property          | Required | Structural rule         |
|-------------------|----------|-------------------------|
| authority.json    | yes      | \$ref="#/\$defs/Digest" |
| project.json      | yes      | \$ref="#/\$defs/Digest" |
| standards.json    | yes      | \$ref="#/\$defs/Digest" |
| technologies.json | yes      | \$ref="#/\$defs/Digest" |

```
{
  "additionalProperties": false,
  "properties": {
    "authority.json": {
      "$ref": "#/$defs/Digest"
    },
    "project.json": {
      "$ref": "#/$defs/Digest"
    },
    "standards.json": {
      "$ref": "#/$defs/Digest"
    },
    "technologies.json": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "project.json",
    "standards.json",
    "technologies.json",
    "authority.json"
  ],
  "type": "object"
}
```

Owner: [core/contracts.schema.json#/\\$defs/InitFileDigests](#) | Validator: [urn:promin:contracts:1#/\\$defs/InitFileDigests](#)

## 52. InventoryInputManifest

Type summary: type="object"; additionalProperties=false

Required: record\_type, candidate\_digest, inventory\_digest, stream\_artifact\_id, stream\_artifact\_record\_digest, stream\_digest, stream\_bytes, entry\_count, row\_schema\_digest, consistency\_mode, snapshot\_provider\_id, snapshot\_digest

| Property                      | Required | Structural rule                                          |
|-------------------------------|----------|----------------------------------------------------------|
| candidate_digest              | yes      | \$ref="#/\$defs/Digest"                                  |
| consistency_mode              | yes      | enum=["immutable-vcs-tree", "observational-best-effort"] |
| entry_count                   | yes      | type="integer"; minimum=0                                |
| inventory_digest              | yes      | \$ref="#/\$defs/Digest"                                  |
| record_type                   | yes      | const="InventoryInputManifest"                           |
| row_schema_digest             | yes      | \$ref="#/\$defs/Digest"                                  |
| snapshot_digest               | yes      | \$ref="#/\$defs/Digest"                                  |
| snapshot_provider_id          | yes      | \$ref="#/\$defs/Id"                                      |
| stream_artifact_id            | yes      | \$ref="#/\$defs/Id"                                      |
| stream_artifact_record_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| stream_bytes                  | yes      | type="integer"; minimum=1                                |
| stream_digest                 | yes      | \$ref="#/\$defs/Digest"                                  |

```
{
  "additionalProperties": false,
  "properties": {
    "candidate_digest": {
```

```

    "$ref": "#/$defs/Digest"
  },
  "consistency_mode": {
    "enum": [
      "immutable-vcs-tree",
      "observational-best-effort"
    ]
  },
  "entry_count": {
    "minimum": 0,
    "type": "integer"
  },
  "inventory_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "InventoryInputManifest"
  },
  "row_schema_digest": {
    "$ref": "#/$defs/Digest"
  },
  "snapshot_digest": {
    "$ref": "#/$defs/Digest"
  },
}

```

```

"snapshot_provider_id": {
  "$ref": "#/$defs/Id"
},
"stream_artifact_id": {
  "$ref": "#/$defs/Id"
},
"stream_artifact_record_digest": {
  "$ref": "#/$defs/Digest"
},
"stream_bytes": {
  "minimum": 1,
  "type": "integer"
},
"stream_digest": {
  "$ref": "#/$defs/Digest"
}
},
"required": [
  "record_type",
  "candidate_digest",
  "inventory_digest",
  "stream_artifact_id",
  "stream_artifact_record_digest",
  "stream_digest",
  "stream_bytes",
  "entry_count",
  "row_schema_digest",
  "consistency_mode",

```

```

    "snapshot_provider_id",
    "snapshot_digest"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "transient-ingress-only"
}

```

Owner: core/contracts.schema.json#/\$defs/InventoryInputManifest | Validator: urn:promin:contracts:1#/\$defs/InventoryInputManifest

## 53. InventoryJsonlRow

Type summary: type="object"; additionalProperties=false

Required: path, digest, size, search\_text

| Property    | Required | Structural rule                            |
|-------------|----------|--------------------------------------------|
| digest      | yes      | \$ref="#/\$defs/Digest"                    |
| path        | yes      | \$ref="#/\$defs/RelativePath"              |
| search_text | yes      | type="string"; minLength=1; maxLength=4096 |
| size        | yes      | type="integer"; minimum=0                  |

```

{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "search_text": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "size": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "path",
    "digest",
    "size",
    "search_text"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/InventoryJsonlRow* | Validator: *urn:promin:contracts:1#/\$defs/InventoryJsonlRow*

## 54. InventoryProjectionRow

Type summary: type="object"; additionalProperties=false

Required: record\_type, path, digest, size, search\_text, semantic\_proxy

| Property       | Required | Structural rule                            |
|----------------|----------|--------------------------------------------|
| digest         | yes      | \$ref="#/\$defs/Digest"                    |
| path           | yes      | \$ref="#/\$defs/RelativePath"              |
| record_type    | yes      | const="InventoryProjectionRow"             |
| search_text    | yes      | type="string"; minLength=1; maxLength=4096 |
| semantic_proxy | yes      | type="object"; additionalProperties=false  |
| size           | yes      | type="integer"; minimum=0                  |

```

{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "record_type": {
      "const": "InventoryProjectionRow"
    },
    "search_text": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "semantic_proxy": {
      "additionalProperties": false,
      "properties": {
        "entity_type": {
          "const": "Artifact"
        },
        "id": {
          "$ref": "#/$defs/Id"
        },
        "payload": {
          "$ref": "#/$defs/Artifact"
        }
      },
      "required": [
        "id",

```

```

        "entity_type",
        "payload"
    ],
    "type": "object"
},
"size": {
    "minimum": 0,
    "type": "integer"
}
},
"required": [
    "record_type",
    "path",
    "digest",
    "size",
    "search_text",
    "semantic_proxy"
],
"type": "object",
"x-promin-ingress-operations": [
    "rebuild"
],
"x-promin-persistence": "derived-rebuild-only"
}

```

Owner: `core/contracts.schema.json#/$defs/InventoryProjectionRow` | Validator: `urn:promin:contracts:1#/$defs/InventoryProjectionRow`

## 55. Invocation

Type summary: `type="object"; additionalProperties=false`

Required: kind, value

| Property | Required | Structural rule                                                            |
|----------|----------|----------------------------------------------------------------------------|
| kind     | yes      | enum=["python-runtime", "python-module", "executable", "platform-service"] |
| value    | yes      | type="string"; minLength=1                                                 |

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
      ]
    },
    "value": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "kind",
    "value"
  ],
  "type": "object"
}

```

Owner: `core/contracts.schema.json#/$defs/Invocation` | Validator: `urn:promin:contracts:1#/$defs/Invocation`

## 56. IssuerGrantTrustProof

Type summary: `type="object"; additionalProperties=false`

Required: kind, issuer\_grant\_id, issuer\_signed\_claim\_digest, signed\_claim\_digest

| Property                   | Required | Structural rule                      |
|----------------------------|----------|--------------------------------------|
| issuer_grant_id            | yes      | <code>\$ref="#/\$defs/Id"</code>     |
| issuer_signed_claim_digest | yes      | <code>\$ref="#/\$defs/Digest"</code> |
| kind                       | yes      | <code>const="issuer-grant"</code>    |
| signed_claim_digest        | yes      | <code>\$ref="#/\$defs/Digest"</code> |

```

{
  "additionalProperties": false,

```

```

"properties": {
  "issuer_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "issuer_signed_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "kind": {
    "const": "issuer-grant"
  },
  "signed_claim_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "kind",
  "issuer_grant_id",
  "issuer_signed_claim_digest",
  "signed_claim_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/IssuerGrantTrustProof | Validator: urn:promin:contracts:1#/\$defs/IssuerGrantTrustProof

## 57. LanguageCapabilityProfile

Type summary: type="object"; additionalProperties=false

Required: schema, profileId, languages, authority\_effect

| Property         | Required | Structural rule                               |
|------------------|----------|-----------------------------------------------|
| authority_effect | yes      | const="none"                                  |
| languages        | yes      | type="array"; minItems=1; uniqueItems=true    |
| profileId        | yes      | \$ref="#/\$defs/Id"                           |
| schema           | yes      | const="promin.language-capability-profile.v1" |

```

{
  "additionalProperties": false,
  "properties": {
    "authority_effect": {
      "const": "none"
    },
    "languages": {
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "profileId": {
      "$ref": "#/$defs/Id"
    },
    "schema": {
      "const": "promin.language-capability-profile.v1"
    }
  },
  "required": [
    "schema",
    "profileId",
    "languages",
    "authority_effect"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/LanguageCapabilityProfile | Validator: urn:promin:contracts:1#/\$defs/LanguageCapabilityProfile

## 58. Lease

Type summary: type="object"; additionalProperties=false; allOf[3]=keys:else,if,then | keys:if,then,title | keys:else,if,then,title

Required: record\_type, lease\_id, task\_id, holder\_grant\_claim\_digest, manager\_grant\_claim\_digest, manager\_grant\_id, manager\_subject\_id, holder\_subject\_id, holder\_grant\_id, generation, fencing\_token, state, acquired\_at, heartbeat\_at, expires\_at, activation\_digest

| Property                   | Required | Structural rule                           |
|----------------------------|----------|-------------------------------------------|
| acquired_at                | yes      | \$ref="#/\$defs/Timestamp"                |
| activation_digest          | yes      | \$ref="#/\$defs/Digest"                   |
| capacity_reconciliation    | no       | \$ref="#/\$defs/CapacityReconciliation"   |
| close_ack                  | no       | \$ref="#/\$defs/CloseAck"                 |
| expires_at                 | yes      | \$ref="#/\$defs/Timestamp"                |
| fencing_token              | yes      | type="integer"; minimum=1                 |
| generation                 | yes      | type="integer"; minimum=1                 |
| heartbeat_at               | yes      | \$ref="#/\$defs/Timestamp"                |
| holder_grant_claim_digest  | yes      | \$ref="#/\$defs/Digest"                   |
| holder_grant_id            | yes      | \$ref="#/\$defs/Id"                       |
| holder_subject_id          | yes      | \$ref="#/\$defs/Id"                       |
| lease_id                   | yes      | \$ref="#/\$defs/Id"                       |
| manager_grant_claim_digest | yes      | \$ref="#/\$defs/Digest"                   |
| manager_grant_id           | yes      | \$ref="#/\$defs/Id"                       |
| manager_subject_id         | yes      | \$ref="#/\$defs/Id"                       |
| record_type                | yes      | const="Lease"                             |
| state                      | yes      | \$ref="#/\$defs/LeaseState"               |
| task_id                    | yes      | \$ref="#/\$defs/Id"                       |
| termination                | no       | type="object"; additionalProperties=false |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "close_ack"
          ]
        }
      },
      "if": {
        "properties": {
          "state": {
            "const": "CLOSED"
          }
        }
      },
      "then": {
        "required": [
          "close_ack"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "state": {
            "not": {
              "enum": [
                "EXPIRED",
                "REVOKED"
              ]
            }
          }
        }
      },
      "then": {
        "not": {
          "required": [
            "capacity_reconciliation"
          ]
        }
      }
    }
  ],
  "title": "capacity reconciliation exists only on reconciled expired or revoked Lease"
},

```

```

{
  "else": {
    "not": {
      "required": [
        "termination"
      ]
    }
  },
  "if": {
    "properties": {
      "state": {
        "enum": [
          "EXPIRED",
          "REVOKED"
        ]
      },
      "required": [
        "state"
      ]
    },
    "then": {
      "required": [
        "termination"
      ]
    },
    "title": "terminal Lease carries exact termination provenance"
  }
},
"properties": {
  "acquired_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "capacity_reconciliation": {
    "$ref": "#/$defs/CapacityReconciliation"
  },
  "close_ack": {
    "$ref": "#/$defs/CloseAck"
  },
  "expires_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "generation": {
    "minimum": 1,
    "type": "integer"
  },
  "heartbeat_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "holder_grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "holder_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "holder_subject_id": {
    "$ref": "#/$defs/Id"
  },
  "lease_id": {
    "$ref": "#/$defs/Id"
  },
  "manager_grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "manager_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "manager_subject_id": {
    "$ref": "#/$defs/Id"
  }
},

```



```

"record_type": {
  "const": "Lease"
},
"state": {
  "$ref": "#/$defs/LeaseState"
},
"task_id": {
  "$ref": "#/$defs/Id"
},
"termination": {
  "additionalProperties": false,
  "properties": {
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "generation": {
      "minimum": 1,
      "type": "integer"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "state": {
      "enum": [
        "EXPIRED",
        "REVOKED"
      ]
    },
    "terminated_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "terminated_by": {
      "$ref": "#/$defs/Id"
    }
  }
},
"required": [
  "state",
  "terminated_by",
  "grant_id",
  "grant_claim_digest",
  "terminated_at",
  "generation",
  "fencing_token"
],
"type": "object"
}
},
"required": [
  "record_type",
  "lease_id",
  "task_id",
  "holder_grant_claim_digest",
  "manager_grant_claim_digest",
  "manager_grant_id",
  "manager_subject_id",
  "holder_subject_id",
  "holder_grant_id",
  "generation",
  "fencing_token",
  "state",
  "acquired_at",
  "heartbeat_at",
  "expires_at",
  "activation_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Lease | Validator: urn:promin:contracts:1#/\$defs/Lease

## 59. LeaseState

Type summary: enum=["ACTIVE", "CLOSED", "CLOSING", "EXPIRED", "REVOKED"]

Required: -

```
{
  "enum": [
    "ACTIVE",
    "CLOSED",
    "CLOSING",
    "EXPIRED",
    "REVOKED"
  ]
}
```

Owner: core/contracts.schema.json#/\$defs/LeaseState | Validator: urn:promin:contracts:1#/\$defs/LeaseState

## 60. LeaseStateMachine

Type summary: const=schema(keys:ACTIVE,CLOSED,CLOSING,EXPIRED,REVOKED)

Required: -

```
{
  "const": {
    "ACTIVE": [
      "CLOSING",
      "EXPIRED",
      "REVOKED"
    ],
    "CLOSED": [],
    "CLOSING": [
      "CLOSED",
      "EXPIRED",
      "REVOKED"
    ],
    "EXPIRED": [],
    "REVOKED": []
  }
}
```

Owner: core/contracts.schema.json#/\$defs/LeaseStateMachine | Validator: urn:promin:contracts:1#/\$defs/LeaseStateMachine

## 61. LicenseBinding

Type summary: type="object"; additionalProperties=false

Required: expression, source\_uris, review\_state

| Property     | Required | Structural rule                                               |
|--------------|----------|---------------------------------------------------------------|
| expression   | yes      | type="string"; minLength=1                                    |
| review_state | yes      | enum=["reviewed", "owner-review-required", "source-verified"] |
| source_uris  | yes      | type="array"; minItems=1; maxItems=8; uniqueItems=true        |

```
{
  "additionalProperties": false,
  "properties": {
    "expression": {
      "minLength": 1,
      "type": "string"
    },
    "review_state": {
      "enum": [
        "reviewed",
        "owner-review-required",
        "source-verified"
      ]
    },
    "source_uris": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  },
  "required": [
    "expression",
    "source_uris",
    "review_state"
  ]
}
```

```

    "review_state"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/LicenseBinding | Validator: urn:promin:contracts:1#/\$defs/LicenseBinding

## 62. LocalActivationProof

Type summary: type="object"; additionalProperties=false

Required: kind, subject\_id, authority\_init\_digest

| Property              | Required | Structural rule         |
|-----------------------|----------|-------------------------|
| authority_init_digest | yes      | \$ref="#/\$defs/Digest" |
| kind                  | yes      | const="local-root"      |
| subject_id            | yes      | \$ref="#/\$defs/Id"     |

```

{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "subject_id",
    "authority_init_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/LocalActivationProof | Validator: urn:promin:contracts:1#/\$defs/LocalActivationProof

## 63. LocalRootCommandProof

Type summary: type="object"; additionalProperties=false

Required: kind, subject\_id, authority\_init\_digest, signed\_intent\_digest

| Property              | Required | Structural rule            |
|-----------------------|----------|----------------------------|
| authority_init_digest | yes      | \$ref="#/\$defs/Digest"    |
| kind                  | yes      | const="local-root-command" |
| signed_intent_digest  | yes      | \$ref="#/\$defs/Digest"    |
| subject_id            | yes      | \$ref="#/\$defs/Id"        |

```

{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root-command"
    },
    "signed_intent_digest": {
      "$ref": "#/$defs/Digest"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "subject_id",
    "authority_init_digest",
    "signed_intent_digest"
  ],
  "type": "object"
}

```

## 64. LocalTrustProof

Type summary: type="object"; additionalProperties=false

Required: kind, root\_subject\_id, authority\_init\_digest, signed\_claim\_digest

| Property              | Required | Structural rule         |
|-----------------------|----------|-------------------------|
| authority_init_digest | yes      | \$ref="#/\$defs/Digest" |
| kind                  | yes      | const="local-root"      |
| root_subject_id       | yes      | \$ref="#/\$defs/Id"     |
| signed_claim_digest   | yes      | \$ref="#/\$defs/Digest" |

```
{
  "additionalProperties": false,
  "properties": {
    "authority_init_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "local-root"
    },
    "root_subject_id": {
      "$ref": "#/$defs/Id"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "root_subject_id",
    "authority_init_digest",
    "signed_claim_digest"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/LocalTrustProof | Validator: urn:promin:contracts:1#/\$defs/LocalTrustProof

## 65. NextResult

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:if,then

Required: record\_type, status, subject\_id, activation\_digest, work\_card, continuation

| Property          | Required | Structural rule                        |
|-------------------|----------|----------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                |
| continuation      | yes      | oneOf[2]=type:null   type:string       |
| record_type       | yes      | const="NextResult"                     |
| status            | yes      | enum=["ready", "empty"]                |
| subject_id        | yes      | \$ref="#/\$defs/Id"                    |
| work_card         | yes      | oneOf[2]=#/\$defs/WorkCard   type:null |

```
{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          },
          "work_card": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "status": {
            "const": "ready"
          }
        }
      }
    }
  ]
}
```

```

    },
    "required": [
        "status"
    ]
},
"then": {
    "properties": {
        "work_card": {
            "$ref": "#/$defs/WorkCard"

```

```

        }
    }
},
{
    "if": {
        "properties": {
            "continuation": {
                "type": "string"
            }
        },
        "required": [
            "continuation"
        ]
    },
    "then": {
        "properties": {
            "work_card": {
                "properties": {
                    "truncated": {
                        "const": true
                    }
                },
                "required": [
                    "truncated"
                ]
            }
        }
    }
}

```

```

    }
},
],
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            },
            {
                "maxLength": 256,
                "minLength": 1,
                "pattern": "^promin-v2\\.[A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
                "type": "string"
            }
        ]
    },
    "record_type": {
        "const": "NextResult"
    },
    "status": {
        "enum": [
            "ready",
            "empty"
        ]
    }
}

```

```

},
"subject_id": {
    "$ref": "#/$defs/Id"
},
"work_card": {
    "oneOf": [
        {
            "$ref": "#/$defs/WorkCard"
        },
        {
            "type": "null"
        }
    ]
}

```

```

    ]
  },
  "required": [
    "record_type",
    "status",
    "subject_id",
    "activation_digest",
    "work_card",
    "continuation"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/NextResult | Validator: urn:promin:contracts:1#/\$defs/NextResult

## 66. NoDegradationResult

Type summary: type="object"; additionalProperties=false

Required: record\_type, status, platform, no\_degradation, source, root, artifact\_binding, candidate\_binding\_digest, test\_manifest, install\_mode, required\_predicates, validation, tests, passed, pass\_credit, acceptance\_pass, product\_acceptance\_pass, product\_public\_approval, scale\_selection, artifact\_binding\_unchanged, standard\_distribution\_gate\_pass, phases, execution\_budget, producer, invocation, raw\_artifact\_manifest\_digest, producer\_attestation, result\_digest

| Property                        | Required | Structural rule                              |
|---------------------------------|----------|----------------------------------------------|
| acceptance_pass                 | yes      | const=false                                  |
| artifact_binding                | yes      | type="object"                                |
| artifact_binding_unchanged      | yes      | const=true                                   |
| candidate_binding_digest        | yes      | \$ref="#/\$defs/Digest"                      |
| execution_budget                | yes      | type="object"; additionalProperties=false    |
| install_mode                    | yes      | const="offline-wheelhouse"                   |
| invocation                      | yes      | \$ref="#/\$defs/ReleaseEvidenceInvocation"   |
| no_degradation                  | yes      | const=true                                   |
| pass_credit                     | yes      | const=false                                  |
| passed                          | yes      | const=true                                   |
| phases                          | yes      | type="array"; minItems=2; maxItems=2         |
| platform                        | yes      | enum=["linux", "windows"]                    |
| producer                        | yes      | \$ref="#/\$defs/ReleaseEvidenceProducer"     |
| producer_attestation            | yes      | \$ref="#/\$defs/EvidenceProducerAttestation" |
| product_acceptance_pass         | yes      | const=false                                  |
| product_public_approval         | yes      | const="not_approved"                         |
| raw_artifact_manifest_digest    | yes      | \$ref="#/\$defs/Digest"                      |
| record_type                     | yes      | const="NoDegradationResult"                  |
| required_predicates             | yes      | type="object"; additionalProperties=false    |
| result_digest                   | yes      | \$ref="#/\$defs/Digest"                      |
| root                            | yes      | type="string"; minLength=1                   |
| scale_selection                 | yes      | type="object"; additionalProperties=false    |
| source                          | yes      | const="live_files_and_executable_tests"      |
| standard_distribution_gate_pass | yes      | const=false                                  |
| status                          | yes      | const="pass"                                 |
| test_manifest                   | yes      | type="object"; additionalProperties=false    |
| tests                           | yes      | type="object"; additionalProperties=false    |
| validation                      | yes      | type="object"; additionalProperties=false    |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },

```

```

"artifact_binding": {
  "type": "object"
},
"artifact_binding_unchanged": {
  "const": true
},
"candidate_binding_digest": {
  "$ref": "#/$defs/Digest"
},
"execution_budget": {
  "additionalProperties": false,
  "properties": {
    "completed_at": {},
    "elapsed_ms": {},
    "started_at": {},
    "test_timeout_seconds": {},
    "total_timeout_seconds": {},
    "validation_timeout_seconds": {}
  },
  "required": [
    "total_timeout_seconds",
    "validation_timeout_seconds",
    "test_timeout_seconds",
    "elapsed_ms",
    "started_at",
    "completed_at"
  ],
  "type": "object"
},
"install_mode": {
  "const": "offline-wheelhouse"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"no_degradation": {
  "const": true
},
"pass_credit": {
  "const": false
},
"passed": {
  "const": true
},
"phases": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "elapsed_ms": {},
      "phase": {},
      "process_group_cleanup": {},
      "returncode": {},
      "status": {},
      "timeout_seconds": {}
    },
    "required": [
      "phase",
      "status",
      "returncode",
      "elapsed_ms",
      "timeout_seconds",
      "process_group_cleanup"
    ],
    "type": "object"
  },
  "maxItems": 2,
  "minItems": 2,
  "type": "array"
},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},

```

```

"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "NoDegradationResult"
},
"required_predicates": {
  "additionalProperties": false,
  "properties": {
    "acceptance_predicates": {},
    "complete": {},
    "missing_acceptance_predicates": {},
    "missing_mutation_families": {},
    "missing_policies": {},
    "mutation_families": {},
    "policies": {}
  },
  "required": [
    "policies",

```

```

    "acceptance_predicates",
    "mutation_families",
    "missing_policies",
    "missing_acceptance_predicates",
    "missing_mutation_families",
    "complete"
  ],
  "type": "object"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"root": {
  "minLength": 1,
  "type": "string"
},
"scale_selection": {
  "additionalProperties": false,
  "properties": {
    "command": {},
    "missing_workspace_is_failure": {},
    "pass_credit": {},
    "runs_separately": {}
  },
  "required": [
    "command",
    "runs_separately",
    "missing_workspace_is_failure",

```

```

    "pass_credit"
  ],
  "type": "object"
},
"source": {
  "const": "live_files_and_executable_tests"
},
"standard_distribution_gate_pass": {
  "const": false
},
"status": {
  "const": "pass"
},
"test_manifest": {
  "additionalProperties": false,
  "properties": {
    "digest": {},
    "file_count": {},
    "files": {}
  },
  "required": [
    "files",

```



```

    "file_count",
    "digest"
  ],
  "type": "object"
},
"tests": {

```

```

  "additionalProperties": false,
  "properties": {
    "command": {},
    "elapsed_ms": {},
    "installed_environment": {
      "additionalProperties": false,
      "properties": {
        "build_dependency_source": {
          "const": "offline-wheelhouse"
        },
        "console_script": {
          "additionalProperties": false,
          "properties": {
            "declared": {
              "const": true
            },
            "installed_wrapper_checked": {
              "const": true
            },
            "path": {
              "minLength": 1,
              "type": "string"
            },
            "posix_execute_bits": {
              "type": [
                "boolean",
                "null"
              ]
            }
          }
        },
        "present": {
          "const": true
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        },
        "size_bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "source_module_invoked": {
          "const": false
        }
      },
      "required": [
        "present",
        "declared",
        "installed_wrapper_checked",
        "source_module_invoked",
        "posix_execute_bits",
        "path",
        "sha256",
        "size_bytes"
      ],
      "type": "object"
    },
    "declared_build_dependencies": {
      "items": {
        "type": "string"
      },
      "minItems": 1,
      "type": "array"
    },
    "declared_python_requirement": {
      "minLength": 1,
      "type": "string"
    },
    "declared_runtime_dependencies": {
      "items": {
        "type": "string"
      },
      "minItems": 1,

```

```

    },
    "present": {
      "const": true
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "source_module_invoked": {
      "const": false
    }
  },
  "required": [
    "present",
    "declared",
    "installed_wrapper_checked",
    "source_module_invoked",
    "posix_execute_bits",
    "path",
    "sha256",
    "size_bytes"
  ],
  "type": "object"
},
"declared_build_dependencies": {

```

```

  "items": {
    "type": "string"
  },
  "minItems": 1,
  "type": "array"
},
"declared_python_requirement": {
  "minLength": 1,
  "type": "string"
},
"declared_runtime_dependencies": {
  "items": {
    "type": "string"
  },
  "minItems": 1,

```

```

    "type": "array"
  },
  "dependency_closure_check": {
    "const": "pass"
  },
  "environment": {
    "const": "clean-venv"
  },
  "installation_performed": {
    "const": true
  },
  "installed_environment_observation": {
    "additionalProperties": false,

```

```

    "properties": {
      "build_backend": {
        "additionalProperties": false,
        "properties": {
          "distribution": {
            "const": "setuptools"
          },
          "name": {
            "const": "setuptools.build_meta"
          },
          "record_sha256": {
            "$ref": "#/$defs/Digest"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        }
      },
      "required": [
        "name",
        "distribution",
        "version",
        "record_sha256"
      ],
      "type": "object"
    },
    "console_wrapper": {
      "additionalProperties": false,

```

```

    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "path",
      "sha256",
      "bytes"
    ],
    "type": "object"
  },
  "license_closure": {
    "additionalProperties": false,
    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"

```

```

    },
    "entries": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "license": {
            "maxLength": 2048,
            "minLength": 1,

```

```

        "not": {
            "const": "UNKNOWN"
        },
        "type": "string"
    },
    "name": {
        "minLength": 1,
        "type": "string"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "name",
    "version",
    "license"
],
"type": "object"

```

```

    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
}
},
"required": [
    "entries",
    "complete",
    "digest"
],
"type": "object"
},
"observation_digest": {
    "$ref": "#/$defs/Digest"
},
"platform": {
    "additionalProperties": false,
    "properties": {
        "machine": {
            "minLength": 1,
            "type": "string"
        },
        "release": {
            "minLength": 1,
            "type": "string"
        }
    },
    "sys_platform": {

```

```

        "minLength": 1,
        "type": "string"
    },
    "system": {
        "enum": [
            "linux",
            "windows"
        ]
    },
    "tags": {
        "items": {
            "minLength": 1,
            "type": "string"
        },
        "maxItems": 64,
        "minItems": 1,
        "type": "array"
    }
},
"required": [
    "system",
    "machine",
    "release",
    "sys_platform",
    "tags"
],
"type": "object"
},

```

```

"promin": {
  "additionalProperties": false,
  "properties": {
    "installed_file_count": {
      "maximum": 4096,
      "minimum": 1,
      "type": "integer"
    },
    "installed_files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 0,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },
      "maxItems": 4096,
      "minItems": 1,
      "type": "array"
    },
    "installed_files_digest": {
      "$ref": "#/$defs/Digest"
    },
    "module_file": {
      "minLength": 1,
      "type": "string"
    },
    "module_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "record": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "path": {
          "minLength": 1,
          "pattern": "\\\\.dist-info/RECORD$",
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "path",
        "sha256",
        "bytes"
      ],
      "type": "object"
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "version",
    "module_file",
    "module_sha256",
    "record",
    "installed_files"
  ]
}

```

```

        "installed_files_digest",
        "installed_file_count"
    ],
    "type": "object"
},
"python": {

```

```

    "additionalProperties": false,
    "properties": {
        "abi_tag": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "base_prefix": {
            "minLength": 1,
            "type": "string"
        },
        "executable": {
            "minLength": 1,
            "type": "string"
        },
        "executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "implementation": {
            "minLength": 1,
            "type": "string"
        }
    },

```

```

        "prefix": {
            "minLength": 1,
            "type": "string"
        },
        "version": {
            "minLength": 1,
            "type": "string"
        }
    },
    "required": [
        "executable",
        "executable_sha256",
        "base_executable",
        "base_executable_sha256",
        "implementation",
        "version",
        "abi_tag",
        "prefix",
        "base_prefix"
    ],
    "type": "object"
},
"sbom": {
    "additionalProperties": false,
    "properties": {
        "components": {
            "items": {
                "additionalProperties": false,

```

```

                "properties": {
                    "name": {
                        "minLength": 1,
                        "type": "string"
                    },
                    "record_sha256": {
                        "$ref": "#/$defs/Digest"
                    },
                    "version": {
                        "minLength": 1,
                        "type": "string"
                    }
                }
            },
            "required": [
                "name",

```

```

        "version",
        "record_sha256"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
},
"digest": {
    "$ref": "#/$defs/Digest"
},
"format": {

```

```

        "const": "promin-spx-lite-v1"
    }
},
"required": [
    "format",
    "components",
    "digest"
],
"type": "object"
},
"site_packages": {
    "minLength": 1,
    "type": "string"
},
"sqlite_version": {
    "minLength": 1,
    "type": "string"
},
"transitive_distribution_digest": {
    "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "license": {
                "maxLength": 2048,
                "minLength": 1,

```

```

            "type": "string"
        },
        "name": {
            "minLength": 1,
            "type": "string"
        },
        "record": {
            "additionalProperties": false,
            "properties": {
                "bytes": {
                    "minimum": 1,
                    "type": "integer"
                },
                "path": {
                    "minLength": 1,
                    "pattern": "\\..dist-info/RECORD$",
                    "type": "string"
                },
                "sha256": {
                    "$ref": "#/$defs/Digest"
                }
            }
        },
        "required": [
            "path",
            "sha256",
            "bytes"
        ],
        "type": "object"

```

```

    },
    "requires": {
        "items": {
            "minLength": 1,
            "type": "string"
        },
        "type": "array"
    },

```

```

        "version": {
            "minLength": 1,
            "type": "string"
        }
    },
    "required": [
        "name",
        "version",
        "record",
        "requires",
        "license"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
}
},
"required": [

```

```

    "python",
    "platform",
    "sqlite_version",
    "site_packages",
    "promin",
    "console_wrapper",
    "transitive_distributions",
    "transitive_distribution_digest",
    "sbom",
    "license_closure",
    "build_backend",
    "observation_digest"
],
"type": "object"
},
"interpreter": {
    "additionalProperties": false,
    "properties": {
        "executable": {
            "minLength": 1,
            "type": "string"
        },
        "nested_venv_created": {
            "const": true
        }
    }
},
"required": [
    "executable",

```

```

    "nested_venv_created"
],
"type": "object"
},
"invocations": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "argv_digest": {
                "$ref": "#/$defs/Digest"
            },
            "command": {
                "maxLength": 128,
                "minLength": 1,
                "type": "string"
            },
            "returncode": {
                "const": 0
            },
            "stderr_bytes": {
                "maximum": 1048576,
                "minimum": 0,
                "type": "integer"
            },
            "stderr_sha256": {
                "$ref": "#/$defs/Digest"
            },
            "stdout_bytes": {

```

```

        "maximum": 1048576,
        "minimum": 0,
        "type": "integer"
    },
    "stdout_sha256": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "command",
    "argv_digest",
    "returncode",
    "stdout_sha256",
    "stdout_bytes",
    "stderr_sha256",
    "stderr_bytes"
],
"type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"mode": {
    "const": "offline-wheelhouse"
},
"nested_venv_created": {
    "const": true

```

```

},
"network_disabled": {
    "const": true
},
"origin_assertion": {
    "additionalProperties": false,
    "properties": {
        "inside_site_packages": {
            "const": true
        },
        "isolated_mode": {
            "const": true
        },
        "module_file": {
            "minLength": 1,
            "type": "string"
        },
        "module_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "outside_install_source": {
            "const": true
        },
        "safe_path": {
            "const": true
        },
        "site_packages": {
            "minLength": 1,

```

```

            "type": "string"
        },
        "source_absent_from_sys_path": {
            "const": true
        },
        "sys_path_digest": {
            "$ref": "#/$defs/Digest"
        }
    },
    "required": [
        "isolated_mode",
        "safe_path",
        "module_file",
        "module_sha256",
        "site_packages",
        "inside_site_packages",
        "outside_install_source",
        "source_absent_from_sys_path",
        "sys_path_digest"
    ],
    "type": "object"
},

```



```

"performed": {
  "const": true
},
"pip_report": {
  "additionalProperties": false,
  "properties": {
    "artifact_digest": {
      "$ref": "#/$defs/Digest"
    },
    "report_count": {
      "const": 2
    },
    "reports": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "artifact_digest": {
            "$ref": "#/$defs/Digest"
          },
          "artifacts": {
            "items": {
              "additionalProperties": false,
              "properties": {
                "hashes": {
                  "items": {
                    "minLength": 3,
                    "type": "string"
                  },
                  "type": "array"
                },
                "is_direct": {
                  "type": "boolean"
                },
                "name": {
                  "minLength": 1,
                  "type": "string"
                },
                "version": {
                  "minLength": 1,
                  "type": "string"
                }
              },
              "required": [
                "name",
                "version",
                "is_direct",
                "hashes"
              ],
              "type": "object"
            },
            "maxItems": 256,
            "minItems": 1,
            "type": "array"
          },
          "bytes": {
            "maximum": 16777216,
            "minimum": 1,
            "type": "integer"
          },
          "install_count": {
            "maximum": 256,
            "minimum": 1,
            "type": "integer"
          },
          "pip_version": {
            "minLength": 1,
            "type": "string"
          },
          "role": {
            "enum": [
              "build-requirements",
              "candidate-install"
            ]
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        }
      }
    }
  }
}

```

```

    },
    "required": [
        "role",
        "sha256",
        "bytes",
        "pip_version",
        "install_count",
        "artifacts",
        "artifact_digest"
    ],
    "type": "object"
},
"maxItems": 2,
    "minItems": 2,
    "type": "array"
}
},
"required": [
    "reports",
    "report_count",
    "artifact_digest"
],
"type": "object"
},
"platform": {
    "additionalProperties": false,
    "properties": {
        "os_name": {
            "enum": [
                "nt",
                "posix"
            ]
        },
        "python": {
            "minLength": 1,
            "type": "string"
        },
        "sys_platform": {
            "minLength": 1,
            "type": "string"
        }
    }
}

```

```

    },
    "required": [
        "os_name",
        "sys_platform",
        "python"
    ],
    "type": "object"
},
"product_acceptance_pass": {
    "const": false
},
"python_distribution_version": {
    "$ref": "#/$defs/SemVer"
},
"resolved_runtime_dependencies": {
    "additionalProperties": false,
    "properties": {
        "cryptography": {
            "additionalProperties": false,
            "properties": {
                "requirement": {
                    "minLength": 1,
                    "type": "string"
                },
                "version": {
                    "minLength": 1,
                    "type": "string"
                }
            }
        }
    }
}

```

```

    },
    "required": [
        "requirement",
        "version"
    ],
    "type": "object"
},
"jsonschema": {

```

```

    "additionalProperties": false,
    "properties": {
      "requirement": {
        "minLength": 1,
        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
  "pypdf": {
    "additionalProperties": false,
    "properties": {

```

```

      "requirement": {
        "minLength": 1,
        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  }
},
"required": [
  "cryptography",
  "jsonschema",
  "pypdf"
],
"type": "object"
},
"runtime_dependency_source": {
  "const": "offline-wheelhouse"
},
"test_dependencies_installed": {
  "const": true

```

```

},
"verified": {
  "const": true
},
"version": {
  "$ref": "#/$defs/SemVer"
},
"wheelhouse_binding": {
  "additionalProperties": false,
  "properties": {
    "file_count": {
      "maximum": 256,
      "minimum": 1,
      "type": "integer"
    },
    "files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {

```

```

        "$ref": "#/$defs/Digest"
    },
    "required": [
        "path",
        "bytes",
        "sha256"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
},
"manifest_sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "files",
    "file_count",
    "manifest_sha256"
],
"type": "object"
},
"wheelhouse_lock": {
    "additionalProperties": false,
    "properties": {

```

```

        "copy_verified": {
            "const": true
        },
        "locked_copy_binding": {
            "additionalProperties": false,
            "properties": {
                "file_count": {
                    "maximum": 256,
                    "minimum": 1,
                    "type": "integer"
                },
                "files": {
                    "items": {
                        "additionalProperties": false,
                        "properties": {
                            "bytes": {
                                "minimum": 1,
                                "type": "integer"
                            },
                            "path": {
                                "minLength": 1,
                                "type": "string"
                            },
                            "sha256": {
                                "$ref": "#/$defs/Digest"
                            }
                        }
                    },
                    "required": [

```

```

                        "path",
                        "bytes",
                        "sha256"
                    ],
                    "type": "object"
                },
                "maxItems": 256,
                "minItems": 1,
                "type": "array"
            },
            "manifest_sha256": {
                "$ref": "#/$defs/Digest"
            }
        },
        "required": [
            "files",
            "file_count",
            "manifest_sha256"
        ],
        "type": "object"
    },
    "source_binding": {

```

```

    "additionalProperties": false,
    "properties": {
      "file_count": {
        "maximum": 256,
        "minimum": 1,
        "type": "integer"
      },
      "files": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "bytes": {
              "minimum": 1,
              "type": "integer"
            },
            "path": {
              "minLength": 1,
              "type": "string"
            },
            "sha256": {
              "$ref": "#/$defs/Digest"
            }
          },
          "required": [
            "path",
            "bytes",
            "sha256"
          ],
          "type": "object"
        },
        "maxItems": 256,
        "minItems": 1,
        "type": "array"
      },
      "manifest_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "required": [
        "files",
        "file_count",
        "manifest_sha256"
      ],
      "type": "object"
    },
    "source_unchanged": {
      "const": true
    },
    "required": [
      "source_binding",
      "locked_copy_binding",
      "source_unchanged",
      "copy_verified"
    ],
    "type": "object"
  },
  "required": [
    "performed",
    "verified",
    "mode",
    "environment",
    "runtime_dependency_source",
    "build_dependency_source",
    "network_disabled",
    "installation_performed",
    "nested_venv_created",
    "wheelhouse_binding",
    "declared_runtime_dependencies",
    "declared_build_dependencies",
    "declared_python_requirement",
    "resolved_runtime_dependencies",
    "dependency_closure_check",
    "installed_environment_observation",
    "pip_report",
    "wheelhouse_lock",

```

```

        "origin_assertion",
        "test_dependencies_installed",
        "version",
        "python_distribution_version",
        "console_script",
        "invocations",
        "interpreter",
        "platform",
        "product_acceptance_pass"
    ],
    "type": "object"
},
"junit_sha256": {},

```

```

"marker_expression": {},
"optional_platform_tests_excluded": {},
"passed": {},
"process_group_cleanup": {},
"ran": {},
"raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"raw_artifacts": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "maximum": 16777216,
                "minimum": 0,
                "type": "integer"
            },
            "data": {
                "type": "string"
            },
            "encoding": {
                "const": "base64"
            },
            "media_type": {
                "enum": [
                    "application/xml",
                    "text/plain; charset=utf-8"
                ]
            }
        }
    }
}

```

```

},
"role": {
    "enum": [
        "required-junit",
        "required-stdout",
        "required-stderr"
    ]
},
"sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "role",
    "media_type",
    "bytes",
    "sha256",
    "encoding",
    "data"
],
"type": "object"
},
"maxItems": 3,
"minItems": 3,
"type": "array"
},
"required_counts": {},
"required_skips_fail_closed": {},

```

```

"returncode": {},
"scale_environment_configured": {},
"selection_expression": {},
"source_tree_shadowing_disabled": {
    "const": true
},
"stderr": {
    "additionalProperties": false,

```

```

"properties": {
  "bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "selection": {
    "enum": [
      "complete",
      "tail"
    ]
  },
  "sha256": {
    "$ref": "#/$defs/Digest"
  },
  "text": {
    "type": "string"
  },
  "truncated": {
    "type": "boolean"
  }
}

```

```

},
"required": [
  "text",
  "bytes",
  "sha256",
  "truncated",
  "selection"
],
"type": "object"
},
"stdout": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "selection": {
      "enum": [
        "complete",
        "tail"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "text": {
      "type": "string"
    },
    "truncated": {
      "type": "boolean"
    }
  },
  "required": [
    "text",
    "bytes",
    "sha256",
    "truncated",
    "selection"
  ],
  "type": "object"
},
"timed_out": {},
"timeout_seconds": {}
},
"required": [
  "ran",
  "command",
  "returncode",
  "stdout",
  "stderr",
  "timed_out",
  "timeout_seconds",
  "elapsed_ms",
  "process_group_cleanup",
  "required_counts",

```

```

        "required_skips_fail_closed",
        "optional_platform_tests_excluded",
        "marker_expression",
        "selection_expression",
        "junit_sha256",
        "passed",
        "scale_environment_configured",
        "raw_artifacts",
        "raw_artifact_manifest_digest",
        "installed_environment",
        "source_tree_shadowing_disabled"
    ],
    "type": "object"
},
"validation": {
    "additionalProperties": false,
    "properties": {
        "checks": {},
        "errors": {},
        "record_type": {},
        "root": {},
        "valid": {}
    },
    "required": [
        "checks",
        "errors",
        "record_type",
        "root",

```

```

        "valid"
    ],
    "type": "object"
}
},
"required": [
    "record_type",
    "status",
    "platform",
    "no_degradation",
    "source",
    "root",
    "artifact_binding",
    "candidate_binding_digest",
    "test_manifest",
    "install_mode",
    "required_predicates",
    "validation",
    "tests",
    "passed",
    "pass_credit",
    "acceptance_pass",
    "product_acceptance_pass",
    "product_public_approval",
    "scale_selection",
    "artifact_binding_unchanged",
    "standard_distribution_gate_pass",
    "phases",

```

```

    "execution_budget",
    "producer",
    "invocation",
    "raw_artifact_manifest_digest",
    "producer_attestation",
    "result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/NoDegradationResult | Validator: urn:promin:contracts:1#/\$defs/NoDegradationResult

## 67. OperationMetrics

Type summary: type="object"; additionalProperties=false

Required: record\_type, claim\_scope, authoritative, duration\_ms, changed\_records, event\_write, projection\_update, runtime\_checkpoint, physical\_payload\_bytes, bytes\_per\_changed\_record

| Property      | Required | Structural rule |
|---------------|----------|-----------------|
| authoritative | yes      | const=false     |



| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required | Structural rule                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------|
| bytes_per_changed_record                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | yes      | type="number"; minimum=0                         |
| changed_records                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | yes      | type="integer"; minimum=0                        |
| claim_scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | const="single-command-observation-only"          |
| duration_ms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | type="number"; minimum=0                         |
| event_write                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | type="object"; additionalProperties=false        |
| physical_payload_bytes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | yes      | type="integer"; minimum=0                        |
| projection_update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | yes      | oneOf[3]=type:object   type:object   type:object |
| record_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | const="OperationMetrics"                         |
| runtime_checkpoint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | yes      | oneOf[2]=type:object   type:object               |
| <pre> {   "additionalProperties": false,   "properties": {     "authoritative": {       "const": false     },     "bytes_per_changed_record": {       "minimum": 0,       "type": "number"     },     "changed_records": {       "minimum": 0,       "type": "integer"     },     "claim_scope": {       "const": "single-command-observation-only"     },     "duration_ms": {       "minimum": 0,       "type": "number"     },     "event_write": {       "additionalProperties": false,       "properties": {         "changed_records": {           "minimum": 0,           "type": "integer"         }, </pre>                                                                                                                                                                                                                           |          |                                                  |
| <pre>         "derived_index_bytes": {           "minimum": 0,           "type": "integer"         },         "head_bytes": {           "minimum": 0,           "type": "integer"         },         "journal_authority_bytes": {           "minimum": 0,           "type": "integer"         },         "journal_checkpoint_bytes": {           "minimum": 0,           "type": "integer"         },         "logical_final_bytes": {           "minimum": 0,           "type": "integer"         },         "physical_payload_bytes": {           "minimum": 0,           "type": "integer"         },         "temporary_staging_bytes": {           "minimum": 0,           "type": "integer"         }       }     },     "required": [       "changed_records",       "journal_authority_bytes",       "temporary_staging_bytes", </pre> |          |                                                  |

```

        "head_bytes",
        "derived_index_bytes",
        "journal_checkpoint_bytes",
        "logical_final_bytes",
        "physical_payload_bytes"
    ],
    "type": "object"
},
"physical_payload_bytes": {
    "minimum": 0,
    "type": "integer"
},
"projection_update": {
    "oneOf": [
        {
            "additionalProperties": false,
            "properties": {
                "bytes_per_changed_record": {
                    "minimum": 0,
                    "type": "number"
                },
                "changed_records": {
                    "minimum": 0,

```

```

                    "type": "integer"
                },
                "changed_shards": {
                    "minimum": 0,
                    "type": "integer"
                },
                "physical_payload_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "projection_authoritative": {
                    "const": false
                },
                "status": {
                    "enum": [
                        "already-current",
                        "missing",
                        "stale-rebuild-required"
                    ]
                }
            }
        },
        "required": [
            "status",
            "projection_authoritative",
            "changed_records",
            "changed_shards",
            "physical_payload_bytes",
            "bytes_per_changed_record"

```

```

    ],
    "type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "bytes_per_changed_record": {
            "minimum": 0,
            "type": "number"
        },
        "changed_records": {
            "minimum": 0,
            "type": "integer"
        },
        "changed_shards": {
            "minimum": 0,
            "type": "integer"
        },
        "error": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
        },
        "physical_payload_bytes": {
            "minimum": 0,
            "type": "integer"

```

```

    },
    "projection_authoritative": {
        "const": false
    },
    "status": {
        "const": "failed-rebuild-required"
    }
},
"required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "error"
],
"type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "bytes_per_changed_record": {
            "minimum": 0,
            "type": "number"
        },
        "changed_records": {
            "minimum": 0,
            "type": "integer"
        },

```

```

        "changed_shards": {
            "minimum": 0,
            "type": "integer"
        },
        "head": {
            "additionalProperties": false,
            "properties": {
                "batch_digest": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Digest"
                        },
                        {
                            "type": "null"
                        }
                    ]
                },
                "batch_id": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Id"
                        },
                        {
                            "type": "null"
                        }
                    ]
                }
            },
            "sequence": {

```

```

                "minimum": 0,
                "type": "integer"
            }
        },
        "required": [
            "sequence",
            "batch_id",
            "batch_digest"
        ],
        "type": "object"
    },
    "logical_payload_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "physical_payload_bytes": {
        "minimum": 0,
        "type": "integer"
    },

```

```

    "projection_authoritative": {
      "const": false
    },
    "projection_growth_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "semantic_digest": {
      "$ref": "#/$defs/Digest"

```

```

    },
    "status": {
      "const": "updated"
    }
  },
  "required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "head",
    "semantic_digest",
    "logical_payload_bytes",
    "projection_growth_bytes"
  ],
  "type": "object"
}
]
},
"record_type": {
  "const": "OperationMetrics"
},
"runtime_checkpoint": {
  "oneOf": [
    {
      "additionalProperties": false,

```

```

    "properties": {
      "authoritative": {
        "const": false
      },
      "batch_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "byte_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "checkpoint_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "checkpoint_count": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_batches": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_bytes": {
        "minimum": 0,
        "type": "integer"
      }
    },

```

```

    "written": {
      "const": false
    }
  },
  "required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",
    "tail_batches",
    "tail_bytes",

```

```

        "written"
      ],
      "type": "object"
    },
    {
      "additionalProperties": false,
      "properties": {
        "authoritative": {
          "const": false
        },
        "batch_threshold": {
          "minimum": 1,
          "type": "integer"
        },
        "byte_threshold": {
          "minimum": 1,

```

```

          "type": "integer"
        },
        "checkpoint_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "checkpoint_count": {
          "minimum": 0,
          "type": "integer"
        },
        "compacted_tail_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "compacted_tail_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "written": {
          "const": true

```

```

      }
    },
    "required": [
      "authoritative",
      "batch_threshold",
      "byte_threshold",
      "checkpoint_bytes",
      "checkpoint_count",
      "tail_batches",
      "tail_bytes",
      "written",
      "compacted_tail_batches",
      "compacted_tail_bytes"
    ],
    "type": "object"
  }
]
}
},
"required": [
  "record_type",
  "claim_scope",
  "authoritative",
  "duration_ms",
  "changed_records",
  "event_write",
  "projection_update",
  "runtime_checkpoint",
  "physical_payload_bytes",
  "bytes_per_changed_record"
],
"type": "object",

```

```

"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/OperationMetrics | Validator: urn:promin:contracts:1#/\$defs/OperationMetrics

## 68. PersistentEntityDef

Type summary: type="object"; additionalProperties=false

Required: kind, purpose, admission\_rule

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Required | Structural rule                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------|
| admission_rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | yes      | type="string"; minLength=1; maxLength=256                                                            |
| kind                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | yes      | enum=["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"] |
| purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | yes      | type="string"; minLength=1                                                                           |
| <pre> {   "additionalProperties": false,   "properties": {     "admission_rule": {       "maxLength": 256,       "minLength": 1,       "type": "string"     },     "kind": {       "enum": [         "Task",         "Candidate",         "Artifact",         "Run",         "Finding",         "Decision",         "GateResult",         "Grant",         "Lease"       ]     },     "purpose": {       "minLength": 1,       "type": "string"     }   },   "required": [     "kind",     "purpose",     "admission_rule"   ],   "type": "object" } </pre> |          |                                                                                                      |

Owner: core/contracts.schema.json#/\$defs/PersistentEntityDef | Validator: urn:promin:contracts:1#/\$defs/PersistentEntityDef

## 69. PlanProposal

Type summary: type="object"; additionalProperties=false

Required: record\_type, proposal\_id, project\_id, project\_mode, goal, source\_plan\_digest, created\_at, tasks, authority, pass\_credit, proposal\_digest

| Property           | Required | Structural rule                                |
|--------------------|----------|------------------------------------------------|
| authority          | yes      | const=false                                    |
| created_at         | yes      | \$ref="#/\$defs/Timestamp"                     |
| goal               | yes      | type="string"; minLength=1; maxLength=8192     |
| pass_credit        | yes      | const=false                                    |
| project_id         | yes      | \$ref="#/\$defs/Id"                            |
| project_mode       | yes      | enum=["greenfield", "existing-code", "hybrid"] |
| proposal_digest    | yes      | \$ref="#/\$defs/Digest"                        |
| proposal_id        | yes      | \$ref="#/\$defs/Id"                            |
| record_type        | yes      | const="PlanProposal"                           |
| source_plan_digest | yes      | \$ref="#/\$defs/Digest"                        |

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required | Structural rule             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|
| tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | yes      | type="array"; maxItems=1024 |
| <pre> {   "additionalProperties": false,   "properties": {     "authority": {       "const": false     },     "created_at": {       "\$ref": "#/\$defs/Timestamp"     },     "goal": {       "maxLength": 8192,       "minLength": 1,       "type": "string"     },     "pass_credit": {       "const": false     },     "project_id": {       "\$ref": "#/\$defs/Id"     },     "project_mode": {       "enum": [         "greenfield",         "existing-code",         "hybrid"       ]     },     "proposal_digest": {       "\$ref": "#/\$defs/Digest"     },     "proposal_id": {       "\$ref": "#/\$defs/Id"     },     "record_type": {       "const": "PlanProposal"     },     "source_plan_digest": {       "\$ref": "#/\$defs/Digest"     },     "tasks": {       "items": {         "additionalProperties": false,         "properties": {           "acceptance_predicate": {             "maxLength": 8192,             "minLength": 1,             "type": "string"           },           "allowed_paths": {             "items": {               "maxLength": 4096,               "minLength": 1,               "type": "string"             },             "maxItems": 256,             "minItems": 1,             "type": "array",             "uniqueItems": true           },           "authority": {             "const": false           },           "depends_on": {             "items": {               "\$ref": "#/\$defs/Id"             },             "maxItems": 256,             "type": "array",             "uniqueItems": true           },           "operation": {             "enum": [               "read",               "mutate"             ]           }         }       }     }   } } </pre> |          |                             |

```

    ],
    "pass_credit": {
      "const": false
    },
    "source_bindings": {
      "items": {
        "maxLength": 4096,
        "minLength": 1,
        "type": "string"
      }
    },

```

```

    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  },
  "title": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  }
},
"required": [
  "task_id",
  "title",
  "operation",
  "depends_on",
  "acceptance_predicate",
  "allowed_paths",
  "source_bindings",
  "authority"
],
"type": "object"
},
"maxItems": 1024,

```

```

    "type": "array"
  }
},
"required": [
  "record_type",
  "proposal_id",
  "project_id",
  "project_mode",
  "goal",
  "source_plan_digest",
  "created_at",
  "tasks",
  "authority",
  "pass_credit",
  "proposal_digest"
],
"type": "object",
"x-promin-persistence": "transient-ingress-only"
}

```

Owner: [core/contracts.schema.json#/\\$defs/PlanProposal](#) | Validator: [urn:promin:contracts:1#/\\$defs/PlanProposal](#)

## 70. PlatformVerificationResult

Type summary: type="object"; additionalProperties=false

Required: record\_type, status, candidate\_binding\_digest, platform, platform\_identity, exact\_candidate\_verified, archive\_sha256, archive\_bytes, portable\_implementation\_closure\_digest, observed\_environment\_closure\_digest, install\_mode, installed\_command\_verified, installation, product\_acceptance\_pass, product\_public\_approval, producer, invocation, raw\_artifact\_manifest\_digest, producer\_attestation, result\_digest

| Property                 | Required | Structural rule           |
|--------------------------|----------|---------------------------|
| archive_bytes            | yes      | type="integer"; minimum=1 |
| archive_sha256           | yes      | \$ref="#/\$defs/Digest"   |
| candidate_binding_digest | yes      | \$ref="#/\$defs/Digest"   |
| exact_candidate_verified | yes      | const=true                |
| install_mode             | yes      | const="online-clean"      |



| Property                               | Required | Structural rule                              |
|----------------------------------------|----------|----------------------------------------------|
| installation                           | yes      | type="object"; additionalProperties=false    |
| installed_command_verified             | yes      | const=true                                   |
| invocation                             | yes      | \$ref="#/\$defs/ReleaseEvidenceInvocation"   |
| observed_environment_closure_digest    | yes      | \$ref="#/\$defs/Digest"                      |
| platform                               | yes      | enum=["linux", "windows"]                    |
| platform_identity                      | yes      | type="object"; additionalProperties=false    |
| portable_implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                      |
| producer                               | yes      | \$ref="#/\$defs/ReleaseEvidenceProducer"     |
| producer_attestation                   | yes      | \$ref="#/\$defs/EvidenceProducerAttestation" |
| product_acceptance_pass                | yes      | const=false                                  |
| product_public_approval                | yes      | const="not_approved"                         |
| raw_artifact_manifest_digest           | yes      | \$ref="#/\$defs/Digest"                      |
| record_type                            | yes      | const="PlatformVerificationResult"           |
| result_digest                          | yes      | \$ref="#/\$defs/Digest"                      |
| status                                 | yes      | const="pass"                                 |

```

{
  "additionalProperties": false,
  "properties": {
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "exact_candidate_verified": {
      "const": true
    },
    "install_mode": {
      "const": "online-clean"
    },
    "installation": {
      "additionalProperties": false,
      "properties": {
        "build_backend": {
          "additionalProperties": false,
          "properties": {
            "distribution": {
              "const": "setuptools"
            }
          }
        },
        "name": {
          "const": "setuptools.build_meta"
        },
        "record_sha256": {
          "$ref": "#/$defs/Digest"
        },
        "version": {
          "minLength": 1,
          "type": "string"
        }
      },
      "required": [
        "name",
        "distribution",
        "version",
        "record_sha256"
      ],
      "type": "object"
    },
    "build_dependency_source": {
      "const": "online-index-explicit"
    },
    "command_invocations": {

```

```

"items": {
  "additionalProperties": false,
  "properties": {
    "argv_digest": {
      "$ref": "#/$defs/Digest"

```

```

    },
    "command": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "returncode": {
      "const": 0
    },
    "stderr_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stderr_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "stdout_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stdout_sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "command",

```

```

    "argv_digest",
    "returncode",
    "stdout_sha256",
    "stdout_bytes",
    "stderr_sha256",
    "stderr_bytes"
  ],
  "type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"console_script": {
  "additionalProperties": false,
  "properties": {
    "declared": {
      "const": true
    },
    "installed_wrapper_checked": {
      "const": true
    },
    "path": {
      "minLength": 1,
      "type": "string"
    },
    "posix_execute_bits": {
      "type": [

```

```

        "boolean",
        "null"
      ]
    },
    "present": {
      "const": true
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "source_module_invoked": {
      "const": false

```

```

    },
    "required": [
        "present",
        "declared",
        "installed_wrapper_checked",
        "source_module_invoked",
        "posix_execute_bits",
        "path",
        "sha256",
        "size_bytes"
    ],

```

```

        "type": "object"
    },
    "declared_python_requirement": {
        "minLength": 1,
        "type": "string"
    },
    "dependency_closure": {
        "additionalProperties": false,
        "properties": {
            "closure_check": {
                "const": "pass"
            },
            "declared_build": {
                "items": {
                    "type": "string"
                },
                "minItems": 1,
                "type": "array"
            },
            "declared_runtime": {
                "items": {
                    "type": "string"
                },
                "minItems": 1,
                "type": "array"
            },
            "dependency_artifact_digest": {
                "$ref": "#/$defs/Digest"
            }
        }
    },
    "digest": {
        "$ref": "#/$defs/Digest"
    },
    "resolved_runtime": {
        "additionalProperties": false,
        "properties": {
            "cryptography": {
                "additionalProperties": false,
                "properties": {
                    "requirement": {
                        "minLength": 1,
                        "type": "string"
                    },
                    "version": {
                        "minLength": 1,
                        "type": "string"
                    }
                }
            },
            "required": [
                "requirement",
                "version"
            ],
            "type": "object"
        },
        "jsonschema": {
            "additionalProperties": false,
            "properties": {
                "requirement": {
                    "minLength": 1,
                    "type": "string"
                },
                "version": {
                    "minLength": 1,
                    "type": "string"
                }
            }
        }
    },

```

```

        "requirement": {
            "minLength": 1,
            "type": "string"
        },
        "version": {
            "minLength": 1,
            "type": "string"
        }
    },

```

```

        "required": [
            "requirement",
            "version"
        ],
        "type": "object"
    },
    "pypdf": {
        "additionalProperties": false,
        "properties": {
            "requirement": {
                "minLength": 1,
                "type": "string"
            },
            "version": {
                "minLength": 1,
                "type": "string"
            }
        }
    },
    "required": [
        "requirement",
        "version"
    ],
    "type": "object"
},
"required": [
    "cryptography",
    "jsonschema",
    "pypdf"
],
"type": "object"
},
"transitive_distribution_digest": {
    "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "license": {
                "maxLength": 2048,
                "minLength": 1,
                "type": "string"
            },
            "name": {
                "minLength": 1,
                "type": "string"
            }
        }
    },
    "record": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "minimum": 1,
                "type": "integer"
            },
            "path": {
                "minLength": 1,
                "pattern": "\\\\.dist-info/RECORD$",
                "type": "string"
            },
            "sha256": {
                "$ref": "#/$defs/Digest"
            }
        }
    },
    "required": [
        "path",
        "sha256",
        "bytes"
    ],
    "type": "object"
},
"requires": {
    "items": {
        "minLength": 1,
        "type": "string"
    }
},
"type": "array"

```

```

        },
        "version": {
            "minLength": 1,
            "type": "string"
        }
    },
    "required": [
        "name",
        "version",
        "record",
        "requires",
        "license"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
}
},
"required": [
    "declared_runtime",
    "declared_build",
    "resolved_runtime",
    "closure_check",
    "transitive_distributions",
    "transitive_distribution_digest",
    "dependency_artifact_digest",
    "digest"
],
"type": "object"
},
"environment": {
    "const": "clean-venv"
},
"installation_performed": {
    "const": true
},
"installed_distribution": {
    "additionalProperties": false,
    "properties": {
        "name": {
            "const": "promin"
        },
        "runtime_version": {
            "$ref": "#/$defs/SemVer"
        },
        "version": {
            "$ref": "#/$defs/SemVer"
        }
    }
},
"required": [
    "name",
    "version",
    "runtime_version"
],
"type": "object"
},
"installed_environment_observation": {
    "additionalProperties": false,
    "properties": {
        "build_backend": {
            "additionalProperties": false,
            "properties": {
                "distribution": {
                    "const": "setuptools"
                },
                "name": {
                    "const": "setuptools.build_meta"
                },
                "record_sha256": {
                    "$ref": "#/$defs/Digest"
                },
                "version": {
                    "minLength": 1,
                    "type": "string"
                }
            }
        }
    }
},

```

```

    "required": [
      "name",
      "distribution",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "console_wrapper": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    }
  },
  "required": [
    "path",
    "sha256",
    "bytes"
  ],
  "type": "object"
},
"license_closure": {
  "additionalProperties": false,

```

```

    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        }
      }
    },

```

```

    "required": [
      "name",
      "version",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}
},
"required": [
  "entries",
  "complete",
  "digest"
],

```

```

    "type": "object"
  },
  "observation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "platform": {
    "additionalProperties": false,
    "properties": {
      "machine": {
        "minLength": 1,
        "type": "string"

```

```

      },
      "release": {
        "minLength": 1,
        "type": "string"
      },
      "sys_platform": {
        "minLength": 1,
        "type": "string"
      },
      "system": {
        "enum": [
          "linux",
          "windows"
        ]
      },
      "tags": {
        "items": {
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 64,
        "minItems": 1,
        "type": "array"
      }
    }
  },
  "required": [
    "system",
    "machine",

```

```

    "release",
    "sys_platform",
    "tags"
  ],
  "type": "object"
},
"promin": {
  "additionalProperties": false,
  "properties": {
    "installed_file_count": {
      "maximum": 4096,
      "minimum": 1,
      "type": "integer"
    },
    "installed_files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 0,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"

```

```

        }
      },
      "required": [
        "path",
        "sha256",
        "bytes"
      ],
      "type": "object"
    },
    "maxItems": 4096,

```

```

        "minItems": 1,
        "type": "array"
    },
    "installed_files_digest": {
        "$ref": "#/$defs/Digest"
    },
    "module_file": {
        "minLength": 1,
        "type": "string"
    },
    "module_sha256": {
        "$ref": "#/$defs/Digest"
    },
    "record": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "minimum": 1,
                "type": "integer"
            },
            "path": {
                "minLength": 1,
                "pattern": "\\\\.dist-info/RECORD$",
                "type": "string"
            },
            "sha256": {
                "$ref": "#/$defs/Digest"
            }
        },
        "required": [
            "path",
            "sha256",
            "bytes"
        ],
        "type": "object"
    },
    "version": {
        "$ref": "#/$defs/SemVer"
    }
},
"required": [
    "version",
    "module_file",
    "module_sha256",
    "record",
    "installed_files",
    "installed_files_digest",
    "installed_file_count"
],
"type": "object"
},
"python": {
    "additionalProperties": false,
    "properties": {
        "abi_tag": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "base_prefix": {
            "minLength": 1,
            "type": "string"
        },
        "executable": {
            "minLength": 1,
            "type": "string"
        },
        "executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "implementation": {

```



```

        "minLength": 1,
        "type": "string"
    },
    "prefix": {
        "minLength": 1,
        "type": "string"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "executable",
    "executable_sha256",
    "base_executable",
    "base_executable_sha256",
    "implementation",
    "version",
    "abi_tag",
    "prefix",
    "base_prefix"
],
"type": "object"
},

```

```

"sbom": {
    "additionalProperties": false,
    "properties": {
        "components": {
            "items": {
                "additionalProperties": false,
                "properties": {
                    "name": {
                        "minLength": 1,
                        "type": "string"
                    },
                    "record_sha256": {
                        "$ref": "#/$defs/Digest"
                    },
                    "version": {
                        "minLength": 1,
                        "type": "string"
                    }
                },
                "required": [
                    "name",
                    "version",
                    "record_sha256"
                ],
                "type": "object"
            },
            "maxItems": 256,
            "minItems": 1,

```

```

            "type": "array"
        },
        "digest": {
            "$ref": "#/$defs/Digest"
        },
        "format": {
            "const": "promin-spx-lite-v1"
        }
    },
    "required": [
        "format",
        "components",
        "digest"
    ],
    "type": "object"
},
"site_packages": {
    "minLength": 1,
    "type": "string"
},
"sqlite_version": {
    "minLength": 1,
    "type": "string"
},
"transitive_distribution_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "transitive_distributions": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "license": {
          "maxLength": 2048,
          "minLength": 1,
          "type": "string"
        },
        "name": {
          "minLength": 1,
          "type": "string"
        },
        "record": {
          "additionalProperties": false,
          "properties": {
            "bytes": {
              "minimum": 1,
              "type": "integer"
            },
            "path": {
              "minLength": 1,
              "pattern": "\\\\.dist-info/RECORD$",
              "type": "string"
            },
            "sha256": {
              "$ref": "#/$defs/Digest"
            }
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },
      "requires": {
        "items": {
          "minLength": 1,
          "type": "string"
        },
        "type": "array"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "name",
      "version",
      "record",
      "requires",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}
},
"required": [
  "python",
  "platform",
  "sqlite_version",
  "site_packages",
  "promin",
  "console_wrapper",
  "transitive_distributions",
  "transitive_distribution_digest",
  "sbom",
  "license_closure",
  "build_backend",
  "observation_digest"

```

```

    ],
    "type": "object"
  },
  "license_closure": {
    "additionalProperties": false,
    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        },
        "required": [
          "name",
          "version",
          "license"
        ]
      },
      "type": "object"
    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  },
  "required": [
    "entries",
    "complete",
    "digest"
  ],
  "type": "object"
},
"nested_venv_created": {
  "const": true
},
"network_disabled": {
  "const": false
},
"pip_report": {
  "additionalProperties": false,
  "properties": {
    "artifact_digest": {
      "$ref": "#/$defs/Digest"
    },
    "report_count": {
      "const": 2
    }
  },
  "reports": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_digest": {
          "$ref": "#/$defs/Digest"
        },
        "artifacts": {
          "items": {
            "additionalProperties": false,

```

```

        "properties": {
          "hashes": {
            "items": {
              "minLength": 3,
              "type": "string"
            },
            "type": "array"
          },
          "is_direct": {
            "type": "boolean"
          },
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "version": {
            "minLength": 1,

```

```

            "type": "string"
          }
        },
        "required": [
          "name",
          "version",
          "is_direct",
          "hashes"
        ],
        "type": "object"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array"
    },
    "bytes": {
      "maximum": 16777216,
      "minimum": 1,
      "type": "integer"
    },
    "install_count": {
      "maximum": 256,
      "minimum": 1,
      "type": "integer"
    },
    "pip_version": {
      "minLength": 1,
      "type": "string"

```

```

    },
    "role": {
      "enum": [
        "build-requirements",
        "candidate-install"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "role",
    "sha256",
    "bytes",
    "pip_version",
    "install_count",
    "artifacts",
    "artifact_digest"
  ],
  "type": "object"
},
"maxItems": 2,
"minItems": 2,
"type": "array"
}
},
"required": [

```

```

  "reports",
  "report_count",
  "artifact_digest"
],

```

```

    "type": "object"
  },
  "runtime_dependency_source": {
    "const": "online-index"
  },
  "sbom": {
    "additionalProperties": false,
    "properties": {
      "components": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "record_sha256": {
              "$ref": "#/$defs/Digest"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        }
      }
    }
  },

```

```

    "required": [
      "name",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
},
"digest": {
  "$ref": "#/$defs/Digest"
},
"format": {
  "const": "promin-spdx-lite-v1"
}
},
"required": [
  "format",
  "components",
  "digest"
],
"type": "object"
}
},
"required": [
  "environment",

```

```

    "installation_performed",
    "nested_venv_created",
    "runtime_dependency_source",
    "build_dependency_source",
    "network_disabled",
    "declared_python_requirement",
    "installed_distribution",
    "console_script",
    "command_invocations",
    "dependency_closure",
    "installed_environment_observation",
    "pip_report",
    "sbom",
    "license_closure",
    "build_backend"
  ],
  "type": "object"
},
"installed_command_verified": {
  "const": true
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"observed_environment_closure_digest": {
  "$ref": "#/$defs/Digest"
}

```

```

},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"platform_identity": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "platform": {
      "enum": [
        "linux",
        "windows"
      ]
    },
    "platform_release": {
      "minLength": 1,
      "type": "string"
    },
    "platform_tags": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "python_abi_tag": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "python_implementation": {
      "minLength": 1,
      "type": "string"
    },
    "python_version": {
      "minLength": 1,
      "type": "string"
    },
    "sys_platform": {
      "minLength": 1,
      "type": "string"
    }
  }
},
"required": [
  "platform",
  "machine",
  "platform_release",
  "sys_platform",
  "platform_tags",
  "python_version",
  "python_implementation",
  "python_executable",
  "python_executable_sha256",
  "python_abi_tag"
],
"type": "object"
},
"portable_implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},

```

```

"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},

```

```

"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "PlatformVerificationResult"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
}

```

```

},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "platform",
  "platform_identity",
  "exact_candidate_verified",
  "archive_sha256",
  "archive_bytes",
  "portable_implementation_closure_digest",
  "observed_environment_closure_digest",
  "install_mode",
  "installed_command_verified",
  "installation",
  "product_acceptance_pass",

```

```

"product_public_approval",
"producer",
"invocation",
"raw_artifact_manifest_digest",
"producer_attestation",
"result_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/PlatformVerificationResult | Validator: urn:promin:contracts:1#/\$defs/PlatformVerificationResult

## 71. PolicyDef

Type summary: type="object"; additionalProperties=false

Required: id, name, validator\_id, severity, statement

| Property     | Required | Structural rule            |
|--------------|----------|----------------------------|
| id           | yes      | enum=[68 items]            |
| name         | yes      | \$ref="#/\$defs/Id"        |
| severity     | yes      | enum=["P0", "P1", "P2"]    |
| statement    | yes      | type="string"; minLength=1 |
| validator_id | yes      | \$ref="#/\$defs/Id"        |

```

{
  "additionalProperties": false,
  "properties": {
    "id": {
      "enum": [
        "P-001",
        "P-002",
        "P-003",
        "P-004",
        "P-005",
        "P-006",
        "P-007",
        "P-008",

```

```
"P-009",  
"P-010",  
"P-011",  
"P-012",  
"P-013",  
"P-014",  
"P-015",  
"P-016",  
"P-017",  
"P-018",  
"P-019",  
"P-020",  
"P-021",  
"P-022",  
"P-023",
```

```
"P-024",  
"P-025",  
"P-026",  
"P-027",  
"P-028",  
"P-029",  
"P-030",  
"P-031",  
"P-032",  
"P-033",  
"P-034",  
"P-035",  
"P-036",  
"P-037",  
"P-038",  
"P-039",  
"P-040",  
"P-041",  
"P-042",  
"P-043",  
"P-044",  
"P-045",  
"P-046",  
"P-047",  
"P-048",  
"P-049",  
"P-050",  
"P-051",
```

```
"P-052",  
"P-053",  
"P-054",  
"P-055",  
"P-056",  
"P-057",  
"P-058",  
"P-059",  
"P-060",  
"P-061",  
"P-062",  
"P-063",  
"P-064",  
"P-065",  
"P-066",  
"P-067",  
"P-068"  
]  
},  
"name": {  
  "$ref": "#/$defs/Id"  
},  
"severity": {  
  "enum": [  
    "P0",  
    "P1",  
    "P2"  
  ]  
}
```

```
},  
"statement": {  
  "minLength": 1,  
  "type": "string"  
},  
"validator_id": {
```



```

    "$ref": "#/$defs/Id"
  },
  "required": [
    "id",
    "name",
    "validator_id",
    "severity",
    "statement"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/PolicyDef | Validator: urn:promin:contracts:1#/\$defs/PolicyDef

## 72. PolicySet

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, document, version, function, validator\_registry\_version, optimization\_gate, state\_machines, policies, alpha4\_standard\_contracts, derived\_result\_contracts, gate\_run\_definition\_contract, research\_draft\_sanitation, semantic\_export\_contract

| Property                     | Required | Structural rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alpha4_standard_contracts    | yes      | const=schema(keys:artifact_policy,authority,claims,documentation_boundary,dynamic_handoff,gate_invalidation,init_profiles,language_analysis,project_package,provider_receipts,schema,static_admission,test_sharding,writer_publication)                                                                                                                                                                                                                                                                                                                                                                     |
| derived_result_contracts     | yes      | const=schema(keys:DoctorResult,OperationMetrics,ReadyFrontier,RetrievalPage,WorkCardProjection)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| document                     | yes      | const="promin.policy-set"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| function                     | yes      | const="Own executable cross-record invariants through stable validator identifiers and no duplicated budget values."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| gate_run_definition_contract | yes      | const=schema(keys:artifact_binding_fields,artifact_record_digest_rule,authorization_scope_contract,command_binding,definition_change,definition_digest_rule,definition_owner,definition_provenance_fields,evidence_class_purpose_pairs,expected_owner_fields,implementation_closure_binding_rule,inline_definition_submission,mixed_evidence_classes,payload_digest_as_artifact_record_binding,persistence,product_credit_required_rule,provider_binding_digest_rule,run_binding_rule,status_outcome_rules,submitted_artifacts_define_expectations,target_binding_rule,task_binding,task_owner_digest_rule) |
| optimization_gate            | yes      | const=schema(keys:base_predicate,formula,stress_predicate)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| policies                     | yes      | const=[68 items]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| record_type                  | yes      | const="PolicySet"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| research_draft_sanitation    | yes      | const=schema(keys:allowed_license_disposition,canonical_bytes_max,citation_required_rule,claim_digest_rule,claim_kinds,evidence_reference_rule,forbidden_normative_claim_classes,forbidden_requested_flag_mutations,metadata_only_rule,migration_rule,normative_candidate_rule,persistence,result_flags,result_persistence,result_record_type,source_binding,source_license_rule,source_payload_bytes_max,source_payload_total_bytes_max,verification_statuses)                                                                                                                                             |
| schema_ref                   | yes      | const="urn:promin:contracts:1#/\$defs/PolicySet"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| semantic_export_contract     | yes      | const=schema(keys:export_kind,input_digests_max,input_digests_rule,output_artifact_kind,output_artifact_record_digest_rule,output_bytes_max,output_bytes_rule,output_media_type,output_record_type,output_records_max,package_or_archive,persistence,product_flags,record_order,required_bindings)                                                                                                                                                                                                                                                                                                          |
| state_machines               | yes      | const=schema(keys:capacity_release,lease,task)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| validator_registry_version   | yes      | const=1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| version                      | yes      | const="1.0.0-alpha.4"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

```

{
  "additionalProperties": false,
  "properties": {
    "alpha4_standard_contracts": {
      "const": {
        "artifact_policy": {
          "classes": [
            "portable-normative-doc",
            "compact-current-report",
            "diagnostic-inventory",
            "forensic-log",
            "runtime-evidence",
            "cache",
            "recovery-backup"
          ]
        }
      }
    }
  }
}

```

```

    ],
    "default_mode": "minimal",
    "detailed_output": "host-local-only",
    "single_current_registry": true,
    "tracked_boundary": ".promin/docs"
  },
  "authority": {
    "dual_authority": "reject",
    "legacy_compatibility": "reject",
    "owner": "core/policy-set.json",
    "project_specific_defaults": "reject"
  },
  "claims": {
    "acceptance_pass": false,

    "pass_credit": false,
    "product_acceptance_pass": false,
    "release_eligible": false,
    "runtime_validated": false
  },
  "documentation_boundary": {
    "extension_root": ".promin/docs/extensions",
    "forbidden_operational_roots": [
      "cache",
      "evidence",
      "init",
      "leases",
      "locks",
      "providers",
      "state"
    ],
    "link_or_reparse": "reject",
    "manifest": "typed-byte-exact",
    "normalization_collision": "reject"
  },
  "dynamic_handoff": {
    "acceptance_promotion": "forbidden",
    "execution": "dynamic-only",
    "required_status": "PENDING_DYNAMIC",
    "shortcuts": [
      "direct-projection-edit",
      "direct-sqlite-edit",
      "synthetic-evidence",
      "bypassed-gate"
    ]
  },
  "gate_invalidation": {
    "classes": [
      "BODY_ONLY",
      "IMPORT_SURFACE",
      "CMAKE_TOPOLOGY",
      "TOOLING_ONLY"
    ],
    "git-head-alone": "not-an-invalidation-input",
    "receipt": "input-digest-scope-timing-status-credit",
    "source_first": true
  },
  "init_profiles": {
    "authority_effect": "none",
    "precedence": [
      "default",
      "host-profile",
      "project-package",
      "cli",
      "interactive-user"
    ],
    "selection": [
      "accept",
      "decline",
      "custom"
    ]
  },
  "unavailable_is_pass": false
},
"language_analysis": {
  "finding_classification": [
    "PROVEN",
    "REVIEW",
    "SAFE"
  ]
}

```

```

    ],
    "interface_weight": "review-only",
    "lexical_mask": "offset-preserving",
    "missing_tool": "UNAVAILABLE",
    "semantic_tool_input": "canonical-compilation-database"
  },
  "project_package": {
    "clean_init": "owner-confirmed-only",
    "member_identity": [
      "path",
      "class",
      "mode",
      "bytes",
      "sha256"
    ],
    "old_operational_state": "quarantine-or-rollback-only",
    "replay_or_progress_import": false,
    "tree_digest": "required"
  },
  "provider_receipts": {
    "byte_authority": "content-addressed-merkle",

    "coverage_modes": [
      "FullScan",
      "Reuse"
    ],
    "metadata_only_reuse": "forbidden",
    "target_scoped": true,
    "transient_inputs": "exclude-with-receipt"
  },
  "schema": "promin.alpha4-standard-contracts.v1",
  "static_admission": {
    "allowed_kinds": [
      "source",
      "documentation",
      "portability",
      "package",
      "handoff"
    ],
    "forbidden_side_effects": [
      "provider",
      "configure",
      "build",
      "runtime",
      "sqlite",
      "projection"
    ],
    "timing_receipts": "required"
  },
  "test_sharding": {

    "aggregate_coverage": "exact-once",
    "duplicate_credit": false,
    "memory_limit": "identical-per-shard",
    "terminal_receipt": "required",
    "timeout": "identical-per-shard"
  },
  "writer_publication": {
    "intent_reuse": "exact-verified-only",
    "liveness": "pid-process-start-lease",
    "post_reservation_revalidation": true,
    "preflight_before_reservation": true,
    "publication": "create-only-overlap-safe"
  }
},
"derived_result_contracts": {
  "const": {
    "DoctorResult": {
      "archival_diagnostics_effect": "none",
      "claim_scope": "runtime-health-only",
      "component_status_values": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ],
    },
    "current_components": [
      "core",

```

```

    "init",
    "providers",
    "recovery",
    "replay",
    "projection"
  ],
  "forbidden_allocation_scopes": [
    "batch",
    "goal",
    "per-run",
    "session",
    "workspace"
  ],
  "forbidden_metric_families": [
    "cost",
    "token",
    "usage"
  ],
  "live_only": true,
  "metrics": [
    "core_artifacts_verified",
    "event_batches",
    "init_records_verified",
    "projection_entities",
    "projection_relations",
    "provider_healthchecks_executed"
  ],
  "pass_credit": false,

```

```

    "product_acceptance_pass": false,
    "product_public_approval": "not_approved",
    "product_tree_scans": 0,
    "report_authoritative": false,
    "rollup_precedence": [
      "failed",
      "incomplete",
      "degraded",
      "healthy"
    ],
    "source_scope": "installed-core-and-live-operational-state-only"
  },
  "OperationMetrics": {
    "authoritative": false,
    "claim_scope": "single-command-observation-only",
    "forbidden_allocation_scopes": [
      "batch",
      "goal",
      "per-run",
      "session",
      "workspace"
    ],
    "forbidden_metric_families": [
      "cost",
      "token",
      "usage"
    ],
    "live_only": true,

```

```

    "metric_groups": [
      "duration_ms",
      "changed_records",
      "event_write",
      "projection_update",
      "runtime_checkpoint",
      "physical_payload_bytes",
      "bytes_per_changed_record"
    ]
  },
  "ReadyFrontier": {
    "blocking_finding_statuses": [
      "OPEN"
    ],
    "cycle": "reject",
    "dependency_relation_kind": "DEPENDS_ON",
    "eligible_task_state": "READY",
    "gate_currentness_bindings": [
      "activation_digest",
      "candidate_digest",
      "definition_digest"
    ],

```

```

"gate_outcome": "pass",
"gate_pass_credit": true,
"live_only": true,
"ordering": [
  "created_at-ascending",
  "task_id-ascending"

```

```

],
"projection_authoritative": false,
"required_dependency_state": "COMPLETED",
"silent_truncation": false
},
"RetrievalPage": {
  "canonical_cursor_fields": [
    "stream_cursor",
    "next_stream_cursor"
  ],
  "forbidden_compatibility_fields": [
    "seed_cursor",
    "next_seed_cursor"
  ],
  "live_only": true,
  "projection_authoritative": false,
  "query_bytes_max": 4096,
  "refinement_hints_max": 8,
  "silent_truncation": false
},
"WorkCardProjection": {
  "continuation_exposed": false,
  "live_only": true,
  "projection_authoritative": false,
  "selected_task_only": true
}
}
},

```

```

"document": {
  "const": "promin.policy-set"
},
"function": {
  "const": "Own executable cross-record invariants through stable validator identifiers and no
duplicated budget values."
},
"gate_run_definition_contract": {
  "const": {
    "artifact_binding_fields": [
      "artifact_id",
      "artifact_record_digest",
      "run_id",
      "run_digest"
    ],
    "artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
    "authorization_scope_contract": {
      "all_selector": "reject",
      "allowed_extra_kinds": [
        "project",
        "task"
      ],
      "lease_bound_task_selector_required": true,
      "project_items_max": 1,
      "target_scope_relation": "subset",
      "task_items_max": 1,
      "task_value_rule": "GateResult.task_id",
      "unrelated_selector": "reject"
    }
  },

```

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"command_binding": "CommandRequest.definition_digest equals the immutable GateRunDefinition digest
and the payload definition_digest",
"definition_change": "reject after Task creation; define a new Task identity for changed
requirements",
"definition_digest_rule": "sha256(canonical GateRunDefinition)",
"definition_owner": "Task.gate_run_definitions",
"definition_provenance_fields": [
  "owner_kind",
  "owner_digest",
  "defined_at_head_digest"
],
"evidence_class_purpose_pairs": {
  "diagnostic": [

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        "validator"
    ],
    "gate": [
        "validator",
        "product-execution"
    ],
    "product": [
        "product-execution"
    ]
},
"expected_owner_fields": [
    "expected_evidence_class",
    "expected_evidence_purpose",
    "product_credit_required",
    "target_kind",
    "target_digest",
    "target_scope",
    "candidate_digest",
    "policy_digest",
    "tool_digest",
    "implementation_closure_digest",
    "provider_binding_digest",
    "input_digests"
],
"implementation_closure_binding_rule": "GateRunDefinition, Run, and every exact bound Artifact
EvidenceBinding carry the same implementation_closure_digest",
"inline_definition_submission": "reject",
"mixed_evidence_classes": "reject",
"payload_digest_as_artifact_record_binding": "forbidden",
"persistence": "immutable-definition-input",
"product_credit_required_rule": "true requires the product purpose and product-execution class;
false may use any declared evidence class-purpose pair",
"provider_binding_digest_rule": "sha256(canonical installed provider-binding evidence array);
actual ProviderInvocationEvidence records remain exact immutable Artifact inputs and are independently
validated against the active provider bindings when present",
"run_binding_rule": "GateResult.run_id and run_digest and every GateEvidenceArtifactBinding.run_id
and run_digest bind the exact canonical Run record",
"status_outcome_rules": {
    "blocked": {
        "evidence_binding": "required",
        "outcome": "blocked",
        "pass_credit": false,
        "reason": "required"
    },
    "fail": {
        "evidence_binding": "required",
        "outcome": "fail",
        "pass_credit": false,
        "reason": "optional"
    }
},
"pass": {
    "diagnostic_pass_with_no_credit": true,
    "evidence_binding": "required",
    "outcome": "pass",
    "pass_credit_rule": "true only when definition product_credit_required is true and every exact
bound Artifact is fresh, resolved, passing, product-execution evidence"
},
"skipped": {
    "evidence_binding": "forbidden",
    "outcome": "skipped",
    "pass_credit": false,
    "reason": "required"
}
},
"submitted_artifacts_define_expectations": false,
"target_binding_rule": "candidate target_digest equals candidate_digest; finding target_digest is
the immutable exact OPEN Finding record digest captured at definition time and is not recomputed from
later mutable Finding state",
"task_binding": "GateResult.task_id and definition_digest resolve one exact embedded Task
definition",
"task_owner_digest_rule": "sha256(canonical immutable Task fields excluding state and
gate_run_definitions)"
}
},
"optimization_gate": {
    "const": {
        "base_predicate": "NetValue>0",
        "formula": "NetValue=3*RiskReduction+2*AutomationGain+1.5*QueryabilityGain+ReuseGain-1.5*Implementa

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tionCost-2*MaintenanceCost-2*CognitiveNoise-1.5*ReferenceCost-MigrationCost",
    "stress_predicate": "NetValue remains >0 when benefits decrease 20 percent and costs increase 20
percent"
    },
    "policies": {
        "const": [
            {
                "id": "P-001",
                "name": "no-false-success",
                "severity": "P0",
                "statement": "Gate pass credit is true exactly when normalized status is pass and every required
gate passes.",
                "validator_id": "no_false_success"
            },
            {
                "id": "P-002",
                "name": "activation-binding",
                "severity": "P0",
                "statement": "Every authoritative record binds the exact active Core, preset, init, and profile
configuration.",
                "validator_id": "activation_binding"
            },
            {
                "id": "P-003",
                "name": "grant-validity",
                "severity": "P0",
                "statement": "Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof
chain, revocation state, and Activation all match.",
                "validator_id": "grant_validity"
            },
            {
                "id": "P-004",
                "name": "lease-fence",
                "severity": "P0",
                "statement": "Every mutation uses the current active Lease generation and monotonic fencing
token.",
                "validator_id": "lease_fence"
            },
            {
                "id": "P-005",
                "name": "candidate-evidence-binding",
                "severity": "P0",
                "statement": "Every evidence Artifact requires an explicit diagnostic, gate, or product purpose
and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event
payload.",
                "validator_id": "candidate_evidence_binding"
            },
            {
                "id": "P-006",
                "name": "atomic-event-commit",
                "severity": "P0",
                "statement": "One command produces one immutable batch under lock, compare-head, sync, atomic
replace, and recovery.",
                "validator_id": "atomic_event_commit"
            },
            {
                "id": "P-007",
                "name": "projection-is-derived",
                "severity": "P0",
                "statement": "Projection deletion and deterministic replay cannot change normative meaning.",
                "validator_id": "projection_derived"
            },
            {
                "id": "P-008",
                "name": "minimal-graph",
                "severity": "P1",
                "statement": "Each verified raw file creates exactly one non-authoritative derived Artifact
proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require
explicit semantic content.",
                "validator_id": "minimal_graph"
            },
            {
                "id": "P-009",
                "name": "bounded-workcard",
                "severity": "P0",
                "statement": "Every WorkCard stays within both Core hard ceilings and its selected profile byte,

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entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.",
  "validator_id": "bounded_workcard"
},
{
  "id": "P-010",
  "name": "size-independent-init",
  "severity": "P0",
  "statement": "Initialization performs zero product-tree scans and creates exactly five init
records.",
  "validator_id": "zero_scan_init"
},
{
  "id": "P-011",
  "name": "explicit-external-bindings",
  "severity": "P1",
  "statement": "External standards and providers exist only in explicit digest-bound init
records.",
  "validator_id": "explicit_bindings"
},
{
  "id": "P-012",
  "name": "canonical-name-only",
  "severity": "P0",
  "statement": "The canonical identity is promin; aliases, redirects, and compatibility names are
rejected.",
  "validator_id": "canonical_name"
},
{
  "id": "P-013",
  "name": "safe-export",
  "severity": "P0",
  "statement": "Export is allowlisted, recursively bounded, symlink-safe, secret-aware, and
path-sanitized.",
  "validator_id": "safe_export"
},
{
  "id": "P-014",
  "name": "closed-acknowledgement",
  "severity": "P0",
  "statement": "Only a valid CLOSED transition with explicit close acknowledgement releases a
mutation slot.",
  "validator_id": "closed_ack"
},
{
  "id": "P-015",
  "name": "fail-closed-critical-fields",
  "severity": "P0",
  "statement": "Missing, unknown, stale, downgraded, or unbound critical fields reject before
mutation.",
  "validator_id": "critical_fields"
},
{
  "id": "P-016",
  "name": "preset-capability-ceiling",
  "severity": "P0",
  "statement": "A preset may configure provider needs and budgets but cannot grant action
authority.",
  "validator_id": "preset_no_authority"
},
{
  "id": "P-017",
  "name": "single-scan-inventory",
  "severity": "P1",
  "statement": "One requested inventory performs at most one full product-tree pass; all
projections derive from its stream.",
  "validator_id": "single_scan"
},
{
  "id": "P-018",
  "name": "idempotent-commands",
  "severity": "P0",
  "statement": "A duplicate idempotency key under the same activation and subject returns the
original result or a conflict.",
  "validator_id": "idempotent_commands"
},
{
  "id": "P-019",
  "name": "core-preset-separation",

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    "severity": "P1",
    "statement": "Core bundle identity excludes replaceable presets; Activation binds the selected
preset separately.",
    "validator_id": "core_preset_separation"
  },
  {
    "id": "P-020",
    "name": "acyclic-bootstrap-authority",
    "severity": "P0",
    "statement": "Init config defines root ceilings, never pre-activation runtime Grants.",
    "validator_id": "bootstrap_authority"
  },
  {
    "id": "P-021",
    "name": "normalization-collision-rejection",
    "severity": "P1",
    "statement": "Duplicate JSON keys and path or ID collisions after NFC/casefold normalization are
rejected.",
    "validator_id": "normalization_collision"
  },
  {
    "id": "P-022",
    "name": "provider-substitution-binding",
    "severity": "P1",
    "statement": "A provider identity or implementation-closure change invalidates the Activation
and every dependent continuation, evidence Artifact, projection, replay result, and current release
closure; it never silently changes authority.",
    "validator_id": "provider_substitution"
  },
  {
    "id": "P-023",
    "name": "exact-init-keyset",
    "severity": "P1",
    "statement": "Activation binds exactly project.json, standards.json, technologies.json, and
authority.json.",
    "validator_id": "exact_init_keyset"
  },
  {
    "id": "P-024",
    "name": "stale-projection-rejection",
    "severity": "P1",
    "statement": "Status, next, and WorkCard generation reject a projection whose source head or
activation differs.",
    "validator_id": "stale_projection"
  },
  {
    "id": "P-025",
    "name": "one-command-one-batch",
    "severity": "P0",
    "statement": "Every committed batch binds exactly one subject, validated command digest, command
intent digest, authorization digest, and idempotency key.",
    "validator_id": "one_command_one_batch"
  },
  {
    "id": "P-026",
    "name": "deterministic-retrieval",
    "severity": "P1",
    "statement": "Ranking ties are deterministic and acceptance-critical facts are added by typed
closure, not fuzzy ranking.",
    "validator_id": "deterministic_retrieval"
  },
  {
    "id": "P-027",
    "name": "strict-time-order",
    "severity": "P1",
    "statement": "Timestamps use canonical UTC seconds and issued, heartbeat, expiry, finish, and
decision order is semantically checked.",
    "validator_id": "strict_time"
  },
  {
    "id": "P-028",
    "name": "exact-standard-files",
    "severity": "P1",
    "statement": "Core and selected preset inputs reject symlinks and unexpected shadow files.",
    "validator_id": "exact_standard_files"
  },

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{
  "id": "P-029",
  "name": "root-authorization-is-bootstrap-only",
  "severity": "P0",
  "statement": "Root command authorization signs one command intent and may issue the first
authority.manage Grant only; it cannot execute ordinary project commands.",
  "validator_id": "root_authorization_limit"
},
{
  "id": "P-030",
  "name": "configuration-exact-idempotency",
  "severity": "P1",
  "statement": "Repeated initialization succeeds only when the requested canonical configuration
digest exactly equals the existing Activation; otherwise it reports conflict.",
  "validator_id": "config_exact_idempotency"
},
{
  "id": "P-031",
  "name": "truthful-observability",
  "severity": "P1",
  "statement": "A command may report a scan, healthcheck, validation, or durability action only
when that action was actually executed and evidence-bound.",
  "validator_id": "truthful_observability"
},
{
  "id": "P-032",
  "name": "depth-aware-retrieval-budget",
  "severity": "P1",
  "statement": "Effective retrieval seed count reserves entity and relation capacity for typed
closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all
remain effective, and no removed or unknown budget field may bypass a ceiling.",
  "validator_id": "depth_aware_retrieval"
},
{
  "id": "P-033",
  "name": "lease-only-for-mutation",
  "severity": "P1",
  "statement": "Mutation WorkCards require the current Lease; read-only WorkCards must not acquire
or carry a mutation Lease.",
  "validator_id": "lease_only_for_mutation"
},
{
  "id": "P-034",
  "name": "command-capability-scope",
  "severity": "P0",
  "statement": "Every command resolves to exactly one Core-owned capability; requested scope
retains every authorization constraint and explicitly contains every command effect target.",
  "validator_id": "command_capability_scope"
},
{
  "id": "P-035",
  "name": "grant-issuance-chain",
  "severity": "P0",
  "statement": "Every Grant proof binds claim_digest and is either the one bootstrap root
issuance, a verified team threshold, or an active authority.manage issuer Grant.",
  "validator_id": "grant_issuance_chain"
},
{
  "id": "P-036",
  "name": "verified-team-signatures",
  "severity": "P0",
  "statement": "Team-signature fields are never valid without threshold evaluation by the
signature adapter resolved from the exact technologies binding; process-local injected adapters are
explicitly non-reconstructable and cannot prove a cross-process CLI workflow.",
  "validator_id": "verified_team_signatures"
},
{
  "id": "P-037",
  "name": "provider-preflight-before-activation",
  "severity": "P1",
  "statement": "Every required provider identity and healthcheck succeeds before initialization is
committed; a declared provider is not evidence of availability.",
  "validator_id": "provider_preflight"
},
{
  "id": "P-038",
  "name": "canonical-state-machines",
  "severity": "P0",

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    "statement": "Task and Lease transitions are accepted only by the policy-owned state machines;
generated lifecycle views cannot invent transitions.",
    "validator_id": "state_machine_transition"
  },
  {
    "id": "P-039",
    "name": "decision-target-binding",
    "severity": "P0",
    "statement": "Every Decision names one typed target and the decision kind, requested scope,
candidate, and authority are consistent with that target.",
    "validator_id": "decision_target_binding"
  },
  {
    "id": "P-040",
    "name": "credit-requires-evidence",
    "severity": "P0",
    "statement": "Pass, fail, and blocked gate results cite at least one exact immutable Artifact
record reference; skipped results never receive pass credit.",
    "validator_id": "credit_requires_evidence"
  },

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  {
    "id": "P-041",
    "name": "command-event-correspondence",
    "severity": "P0",
    "statement": "One command batch binds the exact authorization claim and contains exactly one
primary event equal to the validated command payload; auxiliary events are typed relations only and event
IDs are unique.",
    "validator_id": "command_event_correspondence"
  },
  {
    "id": "P-042",
    "name": "continuation-authorization-and-completeness",
    "severity": "P0",
    "statement": "A continuation is authenticated by a random per-Activation secret and binds the
exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized
query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth,
implementation closure, issue time, and expiry; its complete page union emits every reachable entity and
relation exactly once without loss.",
    "validator_id": "continuation_completeness"
  },
  {
    "id": "P-043",
    "name": "candidate-snapshot-consistency",
    "severity": "P0",
    "statement": "The candidate recipe is applied in the single inventory pass; promin v1 permits
only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and
receive no pass credit.",
    "validator_id": "candidate_snapshot_consistency"
  },
  {
    "id": "P-044",
    "name": "provider-adapter-dispatch",
    "severity": "P0",
    "statement": "Every technologies.json binding resolves to the adapter actually invoked and
carries a recomputed capability-specific implementation closure for the promin runtime, Python
interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject,
and dependent evidence binds the exact implementation_closure_digest.",
    "validator_id": "provider_adapter_dispatch"
  },

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  {
    "id": "P-045",
    "name": "current-release-closure",
    "severity": "P0",
    "statement": "A project-operational release Decision is immutable history; current product
release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence,
implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for
promin distribution.",
    "validator_id": "current_release_closure"
  },
  {
    "id": "P-046",
    "name": "finding-disposition-evidence",
    "severity": "P0",
    "statement": "Every Finding cites exact immutable Artifact record references. Every resolve or
waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an
exception-policy digest, future expiry, and separation of duties.",
    "validator_id": "finding_disposition_evidence"
  },

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{
  "id": "P-047",
  "name": "candidate-delta-scope",
  "severity": "P0",
  "statement": "A diff Artifact binds base and new Candidate digests, the exact WorkCard digest,
and the complete changed-path set; every changed path is contained by both Task allowed_paths and the
authorizing Grant effect scope.",
  "validator_id": "candidate_delta_scope"
},
{
  "id": "P-048",
  "name": "verified-state-checkpoint",
  "severity": "P1",
  "statement": "Events remain the sole authority. A normal command appends 0(batch), incrementally
updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state
compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely
against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal
semantic digest.",
  "validator_id": "verified_state_checkpoint"
},
{
  "id": "P-049",
  "name": "profile-default-depth",
  "severity": "P1",
  "statement": "A profile default_dependency_depth applies only when depth is omitted; explicit
depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all
retrieval budgets.",
  "validator_id": "profile_default_depth"
},
{
  "id": "P-050",
  "name": "canonical-distribution-archive",
  "severity": "P0",
  "statement": "The sole package builder emits immutable candidate bytes and an external
StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is
derived later and never changes the candidate ZIP.",
  "validator_id": "canonical_distribution_archive"
},
{
  "id": "P-051",
  "name": "verified-inventory-provenance",
  "severity": "P0",
  "statement": "Public rebuild accepts only a verified InventoryResult or its persisted manifest;
caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the
exact path and digest.",
  "validator_id": "verified_inventory_provenance"
},
{
  "id": "P-052",
  "name": "supported-command-surface",
  "severity": "P1",
  "statement": "The v1 preset advertises exactly init, doctor, status, next, validate, and
continue; no optional or administrative command catalogue may claim an absent executable path.",
  "validator_id": "supported_command_surface"
},
{
  "id": "P-053",
  "name": "profile-bound-scale",
  "severity": "P0",
  "statement": "Physical 100k evidence binds content-addressed raw inventory, query, continuation,
process-sample, phase-log, and operation-metric artifacts for at least 198999 explicitly recorded
Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and
fails closed on truthful process/invoke status, commit p95/p99, bytes per changed record, checkpoint
count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak
RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or
traversal-pass ceiling.",
  "validator_id": "profile_bound_scale"
},
{
  "id": "P-054",
  "name": "cross-platform-release-evidence",
  "severity": "P0",
  "statement": "The exact candidate ZIP must pass authenticated Windows and Linux platform
observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale,
offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed
PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles,
the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14
supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer

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attestations bind the same candidate and participate in one replay/SOD/chronology graph;
matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain
false.",
  "validator_id": "cross_platform_release"
},
{
  "id": "P-055",
  "name": "standard-release-separation",
  "severity": "P0",
  "statement": "An external StandardReleaseDecision has explicit approve or reject outcome, binds
candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time,
stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability,
independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest,
and signature. The signed decision remains immutable history; current eligibility is derived;
product_acceptance_pass is false and product_public_approval is not_approved.",
  "validator_id": "standard_release_separation"
},
{
  "id": "P-056",
  "name": "required-gates-fail-closed",
  "severity": "P0",
  "statement": "No-degradation runs the complete required suite only through the clean installed
interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion,
zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every
required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor,
offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped,
missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass
credit.",
  "validator_id": "required_gates_fail_closed"
},
{
  "id": "P-057",
  "name": "research-draft-sanitation",
  "severity": "P0",
  "statement": "Research crosses into normative owners only through the non-persistent
MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required
claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution,
acceptance, or public-approval claims cannot become normative text.",
  "validator_id": "research_draft_sanitation"
},
{
  "id": "P-058",
  "name": "doctor-result-is-live-non-authority",
  "severity": "P0",
  "statement": "DoctorResult is an exact live-only derived health report with
report_authoritative=false, pass_credit=false, product_acceptance_pass=false,
product_public_approval=not_approved, zero product scans, and only its Core-owned operational metrics;
token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.",
  "validator_id": "doctor_result_non_authority"
},
{
  "id": "P-059",
  "name": "semantic-inheritance-is-structural",
  "severity": "P0",
  "statement": "The compiled projection preserves the complete SemanticModel owner: admission,
five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation
domain/range/rules, and structured provider forbidden boundaries.",
  "validator_id": "semantic_inheritance"
},
{
  "id": "P-060",
  "name": "authority-owner-is-structural",
  "severity": "P0",
  "statement": "AuthorityModel is the single owner of the capability catalogue, delegation depth,
and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not
substitute hardcoded authority facts.",
  "validator_id": "authority_owner"
},
{
  "id": "P-061",
  "name": "gate-run-definition-is-independent",
  "severity": "P0",
  "statement": "An immutable GateRunDefinition independently owns the expected evidence class,
product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests.
A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.",
  "validator_id": "gate_run_definition_integrity"
},
{

```

```

      "id": "P-062",
      "name": "authority-runtime-policy-is-derived-live-only",
      "severity": "P0",
      "statement": "AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every
mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty
rules and delegation ceiling, and is never persisted or imported as authority.",
      "validator_id": "authority_runtime_policy_projection"
    },
    {
      "id": "P-063",
      "name": "semantic-export-is-export-only",
      "severity": "P0",
      "statement": "SemanticExport is a strict bounded export-only semantic-state record binding the
exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It
is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or
release approval.",
      "validator_id": "semantic_export_integrity"
    },
    {
      "id": "P-064",
      "name": "operation-metrics-are-live-observations",
      "severity": "P0",
      "statement": "OperationMetrics is a strict live-only non-authoritative observation for one
command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and
physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.",
      "validator_id": "operation_metrics_non_authority"
    },
    {
      "id": "P-065",
      "name": "workcard-projection-is-v2-live-only",
      "severity": "P0",
      "statement": "RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively
owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal
selected-Task materialization and exposes no continuation or cursor fields.",
      "validator_id": "workcard_projection_v2"
    },
    {
      "id": "P-066",
      "name": "event-store-policy-is-derived-live-only",
      "severity": "P0",
      "statement": "EventStorePolicy is rebuilt from the installed AuthorityModel before
normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment
fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no
defaults or nullable critical fields.",
      "validator_id": "event_store_policy_projection"
    },
    {
      "id": "P-067",
      "name": "depends-on-graph-is-acyclic",
      "severity": "P0",
      "statement": "Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation
domain and range validity alone cannot admit a cyclic dependency graph.",
      "validator_id": "dependency_graph_acyclicity"
    },
    {
      "id": "P-068",
      "name": "ready-frontier-is-current-derived-state",
      "severity": "P0",
      "statement": "ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A
READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is
current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic
and truncation is explicit.",
      "validator_id": "ready_frontier_projection"
    }
  ]
},
"record_type": {
  "const": "PolicySet"
},
"research_draft_sanitation": {
  "const": {
    "allowed_license_disposition": "allowed",
    "canonical_bytes_max": 1048576,
    "citation_required_rule": "source-refs-or-blocked-with-reason",
    "claim_digest_rule": "sha256(canonical complete claim excluding claim_digest and derived
normative_use_allowed)",
    "claim_kinds": [
      "research_question",

```

```

        "hypothesis",
        "assumption",
        "evidence_claim",
        "blocked_claim"
    ],
    "evidence_reference_rule": "citation_refs identify research sources; evidence_refs identify
independently resolved Artifact records; the sets are semantically distinct",
    "forbidden_normative_claim_classes": [
        "distribution",
        "marketing",
        "proof"
    ],
    "forbidden_requested_flag_mutations": [
        "acceptance_pass",
        "pass_credit",
        "product_acceptance_pass",
        "product_public_approval",
        "public_release_approved"
    ],
    "metadata_only_rule": "missing payload receipt is representable only by source_mode metadata-only
and every referencing claim remains blocked, non-authoritative, and normative_use_allowed=false",
    "migration_rule": "MIGRATE_FACTS_NOT_FILES",
    "normative_candidate_rule": "only a source-bound or verified evidence_claim classified fact with
eligible citations and resolved evidence_refs may become a sanitized normative candidate",
    "persistence": "transient-ingress-only",
    "result_flags": {
        "pass_credit": false,
        "product_acceptance_pass": false,
        "product_public_approval": "not_approved",
        "public_release_approved": false
    },
    "result_persistence": "derived-live-only",
    "result_record_type": "SanitizedResearchDraft",
    "source_binding": "source-id+trusted-source-locator+provenance-digest+artifact-id+artifact-record-d
igest+recomputed-content-digest+size+media-type+active-license-plan-digest",
    "source_license_rule": "allowed only when the active LicensesPlan binding review_state is reviewed
or source-verified",
    "source_payload_bytes_max": 16777216,
    "source_payload_total_bytes_max": 67108864,
    "verification_statuses": [
        "unverified",
        "source-bound",
        "verified",
        "blocked"
    ]
}
},
"schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/PolicySet"
},
"semantic_export_contract": {
    "const": {
        "export_kind": "semantic-state",
        "input_digests_max": 128,
        "input_digests_rule": "sorted unique sha256(canonical exported record) values for the exact
replay-derived record set",
        "output_artifact_kind": "evidence",
        "output_artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
        "output_bytes_max": 1073741824,
        "output_bytes_rule": "exact canonical SemanticStatePayload bytes stored in content-addressed
storage",
        "output_media_type": "application/vnd.promin.semantic-state+json",
        "output_record_type": "SemanticStatePayload",
        "output_records_max": 400000,
        "package_or_archive": false,
        "persistence": "export-only",
        "product_flags": {
            "pass_credit": false,
            "product_acceptance_pass": false,
            "product_public_approval": "not_approved",
            "public_release_approved": false
        },
        "record_order": "record-digest-ascending after exact digest deduplication",
        "required_bindings": [
            "candidate_digest",
            "activation_digest",

```

```

    "head_event_id",
    "head_sequence",

    "head_digest",
    "core_bundle_digest",
    "preset_digest",
    "implementation_closure_digest",
    "provider_binding_digest",
    "input_digests",
    "output_artifact_id",
    "output_artifact_record_digest",
    "output_digest",
    "output_size_bytes",
    "output_media_type"
  ]
}
},
"state_machines": {
  "const": {
    "capacity_release": {
      "closed_release": {
        "requires_field": "close_ack",
        "slot_reusable": true,
        "state": "CLOSED"
      },
      "terminal_reconciliation": {
        "fencing_token_binding": "exact-lease-fencing-token",
        "field": "capacity_reconciliation",
        "generation_binding": "exact-lease-generation",
        "manager_capability": "lease.manage",
        "required_fields": [
          "reconciled_by",
          "grant_id",
          "grant_claim_digest",
          "reconciled_at",
          "generation",
          "fencing_token"
        ],
        "slot_reusable_after_record": true,
        "states": [
          "EXPIRED",
          "REVOKED"
        ]
      },
      "unreconciled_slot_reusable": false
    },
    "lease": {
      "ACTIVE": [
        "CLOSING",
        "EXPIRED",
        "REVOKED"
      ],
      "CLOSED": [],
      "CLOSING": [
        "CLOSED",
        "EXPIRED",
        "REVOKED"
      ],
      "EXPIRED": [],
      "REVOKED": []
    },
    "task": {
      "BLOCKED": [
        "READY",
        "CANCELLED"
      ],
      "CANCELLED": [],
      "COMPLETED": [],
      "LEASED": [
        "RUNNING",
        "READY",
        "BLOCKED",
        "CANCELLED"
      ],
      "PLANNED": [
        "READY",
        "BLOCKED",
        "CANCELLED"
      ]
    }
  }
}

```



```

    ],
    "READY": [
      "LEASED",
      "BLOCKED",
      "CANCELLED"
    ],
    "RUNNING": [
      "COMPLETED",
      "BLOCKED",
      "CANCELLED"
    ]
  }
},
"validator_registry_version": {
  "const": 1
},
"version": {
  "const": "1.0.0-alpha.4"
}
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "validator_registry_version",
  "optimization_gate",
  "state_machines",
  "policies",
  "alpha4_standard_contracts",
  "derived_result_contracts",
  "gate_run_definition_contract",
  "research_draft_sanitation",
  "semantic_export_contract"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/PolicySet | Validator: urn:promin:contracts:1#/\$defs/PolicySet

## 73. Preset

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, document, version, function, preset\_id, default\_profile, base\_user\_commands, required\_provider\_capabilities, optional\_provider\_capabilities, profiles, semantic\_programming, forbidden\_implicit\_behavior

| Property                       | Required | Structural rule                                                                                                                               |
|--------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| base_user_commands             | yes      | type="array"; const=["init", "doctor", "status", "next", "validate", "static-admission", "continue", "audit", "refresh", "context", "skills"] |
| default_profile                | yes      | \$ref="#/\$defs/Id"                                                                                                                           |
| document                       | yes      | const="promin.preset"                                                                                                                         |
| forbidden_implicit_behavior    | yes      | type="array"; minItems=1; maxItems=16; uniqueItems=true                                                                                       |
| function                       | yes      | type="string"; minLength=1                                                                                                                    |
| optional_provider_capabilities | yes      | type="array"; minItems=0; maxItems=16; uniqueItems=true                                                                                       |
| preset_id                      | yes      | \$ref="#/\$defs/Id"                                                                                                                           |
| profiles                       | yes      | type="object"; additionalProperties=schema(keys:\$ref)                                                                                        |
| record_type                    | yes      | const="Preset"                                                                                                                                |
| required_provider_capabilities | yes      | type="array"; minItems=1; maxItems=16; uniqueItems=true                                                                                       |
| schema_ref                     | yes      | const="urn:promin:contracts:1#/\$defs/Preset"                                                                                                 |
| semantic_programming           | yes      | type="object"                                                                                                                                 |
| version                        | yes      | const="1.0.0-alpha.4"                                                                                                                         |

```

{
  "additionalProperties": false,
  "properties": {
    "base_user_commands": {
      "const": [

```

```

        "init",
        "doctor",
        "status",
        "next",
        "validate",
        "static-admission",
        "continue",
        "audit",
        "refresh",
        "context",
        "skills"
    ],
    "type": "array"
},
"default_profile": {
    "$ref": "#/$defs/Id"
},
"document": {
    "const": "promin.preset"
},
"forbidden_implicit_behavior": {
    "items": {
        "$ref": "#/$defs/Id"

```

```

    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"function": {
    "minLength": 1,
    "type": "string"
},
"optional_provider_capabilities": {
    "items": {
        "$ref": "#/$defs/Id"
    },
    "maxItems": 16,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
},
"preset_id": {
    "$ref": "#/$defs/Id"
},
"profiles": {
    "additionalProperties": {
        "$ref": "#/$defs/Profile"
    },
    "minProperties": 1,
    "type": "object"

```

```

},
"record_type": {
    "const": "Preset"
},
"required_provider_capabilities": {
    "items": {
        "$ref": "#/$defs/Id"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/Preset"
},
"semantic_programming": {
    "minProperties": 2,
    "type": "object"
},
"version": {
    "const": "1.0.0-alpha.4"
}
},
"required": [
    "record_type",

```

```

"schema_ref",
"document",

"version",
"function",
"preset_id",
"default_profile",
"base_user_commands",
"required_provider_capabilities",
"optional_provider_capabilities",
"profiles",
"semantic_programming",
"forbidden_implicit_behavior"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Preset | Validator: urn:promin:contracts:1#/\$defs/Preset

## 74. Profile

Type summary: type="object"; additionalProperties=false

Required: model\_tier, max\_context\_bytes, max\_entities, max\_relations, max\_fanout\_per\_entity, top\_k, max\_parallel\_tasks, default\_dependency\_depth

| Property                 | Required | Structural rule                          |
|--------------------------|----------|------------------------------------------|
| default_dependency_depth | yes      | type="integer"; minimum=1; maximum=12    |
| max_context_bytes        | yes      | type="integer"; minimum=1024             |
| max_entities             | yes      | type="integer"; minimum=1                |
| max_fanout_per_entity    | yes      | type="integer"; minimum=1                |
| max_parallel_tasks       | yes      | type="integer"; minimum=1                |
| max_relations            | yes      | type="integer"; minimum=0                |
| model_tier               | yes      | enum=["weak-local", "capable", "strong"] |
| top_k                    | yes      | type="integer"; minimum=1                |

```

{
  "additionalProperties": false,
  "properties": {
    "default_dependency_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "max_context_bytes": {
      "minimum": 1024,
      "type": "integer"
    },
    "max_entities": {
      "minimum": 1,
      "type": "integer"
    },
    "max_fanout_per_entity": {
      "minimum": 1,
      "type": "integer"
    },
    "max_parallel_tasks": {
      "minimum": 1,
      "type": "integer"
    },
    "max_relations": {
      "minimum": 0,
      "type": "integer"
    },
  },

```

```

    "model_tier": {
      "enum": [
        "weak-local",
        "capable",
        "strong"
      ]
    },
    "top_k": {
      "minimum": 1,
      "type": "integer"
    }
  }
}

```

```

},
"required": [
  "model_tier",
  "max_context_bytes",
  "max_entities",
  "max_relations",
  "max_fanout_per_entity",
  "top_k",
  "max_parallel_tasks",
  "default_dependency_depth"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Profile | Validator: urn:promin:contracts:1#/\$defs/Profile

## 75. ProjectInit

Type summary: type="object"; additionalProperties=false

Required: record\_type, project\_id, roots, candidate\_recipe, preset\_id, operating\_profile, intent, resolved\_profile, orchestration, telemetry

| Property          | Required | Structural rule                           |
|-------------------|----------|-------------------------------------------|
| candidate_recipe  | yes      | \$ref="#/\$defs/CandidateRecipe"          |
| intent            | yes      | type="object"; additionalProperties=false |
| operating_profile | yes      | \$ref="#/\$defs/Id"                       |
| orchestration     | yes      | type="object"; additionalProperties=false |
| preset_id         | yes      | \$ref="#/\$defs/Id"                       |
| project_id        | yes      | \$ref="#/\$defs/Id"                       |
| record_type       | yes      | const="ProjectInit"                       |
| resolved_profile  | yes      | type="object"; additionalProperties=false |
| roots             | yes      | type="array"; minItems=1; maxItems=16     |
| telemetry         | yes      | type="object"; additionalProperties=false |

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_recipe": {
      "$ref": "#/$defs/CandidateRecipe"
    },
    "intent": {
      "additionalProperties": false,
      "properties": {
        "autonomy": {
          "enum": [
            "ask",
            "standing-reversible"
          ]
        },
        "constraints": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "deliverables": {
          "items": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 256,
          "type": "array",
          "uniqueItems": true
        },
        "goal": {
          "maxLength": 8192,
          "minLength": 1,

```

```

    "type": "string"
  },
  "non_goals": {
    "items": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 256,
    "type": "array",
    "uniqueItems": true
  },
  "project_mode": {
    "enum": [
      "greenfield",
      "existing-code",
      "hybrid"
    ]
  }
}

```

```

},
"references": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"reporting_language": {
  "enum": [
    "uk",
    "en"
  ]
},
"success_criteria": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"work_sources": {

```

```

  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "project_mode",
  "goal",
  "success_criteria",
  "constraints",
  "non_goals",
  "deliverables",
  "references",
  "work_sources",
  "reporting_language",
  "autonomy"
],
"type": "object"
},
"operating_profile": {
  "$ref": "#/$defs/Id"
},
"orchestration": {

```

```

  "additionalProperties": false,
  "properties": {
    "enabled": {

```

```

    "const": true
  },
  "enforcement": {
    "const": "required"
  },
  "host_source": {
    "enum": [
      "generic-agent-envelope",
      "explicit-promin",
      "inherited-compatible-host"
    ]
  },
  "model_routing": {
    "const": "cheapest-adequate-with-evidence-escalation"
  },
  "unregistered_work": {
    "const": "external-proposal-only"
  }
},
"required": [
  "enabled",
  "enforcement",
  "unregistered_work",
  "model_routing",
  "host_source"

```

```

],
"type": "object"
},
"preset_id": {
  "$ref": "#/$defs/Id"
},
"project_id": {
  "$ref": "#/$defs/Id"
},
"record_type": {
  "const": "ProjectInit"
},
"resolved_profile": {
  "additionalProperties": false,
  "properties": {
    "authority_effect": {
      "const": "none"
    }
  },
  "detected_technologies": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "confidence": {
          "maximum": 1,
          "minimum": 0,
          "type": "number"
        }
      },
      "source_count_complete": {

```

```

    "type": "boolean"
  },
  "sources": {
    "items": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 64,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "sources_truncated": {
    "type": "boolean"
  },
  "technology": {
    "$ref": "#/$defs/Id"
  },
  "total_source_count": {
    "minimum": 1,
    "type": "integer"
  }
},
"required": [

```

```

    "technology",
    "sources",
    "total_source_count",

    "sources_truncated",
    "source_count_complete",
    "confidence"
  ],
  "type": "object"
},
"maxItems": 128,
"type": "array"
},
"init_capability_selection": {
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "authority_granted": {
      "const": false
    },
    "documentation_choice": {
      "enum": [
        "accept",
        "decline",
        "custom",
        "ask"
      ]
    },
    "pass_credit": {
      "const": false
    },
    "profile_digest": {
      "$ref": "#/$defs/Digest"
    },
    "selection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "selection_source": {
      "enum": [
        "default",
        "host-profile",
        "project-package",
        "cli",
        "interactive-user"
      ]
    },
    "status": {
      "enum": [
        "AVAILABLE",
        "PENDING_OWNER_SELECTION",
        "UNAVAILABLE"
      ]
    },
    "verification_choice": {
      "enum": [
        "accept",
        "decline",
        "custom",
        "ask"
      ]
    }
  },
  "required": [
    "status",
    "selection_source",
    "authority_granted",
    "pass_credit",
    "acceptance_pass"
  ],
  "type": "object"
},
"layers": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 32,

```

```

    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "operation_profiles": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "model_tier": {
          "enum": [

```

```

            "tool-only",
            "micro",
            "standard",
            "strong",
            "critical-review"
          ]
        },
        "operation": {
          "$ref": "#/$defs/Id"
        },
        "reason": {
          "maxLength": 2048,
          "minLength": 1,
          "type": "string"
        }
      },
      "required": [
        "operation",
        "model_tier",
        "reason"
      ],
      "type": "object"
    },
    "maxItems": 64,
    "type": "array"
  },
  "profile_digest": {
    "$ref": "#/$defs/Digest"

```

```

  },
  "resolution": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "confidence": {
          "maximum": 1,
          "minimum": 0,
          "type": "number"
        },
        "profile_id": {
          "$ref": "#/$defs/Id"
        },
        "reason": {
          "maxLength": 2048,
          "minLength": 1,
          "type": "string"
        }
      },
      "required": [
        "profile_id",
        "reason",
        "confidence"
      ],
      "type": "object"
    },
    "maxItems": 64,
    "minItems": 1,

```

```

    "type": "array"
  },
  "revision": {
    "minimum": 1,
    "type": "integer"
  }
},
"required": [
  "revision",
  "layers",
  "detected_technologies",

```



```

        "operation_profiles",
        "resolution",
        "init_capability_selection",
        "authority_effect",
        "profile_digest"
    ],
    "type": "object"
},
"roots": {
    "items": {
        "$ref": "#/$defs/ProjectRoot"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array"
},
"telemetry": {

```

```

    "additionalProperties": false,
    "properties": {
        "enabled": {
            "type": "boolean"
        },
        "export_enabled": {
            "const": false
        },
        "local_only": {
            "const": true
        },
        "max_log_bytes": {
            "maximum": 67108864,
            "minimum": 65536,
            "type": "integer"
        },
        "retention_records": {
            "maximum": 100000,
            "minimum": 100,
            "type": "integer"
        },
        "store_full_prompts": {
            "const": false
        },
        "store_secrets_or_pii": {
            "const": false
        },
        "store_source_bodies": {

```

```

            "const": false
        }
    },
    "required": [
        "enabled",
        "local_only",
        "export_enabled",
        "max_log_bytes",
        "retention_records",
        "store_full_prompts",
        "store_source_bodies",
        "store_secrets_or_pii"
    ],
    "type": "object"
}
},
"required": [
    "record_type",
    "project_id",
    "roots",
    "candidate_recipe",
    "preset_id",
    "operating_profile",
    "intent",
    "resolved_profile",
    "orchestration",
    "telemetry"
],
"telemetry": {
    "type": "object"
}
}

```

Owner: core/contracts.schema.json#/\$defs/ProjectInit | Validator: urn:promin:contracts:1#/\$defs/ProjectInit

## 76. ProjectPackageManifest

Type summary: type="object"; additionalProperties=false

Required: schema, package\_id, tree\_digest, members, pass\_credit

| Property    | Required | Structural rule                   |
|-------------|----------|-----------------------------------|
| members     | yes      | type="array"; minItems=1          |
| package_id  | yes      | \$ref="#/\$defs/Id"               |
| pass_credit | yes      | const=false                       |
| schema      | yes      | const="promin.project-package.v1" |
| tree_digest | yes      | \$ref="#/\$defs/Sha256"           |

```
{
  "additionalProperties": false,
  "properties": {
    "members": {
      "minItems": 1,
      "type": "array"
    },
    "package_id": {
      "$ref": "#/$defs/Id"
    },
    "pass_credit": {
      "const": false
    },
    "schema": {
      "const": "promin.project-package.v1"
    },
    "tree_digest": {
      "$ref": "#/$defs/Sha256"
    }
  },
  "required": [
    "schema",
    "package_id",
    "tree_digest",
    "members",
    "pass_credit"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/ProjectPackageManifest | Validator: urn:promin:contracts:1#/\$defs/ProjectPackageManifest

## 77. ProjectRoot

Type summary: type="object"; additionalProperties=false

Required: path, kind

| Property | Required | Structural rule               |
|----------|----------|-------------------------------|
| kind     | yes      | const="product"               |
| path     | yes      | \$ref="#/\$defs/RelativePath" |

```
{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "const": "product"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    }
  },
  "required": [
    "path",
    "kind"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/ProjectRoot | Validator: urn:promin:contracts:1#/\$defs/ProjectRoot

## 78. ProjectionLimits

Type summary: type="object"; additionalProperties=false

Required: record\_type, token\_version, ranking\_algorithm\_id, traversal\_algorithm\_id, dependency\_depth\_hard\_max, continuation\_ttl\_seconds\_max, selected\_profile\_id, selected\_profile\_digest, persistent\_entity\_types, default\_budget, hard\_budget, default\_depth, depth\_min, depth\_max, default\_ttl\_seconds, ttl\_min\_seconds, ttl\_max\_seconds, max\_token\_bytes, max\_continuation\_state\_bytes, max\_query\_bytes, min\_result\_bytes, max\_resume\_binding\_fields, max\_resume\_binding\_key\_chars, max\_resume\_binding\_value\_bytes, max\_resume\_binding\_bytes, required\_resume\_binding\_fields, policy\_digest

| Property                       | Required | Structural rule                                                                                                                                                             |
|--------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| continuation_ttl_seconds_max   | yes      | const=3600                                                                                                                                                                  |
| default_budget                 | yes      | \$ref="#/\$defs/Budget"                                                                                                                                                     |
| default_depth                  | yes      | type="integer"; minimum=1; maximum=12                                                                                                                                       |
| default_ttl_seconds            | yes      | const=900                                                                                                                                                                   |
| dependency_depth_hard_max      | yes      | const=12                                                                                                                                                                    |
| depth_max                      | yes      | const=12                                                                                                                                                                    |
| depth_min                      | yes      | const=1                                                                                                                                                                     |
| hard_budget                    | yes      | \$ref="#/\$defs/Budget"                                                                                                                                                     |
| max_continuation_state_bytes   | yes      | const=16384                                                                                                                                                                 |
| max_query_bytes                | yes      | const=4096                                                                                                                                                                  |
| max_resume_binding_bytes       | yes      | const=4096                                                                                                                                                                  |
| max_resume_binding_fields      | yes      | const=8                                                                                                                                                                     |
| max_resume_binding_key_chars   | yes      | const=64                                                                                                                                                                    |
| max_resume_binding_value_bytes | yes      | const=512                                                                                                                                                                   |
| max_token_bytes                | yes      | const=256                                                                                                                                                                   |
| min_result_bytes               | yes      | const=1024                                                                                                                                                                  |
| persistent_entity_types        | yes      | const=["Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease"]                                                                       |
| policy_digest                  | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                     |
| ranking_algorithm_id           | yes      | const="bm25-v1"                                                                                                                                                             |
| record_type                    | yes      | const="ProjectionLimits"                                                                                                                                                    |
| required_resume_binding_fields | yes      | const=["activation_digest", "capability_id", "grant_claim_digest", "grant_id", "implementation_closure_digest", "requested_scope_digest", "revocation_epoch", "subject_id"] |
| selected_profile_digest        | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                     |
| selected_profile_id            | yes      | \$ref="#/\$defs/Id"                                                                                                                                                         |
| token_version                  | yes      | const=2                                                                                                                                                                     |
| traversal_algorithm_id         | yes      | const="typed-bfs-v2"                                                                                                                                                        |
| ttl_max_seconds                | yes      | const=3600                                                                                                                                                                  |
| ttl_min_seconds                | yes      | const=1                                                                                                                                                                     |

```
{
  "additionalProperties": false,
  "properties": {
    "continuation_ttl_seconds_max": {
      "const": 3600
    },
    "default_budget": {
      "$ref": "#/$defs/Budget"
    },
    "default_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "default_ttl_seconds": {
      "const": 900
    },
    "dependency_depth_hard_max": {
      "const": 12
    },
  },
}
```

```

"depth_max": {
  "const": 12
},
"depth_min": {
  "const": 1
},
"hard_budget": {
  "$ref": "#/$defs/Budget"

```

```

},
"max_continuation_state_bytes": {
  "const": 16384
},
"max_query_bytes": {
  "const": 4096
},
"max_resume_binding_bytes": {
  "const": 4096
},
"max_resume_binding_fields": {
  "const": 8
},
"max_resume_binding_key_chars": {
  "const": 64
},
"max_resume_binding_value_bytes": {
  "const": 512
},
"max_token_bytes": {
  "const": 256
},
"min_result_bytes": {
  "const": 1024
},
"persistent_entity_types": {
  "const": [
    "Task",

```

```

    "Candidate",
    "Artifact",
    "Run",
    "Finding",
    "Decision",
    "GateResult",
    "Grant",
    "Lease"
  ]
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"ranking_algorithm_id": {
  "const": "bm25-v1"
},
"record_type": {
  "const": "ProjectionLimits"
},
"required_resume_binding_fields": {
  "const": [
    "activation_digest",
    "capability_id",
    "grant_claim_digest",
    "grant_id",
    "implementation_closure_digest",
    "requested_scope_digest",
    "revocation_epoch",

```

```

    "subject_id"
  ]
},
"selected_profile_digest": {
  "$ref": "#/$defs/Digest"
},
"selected_profile_id": {
  "$ref": "#/$defs/Id"
},
"token_version": {
  "const": 2
},
"traversal_algorithm_id": {

```

```

    "const": "typed-bfs-v2"
  },
  "ttl_max_seconds": {
    "const": 3600
  },
  "ttl_min_seconds": {
    "const": 1
  }
},
"required": [
  "record_type",
  "token_version",
  "ranking_algorithm_id",
  "traversal_algorithm_id",
  "dependency_depth_hard_max",
  "continuation_ttl_seconds_max",
  "selected_profile_id",
  "selected_profile_digest",
  "persistent_entity_types",
  "default_budget",
  "hard_budget",
  "default_depth",
  "depth_min",
  "depth_max",
  "default_ttl_seconds",
  "ttl_min_seconds",
  "ttl_max_seconds",
  "max_token_bytes",
  "max_continuation_state_bytes",
  "max_query_bytes",
  "min_result_bytes",
  "max_resume_binding_fields",
  "max_resume_binding_key_chars",
  "max_resume_binding_value_bytes",
  "max_resume_binding_bytes",
  "required_resume_binding_fields",
  "policy_digest"
],
"type": "object",
"x-promin-ingress-operations": [
  "rebuild"
],
"x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/ProjectionLimits | Validator: urn:promin:contracts:1#/\$defs/ProjectionLimits

## 79. ProminPlatformNoDegradationMatrix

Type summary: type="object"; additionalProperties=false

Required: record\_type, standard\_name, version, candidate\_binding\_digest, archive\_sha256, archive\_bytes, trust\_configuration\_sha256, lanes, lane\_count, complete, semantic\_validation\_complete, unique\_invocation\_ids, unique\_key\_nonces, matrix\_authoritative, pass\_credit, acceptance\_pass, product\_acceptance\_pass, product\_public\_approval, matrix\_digest

| Property                 | Required | Structural rule                           |
|--------------------------|----------|-------------------------------------------|
| acceptance_pass          | yes      | const=false                               |
| archive_bytes            | yes      | type="integer"; minimum=1                 |
| archive_sha256           | yes      | \$ref="#/\$defs/Digest"                   |
| candidate_binding_digest | yes      | \$ref="#/\$defs/Digest"                   |
| complete                 | yes      | const=true                                |
| lane_count               | yes      | const=8                                   |
| lanes                    | yes      | type="array"; minItems=8; maxItems=8      |
| matrix_authoritative     | yes      | const=false                               |
| matrix_digest            | yes      | \$ref="#/\$defs/Digest"                   |
| pass_credit              | yes      | const=false                               |
| product_acceptance_pass  | yes      | const=false                               |
| product_public_approval  | yes      | const="not_approved"                      |
| record_type              | yes      | const="ProminPlatformNoDegradationMatrix" |

| Property                     | Required | Structural rule         |
|------------------------------|----------|-------------------------|
| semantic_validation_complete | yes      | const=true              |
| standard_name                | yes      | const="promin"          |
| trust_configuration_sha256   | yes      | \$ref="#/\$defs/Digest" |
| unique_invocation_ids        | yes      | const=true              |
| unique_key_nonces            | yes      | const=true              |
| version                      | yes      | const="1.0.0-alpha.4"   |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "lane_count": {
      "const": 8
    },
    "lanes": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "attestation_digest": {
            "$ref": "#/$defs/Digest"
          },
          "evidence_role": {
            "enum": [
              "linux",
              "linux-no-degradation",
              "windows",
              "windows-no-degradation"
            ]
          },
          "install_mode": {
            "enum": [
              "offline-wheelhouse",
              "online-clean"
            ]
          },
          "invocation_id": {
            "$ref": "#/$defs/Id"
          },
          "key_id": {
            "$ref": "#/$defs/Id"
          },
          "lane_id": {
            "enum": [
              "linux-cp313-offline",
              "linux-cp313-online",
              "linux-cp314-offline",
              "linux-cp314-online",
              "windows-cp313-offline",
              "windows-cp313-online",
              "windows-cp314-offline",
              "windows-cp314-online"
            ]
          },
          "nonce": {
            "maxLength": 128,
            "minLength": 24,
            "type": "string"
          }
        }
      }
    }
  }
}

```

```

"os": {
  "enum": [
    "linux",
    "windows"
  ]
},
"path": {
  "$ref": "#/$defs/RelativePath"
},
"python_abi": {
  "maxLength": 128,
  "minLength": 1,
  "type": "string"
},
"python_implementation": {
  "const": "cpython"
},
"python_version": {

```

```

  "maxLength": 128,
  "minLength": 1,
  "type": "string"
},
"record_type": {
  "enum": [
    "NoDegradationResult",
    "PlatformVerificationResult"
  ]
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"semantic_valid": {
  "const": true
},
"sha256": {
  "$ref": "#/$defs/Digest"
},
"size_bytes": {
  "minimum": 1,
  "type": "integer"
},
"status": {
  "const": "pass"
},
"subject_id": {
  "$ref": "#/$defs/Id"

```

```

}
},
"required": [
  "lane_id",
  "path",
  "sha256",
  "size_bytes",
  "result_digest",
  "record_type",
  "status",
  "os",
  "python_version",
  "python_implementation",
  "python_abi",
  "install_mode",
  "evidence_role",
  "key_id",
  "subject_id",
  "attestation_digest",
  "invocation_id",
  "nonce",
  "semantic_valid"
],
"type": "object"
},
"maxItems": 8,
"minItems": 8,
"type": "array"

```

```

},
"matrix_authoritative": {
  "const": false

```

```

    },
    "matrix_digest": {
      "$ref": "#/$defs/Digest"
    },
    },
    "pass_credit": {
      "const": false
    },
    },
    "product_acceptance_pass": {
      "const": false
    },
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    },
    "record_type": {
      "const": "ProminPlatformNoDegradationMatrix"
    },
    },
    "semantic_validation_complete": {
      "const": true
    },
    },
    "standard_name": {
      "const": "promin"
    },
    },
    "trust_configuration_sha256": {
      "$ref": "#/$defs/Digest"
    },
    },

```

```

    "unique_invocation_ids": {
      "const": true
    },
    },
    "unique_key_nonces": {
      "const": true
    },
    },
    "version": {
      "const": "1.0.0-alpha.4"
    }
  },
  "required": [
    "record_type",
    "standard_name",
    "version",
    "candidate_binding_digest",
    "archive_sha256",
    "archive_bytes",
    "trust_configuration_sha256",
    "lanes",
    "lane_count",
    "complete",
    "semantic_validation_complete",
    "unique_invocation_ids",
    "unique_key_nonces",
    "matrix_authoritative",
    "pass_credit",
    "acceptance_pass",
    "product_acceptance_pass",
    "product_public_approval",
    "matrix_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProminPlatformNoDegradationMatrix | Validator:  
urn:promin:contracts:1#/\$defs/ProminPlatformNoDegradationMatrix

## 80. ProviderBinding

Type summary: type="object"; additionalProperties=false; allOf[9]=keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title

Required: capability\_id, provider\_id, version, invocation, purpose, required, healthcheck, license, identity, implementation\_closure, dependency\_receipt

| Property           | Required | Structural rule                                                                                                                                                                     |
|--------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| capability_id      | yes      | enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"] |
| dependency_receipt | yes      | \$ref="#/\$defs/ProviderDependencyReceipt"                                                                                                                                          |



| Property               | Required | Structural rule                        |
|------------------------|----------|----------------------------------------|
| healthcheck            | yes      | \$ref="#/\$defs/Healthcheck"           |
| identity               | yes      | \$ref="#/\$defs/ProviderIdentity"      |
| implementation_closure | yes      | \$ref="#/\$defs/ImplementationClosure" |
| invocation             | yes      | \$ref="#/\$defs/Invocation"            |
| license                | yes      | \$ref="#/\$defs/LicenseBinding"        |
| provider_id            | yes      | \$ref="#/\$defs/Id"                    |
| purpose                | yes      | type="string"; minLength=1             |
| required               | yes      | type="boolean"                         |
| version                | yes      | type="string"; minLength=1             |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "capability_id": {
            "const": "control-runtime"
          }
        },
        "required": [
          "capability_id"
        ]
      },
      "then": {
        "oneOf": [
          {
            "properties": {
              "identity": {
                "properties": {
                  "kind": {
                    "const": "file-digest"
                  }
                },
                "required": [
                  "kind"
                ]
              }
            },
            "required": [
              "kind"
            ]
          }
        ]
      },
      "implementation_closure": {
        "properties": {
          "adapters": {
            "items": {
              "properties": {
                "adapter_id": {
                  "const": "python-runtime-v1"
                }
              },
              "required": [
                "adapter_id"
              ]
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {
            "kind": {
              "const": "python-runtime"
            }
          },
          "required": [
            "kind"
          ]
        }
      }
    }
  ],
  "properties": {

```

```

    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {
                "const": "control-runtime"
              },
              "protocol_id": {
                "const": "promin.control-runtime.v1"
              }
            },
            "required": [
              "capability_id",
              "protocol_id"
            ]
          }
        },
        "required": [
          "adapters"
        ]
      }
    }
  }
}

```

```

  },
  "title": "control-runtime binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "shape-validation"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            },
            "required": [
              "kind"
            ]
          }
        ]
      }
    ]
  }
}

```

```

  },
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      },
      "required": [
        "adapters"
      ]
    },
    "invocation": {
      "properties": {
        "kind": {
          "const": "python-runtime"
        }
      }
    }
  },
}

```

```

        "required": [
            "kind"
        ]
    }
}
],
"properties": {
    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "capability_id": {
                            "const": "shape-validation"
                        },
                        "protocol_id": {
                            "const": "promin.shape-validation.v1"
                        }
                    },
                    "required": [
                        "capability_id",
                        "protocol_id"
                    ]
                }
            }
        },
        "required": [
            "adapters"
        ]
    }
}

```

```

    }
},
"title": "shape-validation binds its exact adapter protocol"
},
{
    "if": {
        "properties": {
            "capability_id": {
                "const": "content-identity"
            }
        },
        "required": [
            "capability_id"
        ]
    },
    "then": {
        "oneOf": [
            {
                "properties": {
                    "identity": {
                        "properties": {
                            "kind": {
                                "const": "file-digest"
                            }
                        },
                        "required": [
                            "kind"
                        ]
                    }
                }
            }
        ]
    }
}

```

```

    },
    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "adapter_id": {
                            "const": "python-runtime-v1"
                        }
                    },
                    "required": [
                        "adapter_id"
                    ]
                }
            }
        },
        "required": [
            "adapters"
        ]
    }
}

```

```

    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    }
  },
  "required": [
    "kind"
  ]
}
},
],
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "content-identity"
            },
            "protocol_id": {
              "const": "promin.content-identity.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    }
  },
  "required": [
    "adapters"
  ]
}

```

```

    ]
  }
},
"properties": {
  "title": "content-identity binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "local-serialization"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            },
            "required": [

```

```

      "kind"
    ]
  },
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          }
        },

```

```

        "required": [
            "adapter_id"
        ]
    }
},
"required": [
    "adapters"
],
"invocation": {
    "properties": {
        "kind": {
            "const": "python-runtime"
        }
    }
},

```

```

        "required": [
            "kind"
        ]
    }
},
],
"properties": {
    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "capability_id": {
                            "const": "local-serialization"
                        },
                        "protocol_id": {
                            "const": "promin.local-serialization.v1"
                        }
                    },
                    "required": [
                        "capability_id",
                        "protocol_id"
                    ]
                }
            }
        },
        "required": [

```

```

            "adapters"
        ]
    }
},
"title": "local-serialization binds its exact adapter protocol"
},
{
    "if": {
        "properties": {
            "capability_id": {
                "const": "query-projection"
            }
        },
        "required": [
            "capability_id"
        ]
    },
    "then": {
        "oneOf": [
            {
                "properties": {
                    "identity": {
                        "properties": {
                            "kind": {
                                "const": "file-digest"
                            }
                        }
                    }
                }
            }
        ]
    }
},

```

```

        "required": [
            "kind"
        ]
    },
    "implementation_closure": {

```

```

    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      },
      "required": [
        "adapters"
      ]
    },
    "invocation": {
      "properties": {
        "kind": {
          "const": "python-runtime"
        }
      }
    }
  },
  "required": [
    "kind"
  ]
}
}
],
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "query-projection"
            },
            "protocol_id": {
              "const": "promin.query-projection.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    }
  }
},

```

```

    "required": [
      "adapters"
    ]
  }
},
"title": "query-projection binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "filesystem-inventory"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {

```

```

    "const": "file-digest"
  }
},
"required": [
  "kind"
]
},
"implementation_closure": {
  "properties": {
    "adapters": {
      "items": {
        "properties": {
          "adapter_id": {
            "const": "git-executable-v1"
          }
        },
        "required": [
          "adapter_id"
        ]
      }
    }
  },
  "required": [
    "adapters"
  ]
},
"invocation": {
  "properties": {
    "kind": {
      "const": "executable"
    }
  }
}

```

```

  }
},
"required": [
  "kind"
]
}
}
},
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "filesystem-inventory"
            },
            "protocol_id": {
              "const": "promin.filesystem-inventory.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    }
  }
}

```

```

  },
  "required": [
    "adapters"
  ]
}
},
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "export-scan"
            }
          },
          "required": [
            "capability_id"
          ]
        }
      }
    }
  }
},
"required": [
  "adapters"
]
}

```

```

    },
    "then": {
      "oneOf": [
        {
          "properties": {
            "identity": {
              "properties": {
                "kind": {
                  "const": "file-digest"
                }
              }
            },
            "required": [
              "kind"
            ]
          },
          "implementation_closure": {
            "properties": {
              "adapters": {
                "items": {
                  "properties": {
                    "adapter_id": {
                      "const": "executable-export-scan-v1"
                    }
                  },
                  "required": [
                    "adapter_id"
                  ]
                }
              },
              "required": [
                "adapters"
              ]
            },
            "invocation": {
              "properties": {
                "kind": {
                  "const": "executable"
                }
              },
              "required": [
                "kind"
              ]
            }
          }
        },
        {
          "properties": {
            "implementation_closure": {
              "properties": {
                "adapters": {
                  "items": {
                    "properties": {
                      "capability_id": {
                        "const": "export-scan"
                      },
                      "protocol_id": {
                        "const": "promin.export-scan.v1"
                      }
                    },
                    "required": [
                      "capability_id",
                      "protocol_id"
                    ]
                  }
                }
              }
            }
          },
          "required": [
            "adapters"
          ]
        }
      ]
    },
    "title": "export-scan binds its exact adapter protocol"
  },
  {
    "if": {

```



```

"properties": {
  "capability_id": {
    "const": "signature"
  }
},
"required": [
  "capability_id"
]
},
"then": {
  "oneOf": [
    {
      "properties": {
        "identity": {
          "properties": {
            "kind": {

```

```

              "const": "file-digest"
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "executable-signature-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {

```

```

            "kind": {
              "const": "executable"
            }
          },
          "required": [
            "kind"
          ]
        }
      ],
    },
    {
      "properties": {
        "identity": {
          "properties": {
            "kind": {
              "const": "module-file-digest"
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {

```

```

                    "const": "python-module-signature-v1"
                  }
                },
                "required": [
                  "adapter_id"

```

```

        ]
      }
    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-module"
      }
    },
    "required": [
      "kind"
    ]
  }
},
],
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "signature"
            },
            "protocol_id": {
              "const": "promin.signature.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      },
      "required": [
        "adapters"
      ]
    }
  },
  "title": "signature binds its exact adapter protocol"
},
{
  "if": {

```

```

    "properties": {
      "capability_id": {
        "const": "build-dependency"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            },
            "required": [
              "kind"
            ]
          }
        },
        "implementation_closure": {
          "properties": {
            "adapters": {

```

```

        "items": {
          "properties": {
            "adapter_id": {
              "const": "executable-build-dependency-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      },
      "required": [
        "adapters"
      ]
    },
    "invocation": {
      "properties": {
        "kind": {
          "const": "executable"
        }
      },
      "required": [
        "kind"
      ]
    }
  }
},
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "build-dependency"
            },
            "protocol_id": {
              "const": "promin.build-dependency.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      },
      "required": [
        "adapters"
      ]
    }
  },
  "title": "build-dependency binds its exact adapter protocol"
}
],

```

```

"properties": {
  "capability_id": {
    "enum": [
      "control-runtime",
      "shape-validation",
      "content-identity",
      "local-serialization",
      "query-projection",
      "filesystem-inventory",
      "export-scan",
      "signature",
      "build-dependency"
    ]
  },
  "dependency_receipt": {
    "$ref": "#/$defs/ProviderDependencyReceipt"
  },
  "healthcheck": {
    "$ref": "#/$defs/Healthcheck"
  }
}

```

```

    },
    "identity": {
      "$ref": "#/$defs/ProviderIdentity"
    },
    "implementation_closure": {
      "$ref": "#/$defs/ImplementationClosure"
    },
    "invocation": {
      "$ref": "#/$defs/Invocation"
    }
  },

```

```

    },
    "license": {
      "$ref": "#/$defs/LicenseBinding"
    },
    "provider_id": {
      "$ref": "#/$defs/Id"
    },
    "purpose": {
      "minLength": 1,
      "type": "string"
    },
    "required": {
      "type": "boolean"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },

```

```

  "required": [
    "capability_id",
    "provider_id",
    "version",
    "invocation",
    "purpose",
    "required",
    "healthcheck",
    "license",

```

```

    "identity",
    "implementation_closure",
    "dependency_receipt"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProviderBinding | Validator: urn:promin:contracts:1#/\$defs/ProviderBinding

## 81. ProviderDependencyReceipt

Type summary: type="object"; additionalProperties=false

Required: complete, provider\_owned\_scope, os\_substrate\_excluded, components, aggregate\_digest

| Property              | Required | Structural rule                                           |
|-----------------------|----------|-----------------------------------------------------------|
| aggregate_digest      | yes      | \$ref="#/\$defs/Digest"                                   |
| complete              | yes      | const=true                                                |
| components            | yes      | type="array"; minItems=1; maxItems=4096; uniqueItems=true |
| os_substrate_excluded | yes      | const=true                                                |
| provider_owned_scope  | yes      | const="configured-provider-root-and-runtime-components"   |

```

{
  "additionalProperties": false,
  "properties": {
    "aggregate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "component_id": {
            "$ref": "#/$defs/Id"
          },
          "component_kind": {

```

```

        "enum": [
            "native-runtime",
            "provider-file",
            "provider-tree",
            "python-distribution"
        ],
        "digest": {
            "$ref": "#/$defs/Digest"
        },
        "size_bytes": {
            "minimum": 1,
            "type": "integer"
        },
        "source": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
        },
        "version": {
            "maxLength": 256,
            "minLength": 1,
            "type": "string"
        }
    },
    "required": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
    ],
    "type": "object"
},
"maxItems": 4096,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"os_substrate_excluded": {
    "const": true
},
"provider_owned_scope": {
    "const": "configured-provider-root-and-runtime-components"
}
},
"required": [
    "complete",
    "provider_owned_scope",
    "os_substrate_excluded",
    "components",
    "aggregate_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProviderDependencyReceipt | Validator: urn:promin:contracts:1#/\$defs/ProviderDependencyReceipt

## 82. ProviderIdentity

Type summary: type="object"; additionalProperties=false

Required: kind, digest, source

| Property                                                                                                                       | Required | Structural rule                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|
| digest                                                                                                                         | yes      | \$ref="#/\$defs/Digest"                                                                                                 |
| kind                                                                                                                           | yes      | enum=["file-digest", "module-file-digest", "distribution-tree-digest", "module-closure-digest", "platform-attestation"] |
| source                                                                                                                         | yes      | type="string"; minLength=1                                                                                              |
| <pre> {   "additionalProperties": false,   "properties": {     "digest": {       "\$ref": "#/\$defs/Digest"     }   } } </pre> |          |                                                                                                                         |

Owner: core/contracts.schema.json#/\$defs/ProviderIdentity | Validator: urn:promin:contracts:1#/\$defs/ProviderIdentity

| Property                  | Required | Structural rule                                  |
|---------------------------|----------|--------------------------------------------------|
| output_size_bytes         | yes      | type="integer"; minimum=0; maximum=1099511627776 |
| output_size_ceiling_bytes | yes      | type="integer"; minimum=0; maximum=1099511627776 |
| pass_credit               | yes      | const=false                                      |
| protocol_id               | yes      | \$ref="#/\$defs/Id"                              |
| provider_id               | yes      | \$ref="#/\$defs/Id"                              |
| started_at                | yes      | \$ref="#/\$defs/Timestamp"                       |
| stderr_capture_digest     | yes      | \$ref="#/\$defs/Digest"                          |
| stderr_capture_size_bytes | yes      | type="integer"; minimum=0; maximum=1048576       |
| stderr_capture_truncated  | yes      | type="boolean"                                   |
| stdout_capture_digest     | yes      | \$ref="#/\$defs/Digest"                          |
| stdout_capture_size_bytes | yes      | type="integer"; minimum=0; maximum=1048576       |
| stdout_capture_truncated  | yes      | type="boolean"                                   |

```
{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "adapter_id": {
          "const": "python-runtime-v1"
        },
        "capability_id": {
          "const": "control-runtime"
        },
        "identity_kind": {
          "const": "file-digest"
        },
        "invocation_kind": {
          "const": "python-runtime"
        },
        "operation": {
          "const": "continue"
        },
        "operation_contract_digest": {
          "const": "a8a728c5c39a2c00083fc3b6463288e8b3797d871675d3d2e27f7feb1be7ff74"
        },
        "protocol_id": {
          "const": "promin.control-runtime.v1"
        }
      }
    }
  ],
}
```

```
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "doctor"
    },
    "operation_contract_digest": {
      "const": "dcf2e435cf7954edc17fd35fc7928951f9a607d2ee91343d1788c114d23cdead"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
```

```

        "const": "python-runtime-v1"
    },
    "capability_id": {
        "const": "control-runtime"
    },
    "identity_kind": {
        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "python-runtime"
    },
    "operation": {
        "const": "healthcheck"
    },
    "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
        "const": "promin.control-runtime.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "init"
        },
        "operation_contract_digest": {
            "const": "9c2e26f6706abd39bef62d2120fcd9785f4f1cf39cdae92c3e4fa8f20ff9dd12"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "next"
        },
        "operation_contract_digest": {
            "const": "b75143a53c7129f54c62fc5c0ebd97927a7732bb8fc6f91918d5858a9578fe5d"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {

```



```

    "const": "control-runtime"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "135f8b5d81f1cecc32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
  },
  "protocol_id": {
    "const": "promin.control-runtime.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "status"
    },
    "operation_contract_digest": {
      "const": "0449eeb95febf5e15ffba4657cdb6cd606c0a12a148bd444384120faa5bc5eb1"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
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    },
    "invocation_kind": {
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    },
    "operation": {
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    },
    "operation_contract_digest": {
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    },
    "protocol_id": {
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    }
  }
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    },
    "capability_id": {
      "const": "shape-validation"
    }
  },

```

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    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
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    },
    "operation_contract_digest": {
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    },
    "protocol_id": {
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    }
  },
  {
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      "capability_id": {
        "const": "shape-validation"
      },
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      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "continue"
      },
      "operation_contract_digest": {
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      },
      "protocol_id": {
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      }
    }
  },
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      "capability_id": {
        "const": "shape-validation"
      },
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      },
      "invocation_kind": {
        "const": "python-runtime"
      },
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        "const": "export"
      },
      "operation_contract_digest": {
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      },
      "protocol_id": {
        "const": "promin.shape-validation.v1"
      }
    }
  },
  {
    "properties": {
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      "capability_id": {
        "const": "shape-validation"
      },
      "identity_kind": {
        "const": "file-digest"
      }
    }
  }
}

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "healthcheck"
    },
    },
    "operation_contract_digest": {
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    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
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    },
  },

```

```

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    },
    },
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    },
    },
    "invocation_kind": {
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    },
    },
    "operation": {
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    },
    },
    "operation_contract_digest": {
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    },
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    }
  }
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  "properties": {
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    },
    "capability_id": {
      "const": "shape-validation"
    },
  },

```

```

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    },
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "provider-binding"
    },
    },
    "operation_contract_digest": {
      "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
    },
    },
    "protocol_id": {
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    }
  }
},
{
  "properties": {
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      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
    "identity_kind": {
      "const": "file-digest"
    },
  },

```

```

      "invocation_kind": {
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      },
      "operation": {
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      },
      "operation_contract_digest": {
        "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
      },
      "protocol_id": {
        "const": "promin.shape-validation.v1"
      }
    }
  },
  {
    "properties": {
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        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "shape-validation"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      }
    },

```

```

      "operation": {
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      },
      "operation_contract_digest": {
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      },
      "protocol_id": {
        "const": "promin.shape-validation.v1"
      }
    }
  },
  {
    "properties": {
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        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "content-identity"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "digest-bytes"
      }
    },

```

```

      "operation_contract_digest": {
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      },
      "protocol_id": {
        "const": "promin.content-identity.v1"
      }
    }
  },
  {
    "properties": {
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        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "content-identity"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      }
    },

```

```

"operation": {
  "const": "digest-file"
},
"operation_contract_digest": {
  "const": "3876865e4a21f0047d92daaca9b6d7177769130a97c781e90ce130ced513f60b"
},

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  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
},
{
  "properties": {
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      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "digest-value"
    },
    "operation_contract_digest": {
      "const": "b0a54275c219f51585f519a4d9c1814478bf072dbeff050a92a80780eb492a39"
    },
    "protocol_id": {
      "const": "promin.content-identity.v1"
    }
  }
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.content-identity.v1"
    }
  }
},
{

```

```

  "properties": {
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    "capability_id": {
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    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "provider-binding"
    }
  }
}

```

```

    },
    "operation_contract_digest": {
      "const": "135f8b5d81f1cec32ec18a36aledcb3455cd186266ed1a72a5b1ecb638102f6f"
    },
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    }
  }
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  "properties": {
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```

```

    },
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    },
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    },
    "operation_contract_digest": {
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    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"

```

```

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    },
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      "const": "directory-sync"
    },
    "operation_contract_digest": {
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    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
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      "const": "python-runtime-v1"
    },
    "capability_id": {
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    },
    "identity_kind": {
      "const": "file-digest"

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "file-sync"
    },
    "operation_contract_digest": {

```

```

        "const": "3d757d7af8c2fcfb955b76db89a8917262aec19de8419305c6617a6f668cbefd"
    },
    "protocol_id": {
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    }
},
{
    "properties": {
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        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        }
    }
},

```

```

    },
    "operation": {
        "const": "healthcheck"
    },
    "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
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            "const": "python-runtime-v1"
        },
        "capability_id": {
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        },
        "identity_kind": {
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        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
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        }
    }
},

```

```

    },
    "operation_contract_digest": {
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    },
    "protocol_id": {
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    }
},
{
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        },
        "capability_id": {
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        },
        "identity_kind": {
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        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "provider-binding"
        },
        "operation_contract_digest": {
            "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
        }
    }
},

```

```

    },
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      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
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      },
      "operation_contract_digest": {
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      "protocol_id": {
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      }
    }
  }
},
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    },
    "invocation_kind": {
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    },
    "operation": {
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    },
    "operation_contract_digest": {
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    "protocol_id": {
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    }
  }
},
{
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    },
    "capability_id": {
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    },
    "identity_kind": {
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    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
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    }
  }
}

```



```

    }
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      "invocation_kind": {
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      },
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      },
      "operation_contract_digest": {
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      "protocol_id": {
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      }
    }
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      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "provider-binding"
      },
      "operation_contract_digest": {
        "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
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      "protocol_id": {
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      }
    }
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  {
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      "capability_id": {
        "const": "query-projection"
      },
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        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "rebuild"
      },
      "operation_contract_digest": {
        "const": "825da07c8c0c880475d1e292bbc0ff059f10e3d47197d93da1cf1fc8c23d58e9"
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      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

},
{
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    "capability_id": {
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    },
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      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
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    },
    "operation_contract_digest": {
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    "protocol_id": {
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    }
  }
},
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    },
    "capability_id": {
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    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
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    "protocol_id": {
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    }
  }
},
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    "capability_id": {
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    "identity_kind": {
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    },
    "invocation_kind": {
      "const": "executable"
    },
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    },
    "operation_contract_digest": {
      "const": "7b9234db382b1728299f568503507fd04df4e1f19433c0e378c21378caad3241"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{

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```

"properties": {
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  "capability_id": {
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  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "immutable-tree-stream"
  },
  "operation_contract_digest": {
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  },
  "protocol_id": {
    "const": "promin.filesystem-inventory.v1"
  }
},
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      "const": "git-executable-v1"
    },
    "capability_id": {
      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "provider-binding"
    },
    "operation_contract_digest": {
      "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{
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      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "bounded-export-scan"
    },
    "operation_contract_digest": {
      "const": "7e0326ca387c256798bc2c37c1d1c3e0fa17440d775c9d0c7deafa7ada33d1c3"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
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    }
  },
  {
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      "adapter_id": {
        "const": "executable-export-scan-v1"
      },

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```

      "capability_id": {
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        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "provider-binding"
      },
      "operation_contract_digest": {
        "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
      },
      "protocol_id": {
        "const": "promin.export-scan.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "executable-signature-v1"
      },
      "capability_id": {
        "const": "signature"
      },

```

```

      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "executable"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
      },
      "protocol_id": {
        "const": "promin.signature.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-module-signature-v1"
      },

```

```

    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
  },
  "invocation_kind": {
    "const": "python-module"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "provider-binding"
    },
  },
  "operation_contract_digest": {
    "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
  },

```

```

    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
  },

```

```

    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
}

```

```

  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "verify-digest"
    },
    "operation_contract_digest": {
      "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {

```

```

    "properties": {
      "adapter_id": {
        "const": "python-module-signature-v1"
      },
      "capability_id": {
        "const": "signature"
      },
      "identity_kind": {

```

```

    "const": "module-file-digest"
  },
  "invocation_kind": {
    "const": "python-module"
  },
  "operation": {
    "const": "verify-digest"
  },
  "operation_contract_digest": {
    "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"

```

```

    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.build-dependency.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"

```

```

    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "import-build-facts"
    },
    "operation_contract_digest": {
      "const": "25063736ddf272a51989981c79fcbb8e9cf2decf040522d3e050b13f0e2457db"
    },
    "protocol_id": {
      "const": "promin.build-dependency.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {
      "const": "file-digest"

```

```

    },
    "invocation_kind": {
      "const": "executable"
    },
    },
    "operation": {
      "const": "provider-binding"
    },
    },
    "operation_contract_digest": {
      "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
    },
    },
    "protocol_id": {
      "const": "promin.build-dependency.v1"
    }
  }
}
],
"properties": {
  "adapter_id": {
    "$ref": "#/$defs/Id"
  },
  },
  "authoritative": {
    "const": false
  },
  },
  "capability_id": {
    "enum": [
      "control-runtime",
      "shape-validation",
      "content-identity",
      "local-serialization",
      "query-projection",
      "filesystem-inventory",
      "export-scan",
      "signature",
      "build-dependency"
    ]
  },
  },
  "completed_at": {
    "$ref": "#/$defs/Timestamp"
  },
  },
  "dependency_receipt_digest": {
    "$ref": "#/$defs/Digest"
  },
  },
  "exit_code": {
    "oneOf": [
      {
        "type": "integer"
      },
      {
        "type": "null"
      }
    ]
  },
  },
  "identity_digest": {
    "$ref": "#/$defs/Digest"
  },
  },
  "identity_kind": {
    "enum": [
      "file-digest",
      "module-file-digest",
      "distribution-tree-digest",
      "module-closure-digest",
      "platform-attestation"
    ]
  },
  },
  "implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  },
  "input_digest": {
    "$ref": "#/$defs/Digest"
  },
  },
  "invocation_kind": {
    "enum": [
      "python-runtime",
      "python-module",
      "executable",
      "platform-service"
    ]
  },
  },

```



```

"invocation_receipt_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation_request_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"invoked": {
  "const": true
},
"operation": {
  "$ref": "#/$defs/Id"
},
"operation_contract_digest": {
  "$ref": "#/$defs/Digest"
},
"outcome": {
  "enum": [
    "success",
    "failure",
    "blocked"
  ]
},
"output_digest": {
  "$ref": "#/$defs/Digest"
},
"output_size_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
},
"output_size_ceiling_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
}

```

```

},
"pass_credit": {
  "const": false
},
"protocol_id": {
  "$ref": "#/$defs/Id"
},
"provider_id": {
  "$ref": "#/$defs/Id"
},
"started_at": {
  "$ref": "#/$defs/Timestamp"
},
"stderr_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stderr_capture_size_bytes": {
  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stderr_capture_truncated": {
  "type": "boolean"
},
"stdout_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stdout_capture_size_bytes": {

```

```

  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stdout_capture_truncated": {
  "type": "boolean"
}
},
"required": [
  "capability_id",
  "provider_id",
  "invocation_kind",
  "identity_kind",
  "identity_digest",
  "adapter_id",

```

```

"protocol_id",
"operation",
"operation_contract_digest",
"dependency_receipt_digest",
"implementation_closure_digest",
"invoked",
"started_at",
"completed_at",
"outcome",
"exit_code",
"invocation_request_digest",
"input_digest",
"output_digest",

"output_size_bytes",
"output_size_ceiling_bytes",
"stdout_capture_digest",
"stdout_capture_size_bytes",
"stdout_capture_truncated",
"stderr_capture_digest",
"stderr_capture_size_bytes",
"stderr_capture_truncated",
"invocation_receipt_digest",
"authoritative",
"pass_credit"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ProviderInvocationEvidence | Validator: urn:promin:contracts:1#/\$defs/ProviderInvocationEvidence

## 84. ProviderReceiptEnvelope

Type summary: type="object"; additionalProperties=false

Required: schema, mode, merkle\_root, coverage\_digest, pass\_credit

| Property        | Required | Structural rule                             |
|-----------------|----------|---------------------------------------------|
| coverage_digest | yes      | \$ref="#/\$defs/Sha256"                     |
| merkle_root     | yes      | \$ref="#/\$defs/Sha256"                     |
| mode            | yes      | enum=["FullScan", "Reuse"]                  |
| pass_credit     | yes      | const=false                                 |
| schema          | yes      | const="promin.provider-receipt-envelope.v1" |

```

{
  "additionalProperties": false,
  "properties": {
    "coverage_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "merkle_root": {
      "$ref": "#/$defs/Sha256"
    },
    "mode": {
      "enum": [
        "FullScan",
        "Reuse"
      ]
    },
    "pass_credit": {
      "const": false
    },
    "schema": {
      "const": "promin.provider-receipt-envelope.v1"
    }
  },
  "required": [
    "schema",
    "mode",
    "merkle_root",
    "coverage_digest",
    "pass_credit"
  ],
  "type": "object"
}

```

## 85. ReactivationIntent

Type summary: type="object"; additionalProperties=false

Required: schema, intent\_digest, owner\_confirmed, rollback

| Property        | Required | Structural rule                       |
|-----------------|----------|---------------------------------------|
| intent_digest   | yes      | \$ref="#/\$defs/Sha256"               |
| owner_confirmed | yes      | const=true                            |
| rollback        | yes      | const="whole-root-only"               |
| schema          | yes      | const="promin.reactivation-intent.v1" |

```

{
  "additionalProperties": false,
  "properties": {
    "intent_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "owner_confirmed": {
      "const": true
    },
    "rollback": {
      "const": "whole-root-only"
    },
    "schema": {
      "const": "promin.reactivation-intent.v1"
    }
  },
  "required": [
    "schema",
    "intent_digest",
    "owner_confirmed",
    "rollback"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ReactivationIntent | Validator: urn:promin:contracts:1#/\$defs/ReactivationIntent

## 86. ReadyFrontier

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: record\_type, activation\_digest, head\_digest, projection\_digest, ordering, ready\_tasks, evaluated\_task\_count, blocked\_task\_count, cycle\_count, truncated, continuation, silent\_truncation, projection\_authoritative

| Property                 | Required | Structural rule                                         |
|--------------------------|----------|---------------------------------------------------------|
| activation_digest        | yes      | \$ref="#/\$defs/Digest"                                 |
| blocked_task_count       | yes      | type="integer"; minimum=0                               |
| continuation             | yes      | oneOf[2]=type:null   type:object                        |
| cycle_count              | yes      | type="integer"; const=0                                 |
| evaluated_task_count     | yes      | type="integer"; minimum=0                               |
| head_digest              | yes      | oneOf[2]=#/\$defs/Digest   type:null                    |
| ordering                 | yes      | const=["created_at-ascending", "task_id-ascending"]     |
| projection_authoritative | yes      | const=false                                             |
| projection_digest        | yes      | \$ref="#/\$defs/Digest"                                 |
| ready_tasks              | yes      | type="array"; minItems=0; maxItems=12; uniqueItems=true |
| record_type              | yes      | const="ReadyFrontier"                                   |
| silent_truncation        | yes      | const=false                                             |
| truncated                | yes      | type="boolean"                                          |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {

```

```

    "properties": {
      "continuation": {
        "type": "null"
      }
    },
    "if": {
      "properties": {
        "truncated": {
          "const": true
        }
      },
      "required": [
        "truncated"
      ]
    },
    "then": {
      "properties": {
        "continuation": {
          "additionalProperties": false,
          "properties": {
            "activation_digest": {
              "$ref": "#/$defs/Digest"
            }
          }
        }
      }
    }
  },
  "budget_digest": {
    "$ref": "#/$defs/Digest"
  },
  "cursor": {
    "minimum": 0,
    "type": "integer"
  },
  "depth": {
    "maximum": 12,
    "minimum": 1,
    "type": "integer"
  },
  "expiry": {
    "$ref": "#/$defs/Timestamp"
  },
  "head_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "projection_digest": {
    "$ref": "#/$defs/Digest"
  },
  "ranking": {
    "const": "bm25-v1"
  },
  "resume_binding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "token": {
    "maxLength": 256,
    "minLength": 1,
    "pattern": "^promin-v2\\.[A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
    "type": "string"
  },
  "traversal": {
    "const": "typed-bfs-v2"
  },
  "version": {
    "const": 2
  }
},
"required": [
  "version",
  "traversal",

```

```

    },
    "budget_digest": {
      "$ref": "#/$defs/Digest"
    },
    "cursor": {
      "minimum": 0,
      "type": "integer"
    },
    "depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "expiry": {
      "$ref": "#/$defs/Timestamp"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking": {
      "const": "bm25-v1"
    },
    "resume_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "token": {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\.[A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
      "type": "string"
    },
    "traversal": {
      "const": "typed-bfs-v2"
    },
    "version": {
      "const": 2
    }
  },
  "required": [
    "version",
    "traversal",

```

```

    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking": {
      "const": "bm25-v1"
    },
    "resume_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "token": {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\.[A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
      "type": "string"
    },
    "traversal": {
      "const": "typed-bfs-v2"
    },
    "version": {
      "const": 2
    }
  },
  "required": [
    "version",
    "traversal",

```

```

        "token",
        "cursor",

        "expiry",
        "activation_digest",
        "head_digest",
        "projection_digest",
        "implementation_closure_digest",
        "ranking",
        "depth",
        "budget_digest",
        "resume_binding_digest"
    ],
    "type": "object"
}
}
}
},
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "blocked_task_count": {
        "minimum": 0,
        "type": "integer"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            }
        ]
    }
}

```

```

    },
    {
        "additionalProperties": false,
        "properties": {
            "activation_digest": {
                "$ref": "#/$defs/Digest"
            },
            "budget_digest": {
                "$ref": "#/$defs/Digest"
            },
            "cursor": {
                "minimum": 0,
                "type": "integer"
            },
            "depth": {
                "maximum": 12,
                "minimum": 1,
                "type": "integer"
            },
            "expiry": {
                "$ref": "#/$defs/Timestamp"
            },
            "head_digest": {
                "oneOf": [
                    {
                        "$ref": "#/$defs/Digest"
                    },
                    {

```

```

                        "type": "null"
                    }
                ]
            },
            "implementation_closure_digest": {
                "$ref": "#/$defs/Digest"
            },
            "projection_digest": {
                "$ref": "#/$defs/Digest"
            },
            "ranking": {
                "const": "bm25-v1"
            },
            "resume_binding_digest": {
                "$ref": "#/$defs/Digest"
            },
            "token": {
                "maxLength": 256,
                "minLength": 1,

```

```

    "pattern": "^promin-v2\\.\\.[A-Za-z0-9_-]{22}\\\\.\\.[0-9a-f]{64}\\\\.\\.[A-Za-z0-9_-]{43}$",
    "type": "string"
  },
  "traversal": {
    "const": "typed-bfs-v2"
  },
  "version": {
    "const": 2
  }
}

```

```

  },
  "required": [
    "version",
    "traversal",
    "token",
    "cursor",
    "expiry",
    "activation_digest",
    "head_digest",
    "projection_digest",
    "implementation_closure_digest",
    "ranking",
    "depth",
    "budget_digest",
    "resume_binding_digest"
  ],
  "type": "object"
}
]
},
"cycle_count": {
  "const": 0,
  "type": "integer"
},
"evaluated_task_count": {
  "minimum": 0,
  "type": "integer"
},
},

```

```

"head_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"ordering": {
  "const": [
    "created_at-ascending",
    "task_id-ascending"
  ]
},
"projection_authoritative": {
  "const": false
},
"projection_digest": {
  "$ref": "#/$defs/Digest"
},
"ready_tasks": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "acceptance_predicate": {
        "maxLength": 256,

```

```

        "minLength": 1,
        "type": "string"
      },
      "candidate_digest": {
        "$ref": "#/$defs/Digest"
      },
      "created_at": {
        "$ref": "#/$defs/Timestamp"
      },
      "dependency_task_ids": {
        "items": {
          "$ref": "#/$defs/Id"

```

```

    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "gate_result_digests": {
    "items": {
      "$ref": "#/$defs/Digest"
    },
    "maxItems": 64,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "task_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "task_id",
  "task_digest",
  "created_at",
  "candidate_digest",
  "acceptance_predicate",
  "dependency_task_ids",
  "gate_result_digests"
],
"type": "object"
},
"maxItems": 12,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"record_type": {
  "const": "ReadyFrontier"
},
"silent_truncation": {
  "const": false
},

```

```

  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "record_type",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "ordering",
  "ready_tasks",
  "evaluated_task_count",
  "blocked_task_count",
  "cycle_count",
  "truncated",
  "continuation",
  "silent_truncation",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: *core/contracts.schema.json#/\$defs/ReadyFrontier* | Validator: *urn:promin:contracts:1#/\$defs/ReadyFrontier*

## 87. Relation

Type summary: type="object"; additionalProperties=false

Required: record\_type, relation\_id, kind, source\_type, source\_id, target\_type, target\_id, activation\_digest, created\_at

| Property          | Required | Structural rule         |
|-------------------|----------|-------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest" |

| Property    | Required | Structural rule                                                          |
|-------------|----------|--------------------------------------------------------------------------|
| created_at  | yes      | \$ref="#/\$defs/Timestamp"                                               |
| kind        | yes      | enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"] |
| record_type | yes      | const="Relation"                                                         |
| relation_id | yes      | \$ref="#/\$defs/Id"                                                      |
| source_id   | yes      | \$ref="#/\$defs/Id"                                                      |
| source_type | yes      | \$ref="#/\$defs/Id"                                                      |
| target_id   | yes      | \$ref="#/\$defs/Id"                                                      |
| target_type | yes      | \$ref="#/\$defs/Id"                                                      |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "record_type": {
      "const": "Relation"
    },
    "relation_id": {
      "$ref": "#/$defs/Id"
    },
    "source_id": {
      "$ref": "#/$defs/Id"
    },
    "source_type": {
      "$ref": "#/$defs/Id"
    },
    "target_id": {
      "$ref": "#/$defs/Id"
    },
    "target_type": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "relation_id",
    "kind",
    "source_type",
    "source_id",
    "target_type",
    "target_id",
    "activation_digest",
    "created_at"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/Relation* | Validator: *urn:promin:contracts:1#/\$defs/Relation*

## 88. RelationDef

Type summary: type="object"; additionalProperties=false

Required: kind, source, target, rules



| Property | Required | Structural rule                                                          |
|----------|----------|--------------------------------------------------------------------------|
| kind     | yes      | enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"] |
| rules    | yes      | type="array"; minItems=1; maxItems=8; uniqueItems=true                   |
| source   | yes      | type="array"; minItems=1; maxItems=4; uniqueItems=true                   |
| target   | yes      | type="array"; minItems=1; maxItems=4; uniqueItems=true                   |

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "rules": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "source": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "target": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  },
  "required": [
    "kind",
    "source",
    "target",
    "rules"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/RelationDef | Validator: urn:promin:contracts:1#/\$defs/RelationDef

## 89. RelativePath

Type summary: type="string"; pattern="^(?!/)(?![A-Za-z:])(?!.\*\\\\)(?!.\*(?:^|/|\\.|\\.(?:/|\$))(?!.\*//).+"; minLength=1; maxLength=4096

Required: -

|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> {   "maxLength": 4096,   "minLength": 1,   "pattern": "^(?!/)(?![A-Za-z:])(?!.*\\\\\\\\)(?!.*(?:^ / \\\\. \\\\.(?:/ \$))(?!.*//).+",   "type": "string" } </pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Owner: core/contracts.schema.json#/\$defs/RelativePath | Validator: urn:promin:contracts:1#/\$defs/RelativePath

## 90. ReleaseEvidenceInvocation

Type summary: type="object"; additionalProperties=false

Required: invocation\_id, operation, arguments\_digest, started\_at, completed\_at, exit\_code, platform\_binding\_digest

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required | Structural rule            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------|
| arguments_digest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | yes      | \$ref="#/\$defs/Digest"    |
| completed_at                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | \$ref="#/\$defs/Timestamp" |
| exit_code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | yes      | type="integer"             |
| invocation_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | yes      | \$ref="#/\$defs/Id"        |
| operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | yes      | \$ref="#/\$defs/Id"        |
| platform_binding_digest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | yes      | \$ref="#/\$defs/Digest"    |
| started_at                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | yes      | \$ref="#/\$defs/Timestamp" |
| <pre>{   "additionalProperties": false,   "properties": {     "arguments_digest": {       "\$ref": "#/\$defs/Digest"     },     "completed_at": {       "\$ref": "#/\$defs/Timestamp"     },     "exit_code": {       "type": "integer"     },     "invocation_id": {       "\$ref": "#/\$defs/Id"     },     "operation": {       "\$ref": "#/\$defs/Id"     },     "platform_binding_digest": {       "\$ref": "#/\$defs/Digest"     },     "started_at": {       "\$ref": "#/\$defs/Timestamp"     }   },   "required": [     "invocation_id",     "operation",     "arguments_digest",     "started_at",     "completed_at",     "exit_code",     "platform_binding_digest"   ],   "type": "object" }</pre> |          |                            |

Owner: core/contracts.schema.json#/\$defs/ReleaseEvidenceInvocation | Validator:  
urn:promin:contracts:1#/\$defs/ReleaseEvidenceInvocation

## 91. ReleaseEvidenceProducer

Type summary: type="object"; additionalProperties=false

Required: tool\_path, tool\_sha256, tool\_version

| Property                                                                                                                               | Required | Structural rule                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tool_path                                                                                                                              | yes      | enum=["tools/generate_human.py", "tools/promin_no_degradation.py", "tools/promin_package.py", "tools/promin_saturation.py", "tools/promin_saturation_audit.py", "tools/promin_validate.py"] |
| tool_sha256                                                                                                                            | yes      | \$ref="#/\$defs/Digest"                                                                                                                                                                     |
| tool_version                                                                                                                           | yes      | \$ref="#/\$defs/SemVer"                                                                                                                                                                     |
| <pre>{   "additionalProperties": false,   "properties": {     "tool_path": {       "enum": [         "tools/generate_human.py", </pre> |          |                                                                                                                                                                                             |

```

        "tools/promin_no_degradation.py",
        "tools/promin_package.py",
        "tools/promin_saturation.py",
        "tools/promin_saturation_audit.py",
        "tools/promin_validate.py"
    ]
},
"tool_sha256": {
    "$ref": "#/$defs/Digest"
},
"tool_version": {
    "$ref": "#/$defs/SemVer"
}
},
"required": [
    "tool_path",
    "tool_sha256",
    "tool_version"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ReleaseEvidenceProducer | Validator:  
urn:promin:contracts:1#/\$defs/ReleaseEvidenceProducer

## 92. ResearchDraftClaim

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Required: claim\_id, claim\_kind, claim\_class, claim\_digest, text, citation\_required, citation\_refs, evidence\_refs, verification\_status, normative\_candidate, normative\_use\_allowed

| Property              | Required | Structural rule                                                                           |
|-----------------------|----------|-------------------------------------------------------------------------------------------|
| blocked_reason        | no       | type="string"; minLength=1; maxLength=2048                                                |
| citation_refs         | yes      | type="array"; maxItems=32; uniqueItems=true                                               |
| citation_required     | yes      | type="boolean"                                                                            |
| claim_class           | yes      | enum=["distribution", "fact", "marketing", "proof"]                                       |
| claim_digest          | yes      | \$ref="#/\$defs/Digest"                                                                   |
| claim_id              | yes      | \$ref="#/\$defs/Id"                                                                       |
| claim_kind            | yes      | enum=["research_question", "hypothesis", "assumption", "evidence_claim", "blocked_claim"] |
| evidence_refs         | yes      | type="array"; maxItems=32; uniqueItems=true                                               |
| normative_candidate   | yes      | type="boolean"                                                                            |
| normative_use_allowed | yes      | const=false                                                                               |
| text                  | yes      | type="string"; minLength=1; maxLength=8192                                                |
| verification_status   | yes      | enum=["unverified", "source-bound", "verified", "blocked"]                                |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "blocked_reason"
          ]
        }
      },
      "if": {
        "anyOf": [
          {
            "properties": {
              "claim_kind": {
                "const": "blocked_claim"
              }
            },
            "required": [
              "claim_kind"
            ]
          },
          {
            "properties": {
              "verification_status": {

```

```

        "const": "blocked"
      }
    },
    "required": [
      "verification_status"
    ]
  }
],
"then": {
  "required": [
    "blocked_reason"
  ]
}
},
"properties": {
  "blocked_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "citation_refs": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },

```

```

  "citation_required": {
    "type": "boolean"
  },
  "claim_class": {
    "enum": [
      "distribution",
      "fact",
      "marketing",
      "proof"
    ]
  },
  "claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "claim_id": {
    "$ref": "#/$defs/Id"
  },
  "claim_kind": {
    "enum": [
      "research_question",
      "hypothesis",
      "assumption",
      "evidence_claim",
      "blocked_claim"
    ]
  },
  "evidence_refs": {
    "items": {

```

```

      "additionalProperties": false,
      "properties": {
        "artifact_id": {
          "$ref": "#/$defs/Id"
        },
        "artifact_record_digest": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "artifact_id",
        "artifact_record_digest"
      ],
      "type": "object"
    },
    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },

```

```

    "normative_candidate": {
      "type": "boolean"
    },
    "normative_use_allowed": {
      "const": false
    },
    "text": {
      "maxLength": 8192,
      "minLength": 1,

```

```

      "type": "string"
    },
    "verification_status": {
      "enum": [
        "unverified",
        "source-bound",
        "verified",
        "blocked"
      ]
    }
  },
  "required": [
    "claim_id",
    "claim_kind",
    "claim_class",
    "claim_digest",
    "text",
    "citation_required",
    "citation_refs",
    "evidence_refs",
    "verification_status",
    "normative_candidate",
    "normative_use_allowed"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ResearchDraftClaim | Validator: urn:promin:contracts:1#/\$defs/ResearchDraftClaim

## 93. ResearchDraftIntake

Type summary: type="object"; additionalProperties=false

Required: record\_type, migration\_rule, license\_plan\_digest, sources, claims, requested\_flag\_mutations

| Property                 | Required | Structural rule                        |
|--------------------------|----------|----------------------------------------|
| claims                   | yes      | type="array"; minItems=1; maxItems=256 |
| license_plan_digest      | yes      | \$ref="#/\$defs/Digest"                |
| migration_rule           | yes      | const="MIGRATE_FACTS_NOT_FILES"        |
| record_type              | yes      | const="ResearchDraftIntake"            |
| requested_flag_mutations | yes      | type="array"; const=[]                 |
| sources                  | yes      | type="array"; minItems=1; maxItems=128 |

```

{
  "additionalProperties": false,
  "properties": {
    "claims": {
      "items": {
        "$ref": "#/$defs/ResearchDraftClaim"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array"
    },
    "license_plan_digest": {
      "$ref": "#/$defs/Digest"
    },
    "migration_rule": {
      "const": "MIGRATE_FACTS_NOT_FILES"
    },
    "record_type": {
      "const": "ResearchDraftIntake"
    },
    "requested_flag_mutations": {
      "const": [],
      "type": "array"
    }
  },

```

```

    "sources": {
      "items": {
        "$ref": "#/$defs/ResearchSourceIntake"
      },
      "maxItems": 128,
      "minItems": 1,
      "type": "array"
    },
    "required": [
      "record_type",
      "migration_rule",
      "license_plan_digest",
      "sources",
      "claims",
      "requested_flag_mutations"
    ],
    "type": "object",
    "x-promin-ingress-operations": [
      "transient"
    ],
    "x-promin-max-canonical-bytes": 1048576,
    "x-promin-persistence": "transient-ingress-only"
  }
}

```

Owner: core/contracts.schema.json#/\$defs/ResearchDraftIntake | Validator: urn:promin:contracts:1#/\$defs/ResearchDraftIntake

## 94. ResearchSourceIntake

Type summary: type="object"; additionalProperties=false; allOf[1]=keys:else;if,then

Required: source\_id, source\_mode, source\_locator, source\_provenance\_digest, media\_type, license\_provider\_id, license\_expression, license\_plan\_digest

| Property                 | Required | Structural rule                            |
|--------------------------|----------|--------------------------------------------|
| artifact_id              | no       | \$ref="#/\$defs/Id"                        |
| artifact_record_digest   | no       | \$ref="#/\$defs/Digest"                    |
| content_digest           | no       | \$ref="#/\$defs/Digest"                    |
| license_expression       | yes      | type="string"; minLength=1; maxLength=512  |
| license_plan_digest      | yes      | \$ref="#/\$defs/Digest"                    |
| license_provider_id      | yes      | \$ref="#/\$defs/Id"                        |
| media_type               | yes      | type="string"; minLength=1; maxLength=256  |
| missing_reason           | no       | type="string"; minLength=1; maxLength=2048 |
| size_bytes               | no       | type="integer"; minimum=0                  |
| source_id                | yes      | \$ref="#/\$defs/Id"                        |
| source_locator           | yes      | type="string"; minLength=1; maxLength=2048 |
| source_mode              | yes      | enum=["captured-payload", "metadata-only"] |
| source_provenance_digest | yes      | \$ref="#/\$defs/Digest"                    |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "artifact_id"
              ]
            },
            {
              "required": [
                "artifact_record_digest"
              ]
            },
            {
              "required": [
                "content_digest"
              ]
            }
          ]
        }
      }
    }
  ]
}

```

```

    {
      "required": [
        "size_bytes"
      ]
    }
  ]
}

```

```

  },
  "required": [
    "missing_reason"
  ],
  "if": {
    "properties": {
      "source_mode": {
        "const": "captured-payload"
      }
    },
    "required": [
      "source_mode"
    ]
  },
  "then": {
    "not": {
      "required": [
        "missing_reason"
      ]
    },
    "required": [
      "artifact_id",
      "artifact_record_digest",
      "content_digest",
      "size_bytes"
    ]
  }
}

```

```

}
],
"properties": {
  "artifact_id": {
    "$ref": "#/$defs/Id"
  },
  "artifact_record_digest": {
    "$ref": "#/$defs/Digest"
  },
  "content_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_expression": {
    "maxLength": 512,
    "minLength": 1,
    "type": "string"
  },
  "license_plan_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "media_type": {
    "maxLength": 256,
    "minLength": 1,
    "type": "string"
  },
},

```

```

"missing_reason": {
  "maxLength": 2048,
  "minLength": 1,
  "type": "string"
},
"size_bytes": {
  "minimum": 0,
  "type": "integer"
},
"source_id": {
  "$ref": "#/$defs/Id"
},
"source_locator": {
  "maxLength": 2048,
  "minLength": 1,

```

```

    "type": "string"
  },
  "source_mode": {
    "enum": [
      "captured-payload",
      "metadata-only"
    ]
  },
  "source_provenance_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "source_id",
  "source_mode",
  "source_locator",
  "source_provenance_digest",
  "media_type",
  "license_provider_id",
  "license_expression",
  "license_plan_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ResearchSourceIntake | Validator: urn:promin:contracts:1#/\$defs/ResearchSourceIntake

## 95. RetrievalPage

Type summary: type="object"; additionalProperties=false

Required: record\_type, query, activation\_digest, depth, ranking, budget, head\_digest, projection\_digest, entities, relations, evidence, truncated, continuation, continuation\_version, stream\_cursor, next\_stream\_cursor, effective\_top\_k, selected\_seed\_count, refinement\_required, refinement\_hints, unselected\_matches\_traversable, selected\_closure\_complete, silent\_truncation, projection\_authoritative

| Property                                        | Required | Structural rule                            |
|-------------------------------------------------|----------|--------------------------------------------|
| activation_digest                               | yes      | \$ref="#/\$defs/Digest"                    |
| budget                                          | yes      | \$ref="#/\$defs/Budget"                    |
| continuation                                    | yes      | oneOf[2]=type:null   type:object           |
| continuation_version                            | yes      | const=2                                    |
| depth                                           | yes      | type="integer"; minimum=1; maximum=12      |
| effective_top_k                                 | yes      | type="integer"; minimum=1; maximum=12      |
| entities                                        | yes      | type="array"; maxItems=32                  |
| evidence                                        | yes      | type="array"; maxItems=32                  |
| head_digest                                     | yes      | oneOf[2]=#/\$defs/Digest   type:null       |
| next_stream_cursor                              | yes      | type="integer"; minimum=0                  |
| projection_authoritative                        | yes      | const=false                                |
| projection_digest                               | yes      | \$ref="#/\$defs/Digest"                    |
| query                                           | yes      | type="string"; minLength=1; maxLength=4096 |
| ranking                                         | yes      | const="bm25-v1"                            |
| record_type                                     | yes      | const="RetrievalPage"                      |
| refinement_hints                                | yes      | type="array"; maxItems=8; uniqueItems=true |
| refinement_required                             | yes      | type="boolean"                             |
| relations                                       | yes      | type="array"; maxItems=48                  |
| selected_closure_complete                       | yes      | type="boolean"                             |
| selected_seed_count                             | yes      | type="integer"; minimum=0                  |
| silent_truncation                               | yes      | const=false                                |
| stream_cursor                                   | yes      | type="integer"; minimum=0                  |
| truncated                                       | yes      | type="boolean"                             |
| unselected_matches_traversable                  | yes      | const=false                                |
| <pre> {   "additionalProperties": false, </pre> |          |                                            |



```

"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "budget": {
    "$ref": "#/$defs/Budget"
  },
  "continuation": {
    "oneOf": [
      {
        "type": "null"
      },
      {
        "additionalProperties": false,
        "properties": {
          "activation_digest": {
            "$ref": "#/$defs/Digest"
          },
          "budget_digest": {
            "$ref": "#/$defs/Digest"
          },
          "cursor": {
            "minimum": 0,
            "type": "integer"
          },
          "depth": {
            "maximum": 12,
            "minimum": 1,
            "type": "integer"
          },
          "expiry": {
            "$ref": "#/$defs/Timestamp"
          },
          "head_digest": {
            "oneOf": [
              {
                "$ref": "#/$defs/Digest"
              },
              {
                "type": "null"
              }
            ]
          },
          "implementation_closure_digest": {
            "$ref": "#/$defs/Digest"
          },
          "projection_digest": {
            "$ref": "#/$defs/Digest"
          },
          "ranking": {
            "const": "bm25-v1"
          },
          "resume_binding_digest": {
            "$ref": "#/$defs/Digest"
          },
          "token": {
            "maxLength": 256,
            "minLength": 1,
            "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\\\. [0-9a-f]{64}\\\\. [A-Za-z0-9_-]{43}$",
            "type": "string"
          },
          "traversal": {
            "const": "typed-bfs-v2"
          },
          "version": {
            "const": 2
          }
        }
      }
    ],
    "required": [
      "version",
      "traversal",
      "token",
      "cursor",
      "expiry",
      "activation_digest",
      "head_digest",
      "projection_digest",
      "implementation_closure_digest",

```

```

        "ranking",
        "depth",
        "budget_digest",
        "resume_binding_digest"
    ],
    "type": "object"
}
],
},
"continuation_version": {
    "const": 2
},
"depth": {
    "maximum": 12,
    "minimum": 1,
    "type": "integer"
},
"effective_top_k": {
    "maximum": 12,
    "minimum": 1,
    "type": "integer"
},
"entities": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "data_class": {
                "enum": [
                    "operational-record",
                    "untrusted-source"
                ]
            }
        }
    },
},

```

```

    "entity_type": {
        "enum": [
            "Task",
            "Candidate",
            "Artifact",
            "Run",
            "Finding",
            "Decision",
            "GateResult",
            "Grant",
            "Lease"
        ]
    },
    "id": {
        "$ref": "#/$defs/Id"
    },
    "payload": {
        "type": "object"
    },
    "source_digest": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "id",
    "entity_type",
    "data_class",
    "payload",

```

```

    "source_digest"
],
"type": "object"
},
"maxItems": 32,
"type": "array"
},
"evidence": {
    "items": {
        "$ref": "#/$defs/Artifact"
    },
    "maxItems": 32,
    "type": "array"
},
"head_digest": {
    "oneOf": [
        {

```

```

        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "next_stream_cursor": {
    "minimum": 0,
    "type": "integer"
  },
},

```

```

"projection_authoritative": {
  "const": false
},
"projection_digest": {
  "$ref": "#/$defs/Digest"
},
"query": {
  "maxLength": 4096,
  "minLength": 1,
  "type": "string"
},
"ranking": {
  "const": "bm25-v1"
},
"record_type": {
  "const": "RetrievalPage"
},
"refinement_hints": {
  "items": {
    "maxLength": 512,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 8,
  "type": "array",
  "uniqueItems": true
},
"refinement_required": {

```

```

  "type": "boolean"
},
"relations": {
  "items": {
    "$ref": "#/$defs/Relation"
  },
  "maxItems": 48,
  "type": "array"
},
"selected_closure_complete": {
  "type": "boolean"
},
"selected_seed_count": {
  "minimum": 0,
  "type": "integer"
},
"silent_truncation": {
  "const": false
},
"stream_cursor": {
  "minimum": 0,
  "type": "integer"
},
"truncated": {
  "type": "boolean"
},
"unselected_matches_traversable": {
  "const": false

```

```

}
},
"required": [
  "record_type",
  "query",
  "activation_digest",
  "depth",
  "ranking",
  "budget",
  "head_digest",

```

```

    "projection_digest",
    "entities",
    "relations",
    "evidence",
    "truncated",
    "continuation",
    "continuation_version",
    "stream_cursor",
    "next_stream_cursor",
    "effective_top_k",
    "selected_seed_count",
    "refinement_required",
    "refinement_hints",
    "unselected_matches_traversable",
    "selected_closure_complete",
    "silent_truncation",
    "projection_authoritative"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: [core/contracts.schema.json#/\\$defs/RetrievalPage](#) | Validator: [urn:promin:contracts:1#/\\$defs/RetrievalPage](#)

## 96. RootAuthority

Type summary: type="object"; additionalProperties=false

Required: subject\_id, capability\_ceiling, scope

| Property           | Required | Structural rule                                         |
|--------------------|----------|---------------------------------------------------------|
| capability_ceiling | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true |
| scope              | yes      | type="array"; minItems=1; maxItems=32                   |
| subject_id         | yes      | \$ref="#/\$defs/Id"                                     |

```

{
  "additionalProperties": false,
  "properties": {
    "capability_ceiling": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "subject_id",
    "capability_ceiling",
    "scope"
  ],
  "type": "object"
}

```

Owner: [core/contracts.schema.json#/\\$defs/RootAuthority](#) | Validator: [urn:promin:contracts:1#/\\$defs/RootAuthority](#)

## 97. RootAuthorization

Type summary: type="object"; additionalProperties=false

Required: kind, subject\_id, proofs

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                              | Required | Structural rule                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------|
| kind                                                                                                                                                                                                                                                                                                                                                                                                                                                  | yes      | const="root"                                            |
| proofs                                                                                                                                                                                                                                                                                                                                                                                                                                                | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true |
| subject_id                                                                                                                                                                                                                                                                                                                                                                                                                                            | yes      | \$ref="#/\$defs/Id"                                     |
| <pre>{   "additionalProperties": false,   "properties": {     "kind": {       "const": "root"     },     "proofs": {       "items": {         "\$ref": "#/\$defs/RootCommandProof"       },       "maxItems": 32,       "minItems": 1,       "type": "array",       "uniqueItems": true     },     "subject_id": {       "\$ref": "#/\$defs/Id"     }   },   "required": [     "kind",     "subject_id",     "proofs"   ],   "type": "object" }</pre> |          |                                                         |
| Owner: core/contracts.schema.json#/\$defs/RootAuthorization   Validator: urn:promin:contracts:1#/\$defs/RootAuthorization                                                                                                                                                                                                                                                                                                                             |          |                                                         |

## 98. RootCommandProof

Type summary: oneOf[2]=#/\$defs/LocalRootCommandProof | #/\$defs/SignatureRootCommandProof

Required: -

|                                                                                                                                                              |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <pre>{   "oneOf": [     {       "\$ref": "#/\$defs/LocalRootCommandProof"     },     {       "\$ref": "#/\$defs/SignatureRootCommandProof"     }   ] }</pre> |  |  |
| Owner: core/contracts.schema.json#/\$defs/RootCommandProof   Validator: urn:promin:contracts:1#/\$defs/RootCommandProof                                      |  |  |

## 99. Run

Type summary: type="object"; additionalProperties=false

Required: record\_type, run\_id, run\_kind, task\_id, candidate\_digest, policy\_digest, tool\_digest, input\_digests, status, started\_at, finished\_at, activation\_digest, definition\_digest, implementation\_closure\_digest, provider\_binding\_digest

| Property                      | Required | Structural rule                                          |
|-------------------------------|----------|----------------------------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"                                  |
| candidate_digest              | yes      | \$ref="#/\$defs/Digest"                                  |
| definition_digest             | yes      | \$ref="#/\$defs/Digest"                                  |
| finished_at                   | yes      | \$ref="#/\$defs/Timestamp"                               |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| input_digests                 | yes      | type="array"; minItems=0; maxItems=128; uniqueItems=true |
| policy_digest                 | yes      | \$ref="#/\$defs/Digest"                                  |
| provider_binding_digest       | yes      | \$ref="#/\$defs/Digest"                                  |
| record_type                   | yes      | const="Run"                                              |
| run_id                        | yes      | \$ref="#/\$defs/Id"                                      |
| run_kind                      | yes      | enum=["execution", "validation"]                         |

| Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required | Structural rule                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|
| started_at                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | yes      | \$ref="#/\$defs/Timestamp"                           |
| status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | yes      | enum=["pass", "fail", "blocked", "skipped", "error"] |
| task_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | yes      | \$ref="#/\$defs/Id"                                  |
| tool_digest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | yes      | \$ref="#/\$defs/Digest"                              |
| <pre> {   "additionalProperties": false,   "properties": {     "activation_digest": {       "\$ref": "#/\$defs/Digest"     },     "candidate_digest": {       "\$ref": "#/\$defs/Digest"     },     "definition_digest": {       "\$ref": "#/\$defs/Digest"     },     "finished_at": {       "\$ref": "#/\$defs/Timestamp"     },     "implementation_closure_digest": {       "\$ref": "#/\$defs/Digest"     },     "input_digests": {       "items": {         "\$ref": "#/\$defs/Digest"       },       "maxItems": 128,       "minItems": 0,       "type": "array",       "uniqueItems": true     },     "policy_digest": {       "\$ref": "#/\$defs/Digest"     },     "provider_binding_digest": {       "\$ref": "#/\$defs/Digest"     },     "record_type": {       "const": "Run"     },     "run_id": {       "\$ref": "#/\$defs/Id"     },     "run_kind": {       "enum": [         "execution",         "validation"       ]     },     "started_at": {       "\$ref": "#/\$defs/Timestamp"     },     "status": {       "enum": [         "pass",         "fail",         "blocked",         "skipped",         "error"       ]     }   },   "task_id": {     "\$ref": "#/\$defs/Id"   },   "tool_digest": {     "\$ref": "#/\$defs/Digest"   } }, "required": [   "record_type",   "run_id",   "run_kind",   "task_id", </pre> |          |                                                      |

```

    "candidate_digest",
    "policy_digest",
    "tool_digest",
    "input_digests",
    "status",
    "started_at",
    "finished_at",
    "activation_digest",
    "definition_digest",
    "implementation_closure_digest",
    "provider_binding_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/Run | Validator: urn:promin:contracts:1#/\$defs/Run

## 100. SanitizedResearchDraft

Type summary: type="object"; additionalProperties=false

Required: record\_type, migration\_rule, license\_plan\_digest, sources, normative\_candidates, excluded\_claims, requested\_flag\_mutations, pass\_credit, product\_acceptance\_pass, product\_public\_approval, public\_release\_approved

| Property                 | Required | Structural rule                                          |
|--------------------------|----------|----------------------------------------------------------|
| excluded_claims          | yes      | type="array"; minItems=0; maxItems=256; uniqueItems=true |
| license_plan_digest      | yes      | \$ref="#/\$defs/Digest"                                  |
| migration_rule           | yes      | const="MIGRATE_FACTS_NOT_FILES"                          |
| normative_candidates     | yes      | type="array"; minItems=0; maxItems=256; uniqueItems=true |
| pass_credit              | yes      | const=false                                              |
| product_acceptance_pass  | yes      | const=false                                              |
| product_public_approval  | yes      | const="not_approved"                                     |
| public_release_approved  | yes      | const=false                                              |
| record_type              | yes      | const="SanitizedResearchDraft"                           |
| requested_flag_mutations | yes      | type="array"; const=[]                                   |
| sources                  | yes      | type="array"; minItems=1; maxItems=128; uniqueItems=true |

```

{
  "additionalProperties": false,
  "properties": {
    "excluded_claims": {
      "items": {
        "additionalProperties": false,
        "allOf": [
          {
            "else": {
              "not": {
                "required": [
                  "blocked_reason"
                ]
              }
            },
            "if": {
              "anyOf": [
                {
                  "properties": {
                    "claim_kind": {
                      "const": "blocked_claim"
                    }
                  },
                  "required": [
                    "claim_kind"
                  ]
                }
              ],
              "required": [
                "verification_status"
              ]
            }
          ],
          {
            "properties": {
              "verification_status": {
                "const": "blocked"
              }
            },
            "required": [
              "verification_status"
            ]
          }
        ]
      }
    }
  }
}

```

```

    }
  ],
  "then": {
    "required": [
      "blocked_reason"
    ]
  }
},
"properties": {
  "blocked_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "citation_refs": {
    "items": {
      "$ref": "#/$defs/Id"
    },

```

```

    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },
  "citation_required": {
    "type": "boolean"
  },
  "claim_class": {
    "enum": [
      "distribution",
      "fact",
      "marketing",
      "proof"
    ]
  },
  "claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "claim_id": {
    "$ref": "#/$defs/Id"
  },
  "claim_kind": {
    "enum": [
      "research_question",
      "hypothesis",
      "assumption",
      "evidence_claim",
      "blocked_claim"
    ]
  },

```

```

  ],
  "evidence_refs": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_id": {
          "$ref": "#/$defs/Id"
        },
        "artifact_record_digest": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "artifact_id",
        "artifact_record_digest"
      ],
      "type": "object"
    },
    "maxItems": 32,
    "type": "array",
    "uniqueItems": true
  },
  "normative_candidate": {
    "type": "boolean"
  },
  "normative_use_allowed": {
    "const": false
  },

```



```

    },
    "reason": {
      "maxLength": 2048,
      "minLength": 1,
      "type": "string"
    },
    "text": {
      "maxLength": 8192,
      "minLength": 1,
      "type": "string"
    },
    "verification_status": {
      "enum": [
        "unverified",
        "source-bound",
        "verified",
        "blocked"
      ]
    }
  },
  "required": [
    "claim_id",
    "claim_kind",
    "claim_class",
    "claim_digest",
    "text",
    "citation_required",
    "citation_refs",
    "evidence_refs",
    "verification_status",
    "normative_candidate",
    "normative_use_allowed",
    "reason"
  ],
  "type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"license_plan_digest": {
  "$ref": "#/$defs/Digest"
},
"migration_rule": {
  "const": "MIGRATE_FACTS_NOT_FILES"
},
"normative_candidates": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "citation_refs": {
        "items": {
          "$ref": "#/$defs/Id"
        },
        "maxItems": 32,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
      },
      "citation_required": {
        "const": true
      },
      "claim_class": {
        "const": "fact"
      },
      "claim_digest": {
        "$ref": "#/$defs/Digest"
      },
      "claim_id": {
        "$ref": "#/$defs/Id"
      },
      "claim_kind": {
        "const": "evidence_claim"
      },
      "evidence_refs": {
        "items": {
          "additionalProperties": false,

```

```

    "properties": {
      "artifact_id": {
        "$ref": "#/$defs/Id"
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "artifact_id",
      "artifact_record_digest"
    ],
    "type": "object"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"normative_candidate": {
  "const": true
},
"normative_use_allowed": {
  "const": true
},
"source_bindings": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "artifact_id": {
        "$ref": "#/$defs/Id"
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "content_digest": {
      "$ref": "#/$defs/Digest"
    },
    "license_expression": {
      "maxLength": 512,
      "minLength": 1,
      "type": "string"
    },
    "license_plan_digest": {
      "$ref": "#/$defs/Digest"
    },
    "license_provider_id": {
      "$ref": "#/$defs/Id"
    },
    "media_type": {
      "maxLength": 256,
      "minLength": 1,
      "type": "string"
    },
    "size_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "source_id": {
      "$ref": "#/$defs/Id"
    },
    "source_locator": {
      "maxLength": 2048,
      "minLength": 1,
      "type": "string"
    },
    "source_mode": {
      "enum": [
        "captured-payload",
        "metadata-only"
      ]
    },
    "source_provenance_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [

```

```

        "source_id",
        "source_mode",
        "source_locator",
        "source_provenance_digest",
        "artifact_id",
        "artifact_record_digest",
        "content_digest",
        "size_bytes",
        "media_type",
        "license_provider_id",
        "license_expression",
        "license_plan_digest"
    ],

```

```

        "type": "object"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"text": {
    "maxLength": 8192,
    "minLength": 1,
    "type": "string"
},
"verification_status": {
    "enum": [
        "source-bound",
        "verified"
    ]
}
},
"required": [
    "claim_id",
    "claim_kind",
    "claim_class",
    "claim_digest",
    "text",
    "citation_required",
    "citation_refs",
    "evidence_refs",

```

```

        "verification_status",
        "normative_candidate",
        "normative_use_allowed",
        "source_bindings"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"pass_credit": {
    "const": false
},
"product_acceptance_pass": {
    "const": false
},
"product_public_approval": {
    "const": "not_approved"
},
"public_release_approved": {
    "const": false
},
"record_type": {
    "const": "SanitizedResearchDraft"
},
"requested_flag_mutations": {

```

```

    "const": [],
    "type": "array"
},
"sources": {
    "items": {
        "additionalProperties": false,
        "allOf": [
            {

```

```

        "else": {
            "not": {
                "anyOf": [
                    {
                        "required": [
                            "artifact_id"
                        ]
                    },
                    {
                        "required": [
                            "artifact_record_digest"
                        ]
                    },
                    {
                        "required": [
                            "content_digest"
                        ]
                    },
                    {
                        "required": [
                            "size_bytes"
                        ]
                    }
                ]
            },
            "required": [
                "missing_reason"
            ]
        },
        "if": {
            "properties": {
                "source_mode": {
                    "const": "captured-payload"
                }
            },
            "required": [
                "source_mode"
            ]
        },
        "then": {
            "not": {
                "required": [
                    "missing_reason"
                ]
            },
            "required": [
                "artifact_id",
                "artifact_record_digest",
                "content_digest",
                "size_bytes"
            ]
        }
    },
    "properties": {
        "artifact_id": {
            "$ref": "#/$defs/Id"
        },
        "artifact_record_digest": {
            "$ref": "#/$defs/Digest"
        },
        "content_digest": {
            "$ref": "#/$defs/Digest"
        },
        "license_expression": {
            "maxLength": 512,
            "minLength": 1,
            "type": "string"
        },
        "license_plan_digest": {
            "$ref": "#/$defs/Digest"
        },
        "license_provider_id": {
            "$ref": "#/$defs/Id"
        },
        "media_type": {

```

```

    "maxLength": 256,
    "minLength": 1,
    "type": "string"
  },
  "missing_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "size_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "source_id": {
    "$ref": "#/$defs/Id"
  },
  "source_locator": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "source_mode": {
    "enum": [
      "captured-payload",
      "metadata-only"
    ]
  },
  "source_provenance_digest": {

```

```

    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "source_id",
  "source_mode",
  "source_locator",
  "source_provenance_digest",
  "media_type",
  "license_provider_id",
  "license_expression",
  "license_plan_digest"
],
"type": "object"
},
"maxItems": 128,
"minItems": 1,
"type": "array",
"uniqueItems": true
}

```

```

"required": [
  "record_type",
  "migration_rule",
  "license_plan_digest",
  "sources",
  "normative_candidates",
  "excluded_claims",

```

```

  "requested_flag_mutations",
  "pass_credit",
  "product_acceptance_pass",
  "product_public_approval",
  "public_release_approved"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Owner: `core/contracts.schema.json#/$defs/SanitizedResearchDraft` | Validator: `urn:promin:contracts:1#/$defs/SanitizedResearchDraft`

## 101. SaturationAudit

Type summary: `type="object"; additionalProperties=false`

Required: `record_type`, `status`, `candidate_binding_digest`, `zero_new_iterations`, `new_findings`, `pass_credit`, `artifact_binding`, `families`, `collection`, `iterations`, `consecutive_full_zero_new`, `requirements`, `predeclared_zero_new`, `acceptance_pass`, `product_acceptance_pass`, `producer`, `invocation`, `raw_artifact_manifest_digest`, `producer_attestation`, `result_digest`

| Property                     | Required | Structural rule                              |
|------------------------------|----------|----------------------------------------------|
| acceptance_pass              | yes      | const=false                                  |
| artifact_binding             | yes      | type="object"                                |
| candidate_binding_digest     | yes      | \$ref="#/\$defs/Digest"                      |
| collection                   | yes      | type="object"; additionalProperties=false    |
| consecutive_full_zero_new    | yes      | const=3                                      |
| families                     | yes      | type="array"; minItems=1; uniqueItems=true   |
| invocation                   | yes      | \$ref="#/\$defs/ReleaseEvidenceInvocation"   |
| iterations                   | yes      | type="array"; minItems=3; maxItems=18        |
| new_findings                 | yes      | const=0                                      |
| pass_credit                  | yes      | const=false                                  |
| predeclared_zero_new         | yes      | const=false                                  |
| producer                     | yes      | \$ref="#/\$defs/ReleaseEvidenceProducer"     |
| producer_attestation         | yes      | \$ref="#/\$defs/EvidenceProducerAttestation" |
| product_acceptance_pass      | yes      | const=false                                  |
| raw_artifact_manifest_digest | yes      | \$ref="#/\$defs/Digest"                      |
| record_type                  | yes      | const="SaturationAudit"                      |
| requirements                 | yes      | type="object"; additionalProperties=false    |
| result_digest                | yes      | \$ref="#/\$defs/Digest"                      |
| status                       | yes      | const="pass"                                 |
| zero_new_iterations          | yes      | const=3                                      |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "collection": {
      "additionalProperties": false,
      "properties": {
        "catalogue_digest": {},
        "families_discovered": {},
        "production_probes_discovered": {}
      },
      "required": [
        "catalogue_digest",
        "families_discovered",
        "production_probes_discovered"
      ],
      "type": "object"
    },
    "consecutive_full_zero_new": {
      "const": 3
    },
    "families": {
      "items": {
        "type": "string"
      },
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "invocation": {
      "$ref": "#/$defs/ReleaseEvidenceInvocation"
    },
    "iterations": {
      "items": {
        "additionalProperties": false,
        "properties": {

```

```

"finding_fingerprints": {},
"focused_and_integration_tests": {
  "additionalProperties": false,
  "properties": {
    "cache_provider_disabled": {},
    "elapsed_ms": {},
    "exit_code": {},
    "log_bytes": {},
    "log_path": {},
    "log_sha256": {},
    "marker_expression": {},
    "output_digest": {},
    "pytest_only_sentinel_executed": {},
    "pytest_only_sentinel_token_digest": {},
    "runner": {},
    "sentinel_bytes": {},
    "sentinel_path": {},
    "sentinel_sha256": {}
  },
  "required": [
    "exit_code",
    "elapsed_ms",
    "output_digest",
    "runner",
    "marker_expression",
    "cache_provider_disabled",
    "pytest_only_sentinel_executed",
    "pytest_only_sentinel_token_digest",
    "log_path",
    "log_sha256",
    "log_bytes",
    "sentinel_path",
    "sentinel_sha256",
    "sentinel_bytes"
  ],
  "type": "object"
},
"full_system_corpus_rerun": {},
"iteration": {},
"mutation_families_run": {},

```

```

"mutation_seed": {},
"new_finding_classes": {},
"physical_runtime_saturation": {
  "additionalProperties": false,
  "properties": {
    "artifact_binding_digest": {},
    "current_release_regression": {},
    "elapsed_ms": {},
    "execution_route": {},
    "exit_code": {},
    "files": {},
    "forced_continuation_chains": {},
    "forced_union_matches": {},
    "fresh_control_state": {
      "additionalProperties": false,
      "properties": {
        "activation_digest": {
          "$ref": "#/defs/Digest"
        },
        "core_bundle_digest": {
          "$ref": "#/defs/Digest"
        },
        "implementation_closure_digest": {
          "$ref": "#/defs/Digest"
        },
        "init_record_count": {
          "const": 5
        }
      },
      "type": "object"
    }
  },
  "required": [
    "activation_digest",
    "core_bundle_digest",
    "implementation_closure_digest",
    "init_record_count"
  ],
  "type": "object"
},

```

```

"physical_corpus_recipe": {
  "additionalProperties": false,
  "properties": {
    "bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "file_count": {
      "const": 100000
    }
  },
  "required": [
    "bytes",
    "file_count"
  ],
  "type": "object"
},

```

```

    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "sha256",
    "bytes",
    "file_count"
  ],
  "type": "object"
},
"preset_digest": {
  "$ref": "#/$defs/Digest"
},
"product_tree_reused": {
  "type": "boolean"
},
"product_tree_scans": {

```

```

  "const": 0
},
"semantic_state_reused": {
  "const": false
},
"status": {
  "const": "created"
}
},
"required": [
  "status",
  "product_tree_scans",
  "init_record_count",
  "activation_digest",
  "implementation_closure_digest",
  "core_bundle_digest",
  "preset_digest",
  "semantic_state_reused",
  "product_tree_reused",
  "physical_corpus_recipe"
],
"type": "object"
},
"immutable_snapshot_creditable": {},
"invocation_id": {},
"log_bytes": {},
"log_path": {},
"log_sha256": {},

```

```

"metrics": {},
"output_digest": {},
"performance_profile": {},
"platform_binding_digest": {},
"pytest_marker": {},
"pytest_selection_command": {},
"pytest_selection_command_executed": {},
"queries": {},
"record_bytes": {
  "minimum": 1,
  "type": "integer"
},
"record_path": {},
"record_sha256": {},
"result_digest": {}
},
"required": [
  "exit_code",
  "elapsed_ms",
  "output_digest",
  "log_path",
  "log_sha256",
  "log_bytes",
  "files",
  "queries",
  "performance_profile",
  "pytest_marker",
  "pytest_selection_command",
  "pytest_selection_command_executed",
  "execution_route",

```



```

    "fresh_control_state",
    "metrics",
    "artifact_binding_digest",
    "immutable_snapshot_creditable",
    "platform_binding_digest",
    "record_path",
    "record_sha256",
    "record_bytes",
    "result_digest",
    "invocation_id",
    "forced_continuation_chains",
    "forced_union_matches",
    "current_release_regression"
  ],
  "type": "object"
},
"silent_truncations": {},
"status": {},
"streak_reset_reason": {},
"unexpected_accepts": {},
"unexpected_crashes": {},
"unexpected_rejects": {},
"zero_new_streak_after": {},
"zero_new_streak_before": {}
},
"required": [

```

```

    "finding_fingerprints",
    "focused_and_integration_tests",
    "full_system_corpus_rerun",
    "iteration",
    "mutation_families_run",
    "mutation_seed",
    "new_finding_classes",
    "physical_runtime_saturation",
    "silent_truncations",
    "status",
    "streak_reset_reason",
    "unexpected_accepts",
    "unexpected_crashes",
    "unexpected_rejects",
    "zero_new_streak_after",
    "zero_new_streak_before"
  ],
  "type": "object"
},
"maxItems": 18,
"minItems": 3,
"type": "array"
},
"new_findings": {
  "const": 0
},
"pass_credit": {
  "const": false
}

```

```

},
"predeclared_zero_new": {
  "const": false
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationAudit"
},
"requirements": {
  "additionalProperties": false,
  "properties": {
    "actual_runtime_queries": {},
    "configured_iteration_limit": {},

```

```

"core_valid_relations": {},
"depths": {},
"focused_pytest_marker": {},
"fresh_control_state_per_iteration": {
    "const": true
},
"init_product_tree_scans": {
    "const": 0
},
"maximum_iterations": {},
"minimum_consecutive_full_zero_new": {},
"missing_scale_workspace_fails": {},
"mixed_query_classes": {},
"performance_profile": {},
"physical_files": {},
"physical_pytest_command": {},
"physical_pytest_marker": {},
"profile_bound_fail_closed": {},
"raw_file_proxy_ratio": {},
"semantic_corpus_relations": {},
"semantic_corpus_tasks": {},
"semantic_state_reused": {
    "const": false
},
"synthetic_task_ratio": {}
},
"required": [
    "minimum_consecutive_full_zero_new",
    "maximum_iterations",
    "configured_iteration_limit",
    "physical_files",
    "core_valid_relations",

```

```

    "actual_runtime_queries",
    "depths",
    "raw_file_proxy_ratio",
    "synthetic_task_ratio",
    "semantic_corpus_tasks",
    "semantic_corpus_relations",
    "mixed_query_classes",
    "performance_profile",
    "profile_bound_fail_closed",
    "focused_pytest_marker",
    "physical_pytest_marker",
    "physical_pytest_command",
    "missing_scale_workspace_fails",
    "fresh_control_state_per_iteration",
    "semantic_state_reused",
    "init_product_tree_scans"
],
"type": "object"
},
"result_digest": {
    "$ref": "#/$defs/Digest"
},
"status": {
    "const": "pass"
},
"zero_new_iterations": {
    "const": 3
}
}

```

```

},
"required": [
    "record_type",
    "status",
    "candidate_binding_digest",
    "zero_new_iterations",
    "new_findings",
    "pass_credit",
    "artifact_binding",
    "families",
    "collection",
    "iterations",
    "consecutive_full_zero_new",
    "requirements",
    "predeclared_zero_new",
    "acceptance_pass",
    "product_acceptance_pass",

```

```

    "producer",
    "invocation",
    "raw_artifact_manifest_digest",
    "producer_attestation",
    "result_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SaturationAudit | Validator: urn:promin:contracts:1#/\$defs/SaturationAudit

## 102. SaturationEvidence

Type summary: type="object"; additionalProperties=false

Required: record\_type, status, candidate\_binding\_digest, artifact\_binding, artifact\_binding\_unchanged, workspace\_initialization, runtime\_binding, physical\_files, core\_valid\_relations, runtime\_queries, silent\_truncations, selected\_closure\_union\_completeness, memory\_amplification\_at\_most\_32, core\_valid\_relations\_exact\_198999, broad\_query\_refinement\_required, high\_cardinality\_terms\_verified, content\_search\_verified, miss\_behavior\_verified, hostile\_proxy\_content\_verified, exact\_artifact\_search\_verified, mixed\_query\_classes\_complete, continuation\_token\_bytes\_at\_most\_256, continuation\_state\_bytes\_at\_most\_16384, continuation\_token\_overhead\_at\_most\_10\_percent, pass\_credit, physical, inventory, projection, query\_authorization, search, resources, performance, contract\_predicates, raw\_artifact\_manifest, current\_release\_regression, claim\_scope, acceptance\_pass, product\_acceptance\_pass, public\_release\_approved, producer, invocation, raw\_artifact\_manifest\_digest, producer\_attestation, result\_digest

| Property                                       | Required | Structural rule                              |
|------------------------------------------------|----------|----------------------------------------------|
| acceptance_pass                                | yes      | const=false                                  |
| artifact_binding                               | yes      | type="object"                                |
| artifact_binding_unchanged                     | yes      | const=true                                   |
| broad_query_refinement_required                | yes      | type="boolean"                               |
| candidate_binding_digest                       | yes      | \$ref="#/\$defs/Digest"                      |
| claim_scope                                    | yes      | type="string"; minLength=1                   |
| content_search_verified                        | yes      | type="boolean"                               |
| continuation_state_bytes_at_most_16384         | yes      | type="boolean"                               |
| continuation_token_bytes_at_most_256           | yes      | type="boolean"                               |
| continuation_token_overhead_at_most_10_percent | yes      | type="boolean"                               |
| contract_predicates                            | yes      | type="object"; additionalProperties=false    |
| core_valid_relations                           | yes      | const=198999                                 |
| core_valid_relations_exact_198999              | yes      | const=true                                   |
| current_release_regression                     | yes      | type="object"; additionalProperties=false    |
| exact_artifact_search_verified                 | yes      | type="boolean"                               |
| high_cardinality_terms_verified                | yes      | type="boolean"                               |
| hostile_proxy_content_verified                 | yes      | type="boolean"                               |
| inventory                                      | yes      | type="object"; additionalProperties=false    |
| invocation                                     | yes      | \$ref="#/\$defs/ReleaseEvidenceInvocation"   |
| memory_amplification_at_most_32                | yes      | type="boolean"                               |
| miss_behavior_verified                         | yes      | type="boolean"                               |
| mixed_query_classes_complete                   | yes      | type="boolean"                               |
| pass_credit                                    | yes      | const=false                                  |
| performance                                    | yes      | type="object"; additionalProperties=false    |
| physical                                       | yes      | type="object"; additionalProperties=false    |
| physical_files                                 | yes      | const=100000                                 |
| producer                                       | yes      | \$ref="#/\$defs/ReleaseEvidenceProducer"     |
| producer_attestation                           | yes      | \$ref="#/\$defs/EvidenceProducerAttestation" |
| product_acceptance_pass                        | yes      | const=false                                  |

| Property                            | Required | Structural rule                           |
|-------------------------------------|----------|-------------------------------------------|
| projection                          | yes      | type="object"; additionalProperties=false |
| public_release_approved             | yes      | const=false                               |
| query_authorization                 | yes      | type="object"; additionalProperties=false |
| raw_artifact_manifest               | yes      | type="object"; additionalProperties=false |
| raw_artifact_manifest_digest        | yes      | \$ref="#/\$defs/Digest"                   |
| record_type                         | yes      | const="SaturationEvidence"                |
| resources                           | yes      | type="object"; additionalProperties=false |
| result_digest                       | yes      | \$ref="#/\$defs/Digest"                   |
| runtime_binding                     | yes      | type="object"; additionalProperties=false |
| runtime_queries                     | yes      | const=600                                 |
| search                              | yes      | type="object"; additionalProperties=false |
| selected_closure_union_completeness | yes      | type="number"; minimum=0.0; maximum=1.0   |
| silent_truncations                  | yes      | const=0                                   |
| status                              | yes      | enum=["pass", "fail"]                     |
| workspace_initialization            | yes      | type="object"; additionalProperties=false |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },
    "artifact_binding_unchanged": {
      "const": true
    },
    "broad_query_refinement_required": {
      "type": "boolean"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "claim_scope": {
      "minLength": 1,
      "type": "string"
    },
    "content_search_verified": {
      "type": "boolean"
    },
    "continuation_state_bytes_at_most_16384": {
      "type": "boolean"
    },
    "continuation_token_bytes_at_most_256": {
      "type": "boolean"
    },
    "continuation_token_overhead_at_most_10_percent": {
      "type": "boolean"
    },
    "contract_predicates": {
      "additionalProperties": false,
      "properties": {
        "broad_query_refinement_required": {},
        "content_search_verified": {},
        "continuation_state_bytes_at_most_16384": {},
        "continuation_token_bytes_at_most_256": {},
        "continuation_token_overhead_at_most_10_percent": {},
        "continuation_union_complete": {},
        "core_valid_relations_exact": {
          "const": true
        },
        "exact_artifact_binding_unchanged": {},
        "exact_artifact_search_verified": {},
        "high_cardinality_terms_verified": {},
        "hostile_proxy_content_verified": {},
        "inventory_incremental_memory_amplification_at_most_32": {}
      }
    }
  }
}

```

```

"inventory_passes_exact": {},
"miss_behavior_verified": {},
"mixed_query_classes_complete": {},
"physical_relation_artifact_coverage_complete": {},
"raw_file_proxy_ratio_exact": {},

"rebuild_digest_equal": {},
"rebuild_product_passes_zero": {},
"runtime_depths_1_through_12": {},
"runtime_queries_exact": {
  "const": true
},
"runtime_query_budget_bounded": {},
"selected_closure_union_complete": {},
"semantic_commit_count_exact": {},
"silent_truncations_zero": {},
"synthetic_task_ratio_zero": {}
},
"required": [
  "broad_query_refinement_required",
  "content_search_verified",
  "continuation_state_bytes_at_most_16384",
  "continuation_token_bytes_at_most_256",
  "continuation_token_overhead_at_most_10_percent",
  "continuation_union_complete",
  "core_valid_relations_exact",
  "exact_artifact_binding_unchanged",
  "exact_artifact_search_verified",
  "high_cardinality_terms_verified",
  "hostile_proxy_content_verified",
  "inventory_incremental_memory_amplification_at_most_32",
  "inventory_passes_exact",
  "miss_behavior_verified",
  "mixed_query_classes_complete",

```

```

"physical_relation_artifact_coverage_complete",
"raw_file_proxy_ratio_exact",
"rebuild_digest_equal",
"rebuild_product_passes_zero",
"runtime_depths_1_through_12",
"runtime_queries_exact",
"runtime_query_budget_bounded",
"selected_closure_union_complete",
"semantic_commit_count_exact",
"silent_truncations_zero",
"synthetic_task_ratio_zero"
],
"type": "object"
},
"core_valid_relations": {
  "const": 198999
},
"core_valid_relations_exact_198999": {
  "const": true
},
"current_release_regression": {
  "additionalProperties": false,
  "properties": {
    "after": {},
    "before": {},
    "eligibility_remained_false": {},
    "historical_decision_preserved": {}
  }
},

```

```

"required": [
  "before",
  "after",
  "eligibility_remained_false",
  "historical_decision_preserved"
],
"type": "object"
},
"exact_artifact_search_verified": {
  "type": "boolean"
},
"high_cardinality_terms_verified": {
  "type": "boolean"
},
"hostile_proxy_content_verified": {
  "type": "boolean"
}

```

```

},
"inventory": {
  "additionalProperties": false,
  "properties": {
    "candidate_digest": {},
    "elapsed_ms": {},
    "entries": {},
    "passes": {},
    "snapshot": {},
    "stream_bytes": {}
  },
  "required": [
    "candidate_digest",
    "elapsed_ms",
    "entries",
    "passes",
    "snapshot",
    "stream_bytes"
  ],
  "type": "object"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"memory_amplification_at_most_32": {
  "type": "boolean"
},
"miss_behavior_verified": {
  "type": "boolean"
},
"mixed_query_classes_complete": {
  "type": "boolean"
},
"pass_credit": {
  "const": false
},
"performance": {
  "additionalProperties": false,
  "properties": {
    "all_within_profile": {},
    "compatible_platforms": {
      "const": [
        "linux",
        "windows"
      ]
    },
    "observed": {},
    "platform_binding": {
      "additionalProperties": false,
      "properties": {
        "binding_digest": {
          "pattern": "^sha256:[0-9a-f]{64}$",
          "type": "string"
        },
        "machine": {
          "minLength": 1,
          "type": "string"
        },
        "profile_key": {
          "minLength": 1,
          "type": "string"
        },
        "python_executable_sha256": {
          "pattern": "^sha256:[0-9a-f]{64}$",
          "type": "string"
        },
        "python_implementation": {
          "const": "CPython"
        },
        "python_version": {
          "pattern": "^[0-9]+\\.?[0-9]+\\.?[0-9]+$",
          "type": "string"
        },
        "release": {
          "minLength": 1,
          "type": "string"
        }
      }
    }
  }
}

```

```

    "sqlite_version": {
      "pattern": "^[0-9]+\\.\\.[0-9]+\\.\\.[0-9]+$",
      "type": "string"
    },
    "system": {
      "enum": [
        "darwin",
        "linux",
        "windows"
      ]
    }
  },
  "required": [
    "binding_digest",
    "system",
    "release",
    "machine",
    "python_implementation",
    "python_version",

```

```

    "python_executable_sha256",
    "sqlite_version",
    "profile_key"
  ],
  "type": "object"
},
"predicates": {},
"profile_id": {
  "const": "portable-local-v1"
},
"requires_same_runner_no_degradation": {
  "const": true
},
"thresholds": {
  "additionalProperties": false,
  "properties": {
    "commit_bytes_per_changed_record_max": {
      "const": 32768
    },
    "commit_p95_ms_max": {
      "const": 500
    },
    "commit_p99_ms_max": {
      "const": 1000
    },
    "database_bytes_max": {
      "const": 402653184
    }
  },

```

```

    "p50_ms_max": {
      "const": 100
    },
    "p95_ms_max": {
      "const": 150
    },
    "p99_ms_max": {
      "const": 500
    },
    "peak_rss_bytes_max": {
      "const": 805306368
    },
    "projection_amplification_max": {
      "const": 20
    },
    "runtime_checkpoint_count_max": {
      "const": 24
    },
    "semantic_inflation_max": {
      "const": 3.25
    },
    "semantic_ingestion_seconds_max": {
      "const": 600
    }
  },
  "required": [
    "p50_ms_max",
    "p95_ms_max",

```

```

    "p99_ms_max",
    "peak_rss_bytes_max",

```

```

        "database_bytes_max",
        "projection_amplification_max",
        "semantic_inflation_max",
        "commit_p95_ms_max",
        "commit_p99_ms_max",
        "commit_bytes_per_changed_record_max",
        "runtime_checkpoint_count_max",
        "semantic_ingestion_seconds_max"
    ],
    "type": "object"
}
},
"required": [
    "all_within_profile",
    "compatible_platforms",
    "observed",
    "platform_binding",
    "predicates",
    "profile_id",
    "requires_same_runner_no_degradation",
    "thresholds"
],
"type": "object"
},
"physical": {
    "additionalProperties": false,

```

```

    "properties": {
        "explicit_semantic_corpus": {
            "additionalProperties": false,
            "properties": {
                "conflicting_exact_id_text": {},
                "continuation_query_ids": {},
                "depths": {},
                "generation": {},
                "harness_generated": {
                    "const": true
                },
            },
            "high_fanout": {},
            "physical_relation_fixture": {
                "additionalProperties": false,
                "properties": {
                    "artifact_target_count": {},
                    "artifact_target_coverage": {},
                    "relation_count": {},
                    "relation_kind": {},
                    "relations_per_atomic_batch_max": {},
                    "target_type": {},
                    "task_count": {}
                },
                "required": [
                    "task_count",
                    "relation_count",
                    "relation_kind",
                    "target_type",

```

```

                    "artifact_target_count",
                    "artifact_target_coverage",
                    "relations_per_atomic_batch_max"
                ],
                "type": "object"
            },
            "physical_relation_fixture_reused": {
                "const": false
            },
            "product_acceptance_credit": {
                "const": false
            },
            "query_ids": {},
            "record_type": {
                "const": "SaturationSemanticCorpus"
            },
            "relation_count": {},
            "reused": {
                "const": false
            },
            "search_fixture": {
                "additionalProperties": false,
                "properties": {
                    "depths": {},

```



```

    "high_fanout": {},
    "relation_count": {},
    "task_count": {}
  },

```

```

    "required": [
      "task_count",
      "relation_count",
      "depths",
      "high_fanout"
    ],
    "type": "object"
  },
  "search_fixture_reused": {
    "const": false
  },
  "task_count": {}
},
"required": [
  "record_type",
  "generation",
  "harness_generated",
  "product_acceptance_credit",
  "task_count",
  "relation_count",
  "depths",
  "high_fanout",
  "conflicting_exact_id_text",
  "query_ids",
  "continuation_query_ids",
  "reused",
  "search_fixture",
  "physical_relation_fixture",

```

```

  "search_fixture_reused",
  "physical_relation_fixture_reused"
],
"type": "object"
},
"files": {},
"generation_elapsed_ms": {},
"inventory_relations": {},
"raw_file_proxies": {},
"raw_file_proxy_ratio": {},
"raw_files": {},
"relations": {},
"reused_product": {},
"semantic_proxies": {},
"synthetic_task_count": {},
"synthetic_task_ratio": {},
"synthetic_tasks": {},
"vcs_commit": {},
"vcs_provider_version": {},
"vcs_tree_digest": {},
"vcs_tree_files": {}
},
"required": [
  "explicit_semantic_corpus",
  "files",
  "generation_elapsed_ms",
  "inventory_relations",
  "raw_file_proxies",

```

```

  "raw_file_proxy_ratio",
  "raw_files",
  "relations",
  "reused_product",
  "semantic_proxies",
  "synthetic_task_count",
  "synthetic_task_ratio",
  "synthetic_tasks",
  "vcs_commit",
  "vcs_provider_version",
  "vcs_tree_digest",
  "vcs_tree_files"
],
"type": "object"
},
"physical_files": {
  "const": 100000

```

```

},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"projection": {

```

```

  "additionalProperties": false,
  "properties": {
    "database_bytes": {},
    "elapsed_ms": {},
    "entity_count": {},
    "equal_semantic_digest": {},
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "initial_inventory_passes": {},
    "initial_product_passes": {},
    "inventory_integrity": {},
    "inventory_projection_amplification": {},
    "projection_amplification": {},
    "rebuild_inventory_passes": {},
    "rebuild_product_passes": {},
    "relation_count": {},
    "semantic_digest": {},
    "semantic_inflation": {}
  },
  "required": [
    "database_bytes",
    "elapsed_ms",
    "entity_count",
    "equal_semantic_digest",
    "implementation_closure_digest",
    "initial_inventory_passes",
    "initial_product_passes",

```

```

    "inventory_integrity",
    "inventory_projection_amplification",
    "projection_amplification",
    "rebuild_inventory_passes",
    "rebuild_product_passes",
    "relation_count",
    "semantic_digest",
    "semantic_inflation"
  ],
  "type": "object"
},
"public_release_approved": {
  "const": false
},
"query_authorization": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capability_id": {
      "const": "projection.read"
    },
    "claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"

```

```

  },
  "grant_id": {
    "$ref": "#/$defs/Id"
  },
  "scope": {
    "items": {
      "type": "object"
    },
    "maxItems": 128,
    "minItems": 1,

```

```

    "type": "array"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "activation_digest",
  "capability_id",
  "claim_digest",
  "expires_at",
  "grant_id",
  "scope",
  "subject_id"
],
"type": "object"
},
"raw_artifact_manifest": {

```

```

  "additionalProperties": false,
  "properties": {
    "artifact_count": {
      "const": 6
    },
    "artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "maximum": 268435456,
            "minimum": 1,
            "type": "integer"
          },
          "media_type": {
            "enum": [
              "application/json",
              "application/x-ndjson"
            ]
          },
          "path": {
            "pattern": "^raw/",
            "type": "string"
          },
          "records": {
            "maximum": 100000,
            "minimum": 1,
            "type": "integer"
          }
        }
      }
    }
  }
},

```

```

    "role": {
      "enum": [
        "inventory-stream",
        "query-results",
        "process-samples",
        "continuation-state-manifest",
        "phase-log",
        "operation-metrics"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "role",
    "path",
    "media_type",
    "sha256",
    "bytes",
    "records"
  ],
  "type": "object"
},
"maxItems": 6,
"minItems": 6,
"type": "array"

```

```

},
"evidence_class": {
  "const": "harness_generated"
}

```

```

    },
    "inventory_identity_digest": {
      "$ref": "#/$defs/Digest"
    },
    "inventory_stream_digest": {
      "$ref": "#/$defs/Digest"
    },
    "manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "path_scope": {
      "const": "saturation-result-directory"
    },
    "product_acceptance_credit": {
      "const": false
    },
    "record_type": {
      "const": "SaturationRawArtifactManifest"
    }
  },
  "required": [
    "record_type",
    "path_scope",
    "evidence_class",
    "product_acceptance_credit",

```

```

    "artifacts",
    "artifact_count",
    "inventory_stream_digest",
    "inventory_identity_digest",
    "manifest_digest"
  ],
  "type": "object"
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationEvidence"
},
"resources": {
  "additionalProperties": false,
  "properties": {
    "inventory_absolute_rss_amplification": {},
    "inventory_incremental_memory_amplification": {},
    "inventory_pipeline_incremental_peak_bytes": {},
    "inventory_pipeline_peak_rss_bytes": {},
    "inventory_stage_rss": {},
    "memory_amplification_metric": {},
    "peak_rss_bytes": {},
    "projection_stage_rss": {}
  }
},
"required": [
  "inventory_absolute_rss_amplification",

```

```

    "inventory_incremental_memory_amplification",
    "inventory_pipeline_incremental_peak_bytes",
    "inventory_pipeline_peak_rss_bytes",
    "inventory_stage_rss",
    "memory_amplification_metric",
    "peak_rss_bytes",
    "projection_stage_rss"
  ],
  "type": "object"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"runtime_binding": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "platform_binding_digest": {
    "pattern": "^sha256:[0-9a-f]{64}$",
    "type": "string"
  }
}

```

```

    }
  },
  "required": [
    "activation_digest",
    "implementation_closure_digest",
    "platform_binding_digest"
  ],
  "type": "object"
},
"runtime_queries": {
  "const": 600
},
"search": {
  "additionalProperties": false,
  "properties": {
    "actual_runtime_queries": {},
    "broad_query_refinement_required": {},
    "content_search_verified": {},
    "continuation_state": {},
    "continuation_token_overhead_at_most_10_percent": {},
    "continuation_union_complete": {},
    "continuation_union_completeness": {},
    "continuations_checked": {},
    "depth_counts": {},
    "depth_max": {},
    "depth_min": {},
    "exact_artifact_search_verified": {},
    "explicit_truncations": {},
    "forced_continuation_chains": {},
    "forced_depths": {},
    "forced_query_budget": {},
    "forced_union_matches": {},
    "high_cardinality_terms_verified": {},
    "hostile_proxy_content_verified": {},
    "maximum_continuation_token_bytes": {},
    "miss_behavior_verified": {},
    "mixed_query_classes_complete": {},
    "p50_ms": {},
    "p95_ms": {},
    "p99_ms": {},
    "pages_observed": {},
    "promin_service_continuation_calls": {},
    "promin_service_search_calls": {},
    "query_class_depths": {},
    "query_class_latency_ms": {},
    "query_mix": {},
    "result_digest": {},
    "runtime_ingress": {},
    "runtime_query_budget": {},
    "selected_closure_chains": {},
    "selected_closure_union_completeness": {},
    "silent_truncations": {}
  },
  "required": [
    "actual_runtime_queries",
    "broad_query_refinement_required",
    "content_search_verified",
    "continuation_state",
    "continuation_token_overhead_at_most_10_percent",
    "continuation_union_complete",
    "continuation_union_completeness",
    "continuations_checked",
    "depth_counts",
    "depth_max",
    "depth_min",
    "exact_artifact_search_verified",
    "explicit_truncations",
    "forced_continuation_chains",
    "forced_depths",
    "forced_query_budget",
    "forced_union_matches",
    "high_cardinality_terms_verified",
    "hostile_proxy_content_verified",
    "maximum_continuation_token_bytes",
    "miss_behavior_verified",
    "mixed_query_classes_complete",
    "p50_ms",

```

```

    "p95_ms",
    "p99_ms",
    "pages_observed",
    "promin_service_continuation_calls",
    "promin_service_search_calls",
    "query_class_depths",
    "query_class_latency_ms",
    "query_mix",
    "result_digest",
    "runtime_ingress",

    "runtime_query_budget",
    "selected_closure_chains",
    "selected_closure_union_completeness",
    "silent_truncations"
  ],
  "type": "object"
},
"selected_closure_union_completeness": {
  "maximum": 1.0,
  "minimum": 0.0,
  "type": "number"
},
"silent_truncations": {
  "const": 0
},
"status": {
  "enum": [
    "pass",
    "fail"
  ]
},
"workspace_initialization": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "init_record_count": {
      "const": 5
    },
    "init_records": {
      "const": [
        "activation.json",
        "authority.json",
        "project.json",
        "standards.json",
        "technologies.json"
      ]
    },
    "product_tree_scans": {
      "const": 0
    },
    "project_id": {
      "const": "promin-physical-saturation"
    },
    "record_type": {
      "const": "SaturationWorkspaceInitialization"
    },
    "snapshot_provider_id": {
      "const": "git-filesystem-inventory"
    },
    "snapshot_provider_version": {
      "minLength": 1,
      "type": "string"
    },
    "status": {
      "enum": [
        "created",
        "reused"
      ]
    }
  },
  "required": [
    "activation_digest",

```

```

        "implementation_closure_digest",
        "init_record_count",
        "init_records",
        "product_tree_scans",
        "project_id",
        "record_type",
        "snapshot_provider_id",
        "snapshot_provider_version",
        "status"
    ],
    "type": "object"
}
},
"required": [
    "record_type",
    "status",
    "candidate_binding_digest",

    "artifact_binding",
    "artifact_binding_unchanged",
    "workspace_initialization",
    "runtime_binding",
    "physical_files",
    "core_valid_relations",
    "runtime_queries",
    "silent_truncations",
    "selected_closure_union_completeness",
    "memory_amplification_at_most_32",
    "core_valid_relations_exact_198999",
    "broad_query_refinement_required",
    "high_cardinality_terms_verified",
    "content_search_verified",
    "miss_behavior_verified",
    "hostile_proxy_content_verified",
    "exact_artifact_search_verified",
    "mixed_query_classes_complete",
    "continuation_token_bytes_at_most_256",
    "continuation_state_bytes_at_most_16384",
    "continuation_token_overhead_at_most_10_percent",
    "pass_credit",
    "physical",
    "inventory",
    "projection",
    "query_authorization",
    "search",
    "resources",

    "performance",
    "contract_predicates",
    "raw_artifact_manifest",
    "current_release_regression",
    "claim_scope",
    "acceptance_pass",
    "product_acceptance_pass",
    "public_release_approved",
    "producer",
    "invocation",
    "raw_artifact_manifest_digest",
    "producer_attestation",
    "result_digest"
],
    "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SaturationEvidence | Validator: urn:promin:contracts:1#/\$defs/SaturationEvidence

## 103. ScopeSelector

Type summary: type="object"; additionalProperties=false

Required: kind, value

| Property | Required | Structural rule                                                             |
|----------|----------|-----------------------------------------------------------------------------|
| kind     | yes      | enum=["project", "path", "task", "candidate", "artifact", "finding", "all"] |
| value    | yes      | type="string"; minLength=1; maxLength=4096                                  |

```

{
  "additionalProperties": false,
  "properties": {

```

```

    "kind": {
      "enum": [
        "project",
        "path",
        "task",
        "candidate",
        "artifact",
        "finding",
        "all"
      ]
    },
    "value": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "kind",
    "value"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ScopeSelector | Validator: urn:promin:contracts:1#/\$defs/ScopeSelector

## 104. SelectorShardAggregate

Type summary: type="object"; additionalProperties=false

Required: status, selector\_digest, terminal\_receipt, pass\_credit

| Property         | Required | Structural rule                                                    |
|------------------|----------|--------------------------------------------------------------------|
| pass_credit      | yes      | const=false                                                        |
| selector_digest  | yes      | \$ref="#/\$defs/Sha256"                                            |
| status           | yes      | enum=["PASS", "FAIL", "TIMEOUT", "UNAVAILABLE", "INVALID_HARNESS"] |
| terminal_receipt | yes      | type="boolean"                                                     |

```

{
  "additionalProperties": false,
  "properties": {
    "pass_credit": {
      "const": false
    },
    "selector_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "status": {
      "enum": [
        "PASS",
        "FAIL",
        "TIMEOUT",
        "UNAVAILABLE",
        "INVALID_HARNESS"
      ]
    },
    "terminal_receipt": {
      "type": "boolean"
    }
  },
  "required": [
    "status",
    "selector_digest",
    "terminal_receipt",
    "pass_credit"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SelectorShardAggregate | Validator: urn:promin:contracts:1#/\$defs/SelectorShardAggregate

## 105. SemVer

Type summary: type="string"; pattern="^(0|[1-9][0-9]\*)\\.0|[1-9][0-9]\*\\.0|[1-9][0-9]\*\\.0(?:-((?:0|[1-9][0-9]\*|[0-9]\*[A-Za-z-][0-9A-Za-z-]\*)?(?:\\.(?:0|[1-9][0-9]\*|[0-9]\*[A-Za-z-][0-9A-Za-z-]\*)\*))?(?:\\+([0-9A-Za-z-]+(?:\\.0-9A-Za-z-]+)\*)?)?\$"; maxLength=128



Required: -

```
{
  "maxLength": 128,
  "pattern": "^((0|[1-9][0-9]*)\\.(0|[1-9][0-9]*)\\.(0|[1-9][0-9]*)((?:-((?:0|[1-9][0-9]*)[0-9]*[A-Za-z-][0-9A-Za-z-]*)|(?:\\.(?:0|[1-9][0-9]*)[0-9]*[A-Za-z-][0-9A-Za-z-]*)*))?(?:\\.(0-9A-Za-z-)+(?:\\.[0-9A-Za-z-]+)?)?)?$",
  "type": "string"
}
```

Owner: core/contracts.schema.json#/\$defs/SemVer | Validator: urn:promin:contracts:1#/\$defs/SemVer

## 106. SemanticExport

Type summary: type="object"; additionalProperties=false

Required: record\_type, export\_kind, package\_or\_archive, candidate\_digest, activation\_digest, head\_event\_id, head\_sequence, head\_digest, core\_bundle\_digest, preset\_digest, implementation\_closure\_digest, provider\_binding\_digest, input\_digests, output\_artifact\_id, output\_artifact\_record\_digest, output\_digest, output\_size\_bytes, output\_media\_type, pass\_credit, product\_acceptance\_pass, product\_public\_approval, public\_release\_approved

| Property                      | Required | Structural rule                                          |
|-------------------------------|----------|----------------------------------------------------------|
| activation_digest             | yes      | \$ref="#/\$defs/Digest"                                  |
| candidate_digest              | yes      | \$ref="#/\$defs/Digest"                                  |
| core_bundle_digest            | yes      | \$ref="#/\$defs/Digest"                                  |
| export_kind                   | yes      | const="semantic-state"                                   |
| head_digest                   | yes      | oneOf[2]=#/\$defs/Digest   type:null                     |
| head_event_id                 | yes      | oneOf[2]=#/\$defs/Id   type:null                         |
| head_sequence                 | yes      | type="integer"; minimum=0                                |
| implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| input_digests                 | yes      | type="array"; minItems=0; maxItems=128; uniqueItems=true |
| output_artifact_id            | yes      | \$ref="#/\$defs/Id"                                      |
| output_artifact_record_digest | yes      | \$ref="#/\$defs/Digest"                                  |
| output_digest                 | yes      | \$ref="#/\$defs/Digest"                                  |
| output_media_type             | yes      | const="application/vnd.promin.semantic-state+json"       |
| output_size_bytes             | yes      | type="integer"; minimum=1; maximum=1073741824            |
| package_or_archive            | yes      | const=false                                              |
| pass_credit                   | yes      | const=false                                              |
| preset_digest                 | yes      | \$ref="#/\$defs/Digest"                                  |
| product_acceptance_pass       | yes      | const=false                                              |
| product_public_approval       | yes      | const="not_approved"                                     |
| provider_binding_digest       | yes      | \$ref="#/\$defs/Digest"                                  |
| public_release_approved       | yes      | const=false                                              |
| record_type                   | yes      | const="SemanticExport"                                   |

```
{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "export_kind": {
      "const": "semantic-state"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {

```

```

        "type": "null"
    }
}
},
"head_event_id": {
    "oneOf": [
        {
            "$ref": "#/$defs/Id"
        },
        {
            "type": "null"
        }
    ]
},
"head_sequence": {
    "minimum": 0,
    "type": "integer"
},
"implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
},
"input_digests": {
    "items": {
        "$ref": "#/$defs/Digest"
    },
    "maxItems": 128,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
},
"output_artifact_id": {
    "$ref": "#/$defs/Id"
},
"output_artifact_record_digest": {
    "$ref": "#/$defs/Digest"
},
"output_digest": {
    "$ref": "#/$defs/Digest"
},
"output_media_type": {
    "const": "application/vnd.promin.semantic-state+json"
},
"output_size_bytes": {
    "maximum": 1073741824,
    "minimum": 1,
    "type": "integer"
},
"package_or_archive": {
    "const": false
},
"pass_credit": {
    "const": false
},
"preset_digest": {
    "$ref": "#/$defs/Digest"
},
"product_acceptance_pass": {
    "const": false
},
"product_public_approval": {
    "const": "not_approved"
},
"provider_binding_digest": {
    "$ref": "#/$defs/Digest"
},
"public_release_approved": {
    "const": false
},
"record_type": {
    "const": "SemanticExport"
},
"required": [
    "record_type",
    "export_kind",
    "package_or_archive",
    "candidate_digest",

```

```

    "activation_digest",
    "head_event_id",
    "head_sequence",
    "head_digest",
    "core_bundle_digest",
    "preset_digest",
    "implementation_closure_digest",
    "provider_binding_digest",
    "input_digests",
    "output_artifact_id",
    "output_artifact_record_digest",
    "output_digest",
    "output_size_bytes",
    "output_media_type",

    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "public_release_approved"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "export"
  ],
  "x-promin-persistence": "export-only"
}

```

Owner: core/contracts.schema.json#/\$defs/SemanticExport | Validator: urn:promin:contracts:1#/\$defs/SemanticExport

## 107. SemanticModel

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, document, version, function, persistent\_entities, relations, derived\_projection\_kinds, value\_objects, design\_rules, technology\_capabilities, admission

| Property                 | Required | Structural rule                                                                                                                                                                                                                             |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| admission                | yes      | const=schema(keys:derived_by_default,entity,relation)                                                                                                                                                                                       |
| derived_projection_kinds | yes      | const=[16 items]                                                                                                                                                                                                                            |
| design_rules             | yes      | const=schema(keys:alpha4_standard_authority,artifact_subtypes,candidate_consistency,granularities_are_independent,implementation_closure,inventory_projection,promotion_test,provider_invocation_evidence,semantic_objects_default)         |
| document                 | yes      | const="promin.semantic-model"                                                                                                                                                                                                               |
| function                 | yes      | const="Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and provider capability contracts."                                                                                                            |
| persistent_entities      | yes      | const=[9 items]; maxItems=9                                                                                                                                                                                                                 |
| record_type              | yes      | const="SemanticModel"                                                                                                                                                                                                                       |
| relations                | yes      | const=[6 items]; maxItems=6                                                                                                                                                                                                                 |
| schema_ref               | yes      | const="urn:promin:contracts:1#/\$defs/SemanticModel"                                                                                                                                                                                        |
| technology_capabilities  | yes      | const=[9 items]                                                                                                                                                                                                                             |
| value_objects            | yes      | const=schema(keys:CandidateDelta,Command,Comparator,ContinuationToken,CurrentReleaseStatus,ImplementationClosure,Operation,PlanProposal,Policy,Predicate,ProviderInvocationEvidence,Query,StandardReleaseDecision,StateCheckpoint,WorkCard) |
| version                  | yes      | const="1.0.0-alpha.4"                                                                                                                                                                                                                       |

```

{
  "additionalProperties": false,
  "properties": {
    "admission": {
      "const": {
        "derived_by_default": "Each raw inventory file yields exactly one derived Artifact proxy. Raw-file proxies never create a persistent Task or relation; persistence still requires an independent invariant.",
        "entity": "EntityUtility=QueriesEnabled+RulesEnabled+AutomationEnabled-StorageCost-MaintenanceCost-NavigationNoise > 0",
        "relation": "RelationUtility=QueriesEnabled+RulesEnabled+SchedulingGain-WriteCost-FanoutNoise-Ambiguity > 0"
      }
    },
    "derived_projection_kinds": {
      "const": [
        "InventoryEntry",
        "SearchDocument",

```

```

    "InventoryProjectionRow",
    "ProjectionLimits",
    "RetrievalPage",
    "ReadyFrontier",
    "WorkCardProjection",
    "WorkCard",
    "NextResult",
    "ContinueResult",
    "NavigationView",
    "StateCheckpoint",
    "CurrentReleaseStatus",
    "DoctorResult",
    "OperationMetrics",
    "Report"
  ],
  },
  "design_rules": {
    "const": {
      "alpha4_standard_authority": {
        "canonical_policy_owner": "policy-set.json.alpha4_standard_contracts",
        "derived_outputs": "non-authoritative",
        "project_specific_defaults": "reject",
        "standard_only": true
      },
      "artifact_subtypes": {
        "candidate-delta": "An Artifact with kind diff binds one base Candidate, one new Candidate, one WorkCard, and the complete normalized changed-path set.",
        "evidence": "An evidence Artifact binds exact Candidate, policy, tool, inputs, and every invoked provider adapter; Finding disposition evidence additionally binds the canonical Finding digest."
      },
      "candidate_consistency": {
        "credible": "Only an immutable-vcs-tree provider may produce a credible Candidate in promin v1.",
        "observational": "A best-effort filesystem observation is explicit, non-credible, and cannot receive product or release pass credit.",
        "recipe_binding": "Candidate identity binds the exact candidate_recipe digest, including include, exclude, path identity, collision, and symlink rules."
      },
      "granularities_are_independent": [
        "semantic",
        "public",
        "compilation",
        "runtime-dispatch",
        "documentation"
      ],
      "implementation_closure": {
        "activation_binding": "Activation binds the aggregate digest of every capability-specific implementation closure.",
        "binding_shape": "Each TechnologiesInit binding carries runtime, interpreter, zero or more capability components, exactly one matching adapter, and a recomputed closure_digest.",
        "required_components": {
          "control-runtime": [
            "promin-runtime",
            "python-interpreter"
          ],
          "query-projection": [
            "promin-runtime",
            "python-interpreter",
            "sqlite"
          ],
          "shape-validation": [
            "promin-runtime",
            "python-interpreter",
            "jsonschema"
          ]
        },
        "stale_rule": "A runtime, interpreter, component, adapter, or closure digest change invalidates the Activation and every dependent evidence Artifact, projection, continuation, and current release closure."
      },
      "inventory_projection": {
        "current_inventory_persistence": "derived-rebuild-only",
        "current_inventory_record_type": "CurrentInventory",
        "data_class": "untrusted-source",
        "derived_entity_type": "Artifact",
        "input": "Only a verified InventoryResult or its persisted manifest may enter the public rebuild path.",
        "input_manifest_persistence": "transient-ingress-only",

```

```



```

```

        "bounded-context-local"
    }
},
"document": {
    "const": "promin.semantic-model"
},
"function": {
    "const": "Own the minimal persistent vocabulary, relation grammar, derived-projection boundary, and
provider capability contracts."
},
"persistent_entities": {
    "const": [
        {
            "admission_rule": "independent-scheduling-and-acceptance-boundary",
            "kind": "Task",
            "purpose": "Represent one schedulable unit with one acceptance predicate and bounded mutation
scope."
        },
        {
            "admission_rule": "independent-candidate-identity-and-consistency-boundary",
            "kind": "Candidate",
            "purpose": "Identify either one provider-proven immutable product snapshot or one explicitly
non-creditable observation, independently from control state and bound to the exact candidate recipe."
        },
        {
            "admission_rule": "independent-content-identity-provenance-and-retention-boundary",
            "kind": "Artifact",
            "purpose": "Identify immutable product, evidence, log, report, or diff bytes with provenance and
retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the
canonical Candidate-delta change manifest without creating another persistent entity."
        },
        {
            "admission_rule": "independent-execution-attempt-and-evidence-boundary",
            "kind": "Run",
            "purpose": "Record one execution or validation attempt against exact candidate, policy, tool,
and input digests."
        },
        {
            "admission_rule": "independent-defect-lifecycle-and-blocking-boundary",
            "kind": "Finding",
            "purpose": "Record an observed defect or policy violation with explicit effective-blocking
semantics."
        },
        {
            "admission_rule": "independent-human-judgement-and-target-binding-boundary",
            "kind": "Decision",
            "purpose": "Record a project-operational acceptance, rejection, resolution, waiver, promotion,
or release judgement. It is distinct from the external StandardReleaseDecision for the standard
distribution."
        },
        {
            "admission_rule": "immutable-task-owned-gate-definition-plus-current-run-and-finalized-artifact-r
ecord-bindings",
            "kind": "GateResult",
            "purpose": "Record one gate outcome against the exact immutable Task-owned GateRunDefinition,
current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved
evidence never receives pass credit."
        },
        {
            "admission_rule": "independent-capability-claim-and-revocation-boundary",
            "kind": "Grant",
            "purpose": "Prove one subject capability within one activation, time interval, nonce, and scope."
        },
        {
            "admission_rule": "independent-mutation-ownership-and-fencing-boundary",
            "kind": "Lease",
            "purpose": "Coordinate temporary mutation ownership through generation, fencing, expiry,
heartbeat, and close acknowledgement."
        }
    ],
    "maxItems": 9
},
"record_type": {
    "const": "SemanticModel"
},
"relations": {
    "const": [

```

```

{
  "kind": "DEPENDS_ON",
  "rules": [
    "acyclic-for-ready-frontier",
    "accepted-current-outcome-required"
  ],
  "source": [
    "Task"
  ],
  "target": [
    "Task"
  ]
},
{
  "kind": "READS",
  "rules": [
    "read-does-not-imply-authority"
  ],
  "source": [
    "Task",
    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact"
  ]
},
{
  "kind": "PRODUCES",
  "rules": [
    "content-or-record-identity-required"
  ],
  "source": [
    "Task",
    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact",
    "Finding",
    "Decision"
  ]
},
{
  "kind": "VALIDATES",
  "rules": [
    "candidate-policy-tool-input-binding-required"
  ],
  "source": [
    "Run"
  ],
  "target": [
    "Candidate",
    "Artifact",
    "Task"
  ]
},
{
  "kind": "CITES",
  "rules": [
    "selector-and-content-resolution-required"
  ],
  "source": [
    "Finding",
    "Decision"
  ],
  "target": [
    "Artifact",
    "Run",
    "Finding"
  ]
},
{
  "kind": "BLOCKS",
  "rules": [
    "effective-only-while-open-and-blocking"
  ],

```

```

    "source": [
      "Finding"
    ],
    "target": [
      "Task",
      "Candidate",
      "Decision"
    ]
  }
],
"maxItems": 6
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/SemanticModel"
},
"technology_capabilities": {
  "const": [

```

```

{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "continue": {
        "request": "validated-continuation-token",
        "response": "WorkCardProjection"
      },
      "doctor": {
        "request": "installed-activation",
        "response": "DoctorResult"
      },
      "healthcheck": {

```

```

        "request": "configured-healthcheck",
        "response": "bounded-process-observation"
      },
      "init": {
        "request": "explicit-init-plans",
        "response": "InitResult"
      },
      "next": {
        "request": "authorized-bounded-query",
        "response": "WorkCardProjection"
      },
      "provider_binding": {
        "request": "configured-provider-binding",
        "response": "content-addressed-receipt"
      },
      "status": {
        "request": "installed-activation",
        "response": "StatusResult"
      },
      "validate": {
        "request": "installed-activation",
        "response": "ValidationResult"
      }
    },
    "protocol_id": "promin.control-runtime.v1",
    "receipt_persistence": "content-addressed-receipt",
    "supported_bindings": [

```

```

{
  "adapter_id": "python-runtime-v1",
  "identity_kind": "file-digest",
  "invocation_kind": "python-runtime"
}

```



```

    ],
    "supported_commands": [
      "continue",
      "doctor",
      "healthcheck",
      "init",
      "next",
      "provider-binding",
      "status",
      "validate"
    ],
    "unknown_binding": "reject"
  },
  "forbidden_boundary": {
    "effect": "reject",
    "forbids": [
      "authority-invention",
      "policy-invention",
      "provider-binding-invention"
    ],
    "normative_authority": false
  },
  "id": "control-runtime",
  "task": "Parse explicit configuration, canonicalize JSON, and run local CLI orchestration."
}

```

```

{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "command": {
        "request": "strict-json-record",
        "response": "Draft-2020-12-plus-semantic-result"
      },
      "continue": {
        "request": "strict-json-record",
        "response": "Draft-2020-12-plus-semantic-result"
      }
    }
  },
}

```

```

    "export": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    },
    "healthcheck": {
      "request": "configured-healthcheck",
      "response": "bounded-process-observation"
    },
    "import": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    },
    "provider-binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    },
    "rebuild": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    },
    "replay": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    }
  },
  "protocol_id": "promin.shape-validation.v1",
}

```

```

"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
  {
    "adapter_id": "python-runtime-v1",
    "identity_kind": "file-digest",
    "invocation_kind": "python-runtime"
  }
],
"supported_commands": [
  "command",
  "continue",
  "export",
  "healthcheck",
  "import",
  "provider-binding",
  "rebuild",
  "replay"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "authority-decision",
    "cross-record-semantic-decision"
  ],
  "normative_authority": false
},
"id": "shape-validation",
"task": "Validate the single Draft 2020-12 contract bundle with enforced formats and local
references."

```

```

},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "digest-bytes": {
        "request": "bounded-bytes",
        "response": "lowercase-sha256"
      },
      "digest-file": {
        "request": "no-follow-regular-file",
        "response": "lowercase-sha256"
      }
    },

```

```

    "digest-value": {
      "request": "canonical-json-value",
      "response": "lowercase-sha256"
    },
    "healthcheck": {
      "request": "configured-healthcheck",
      "response": "bounded-process-observation"
    },
    "provider-binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    }
  },
  "protocol_id": "promin.content-identity.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "python-runtime-v1",

```

```

        "identity_kind": "file-digest",
        "invocation_kind": "python-runtime"
    }
],
"supported_commands": [
    "digest-bytes",
    "digest-file",
    "digest-value",
    "healthcheck",
    "provider-binding"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "digest-as-action-authority"
    ],
    "normative_authority": false
},
"id": "content-identity",
"task": "Compute content digests and content-addressed object identities."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "atomic-replace": {
                "request": "same-directory-staged-bytes",
                "response": "durable-replacement"
            },
            "directory-sync": {
                "request": "owning-directory",
                "response": "durability-observation"
            },
            "file-sync": {
                "request": "open-regular-file",
                "response": "durability-observation"
            },
            "healthcheck": {
                "request": "configured-healthcheck",
                "response": "bounded-process-observation"
            },
            "local-writer-lock": {
                "request": "activation-bound-lock",
                "response": "exclusive-lease"
            },
            "provider-binding": {
                "request": "configured-provider-binding",
                "response": "content-addressed-receipt"
            }
        },
        "recover": {
            "request": "durable-journal-state",
            "response": "verified-head"
        }
    },
    "protocol_id": "promin.local-serialization.v1",
    "receipt_persistence": "content-addressed-receipt",
    "supported_bindings": [
        {
            "adapter_id": "python-runtime-v1",
            "identity_kind": "file-digest",

```

```

        "invocation_kind": "python-runtime"
    }
],
"supported_commands": [
    "atomic-replace",
    "directory-sync",
    "file-sync",
    "healthcheck",
    "local-writer-lock",
    "provider-binding",
    "recover"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "distributed-consensus-claim"
    ],
    "normative_authority": false
},
"id": "local-serialization",
"task": "Provide one local-writer lock, atomic replace, file sync, directory sync, and recovery
primitives."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "continue": {
                "request": "authenticated-ready-frontier-continuation",
                "response": "ContinueResult"
            },
            "healthcheck": {
                "request": "configured-healthcheck",
                "response": "bounded-process-observation"
            },
            "next": {
                "request": "authorized-bounded-ready-query",
                "response": "NextResult"
            },
            "provider-binding": {
                "request": "configured-provider-binding",
                "response": "content-addressed-receipt"
            },
            "rebuild": {
                "request": "verified-event-and-inventory-input",
                "response": "current-projection-state"
            },
            "status": {
                "request": "projection-file",
                "response": "current-projection-state"
            }
        },
        "protocol_id": "promin.query-projection.v1",
        "receipt_persistence": "content-addressed-receipt",
        "supported_bindings": [
            {
                "adapter_id": "python-runtime-v1",
                "identity_kind": "file-digest",
                "invocation_kind": "python-runtime"
            }
        ]
    }
}

```

```

    ],
    "supported_commands": [
        "continue",
        "healthcheck",
        "next",
        "provider-binding",
        "rebuild",
        "status"
    ],
    "unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "projection-as-normative-authority"
    ],
    "normative_authority": false
},
"id": "query-projection",
"task": "Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier,
materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for
the six supported CLI commands."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "healthcheck": {
                "request": "configured-bounded-git-healthcheck",
                "response": "exit-zero-captured-stdout-stderr"
            },
            "immutable-tree-object": {
                "request": "argv:rev-parse --verify HEAD^{tree};cwd:configured-root;stdin:none",
                "response": "exit-zero;stdout:one-lowercase-hex-object-id-plus-LF;stderr:captured"
            },
            "immutable-tree-stream": {
                "request": "argv:archive --format=tar <verified-tree-object>;cwd:configured-root;stdin:none",
                "response": "exit-zero;stdout:raw-git-tar-stream;stderr:captured"
            },
            "provider-binding": {
                "request": "configured-executable-file-digest",
                "response": "content-addressed-receipt"
            }
        },
        "protocol_id": "promin.filesystem-inventory.v1",
        "receipt_persistence": "content-addressed-receipt",
        "supported_bindings": [
            {
                "adapter_id": "git-executable-v1",
                "identity_kind": "file-digest",
                "invocation_kind": "executable"
            }
        ],
        "supported_commands": [
            "healthcheck",
            "immutable-tree-object",
            "immutable-tree-stream",
            "provider-binding"
        ],
        "unknown_binding": "reject"
    },
    "forbidden_boundary": {

```

```

    "effect": "reject",
    "forbids": [
        "implicit-initialization-scan"
    ],
    "normative_authority": false
},

```

```

    "id": "filesystem-inventory",
    "task": "Apply the exact candidate recipe in one product pass, use no-follow reads, and publish
either a provider-proven immutable VCS tree stream or an explicit non-creditable observation."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "bounded-export-scan": {
                "request": "canonical-json-stdin:{operation,root,allowlist_digest,max_files,max_bytes}",
                "response": "exit-zero;canonical-json-stdout:{status,files_scanned,bytes_scanned,violations,
output_digest};stderr:captured"
            },
            "healthcheck": {
                "request": "configured-bounded-healthcheck",

```

```

                "response": "exit-zero-captured-stdout-stderr"
            },
            "provider-binding": {
                "request": "configured-executable-file-digest",
                "response": "content-addressed-receipt"
            }
        },
        "protocol_id": "promin.export-scan.v1",
        "receipt_persistence": "content-addressed-receipt",
        "supported_bindings": [
            {
                "adapter_id": "executable-export-scan-v1",
                "identity_kind": "file-digest",
                "invocation_kind": "executable"
            }
        ],
        "supported_commands": [
            "bounded-export-scan",
            "healthcheck",
            "provider-binding"
        ],
        "unknown_binding": "reject"
    },
    "forbidden_boundary": {
        "effect": "reject",
        "forbids": [
            "package-as-authorization-boundary"
        ],

```

```

        "normative_authority": false
    },
    "id": "export-scan",
    "task": "Perform bounded recursive export inspection, allowlisting, secret classification, and
symlink rejection."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",

```

```

        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
    ],
    "component_order": "component-id-ascending",
    "os_substrate_excluded": true,
    "provider_owned_scope": "configured-provider-root-and-runtime-components",
    "unresolved_dependencies": "reject"
},
"operation_contracts": {
    "healthcheck": {
        "request": "configured-bounded-signature-healthcheck",
        "response": "exit-zero-captured-stdout-stderr"
    },
    "provider-binding": {
        "request": "configured-module-or-executable-digest",
        "response": "content-addressed-receipt"
    },
    "sign-digest": {
        "request": "domain-separated-canonical-claim-digest",
        "response": "configured-adapter-signature"
    },
    "verify-digest": {
        "request": "domain-separated-canonical-claim-digest-plus-signature",
        "response": "strict-boolean-verification"
    }
},
"protocol_id": "promin.signature.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
    {
        "adapter_id": "executable-signature-v1",
        "identity_kind": "file-digest",
        "invocation_kind": "executable"
    },
    {
        "adapter_id": "python-module-signature-v1",
        "identity_kind": "module-file-digest",
        "invocation_kind": "python-module"
    }
],
"supported_commands": [
    "healthcheck",
    "provider-binding",
    "sign-digest",
    "verify-digest"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "core-signer-implementation-lock-in"
    ],
    "normative_authority": false
},
"id": "signature",
"task": "Dispatch the exact configured signature adapter to sign or verify activation, Grant, and Decision digests for team trust mode, and evidence the invoked adapter identity."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,

```

```

    "provider_owned_scope": "configured-provider-root-and-runtime-components",
    "unresolved_dependencies": "reject"
  },
  "operation_contracts": {
    "healthcheck": {
      "request": "configured-bounded-healthcheck",
      "response": "exit-zero-captured-stdout-stderr"
    },
    "import-build-facts": {
      "request": "canonical-json-stdin:{operation,project_root,input_digests,max_files,max_bytes}"
    },
    "response": "exit-zero;canonical-json-stdout:{status,facts,input_digests,output_digest};std
err:captured"
  },
  "provider-binding": {
    "request": "configured-executable-file-digest",
    "response": "content-addressed-receipt"
  }
},
"protocol_id": "promin.build-dependency.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
  {
    "adapter_id": "executable-build-dependency-v1",
    "identity_kind": "file-digest",
    "invocation_kind": "executable"
  }
],
"supported_commands": [
  "healthcheck",
  "import-build-facts",
  "provider-binding"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "semantic-model-redefinition"
  ],
  "normative_authority": false
},
"id": "build-dependency",
"task": "Import project-specific build and dependency facts as derived projections."
}
]
},
"value_objects": {
  "const": {
    "CandidateDelta": "A diff Artifact payload binding base and new Candidate digests, WorkCard
digest, and the complete normalized changed-path set.",
    "Command": "A state-changing request validated before commit.",
    "Comparator": "A pure ordering or equality rule.",
    "ContinuationToken": "An authenticated bounded runtime value signed with the random per-Activation
secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized
query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure,
issue time, and expiry; it never grants authority by itself.",
    "CurrentReleaseStatus": "A derived evaluation of an immutable historical release Decision against
the current closure, blockers, evidence, providers, and Activation.",
    "ImplementationClosure": "A canonical capability-specific binding of the promin runtime, Python
interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate
closure digest.",
    "Operation": "One completed semantic step with explicit input, output, failure, and evidence
semantics.",
    "PlanProposal": "A non-authoritative bounded proposal that derives candidate Tasks and relations
from project intent, repository facts, backlog, and references; it cannot mutate operational state or
grant authority.",
    "Policy": "A machine-addressable invariant owned by policy-set.json.",
    "Predicate": "A pure decision rule.",
    "ProviderInvocationEvidence": "The exact capability, provider, invocation kind, identity kind and
digest, and adapter ID actually used by one operation.",
    "Query": "A read-only request that cannot mutate authoritative state.",
    "StandardReleaseDecision": "An external human-reviewable distribution record bound to the exact
promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and
evidence manifest. It never grants product acceptance.",
    "StateCheckpoint": "A disposable self-digested state or journal index bound to an exact
authoritative HEAD; it cannot grant authority or acceptance.",
    "WorkCard": "A bounded generated context package for one Task that separately binds the

```



```

projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance."
    }
  },
  "version": {
    "const": "1.0.0-alpha.4"
  }
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "persistent_entities",
  "relations",
  "derived_projection_kinds",
  "value_objects",
  "design_rules",
  "technology_capabilities",
  "admission"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SemanticModel | Validator: urn:promin:contracts:1#/\$defs/SemanticModel

## 108. SemanticStatePayload

Type summary: type="object"; additionalProperties=false

Required: record\_type, candidate\_digest, activation\_digest, head\_event\_id, head\_sequence, head\_digest, records

| Property          | Required | Structural rule                           |
|-------------------|----------|-------------------------------------------|
| activation_digest | yes      | \$ref="#/\$defs/Digest"                   |
| candidate_digest  | yes      | \$ref="#/\$defs/Digest"                   |
| head_digest       | yes      | oneOf[2]=#/\$defs/Digest   type:null      |
| head_event_id     | yes      | oneOf[2]=#/\$defs/Id   type:null          |
| head_sequence     | yes      | type="integer"; minimum=0                 |
| record_type       | yes      | const="SemanticStatePayload"              |
| records           | yes      | type="array"; minItems=0; maxItems=400000 |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_event_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_sequence": {
      "minimum": 0,
      "type": "integer"
    },
    "records": {
      "type": "array",
      "minItems": 0,
      "maxItems": 400000
    }
  }
}

```

```

"record_type": {
  "const": "SemanticStatePayload"
},
"records": {
  "items": {
    "oneOf": [
      {
        "$ref": "#/$defs/Activation"
      },
      {
        "$ref": "#/$defs/Task"
      },
      {
        "$ref": "#/$defs/Candidate"
      },
      {
        "$ref": "#/$defs/Artifact"
      },
      {
        "$ref": "#/$defs/Run"
      },
      {
        "$ref": "#/$defs/Finding"
      },
      {
        "$ref": "#/$defs/Decision"
      },
      {
        "$ref": "#/$defs/Grant"
      },
      {
        "$ref": "#/$defs/Lease"
      },
      {
        "$ref": "#/$defs/GateResult"
      },
      {
        "$ref": "#/$defs/GrantRevocation"
      },
      {
        "$ref": "#/$defs/Relation"
      }
    ]
  },
  "maxItems": 400000,
  "minItems": 0,
  "type": "array"
}
},
"required": [
  "record_type",
  "candidate_digest",
  "activation_digest",
  "head_event_id",
  "head_sequence",
  "head_digest",
  "records"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SemanticStatePayload | Validator: urn:promin:contracts:1#/\$defs/SemanticStatePayload

## 109. SignatureActivationProof

Type summary: type="object"; additionalProperties=false

Required: kind, key\_id, algorithm, signed\_digest, signature

| Property      | Required | Structural rule            |
|---------------|----------|----------------------------|
| algorithm     | yes      | type="string"; minLength=1 |
| key_id        | yes      | \$ref="#/\$defs/Id"        |
| kind          | yes      | const="signature"          |
| signature     | yes      | type="string"; minLength=1 |
| signed_digest | yes      | \$ref="#/\$defs/Digest"    |

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/SignatureActivationProof | Validator: urn:promin:contracts:1#/\$defs/SignatureActivationProof

## 110. SignatureRootCommandProof

Type summary: type="object"; additionalProperties=false

Required: kind, key\_id, algorithm, signed\_intent\_digest, signature

| Property             | Required | Structural rule            |
|----------------------|----------|----------------------------|
| algorithm            | yes      | type="string"; minLength=1 |
| key_id               | yes      | \$ref="#/\$defs/Id"        |
| kind                 | yes      | const="signature-command"  |
| signature            | yes      | type="string"; minLength=1 |
| signed_intent_digest | yes      | \$ref="#/\$defs/Digest"    |

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature-command"
    },
    "signature": {
      "minLength": 1,
      "type": "string"
    },
    "signed_intent_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_intent_digest",
    "signature"
  ],
}

```

```
"type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/SignatureRootCommandProof | Validator: urn:promin:contracts:1#/\$defs/SignatureRootCommandProof

## 111. SignatureTrustProof

Type summary: type="object"; additionalProperties=false

Required: kind, key\_id, algorithm, signed\_digest, signature

| Property      | Required | Structural rule                            |
|---------------|----------|--------------------------------------------|
| algorithm     | yes      | type="string"; minLength=1; maxLength=128  |
| key_id        | yes      | \$ref="#/\$defs/Id"                        |
| kind          | yes      | const="signature"                          |
| signature     | yes      | type="string"; minLength=1; maxLength=8192 |
| signed_digest | yes      | \$ref="#/\$defs/Digest"                    |

```
{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "maxLength": 8192,
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
  "type": "object"
}
```

Owner: core/contracts.schema.json#/\$defs/SignatureTrustProof | Validator: urn:promin:contracts:1#/\$defs/SignatureTrustProof

## 112. StandardBinding

Type summary: type="object"; additionalProperties=false

Required: standard\_id, version, source\_uri, source\_digest, purpose

| Property      | Required | Structural rule            |
|---------------|----------|----------------------------|
| purpose       | yes      | type="string"; minLength=1 |
| source_digest | yes      | \$ref="#/\$defs/Digest"    |
| source_uri    | yes      | type="string"; minLength=1 |
| standard_id   | yes      | \$ref="#/\$defs/Id"        |
| version       | yes      | type="string"; minLength=1 |

```
{
  "additionalProperties": false,
  "properties": {
    "purpose": {
      "minLength": 1,
      "type": "string"
    },

```

```

    "source_digest": {
      "$ref": "#/$defs/Digest"
    },
    "source_uri": {
      "minLength": 1,
      "type": "string"
    },
    "standard_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "standard_id",
    "version",
    "source_uri",
    "source_digest",
    "purpose"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardBinding | Validator: urn:promin:contracts:1#/\$defs/StandardBinding

## 113. StandardManifest

Type summary: type="object"; additionalProperties=false

Required: record\_type, schema\_ref, canonical\_name, aliases, canonical\_name\_only, version, function, core\_components, bundle\_digest, core\_artifact\_count, schema\_definition\_count, selected\_preset\_count, persistent\_entity\_kind\_count, relation\_kind\_count, technology\_capability\_count, action\_capability\_count, policy\_count, acceptance\_predicate\_count, mutation\_family\_count

| Property                     | Required | Structural rule                                         |
|------------------------------|----------|---------------------------------------------------------|
| acceptance_predicate_count   | yes      | type="integer"; const=102                               |
| action_capability_count      | yes      | type="integer"; const=16                                |
| aliases                      | yes      | type="array"; maxItems=0                                |
| bundle_digest                | yes      | \$ref="#/\$defs/Digest"                                 |
| canonical_name               | yes      | const="promin"                                          |
| canonical_name_only          | yes      | const=true                                              |
| core_artifact_count          | yes      | type="integer"; const=6                                 |
| core_components              | yes      | type="array"; minItems=5; maxItems=5                    |
| function                     | yes      | type="string"; minLength=1                              |
| mutation_family_count        | yes      | type="integer"; const=77                                |
| persistent_entity_kind_count | yes      | type="integer"; const=9                                 |
| policy_count                 | yes      | type="integer"; const=68                                |
| record_type                  | yes      | const="StandardManifest"                                |
| relation_kind_count          | yes      | type="integer"; const=6                                 |
| schema_definition_count      | yes      | type="integer"; const=141                               |
| schema_ref                   | yes      | const="urn:promin:contracts:1#/\$defs/StandardManifest" |
| selected_preset_count        | yes      | type="integer"; const=1                                 |
| technology_capability_count  | yes      | type="integer"; const=9                                 |
| version                      | yes      | const="1.0.0-alpha.4"                                   |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_predicate_count": {
      "const": 102,
      "type": "integer"
    },
    "action_capability_count": {
      "const": 16,
      "type": "integer"
    }
  }
}

```

```

},
"aliases": {
  "maxItems": 0,
  "type": "array"
},
"bundle_digest": {
  "$ref": "#/$defs/Digest"
},
"canonical_name": {
  "const": "promin"
},
"canonical_name_only": {
  "const": true
},
"core_artifact_count": {
  "const": 6,
  "type": "integer"
},

```

```

"core_components": {
  "items": {
    "$ref": "#/$defs/CoreComponent"
  },
  "maxItems": 5,
  "minItems": 5,
  "type": "array"
},
"function": {
  "minLength": 1,
  "type": "string"
},
"mutation_family_count": {
  "const": 77,
  "type": "integer"
},
"persistent_entity_kind_count": {
  "const": 9,
  "type": "integer"
},
"policy_count": {
  "const": 68,
  "type": "integer"
},
"record_type": {
  "const": "StandardManifest"
},
"relation_kind_count": {

```

```

  "const": 6,
  "type": "integer"
},
"schema_definition_count": {
  "const": 141,
  "type": "integer"
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/StandardManifest"
},
"selected_preset_count": {
  "const": 1,
  "type": "integer"
},
"technology_capability_count": {
  "const": 9,
  "type": "integer"
},
"version": {
  "const": "1.0.0-alpha.4"
}
},
"required": [
  "record_type",
  "schema_ref",
  "canonical_name",
  "aliases",
  "canonical_name_only",

```

```

"version",
"function",
"core_components",

```

```

    "bundle_digest",
    "core_artifact_count",
    "schema_definition_count",
    "selected_preset_count",
    "persistent_entity_kind_count",
    "relation_kind_count",
    "technology_capability_count",
    "action_capability_count",
    "policy_count",
    "acceptance_predicate_count",
    "mutation_family_count"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardManifest | Validator: urn:promin:contracts:1#/\$defs/StandardManifest

## 114. StandardReleaseCandidateBinding

Type summary: type="object"; additionalProperties=false

Required: record\_type, standard\_name, version, archive\_sha256, archive\_bytes, archive\_member\_manifest\_digest, package\_manifest\_digest, checksums\_digest, core\_bundle\_digest, preset\_digest, package\_tool\_digest, validator\_digest, test\_manifest\_digest, portable\_implementation\_closure\_digest, evidence\_tool\_digests, candidate\_binding\_digest

| Property                               | Required | Structural rule                           |
|----------------------------------------|----------|-------------------------------------------|
| archive_bytes                          | yes      | type="integer"; minimum=1                 |
| archive_member_manifest_digest         | yes      | \$ref="#/\$defs/Digest"                   |
| archive_sha256                         | yes      | \$ref="#/\$defs/Digest"                   |
| candidate_binding_digest               | yes      | \$ref="#/\$defs/Digest"                   |
| checksums_digest                       | yes      | \$ref="#/\$defs/Digest"                   |
| core_bundle_digest                     | yes      | \$ref="#/\$defs/Digest"                   |
| evidence_tool_digests                  | yes      | type="object"; additionalProperties=false |
| package_manifest_digest                | yes      | \$ref="#/\$defs/Digest"                   |
| package_tool_digest                    | yes      | \$ref="#/\$defs/Digest"                   |
| portable_implementation_closure_digest | yes      | \$ref="#/\$defs/Digest"                   |
| preset_digest                          | yes      | \$ref="#/\$defs/Digest"                   |
| record_type                            | yes      | const="StandardReleaseCandidateBinding"   |
| standard_name                          | yes      | const="promin"                            |
| test_manifest_digest                   | yes      | \$ref="#/\$defs/Digest"                   |
| validator_digest                       | yes      | \$ref="#/\$defs/Digest"                   |
| version                                | yes      | \$ref="#/\$defs/SemVer"                   |

```

{
  "additionalProperties": false,
  "properties": {
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_member_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "checksums_digest": {
      "$ref": "#/$defs/Digest"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_tool_digests": {
      "additionalProperties": false,
      "properties": {

```

```

    "tools/generate_human.py": {
      "$ref": "#/$defs/Digest"
    },
    "tools/promin_no_degradation.py": {
      "$ref": "#/$defs/Digest"
    },
    "tools/promin_package.py": {
      "$ref": "#/$defs/Digest"
    },
    "tools/promin_saturation.py": {
      "$ref": "#/$defs/Digest"
    },
    "tools/promin_saturation_audit.py": {
      "$ref": "#/$defs/Digest"
    },
    "tools/promin_validate.py": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "tools/generate_human.py",
    "tools/promin_no_degradation.py",
    "tools/promin_package.py",
    "tools/promin_saturation.py",
    "tools/promin_saturation_audit.py",
    "tools/promin_validate.py"
  ],
  "type": "object"
},
"package_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"package_tool_digest": {
  "$ref": "#/$defs/Digest"
},
"portable_implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"preset_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "StandardReleaseCandidateBinding"
},
"standard_name": {
  "const": "promin"
},
"test_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"validator_digest": {
  "$ref": "#/$defs/Digest"
},
"version": {
  "$ref": "#/$defs/SemVer"
}
},
"required": [
  "record_type",
  "standard_name",
  "version",
  "archive_sha256",
  "archive_bytes",
  "archive_member_manifest_digest",
  "package_manifest_digest",
  "checksums_digest",
  "core_bundle_digest",
  "preset_digest",
  "package_tool_digest",
  "validator_digest",
  "test_manifest_digest",
  "portable_implementation_closure_digest",
  "evidence_tool_digests",
  "candidate_binding_digest"
],
"type": "object"
}

```



## 115. StandardReleaseDecision

Type summary: type="object"; additionalProperties=false

Required: record\_type, decision\_id, standard\_name, version, candidate\_binding\_digest, evidence\_manifest\_digest, outcome, decider\_id, release\_capability, trust\_root\_id, signature\_provider\_id, key\_id, nonce, decided\_at, signed\_claim\_digest, signature

| Property                 | Required | Structural rule                            |
|--------------------------|----------|--------------------------------------------|
| candidate_binding_digest | yes      | \$ref="#/\$defs/Digest"                    |
| decided_at               | yes      | \$ref="#/\$defs/Timestamp"                 |
| decider_id               | yes      | \$ref="#/\$defs/Id"                        |
| decision_id              | yes      | \$ref="#/\$defs/Id"                        |
| evidence_manifest_digest | yes      | \$ref="#/\$defs/Digest"                    |
| key_id                   | yes      | \$ref="#/\$defs/Id"                        |
| nonce                    | yes      | type="string"; minLength=24; maxLength=128 |
| outcome                  | yes      | enum=["approve", "reject"]                 |
| record_type              | yes      | const="StandardReleaseDecision"            |
| release_capability       | yes      | const="standard.distribute"                |
| signature                | yes      | type="string"; minLength=80; maxLength=128 |
| signature_provider_id    | yes      | const="cryptography-ed25519-v1"            |
| signed_claim_digest      | yes      | \$ref="#/\$defs/Digest"                    |
| standard_name            | yes      | const="promin"                             |
| trust_root_id            | yes      | \$ref="#/\$defs/Id"                        |
| version                  | yes      | \$ref="#/\$defs/SemVer"                    |

```
{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "decided_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "decider_id": {
      "$ref": "#/$defs/Id"
    },
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "nonce": {
      "maxLength": 128,
      "minLength": 24,
      "type": "string"
    },
    "outcome": {
      "enum": [
        "approve",
        "reject"
      ]
    },
    "record_type": {
      "const": "StandardReleaseDecision"
    },
    "release_capability": {
      "const": "standard.distribute"
    },
    "signature": {
      "maxLength": 128,
```

```

    "minLength": 80,
    "type": "string"
  },
  "signature_provider_id": {
    "const": "cryptography-ed25519-v1"
  },
  "signed_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "standard_name": {
    "const": "promin"
  },
  "trust_root_id": {
    "$ref": "#/$defs/Id"
  },
  "version": {
    "$ref": "#/$defs/SemVer"
  }
},
"required": [
  "record_type",
  "decision_id",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "evidence_manifest_digest",
  "outcome",
  "decider_id",
  "release_capability",
  "trust_root_id",
  "signature_provider_id",
  "key_id",
  "nonce",
  "decided_at",
  "signed_claim_digest",
  "signature"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseDecision | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseDecision

## 116. StandardReleaseEvidenceEntry

Type summary: type="object"; additionalProperties=false

Required: evidence\_id, evidence\_role, path, sha256, size\_bytes, record\_type, status, candidate\_binding\_digest, predicates

| Property                 | Required | Structural rule                                                                                                                      |
|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| candidate_binding_digest | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| evidence_id              | yes      | \$ref="#/\$defs/Id"                                                                                                                  |
| evidence_role            | yes      | enum=["linux", "windows", "physical-scale", "saturation-audit", "human-documents", "linux-no-degradation", "windows-no-degradation"] |
| path                     | yes      | \$ref="#/\$defs/RelativePath"                                                                                                        |
| predicates               | yes      | type="array"; minItems=1; maxItems=64                                                                                                |
| record_type              | yes      | \$ref="#/\$defs/Id"                                                                                                                  |
| sha256                   | yes      | \$ref="#/\$defs/Digest"                                                                                                              |
| size_bytes               | yes      | type="integer"; minimum=1                                                                                                            |
| status                   | yes      | const="pass"                                                                                                                         |

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_role": {
      "enum": [
        "linux",

```

```

        "windows",
        "physical-scale",
        "saturation-audit",
        "human-documents",
        "linux-no-degradation",
        "windows-no-degradation"
    ]
},
"path": {
    "$ref": "#/$defs/RelativePath"
},
"predicates": {
    "items": {
        "$ref": "#/$defs/StandardReleaseEvidencePredicate"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
},
"record_type": {
    "$ref": "#/$defs/Id"
},
"sha256": {
    "$ref": "#/$defs/Digest"
},
"size_bytes": {
    "minimum": 1,
    "type": "integer"
},
"status": {
    "const": "pass"
}
},
"required": [
    "evidence_id",
    "evidence_role",
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "status",
    "candidate_binding_digest",
    "predicates"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseEvidenceEntry | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseEvidenceEntry

## 117. StandardReleaseEvidenceManifest

Type summary: type="object"; additionalProperties=false

Required: record\_type, standard\_name, version, candidate\_binding\_digest, max\_evidence\_completed\_at, entries, matrix\_aggregate, supplemental\_lanes, evidence\_manifest\_digest

| Property                  | Required | Structural rule                               |
|---------------------------|----------|-----------------------------------------------|
| candidate_binding_digest  | yes      | \$ref="#/\$defs/Digest"                       |
| entries                   | yes      | type="array"; minItems=7; maxItems=7          |
| evidence_manifest_digest  | yes      | \$ref="#/\$defs/Digest"                       |
| matrix_aggregate          | yes      | \$ref="#/\$defs/StandardReleaseMatrixBinding" |
| max_evidence_completed_at | yes      | \$ref="#/\$defs/Timestamp"                    |
| record_type               | yes      | const="StandardReleaseEvidenceManifest"       |
| standard_name             | yes      | const="promin"                                |
| supplemental_lanes        | yes      | type="array"; minItems=4; maxItems=4          |
| version                   | yes      | \$ref="#/\$defs/SemVer"                       |

```

{
    "additionalProperties": false,
    "properties": {
        "candidate_binding_digest": {
            "$ref": "#/$defs/Digest"

```

```

    },
    "entries": {
      "items": {
        "$ref": "#/$defs/StandardReleaseEvidenceEntry"
      },
      "maxItems": 7,
      "minItems": 7,
      "type": "array"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "matrix_aggregate": {
      "$ref": "#/$defs/StandardReleaseMatrixBinding"
    },
    "max_evidence_completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "record_type": {
      "const": "StandardReleaseEvidenceManifest"
    },
    "standard_name": {
      "const": "promin"
    }
  },
  "supplemental_lanes": {
    "items": {
      "$ref": "#/$defs/StandardReleaseSupplementalLane"
    },
    "maxItems": 4,
    "minItems": 4,
    "type": "array"
  },
  "version": {
    "$ref": "#/$defs/SemVer"
  }
},
"required": [
  "record_type",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "max_evidence_completed_at",
  "entries",
  "matrix_aggregate",
  "supplemental_lanes",
  "evidence_manifest_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseEvidenceManifest | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseEvidenceManifest

## 118. StandardReleaseEvidencePredicate

Type summary: type="object"; additionalProperties=false

Required: pointer, equals

| Property | Required | Structural rule                             |
|----------|----------|---------------------------------------------|
| equals   | yes      | composed structural definition              |
| pointer  | yes      | type="string"; pattern="^/"; maxLength=1024 |

```

{
  "additionalProperties": false,
  "properties": {
    "equals": {},
    "pointer": {
      "maxLength": 1024,
      "pattern": "^/",
      "type": "string"
    }
  },
  "required": [
    "pointer",
    "equals"
  ],
}

```

```

    "type": "object"
  }

```

Owner: [core/contracts.schema.json#/\\$defs/StandardReleaseEvidencePredicate](#) | Validator: [urn:promin:contracts:1#/\\$defs/StandardReleaseEvidencePredicate](#)

## 119. StandardReleaseMatrixBinding

Type summary: type="object"; additionalProperties=false

Required: path, sha256, size\_bytes, record\_type, matrix\_digest, lane\_count, matrix\_authoritative, pass\_credit, acceptance\_pass, product\_acceptance\_pass, product\_public\_approval

| Property                | Required | Structural rule                                            |
|-------------------------|----------|------------------------------------------------------------|
| acceptance_pass         | yes      | const=false                                                |
| lane_count              | yes      | const=8                                                    |
| matrix_authoritative    | yes      | const=false                                                |
| matrix_digest           | yes      | \$ref="#/\$defs/Digest"                                    |
| pass_credit             | yes      | const=false                                                |
| path                    | yes      | const="matrix-current/platform-no-degradation-matrix.json" |
| product_acceptance_pass | yes      | const=false                                                |
| product_public_approval | yes      | const="not_approved"                                       |
| record_type             | yes      | const="ProminPlatformNoDegradationMatrix"                  |
| sha256                  | yes      | \$ref="#/\$defs/Digest"                                    |
| size_bytes              | yes      | type="integer"; minimum=1                                  |

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "lane_count": {
      "const": 8
    },
    "matrix_authoritative": {
      "const": false
    },
    "matrix_digest": {
      "$ref": "#/$defs/Digest"
    },
    "pass_credit": {
      "const": false
    },
    "path": {
      "const": "matrix-current/platform-no-degradation-matrix.json"
    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "record_type": {
      "const": "ProminPlatformNoDegradationMatrix"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "matrix_digest",
    "lane_count",
    "matrix_authoritative",
    "pass_credit",

```

```

    "acceptance_pass",
    "product_acceptance_pass",
    "product_public_approval"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseMatrixBinding | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseMatrixBinding

## 120. StandardReleaseSupplementalLane

Type summary: type="object"; additionalProperties=false

Required: lane\_id, path, sha256, size\_bytes, record\_type, status, candidate\_binding\_digest, result\_digest, evidence\_role, producer\_attestation\_digest

| Property                    | Required | Structural rule                                                                                     |
|-----------------------------|----------|-----------------------------------------------------------------------------------------------------|
| candidate_binding_digest    | yes      | \$ref="#/\$defs/Digest"                                                                             |
| evidence_role               | yes      | enum=["linux", "linux-no-degradation", "windows", "windows-no-degradation"]                         |
| lane_id                     | yes      | enum=["linux-cp314-offline", "linux-cp314-online", "windows-cp314-offline", "windows-cp314-online"] |
| path                        | yes      | \$ref="#/\$defs/RelativePath"                                                                       |
| producer_attestation_digest | yes      | \$ref="#/\$defs/Digest"                                                                             |
| record_type                 | yes      | enum=["NoDegradationResult", "PlatformVerificationResult"]                                          |
| result_digest               | yes      | \$ref="#/\$defs/Digest"                                                                             |
| sha256                      | yes      | \$ref="#/\$defs/Digest"                                                                             |
| size_bytes                  | yes      | type="integer"; minimum=1                                                                           |
| status                      | yes      | const="pass"                                                                                        |

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "linux-no-degradation",
        "windows",
        "windows-no-degradation"
      ]
    },
    "lane_id": {
      "enum": [
        "linux-cp314-offline",
        "linux-cp314-online",
        "windows-cp314-offline",
        "windows-cp314-online"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "producer_attestation_digest": {
      "$ref": "#/$defs/Digest"
    },

```

```

    "record_type": {
      "enum": [
        "NoDegradationResult",
        "PlatformVerificationResult"
      ]
    },
    "result_digest": {
      "$ref": "#/$defs/Digest"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,

```

```

    "type": "integer"
  },
  "status": {
    "const": "pass"
  }
},
"required": [
  "lane_id",
  "path",
  "sha256",
  "size_bytes",
  "record_type",
  "status",
  "candidate_binding_digest",
  "result_digest",
  "evidence_role",
  "producer_attestation_digest"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StandardReleaseSupplementalLane | Validator: urn:promin:contracts:1#/\$defs/StandardReleaseSupplementalLane

## 121. StandardReleaseTrustConfiguration

Type summary: type="object"; additionalProperties=false; allOf[7]=keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties

Required: record\_type, trust\_root\_id, signature\_provider\_id, algorithm, keys

| Property              | Required | Structural rule                                         |
|-----------------------|----------|---------------------------------------------------------|
| algorithm             | yes      | const="Ed25519"                                         |
| keys                  | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true |
| record_type           | yes      | const="StandardReleaseTrustConfiguration"               |
| signature_provider_id | yes      | const="cryptography-ed25519-v1"                         |
| trust_root_id         | yes      | \$ref="#/\$defs/Id"                                     |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "properties": {
        "keys": {
          "contains": {
            "properties": {
              "evidence_roles": {
                "contains": {
                  "const": "human-documents"
                }
              }
            }
          },
          "required": [
            "evidence_roles"
          ]
        },
        "maxContains": 1,
        "minContains": 1
      }
    },
    {
      "properties": {
        "keys": {
          "contains": {
            "properties": {
              "evidence_roles": {
                "contains": {
                  "const": "linux"
                }
              }
            }
          },
          "required": [
            "evidence_roles"
          ]
        }
      }
    }
  ]
}

```





```

        "required": [
            "evidence_roles"
        ],
        "maxContains": 1,
        "minContains": 1
    }
}
},
{
    "properties": {
        "keys": {
            "contains": {
                "properties": {
                    "evidence_roles": {
                        "contains": {
                            "const": "windows-no-degradation"
                        }
                    }
                },
                "required": [
                    "evidence_roles"
                ]
            },
            "maxContains": 1,
            "minContains": 1
        }
    }
}

```

```

    }
}
},
"properties": {
    "algorithm": {
        "const": "Ed25519"
    },
    "keys": {
        "items": {
            "additionalProperties": false,
            "allOf": [
                {
                    "if": {
                        "properties": {
                            "capabilities": {
                                "const": [
                                    "evidence.produce"
                                ]
                            }
                        }
                    },
                    "then": {
                        "properties": {
                            "evidence_roles": {
                                "maxItems": 1,
                                "minItems": 1
                            }
                        }
                    }
                }
            ]
        }
    }
}

```

```

        "platforms": {
            "minItems": 1
        }
    }
},
{
    "if": {
        "properties": {
            "capabilities": {
                "const": [
                    "standard.distribute"
                ]
            }
        }
    },
    "then": {
        "properties": {
            "evidence_roles": {
                "maxItems": 0
            },
            "platforms": {
                "maxItems": 0
            }
        }
    }
}

```

```

    }
  }
},
{

```

```

  "if": {
    "properties": {
      "evidence_roles": {
        "const": [
          "human-documents"
        ]
      }
    }
  },
  "then": {
    "properties": {
      "platforms": {
        "const": [
          "windows"
        ]
      }
    }
  }
},
{
  "if": {
    "properties": {
      "evidence_roles": {
        "const": [
          "linux"
        ]
      }
    }
  }
}

```

```

},
"then": {
  "properties": {
    "platforms": {
      "const": [
        "linux"
      ]
    }
  }
}
},
{
  "if": {
    "properties": {
      "evidence_roles": {
        "const": [
          "linux-no-degradation"
        ]
      }
    }
  },
  "then": {
    "properties": {
      "platforms": {
        "const": [
          "linux"
        ]
      }
    }
  }
}

```

```

}
}
},
{
  "if": {
    "properties": {
      "evidence_roles": {
        "const": [
          "physical-scale"
        ]
      }
    }
  },
  "then": {
    "properties": {
      "platforms": {
        "const": [

```

```

        "windows"
      ]
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [

```

```

        "saturation-audit"
      ]
    }
  },
  {
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "windows"
          ]
        }
      }
    }
  },
  {
    "then": {
      "properties": {
        "platforms": {

```

```

        "const": [
          "windows"
        ]
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "windows-no-degradation"
          ]
        }
      }
    }
  },
  {
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  }
},
1,

```

```

"properties": {
  "capabilities": {
    "items": {
      "enum": [
        "evidence.produce",
        "standard.distribute"
      ]
    },
    "maxItems": 1,
    "minItems": 1,

```

```

    "type": "array",
    "uniqueItems": true
  },
  "evidence_roles": {
    "items": {
      "enum": [
        "human-documents",
        "linux",
        "linux-no-degradation",
        "physical-scale",
        "saturation-audit",
        "windows",
        "windows-no-degradation"
      ]
    },
    "maxItems": 1,
    "type": "array",
    "uniqueItems": true
  }
}

```

```

  },
  "key_id": {
    "$ref": "#/$defs/Id"
  },
  "not_after": {
    "$ref": "#/$defs/Timestamp"
  },
  "not_before": {
    "$ref": "#/$defs/Timestamp"
  },
  "platforms": {
    "items": {
      "enum": [
        "cross-platform",
        "darwin",
        "linux",
        "windows"
      ]
    },
    "maxItems": 4,
    "type": "array",
    "uniqueItems": true
  },
  "public_key": {
    "maxLength": 44,
    "minLength": 44,
    "pattern": "^[A-Za-z0-9+/{43}=$",
    "type": "string"
  }
}

```

```

  },
  "revoked": {
    "type": "boolean"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "key_id",
  "subject_id",
  "capabilities",
  "evidence_roles",
  "platforms",
  "public_key",
  "not_before",
  "not_after",
  "revoked"
],
"type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"record_type": {
  "const": "StandardReleaseTrustConfiguration"
},
"signature_provider_id": {
  "const": "cryptography-ed25519-v1"
}
}

```

```

    },
    "trust_root_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "trust_root_id",
    "signature_provider_id",
    "algorithm",
    "keys"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StandardReleaseTrustConfiguration* | Validator: *urn:promin:contracts:1#/\$defs/StandardReleaseTrustConfiguration*

## 122. StandardsInit

Type summary: type="object"; additionalProperties=false

Required: record\_type, bindings

| Property    | Required | Structural rule                       |
|-------------|----------|---------------------------------------|
| bindings    | yes      | type="array"; minItems=0; maxItems=64 |
| record_type | yes      | const="StandardsInit"                 |

```

{
  "additionalProperties": false,
  "properties": {
    "bindings": {
      "items": {
        "$ref": "#/$defs/StandardBinding"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array"
    },
    "record_type": {
      "const": "StandardsInit"
    }
  },
  "required": [
    "record_type",
    "bindings"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StandardsInit* | Validator: *urn:promin:contracts:1#/\$defs/StandardsInit*

## 123. StateBindingUpdate

Type summary: type="object"; additionalProperties=false; oneOf[2]=keys:properties,required | keys:not,properties

Required: leaf\_type, leaf\_id, operation

| Property     | Required | Structural rule                                                                                                                |
|--------------|----------|--------------------------------------------------------------------------------------------------------------------------------|
| leaf_id      | yes      | \$ref="#/\$defs/Id"                                                                                                            |
| leaf_type    | yes      | enum=["Activation", "Task", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Lease", "Relation"] |
| operation    | yes      | enum=["delete", "set"]                                                                                                         |
| value_digest | no       | \$ref="#/\$defs/Digest"                                                                                                        |

```

{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "operation": {
          "const": "set"
        }
      },
      "required": [
        "value_digest"
      ]
    }
  ]
}

```

```

    },
    {
      "not": {
        "required": [
          "value_digest"
        ]
      },
      "properties": {
        "operation": {
          "const": "delete"
        }
      }
    }
  ],
  "properties": {
    "leaf_id": {
      "$ref": "#/$defs/Id"
    },
    "leaf_type": {
      "enum": [
        "Activation",
        "Task",
        "Candidate",
        "Artifact",
        "Run",
        "Finding",
        "Decision",
        "GateResult",
        "Grant",
        "Lease",
        "Relation"
      ]
    },
    "operation": {
      "enum": [
        "delete",
        "set"
      ]
    },
    "value_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "leaf_type",
    "leaf_id",
    "operation"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/StateBindingUpdate | Validator: urn:promin:contracts:1#/\$defs/StateBindingUpdate

## 124. StateMachines

Type summary: const=schema(keys:capacity\_release,lease,task)

Required: -

```

{
  "const": {
    "capacity_release": {
      "closed_release": {
        "requires_field": "close_ack",
        "slot_reusable": true,
        "state": "CLOSED"
      },
      "terminal_reconciliation": {
        "fencing_token_binding": "exact-lease-fencing-token",
        "field": "capacity_reconciliation",
        "generation_binding": "exact-lease-generation",
        "manager_capability": "lease.manage",
        "required_fields": [
          "reconciled_by",
          "grant_id",
          "grant_claim_digest",
          "reconciled_at",
          "generation",

```

```

        "fencing_token"
      ],
      "slot_reusable_after_record": true,
      "states": [
        "EXPIRED",
        "REVOKED"
      ]
    },
    "unreconciled_slot_reusable": false
  }
}

```

```

},
"lease": {
  "ACTIVE": [
    "CLOSING",
    "EXPIRED",
    "REVOKED"
  ],
  "CLOSED": [],
  "CLOSING": [
    "CLOSED",
    "EXPIRED",
    "REVOKED"
  ],
  "EXPIRED": [],
  "REVOKED": []
},
"task": {
  "BLOCKED": [
    "READY",
    "CANCELLED"
  ],
  "CANCELLED": [],
  "COMPLETED": [],
  "LEASED": [
    "RUNNING",
    "READY",
    "BLOCKED",
    "CANCELLED"
  ]
}
}
}

```

```

},
"PLANNED": [
  "READY",
  "BLOCKED",
  "CANCELLED"
],
"READY": [
  "LEASED",
  "BLOCKED",
  "CANCELLED"
],
"RUNNING": [
  "COMPLETED",
  "BLOCKED",
  "CANCELLED"
]
}
}
}

```

Owner: core/contracts.schema.json#/\$defs/StateMachines | Validator: urn:promin:contracts:1#/\$defs/StateMachines

## 125. StructuralBudgets

Type summary: type="object"; additionalProperties=false

Required: core\_normative\_artifacts, selected\_presets, authority\_bearing\_init\_records, persistent\_entity\_kinds\_max, relation\_kinds\_max, base\_user\_commands\_max

| Property                       | Required | Structural rule           |
|--------------------------------|----------|---------------------------|
| authority_bearing_init_records | yes      | type="integer"; minimum=1 |
| base_user_commands_max         | yes      | type="integer"; minimum=1 |
| core_normative_artifacts       | yes      | type="integer"; minimum=1 |
| persistent_entity_kinds_max    | yes      | type="integer"; minimum=1 |
| relation_kinds_max             | yes      | type="integer"; minimum=1 |
| selected_presets               | yes      | type="integer"; minimum=1 |

```

{
  "additionalProperties": false,
  "properties": {
    "authority_bearing_init_records": {
      "minimum": 1,
      "type": "integer"
    },
    "base_user_commands_max": {
      "minimum": 1,
      "type": "integer"
    },
    "core_normative_artifacts": {
      "minimum": 1,
      "type": "integer"
    },
    "persistent_entity_kinds_max": {
      "minimum": 1,
      "type": "integer"
    },
    "relation_kinds_max": {
      "minimum": 1,
      "type": "integer"
    },
    "selected_presets": {
      "minimum": 1,
      "type": "integer"
    }
  }
},
{
  "required": [
    "core_normative_artifacts",
    "selected_presets",
    "authority_bearing_init_records",
    "persistent_entity_kinds_max",
    "relation_kinds_max",
    "base_user_commands_max"
  ],
  "type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/StructuralBudgets* | Validator: *urn:promin:contracts:1#/\$defs/StructuralBudgets*

## 126. SubjectInit

Type summary: type="object"; additionalProperties=false

Required: subject\_id, kind, display\_name

| Property     | Required | Structural rule                    |
|--------------|----------|------------------------------------|
| display_name | yes      | type="string"; minLength=1         |
| kind         | yes      | enum=["human", "service", "agent"] |
| subject_id   | yes      | \$ref="#/\$defs/Id"                |

```

{
  "additionalProperties": false,
  "properties": {
    "display_name": {
      "minLength": 1,
      "type": "string"
    },
    "kind": {
      "enum": [
        "human",
        "service",
        "agent"
      ]
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "subject_id",
    "kind",
    "display_name"
  ],
  "type": "object"
}

```



## 127. Task

Type summary: type="object"; additionalProperties=false

Required: record\_type, task\_id, state, required\_capability, acceptance\_predicate, allowed\_paths, activation\_digest, candidate\_digest, created\_at, gate\_run\_definitions

| Property               | Required | Structural rule                                                      |
|------------------------|----------|----------------------------------------------------------------------|
| acceptance_predicate   | yes      | type="string"; minLength=1                                           |
| activation_digest      | yes      | \$ref="#/\$defs/Digest"                                              |
| allowed_paths          | yes      | type="array"; minItems=0; maxItems=64; uniqueItems=true              |
| candidate_digest       | yes      | \$ref="#/\$defs/Digest"                                              |
| created_at             | yes      | \$ref="#/\$defs/Timestamp"                                           |
| gate_run_definitions   | yes      | type="array"; minItems=1; maxItems=64                                |
| operation_profile_id   | no       | \$ref="#/\$defs/Id"                                                  |
| orchestration_required | no       | const=true                                                           |
| recommended_model_tier | no       | enum=["tool-only", "micro", "standard", "strong", "critical-review"] |
| record_type            | yes      | const="Task"                                                         |
| required_capability    | yes      | \$ref="#/\$defs/Id"                                                  |
| state                  | yes      | \$ref="#/\$defs/TaskState"                                           |
| task_id                | yes      | \$ref="#/\$defs/Id"                                                  |

```
{
  "additionalProperties": false,
  "properties": {
    "acceptance_predicate": {
      "minLength": 1,
      "type": "string"
    },
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "allowed_paths": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "gate_run_definitions": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "definition": {
            "$ref": "#/$defs/GateRunDefinition"
          },
          "definition_digest": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "definition_digest",
          "definition"
        ],
        "type": "object"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "operation_profile_id": {
```

```

    "$ref": "#/$defs/Id"
  },
  "orchestration_required": {
    "const": true
  },
  "recommended_model_tier": {
    "enum": [
      "tool-only",
      "micro",
      "standard",
      "strong",
      "critical-review"
    ]
  },
  "record_type": {
    "const": "Task"
  },
  "required_capability": {
    "$ref": "#/$defs/Id"
  },
  "state": {
    "$ref": "#/$defs/TaskState"
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "record_type",
  "task_id",
  "state",
  "required_capability",
  "acceptance_predicate",
  "allowed_paths",
  "activation_digest",
  "candidate_digest",
  "created_at",
  "gate_run_definitions"
],
"type": "object"
}

```

Owner: *core/contracts.schema.json#/\$defs/Task* | Validator: *urn:promin:contracts:1#/\$defs/Task*

## 128. TaskState

Type summary: enum=["BLOCKED", "CANCELLED", "COMPLETED", "LEASED", "PLANNED", "READY", "RUNNING"]

Required: -

```

{
  "enum": [
    "BLOCKED",
    "CANCELLED",
    "COMPLETED",
    "LEASED",
    "PLANNED",
    "READY",
    "RUNNING"
  ]
}

```

Owner: *core/contracts.schema.json#/\$defs/TaskState* | Validator: *urn:promin:contracts:1#/\$defs/TaskState*

## 129. TaskStateMachine

Type summary: const=schema(keys:BLOCKED,CANCELLED,COMPLETED,LEASED,PLANNED,READY,RUNNING)

Required: -

```

{
  "const": {
    "BLOCKED": [
      "READY",
      "CANCELLED"
    ],
    "CANCELLED": [],
    "COMPLETED": [],
    "LEASED": [

```

```

    "RUNNING",
    "READY",
    "BLOCKED",
    "CANCELLED"
  ],
  "PLANNED": [
    "READY",
    "BLOCKED",
    "CANCELLED"
  ],
  "READY": [
    "LEASED",
    "BLOCKED",
    "CANCELLED"
  ],
  "RUNNING": [
    "COMPLETED",
    "BLOCKED",
    "CANCELLED"
  ]
}
}

```

Owner: core/contracts.schema.json#/\$defs/TaskStateMachine | Validator: urn:promin:contracts:1#/\$defs/TaskStateMachine

## 130. TaskTransition

Type summary: type="object"; additionalProperties=false

Required: task\_id, from\_state, to\_state, reason

| Property   | Required | Structural rule            |
|------------|----------|----------------------------|
| from_state | yes      | \$ref="#/\$defs/TaskState" |
| reason     | yes      | type="string"; minLength=1 |
| task_id    | yes      | \$ref="#/\$defs/Id"        |
| to_state   | yes      | \$ref="#/\$defs/TaskState" |

```

{
  "additionalProperties": false,
  "properties": {
    "from_state": {
      "$ref": "#/$defs/TaskState"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    "to_state": {
      "$ref": "#/$defs/TaskState"
    }
  },
  "required": [
    "task_id",
    "from_state",
    "to_state",
    "reason"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TaskTransition | Validator: urn:promin:contracts:1#/\$defs/TaskTransition

## 131. TeamPolicy

Type summary: type="object"; additionalProperties=false

Required: threshold, key\_ids, signature\_provider\_capability

| Property                      | Required | Structural rule                                         |
|-------------------------------|----------|---------------------------------------------------------|
| key_ids                       | yes      | type="array"; minItems=1; maxItems=32; uniqueItems=true |
| signature_provider_capability | yes      | const="signature"                                       |
| threshold                     | yes      | type="integer"; minimum=1                               |

```

{
  "additionalProperties": false,
  "properties": {
    "key_ids": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "signature_provider_capability": {
      "const": "signature"
    },
    "threshold": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "threshold",
    "key_ids",
    "signature_provider_capability"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TeamPolicy | Validator: urn:promin:contracts:1#/\$defs/TeamPolicy

## 132. TechnologiesInit

Type summary: type="object"; additionalProperties=false

Required: record\_type, bindings

| Property    | Required | Structural rule                       |
|-------------|----------|---------------------------------------|
| bindings    | yes      | type="array"; minItems=1; maxItems=32 |
| record_type | yes      | const="TechnologiesInit"              |

```

{
  "additionalProperties": false,
  "properties": {
    "bindings": {
      "items": {
        "$ref": "#/$defs/ProviderBinding"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "record_type": {
      "const": "TechnologiesInit"
    }
  },
  "required": [
    "record_type",
    "bindings"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TechnologiesInit | Validator: urn:promin:contracts:1#/\$defs/TechnologiesInit

## 133. TechnologyCapabilityDef

Type summary: type="object"; additionalProperties=false

Required: id, task, forbidden\_boundary, adapter\_protocol

| Property           | Required | Structural rule                                                                                                                                                                     |
|--------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| adapter_protocol   | yes      | type="object"; additionalProperties=false                                                                                                                                           |
| forbidden_boundary | yes      | type="object"; additionalProperties=false                                                                                                                                           |
| id                 | yes      | enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"] |
| task               | yes      | type="string"; minLength=1                                                                                                                                                          |

```

{
  "additionalProperties": false,
  "properties": {
    "adapter_protocol": {
      "additionalProperties": false,
      "properties": {
        "dependency_receipt_rule": {
          "const": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
              "component_id",
              "component_kind",
              "source",
              "version",
              "digest",
              "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
          }
        },
        "operation_contracts": {
          "additionalProperties": {
            "additionalProperties": false,
            "properties": {
              "request": {
                "maxLength": 1024,
                "minLength": 1,
                "type": "string"
              },
              "response": {
                "maxLength": 1024,
                "minLength": 1,
                "type": "string"
              }
            },
            "required": [
              "request",
              "response"
            ],
            "type": "object"
          },
          "maxProperties": 32,
          "minProperties": 1,
          "propertyNames": {
            "$ref": "#/$defs/Id"
          },
          "type": "object"
        },
        "protocol_id": {
          "$ref": "#/$defs/Id"
        },
        "receipt_persistence": {
          "const": "content-addressed-receipt"
        },
        "supported_bindings": {
          "items": {
            "additionalProperties": false,
            "properties": {
              "adapter_id": {
                "$ref": "#/$defs/Id"
              },
              "identity_kind": {
                "enum": [
                  "file-digest",
                  "module-file-digest",
                  "distribution-tree-digest",
                  "module-closure-digest",
                  "platform-attestation"
                ]
              },
              "invocation_kind": {
                "enum": [
                  "python-runtime",
                  "python-module",

```

```

        "executable",
        "platform-service"
    ]
}
},
"required": [

```

```

    "adapter_id",
    "identity_kind",
    "invocation_kind"
],
"type": "object"
},
"maxItems": 8,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"supported_commands": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"unknown_binding": {
  "const": "reject"
}
},
"required": [
  "dependency_receipt_rule",
  "operation_contracts",
  "protocol_id",

```

```

  "receipt_persistence",
  "supported_bindings",
  "supported_commands",
  "unknown_binding"
],
"type": "object"
},
"forbidden_boundary": {
  "additionalProperties": false,
  "properties": {
    "effect": {
      "const": "reject"
    },
    "forbids": {
      "items": {
        "maxLength": 128,
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "normative_authority": {
      "const": false
    }
  }
},

```

```

"required": [
  "effect",
  "forbids",
  "normative_authority"
],
"type": "object"
},
"id": {
  "enum": [
    "control-runtime",
    "shape-validation",
    "content-identity",
    "local-serialization",
    "query-projection",
    "filesystem-inventory",

```

```

        "export-scan",
        "signature",
        "build-dependency"
    ]
},
"task": {
    "minLength": 1,
    "type": "string"
}
},
"required": [
    "id",
    "task",
    "forbidden_boundary",
    "adapter_protocol"
],
"type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TechnologyCapabilityDef | Validator: urn:promin:contracts:1#/\$defs/TechnologyCapabilityDef

## 134. Timestamp

Type summary: type="string"; format="date-time"; pattern="^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z\$"

Required: -

```

{
    "format": "date-time",
    "pattern": "^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z$",
    "type": "string"
}

```

Owner: core/contracts.schema.json#/\$defs/Timestamp | Validator: urn:promin:contracts:1#/\$defs/Timestamp

## 135. ToolOutcome

Type summary: type="object"; additionalProperties=false

Required: availability, status, pass\_credit, acceptance\_pass

| Property        | Required | Structural rule                                            |
|-----------------|----------|------------------------------------------------------------|
| acceptance_pass | yes      | const=false                                                |
| availability    | yes      | enum=["AVAILABLE", "UNAVAILABLE"]                          |
| pass_credit     | yes      | const=false                                                |
| status          | yes      | enum=["PASS", "FAIL", "SKIPPED", "UNAVAILABLE", "TIMEOUT"] |

```

{
    "additionalProperties": false,
    "properties": {
        "acceptance_pass": {
            "const": false
        },
        "availability": {
            "enum": [
                "AVAILABLE",
                "UNAVAILABLE"
            ]
        },
        "pass_credit": {
            "const": false
        },
        "status": {
            "enum": [
                "PASS",
                "FAIL",
                "SKIPPED",
                "UNAVAILABLE",
                "TIMEOUT"
            ]
        }
    },
    "required": [
        "availability",
        "status",

```

```

    "pass_credit",
    "acceptance_pass"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/ToolOutcome | Validator: urn:promin:contracts:1#/\$defs/ToolOutcome

## 136. TrackedDocumentationBoundary

Type summary: type="object"; additionalProperties=false

Required: record\_type, root, manifest\_digest, pass\_credit

| Property        | Required | Structural rule                      |
|-----------------|----------|--------------------------------------|
| manifest_digest | yes      | \$ref="#/\$defs/Sha256"              |
| pass_credit     | yes      | const=false                          |
| record_type     | yes      | const="TrackedDocumentationBoundary" |
| root            | yes      | const=".promin/docs"                 |

```

{
  "additionalProperties": false,
  "properties": {
    "manifest_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "pass_credit": {
      "const": false
    },
    "record_type": {
      "const": "TrackedDocumentationBoundary"
    },
    "root": {
      "const": ".promin/docs"
    }
  },
  "required": [
    "record_type",
    "root",
    "manifest_digest",
    "pass_credit"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TrackedDocumentationBoundary | Validator: urn:promin:contracts:1#/\$defs/TrackedDocumentationBoundary

## 137. TrackedExtensionSet

Type summary: type="object"; additionalProperties=false

Required: root, members, byte\_exact

| Property   | Required | Structural rule                 |
|------------|----------|---------------------------------|
| byte_exact | yes      | const=true                      |
| members    | yes      | type="array"; uniqueItems=true  |
| root       | yes      | const=".promin/docs/extensions" |

```

{
  "additionalProperties": false,
  "properties": {
    "byte_exact": {
      "const": true
    },
    "members": {
      "items": {
        "$ref": "#/$defs/Sha256"
      },
      "type": "array",
      "uniqueItems": true
    },
    "root": {
      "const": ".promin/docs/extensions"
    }
  },
  "required": [

```



```

    "root",
    "members",
    "byte_exact"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TrackedExtensionSet | Validator: urn:promin:contracts:1#/\$defs/TrackedExtensionSet

## 138. TrustKey

Type summary: type="object"; additionalProperties=false

Required: key\_id, subject\_id, algorithm, public\_key, fingerprint

| Property    | Required | Structural rule            |
|-------------|----------|----------------------------|
| algorithm   | yes      | type="string"; minLength=1 |
| fingerprint | yes      | \$ref="#/\$defs/Digest"    |
| key_id      | yes      | \$ref="#/\$defs/Id"        |
| public_key  | yes      | type="string"; minLength=1 |
| subject_id  | yes      | \$ref="#/\$defs/Id"        |

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "fingerprint": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "public_key": {
      "minLength": 1,
      "type": "string"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "key_id",
    "subject_id",
    "algorithm",
    "public_key",
    "fingerprint"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/TrustKey | Validator: urn:promin:contracts:1#/\$defs/TrustKey

## 139. WorkCard

Type summary: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:else,if,then,title

Required: record\_type, task\_id, operation\_mode, operation, acceptance\_predicate, allowed\_paths, holder\_grant\_id, candidate\_digest, activation\_digest, context\_digest, stop\_conditions, budget, truncated, query\_grant\_id, query\_grant\_claim\_digest

| Property             | Required | Structural rule                                         |
|----------------------|----------|---------------------------------------------------------|
| acceptance_predicate | yes      | type="string"; minLength=1                              |
| activation_digest    | yes      | \$ref="#/\$defs/Digest"                                 |
| allowed_paths        | yes      | type="array"; minItems=0; maxItems=64; uniqueItems=true |
| budget               | yes      | \$ref="#/\$defs/Budget"                                 |
| candidate_digest     | yes      | \$ref="#/\$defs/Digest"                                 |
| context_digest       | yes      | \$ref="#/\$defs/Digest"                                 |
| continuation_query   | no       | type="string"; minLength=1                              |

| Property                 | Required | Structural rule                                                      |
|--------------------------|----------|----------------------------------------------------------------------|
| fencing_token            | no       | type="integer"; minimum=1                                            |
| holder_grant_id          | yes      | \$ref="#/\$defs/Id"                                                  |
| lease_generation         | no       | type="integer"; minimum=1                                            |
| lease_id                 | no       | \$ref="#/\$defs/Id"                                                  |
| operation                | yes      | type="string"; minLength=1                                           |
| operation_mode           | yes      | enum=["read", "mutate"]                                              |
| operation_profile_id     | no       | \$ref="#/\$defs/Id"                                                  |
| orchestration_required   | no       | const=true                                                           |
| query_grant_claim_digest | yes      | \$ref="#/\$defs/Digest"                                              |
| query_grant_id           | yes      | \$ref="#/\$defs/Id"                                                  |
| recommended_model_tier   | no       | enum=["tool-only", "micro", "standard", "strong", "critical-review"] |
| record_type              | yes      | const="WorkCard"                                                     |
| stop_conditions          | yes      | type="array"; minItems=1; maxItems=16; uniqueItems=true              |
| task_id                  | yes      | \$ref="#/\$defs/Id"                                                  |
| truncated                | yes      | type="boolean"                                                       |

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "continuation_query"
          ]
        }
      },
      "if": {
        "properties": {
          "truncated": {
            "const": true
          }
        }
      },
      "then": {
        "required": [
          "continuation_query"
        ]
      }
    },
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "lease_id"
              ]
            },
            {
              "required": [
                "lease_generation"
              ]
            },
            {
              "required": [
                "fencing_token"
              ]
            }
          ]
        }
      },
      "if": {
        "properties": {
          "operation_mode": {
            "const": "mutate"
          }
        }
      }
    }
  ]
}

```

```

    "required": [
      "operation_mode"
    ]
  },
  "then": {
    "required": [
      "lease_id",
      "lease_generation",
      "fencing_token"
    ]
  },
  "title": "require current Lease generation and fence for mutation"
}
],
"properties": {
  "acceptance_predicate": {
    "minLength": 1,
    "type": "string"
  },
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "allowed_paths": {
    "items": {
      "$ref": "#/$defs/RelativePath"
    },
    "maxItems": 64,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "budget": {

```

```

    "$ref": "#/$defs/Budget"
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "context_digest": {
    "$ref": "#/$defs/Digest"
  },
  "continuation_query": {
    "minLength": 1,
    "type": "string"
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "holder_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "lease_generation": {
    "minimum": 1,
    "type": "integer"
  },
  "lease_id": {
    "$ref": "#/$defs/Id"
  },
  "operation": {
    "minLength": 1,

```

```

    "type": "string"
  },
  "operation_mode": {
    "enum": [
      "read",
      "mutate"
    ]
  },
  "operation_profile_id": {
    "$ref": "#/$defs/Id"
  },
  "orchestration_required": {
    "const": true
  },
  "query_grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
}

```

```

    "query_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "recommended_model_tier": {
      "enum": [
        "tool-only",
        "micro",
        "standard",
        "strong",
        "critical-review"
      ]
    },
    "record_type": {
      "const": "WorkCard"
    },
    "stop_conditions": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    "truncated": {
      "type": "boolean"
    }
  },
  "required": [
    "record_type",
    "task_id",
    "operation_mode",
    "operation",
    "acceptance_predicate",
    "allowed_paths",
    "holder_grant_id",
    "candidate_digest",
    "activation_digest",
    "context_digest",
    "stop_conditions",
    "budget",
    "truncated",
    "query_grant_id",
    "query_grant_claim_digest"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/WorkCard | Validator: urn:promin:contracts:1#/\$defs/WorkCard

## 140. WorkCardProjection

Type summary: type="object"; additionalProperties=false

Required: record\_type, activation\_digest, head\_digest, projection\_digest, task\_id, task\_digest, task, budget, projection\_authoritative

| Property                 | Required | Structural rule                           |
|--------------------------|----------|-------------------------------------------|
| activation_digest        | yes      | \$ref="#/\$defs/Digest"                   |
| budget                   | yes      | \$ref="#/\$defs/Budget"                   |
| head_digest              | yes      | oneOf[2]=#/\$defs/Digest   type:null      |
| projection_authoritative | yes      | const=false                               |
| projection_digest        | yes      | \$ref="#/\$defs/Digest"                   |
| record_type              | yes      | const="WorkCardProjection"                |
| task                     | yes      | type="object"; additionalProperties=false |
| task_digest              | yes      | \$ref="#/\$defs/Digest"                   |
| task_id                  | yes      | \$ref="#/\$defs/Id"                       |

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "budget": {
      "$ref": "#/$defs/Budget"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "projection_authoritative": {
      "const": false
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "record_type": {
      "const": "WorkCardProjection"
    }
  },

```

```

    "task": {
      "additionalProperties": false,
      "properties": {
        "data_class": {
          "enum": [
            "operational-record",
            "untrusted-source"
          ]
        },
        "entity_type": {
          "enum": [
            "Task",
            "Candidate",
            "Artifact",
            "Run",
            "Finding",
            "Decision",
            "GateResult",
            "Grant",
            "Lease"
          ]
        },
        "id": {
          "$ref": "#/$defs/Id"
        },
        "payload": {
          "type": "object"
        }
      },

```

```

      "source_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "id",
      "entity_type",
      "data_class",
      "payload",
      "source_digest"
    ],
    "type": "object"
  },
  "task_digest": {
    "$ref": "#/$defs/Digest"
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "record_type",

```

```

    "activation_digest",
    "head_digest",
    "projection_digest",
    "task_id",
    "task_digest",
    "task",
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Owner: core/contracts.schema.json#/\$defs/WorkCardProjection | Validator: urn:promin:contracts:1#/\$defs/WorkCardProjection

## 141. WriterIdentity

Type summary: type="object"; additionalProperties=false

Required: pid, process\_start\_identity, lease\_digest, state

| Property               | Required | Structural rule                                  |
|------------------------|----------|--------------------------------------------------|
| lease_digest           | yes      | \$ref="#/\$defs/Sha256"                          |
| pid                    | yes      | type="integer"; minimum=1                        |
| process_start_identity | yes      | \$ref="#/\$defs/Sha256"                          |
| state                  | yes      | enum=["LIVE", "DEAD", "EXPIRED", "UNVERIFIABLE"] |

```

{
  "additionalProperties": false,
  "properties": {
    "lease_digest": {
      "$ref": "#/$defs/Sha256"
    },
    "pid": {
      "minimum": 1,
      "type": "integer"
    },
    "process_start_identity": {
      "$ref": "#/$defs/Sha256"
    },
    "state": {
      "enum": [
        "LIVE",
        "DEAD",
        "EXPIRED",
        "UNVERIFIABLE"
      ]
    }
  },
  "required": [
    "pid",
    "process_start_identity",
    "lease_digest",
    "state"
  ],
  "type": "object"
}

```

Owner: core/contracts.schema.json#/\$defs/WriterIdentity | Validator: urn:promin:contracts:1#/\$defs/WriterIdentity

# Policy registry - 68 policies

| ID    | Severity | Name                              | Statement                                                                                                                                                                                                   | Validator ID               | Owner                |
|-------|----------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------|
| P-001 | P0       | no-false-success                  | Gate pass credit is true exactly when normalized status is pass and every required gate passes.                                                                                                             | no_false_success           | core/policy-set.json |
| P-002 | P0       | activation-binding                | Every authoritative record binds the exact active Core, preset, init, and profile configuration.                                                                                                            | activation_binding         | core/policy-set.json |
| P-003 | P0       | grant-validity                    | Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof chain, revocation state, and Activation all match.                                                                       | grant_validity             | core/policy-set.json |
| P-004 | P0       | lease-fence                       | Every mutation uses the current active Lease generation and monotonic fencing token.                                                                                                                        | lease_fence                | core/policy-set.json |
| P-005 | P0       | candidate-evidence-binding        | Every evidence Artifact requires an explicit diagnostic, gate, or product purpose and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event payload.             | candidate_evidence_binding | core/policy-set.json |
| P-006 | P0       | atomic-event-commit               | One command produces one immutable batch under lock, compare-head, sync, atomic replace, and recovery.                                                                                                      | atomic_event_commit        | core/policy-set.json |
| P-007 | P0       | projection-is-derived             | Projection deletion and deterministic replay cannot change normative meaning.                                                                                                                               | projection_derived         | core/policy-set.json |
| P-008 | P1       | minimal-graph                     | Each verified raw file creates exactly one non-authoritative derived Artifact proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require explicit semantic content. | minimal_graph              | core/policy-set.json |
| P-009 | P0       | bounded-workcard                  | Every WorkCard stays within both Core hard ceilings and its selected profile byte, entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.                      | bounded_workcard           | core/policy-set.json |
| P-010 | P0       | size-independent-init             | Initialization performs zero product-tree scans and creates exactly five init records.                                                                                                                      | zero_scan_init             | core/policy-set.json |
| P-011 | P1       | explicit-external-bindings        | External standards and providers exist only in explicit digest-bound init records.                                                                                                                          | explicit_bindings          | core/policy-set.json |
| P-012 | P0       | canonical-name-only               | The canonical identity is promin; aliases, redirects, and compatibility names are rejected.                                                                                                                 | canonical_name             | core/policy-set.json |
| P-013 | P0       | safe-export                       | Export is allowlisted, recursively bounded, symlink-safe, secret-aware, and path-sanitized.                                                                                                                 | safe_export                | core/policy-set.json |
| P-014 | P0       | closed-acknowledgment             | Only a valid CLOSED transition with explicit close acknowledgement releases a mutation slot.                                                                                                                | closed_ack                 | core/policy-set.json |
| P-015 | P0       | fail-closed-critical-fields       | Missing, unknown, stale, downgraded, or unbound critical fields reject before mutation.                                                                                                                     | critical_fields            | core/policy-set.json |
| P-016 | P0       | preset-capability-ceiling         | A preset may configure provider needs and budgets but cannot grant action authority.                                                                                                                        | preset_no_authority        | core/policy-set.json |
| P-017 | P1       | single-scan-inventory             | One requested inventory performs at most one full product-tree pass; all projections derive from its stream.                                                                                                | single_scan                | core/policy-set.json |
| P-018 | P0       | idempotent-commands               | A duplicate idempotency key under the same activation and subject returns the original result or a conflict.                                                                                                | idempotent_commands        | core/policy-set.json |
| P-019 | P1       | core-preset-separation            | Core bundle identity excludes replaceable presets; Activation binds the selected preset separately.                                                                                                         | core_preset_separation     | core/policy-set.json |
| P-020 | P0       | acyclic-bootstrap-authority       | Init config defines root ceilings, never pre-activation runtime Grants.                                                                                                                                     | bootstrap_authority        | core/policy-set.json |
| P-021 | P1       | normalization-collision-rejection | Duplicate JSON keys and path or ID collisions after NFC/casefold normalization are rejected.                                                                                                                | normalization_collision    | core/policy-set.json |

| ID    | Severity | Name                                 | Statement                                                                                                                                                                                                                                                            | Validator ID             | Owner                |
|-------|----------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------|
| P-022 | P1       | provider-substitution-binding        | A provider identity or implementation-closure change invalidates the Activation and every dependent continuation, evidence Artifact, projection, replay result, and current release closure; it never silently changes authority.                                    | provider_substitution    | core/policy-set.json |
| P-023 | P1       | exact-init-keyset                    | Activation binds exactly project.json, standards.json, technologies.json, and authority.json.                                                                                                                                                                        | exact_init_keyset        | core/policy-set.json |
| P-024 | P1       | stale-projection-rejection           | Status, next, and WorkCard generation reject a projection whose source head or activation differs.                                                                                                                                                                   | stale_projection         | core/policy-set.json |
| P-025 | P0       | one-command-one-batch                | Every committed batch binds exactly one subject, validated command digest, command intent digest, authorization digest, and idempotency key.                                                                                                                         | one_command_one_batch    | core/policy-set.json |
| P-026 | P1       | deterministic-retrieval              | Ranking ties are deterministic and acceptance-critical facts are added by typed closure, not fuzzy ranking.                                                                                                                                                          | deterministic_retrieval  | core/policy-set.json |
| P-027 | P1       | strict-time-order                    | Timestamps use canonical UTC seconds and issued, heartbeat, expiry, finish, and decision order is semantically checked.                                                                                                                                              | strict_time              | core/policy-set.json |
| P-028 | P1       | exact-standard-files                 | Core and selected preset inputs reject symlinks and unexpected shadow files.                                                                                                                                                                                         | exact_standard_files     | core/policy-set.json |
| P-029 | P0       | root-authorization-is-bootstrap-only | Root command authorization signs one command intent and may issue the first authority.manage Grant only; it cannot execute ordinary project commands.                                                                                                                | root_authorization_limit | core/policy-set.json |
| P-030 | P1       | configuration-exact-idempotency      | Repeated initialization succeeds only when the requested canonical configuration digest exactly equals the existing Activation; otherwise it reports conflict.                                                                                                       | config_exact_idempotency | core/policy-set.json |
| P-031 | P1       | truthful-observability               | A command may report a scan, healthcheck, validation, or durability action only when that action was actually executed and evidence-bound.                                                                                                                           | truthful_observability   | core/policy-set.json |
| P-032 | P1       | depth-aware-retrieval-budget         | Effective retrieval seed count reserves entity and relation capacity for typed closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all remain effective, and no removed or unknown budget field may bypass a ceiling. | depth_aware_retrieval    | core/policy-set.json |
| P-033 | P1       | lease-only-for-mutation              | Mutation WorkCards require the current Lease; read-only WorkCards must not acquire or carry a mutation Lease.                                                                                                                                                        | lease_only_for_mutation  | core/policy-set.json |
| P-034 | P0       | command-capability-scope             | Every command resolves to exactly one Core-owned capability; requested scope retains every authorization constraint and explicitly contains every command effect target.                                                                                             | command_capability_scope | core/policy-set.json |
| P-035 | P0       | grant-issuance-chain                 | Every Grant proof binds claim_digest and is either the one bootstrap root issuance, a verified team threshold, or an active authority.manage issuer Grant.                                                                                                           | grant_issuance_chain     | core/policy-set.json |
| P-036 | P0       | verified-team-signatures             | Team-signature fields are never valid without threshold evaluation by the signature adapter resolved from the exact technologies binding; process-local injected adapters are explicitly non-reconstructable and cannot prove a cross-process CLI workflow.          | verified_team_signatures | core/policy-set.json |
| P-037 | P1       | provider-preflight-before-activation | Every required provider identity and healthcheck succeeds before initialization is committed; a declared provider is not evidence of availability.                                                                                                                   | provider_preflight       | core/policy-set.json |
| P-038 | P0       | canonical-state-machines             | Task and Lease transitions are accepted only by the policy-owned state machines; generated lifecycle views cannot invent transitions.                                                                                                                                | state_machine_transition | core/policy-set.json |



| ID    | Severity | Name                                        | Statement                                                                                                                                                                                                                                                                                                                                                                                                                       | Validator ID                   | Owner                |
|-------|----------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------|
| P-039 | P0       | decision-target-binding                     | Every Decision names one typed target and the decision kind, requested scope, candidate, and authority are consistent with that target.                                                                                                                                                                                                                                                                                         | decision_target_binding        | core/policy-set.json |
| P-040 | P0       | credit-requires-evidence                    | Pass, fail, and blocked gate results cite at least one exact immutable Artifact record reference; skipped results never receive pass credit.                                                                                                                                                                                                                                                                                    | credit_requires_evidence       | core/policy-set.json |
| P-041 | P0       | command-event-correspondence                | One command batch binds the exact authorization claim and contains exactly one primary event equal to the validated command payload; auxiliary events are typed relations only and event IDs are unique.                                                                                                                                                                                                                        | command_event_correspondence   | core/policy-set.json |
| P-042 | P0       | continuation-authorization-and-completeness | A continuation is authenticated by a random per-Activation secret and binds the exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; its complete page union emits every reachable entity and relation exactly once without loss. | continuation_completeness      | core/policy-set.json |
| P-043 | P0       | candidate-snapshot-consistency              | The candidate recipe is applied in the single inventory pass; promin v1 permits only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and receive no pass credit.                                                                                                                                                                                                                    | candidate_snapshot_consistency | core/policy-set.json |
| P-044 | P0       | provider-adapter-dispatch                   | Every technologies.json binding resolves to the adapter actually invoked and carries a recomputed capability-specific implementation closure for the promin runtime, Python interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject, and dependent evidence binds the exact implementation_closure_digest.                                                                    | provider_adapter_dispatch      | core/policy-set.json |
| P-045 | P0       | current-release-closure                     | A project-operational release Decision is immutable history; current product release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence, implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for promin distribution.                                                                                                             | current_release_closure        | core/policy-set.json |
| P-046 | P0       | finding-disposition-evidence                | Every Finding cites exact immutable Artifact record references. Every resolve or waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an exception-policy digest, future expiry, and separation of duties.                                                                                                                                                                           | finding_disposition_evidence   | core/policy-set.json |
| P-047 | P0       | candidate-delta-scope                       | A diff Artifact binds base and new Candidate digests, the exact WorkCard digest, and the complete changed-path set; every changed path is contained by both Task allowed_paths and the authorizing Grant effect scope.                                                                                                                                                                                                          | candidate_delta_scope          | core/policy-set.json |
| P-048 | P1       | verified-state-checkpoint                   | Events remain the sole authority. A normal command appends O(batch), incrementally updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal semantic digest.   | verified_state_checkpoint      | core/policy-set.json |
| P-049 | P1       | profile-default-depth                       | A profile default_dependency_depth applies only when depth is omitted; explicit depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all retrieval budgets.                                                                                                                                                                                                                         | profile_default_depth          | core/policy-set.json |

| ID    | Severity | Name                            | Statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Validator ID                   | Owner                |
|-------|----------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------|
| P-050 | P0       | canonical-distribution-archive  | The sole package builder emits immutable candidate bytes and an external StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is derived later and never changes the candidate ZIP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | canonical_distribution_archive | core/policy-set.json |
| P-051 | P0       | verified-inventory-provenance   | Public rebuild accepts only a verified InventoryResult or its persisted manifest; caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the exact path and digest.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | verified_inventory_provenance  | core/policy-set.json |
| P-052 | P1       | supported-command-surface       | The v1 preset advertises exactly init, doctor, status, next, validate, and continue; no optional or administrative command catalogue may claim an absent executable path.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | supported_command_surface      | core/policy-set.json |
| P-053 | P0       | profile-bound-scale             | Physical 100k evidence binds content-addressed raw inventory, query, continuation, process-sample, phase-log, and operation-metric artifacts for at least 198999 explicitly recorded Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and fails closed on truthful process/invoke status, commit p95/p99, bytes per changed record, checkpoint count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or traversal-pass ceiling.                                                                                                                                                                                     | profile_bound_scale            | core/policy-set.json |
| P-054 | P0       | cross-platform-release-evidence | The exact candidate ZIP must pass authenticated Windows and Linux platform observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale, offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles, the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14 supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer attestations bind the same candidate and participate in one replay/SOD/chronology graph; matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain false. | cross_platform_release         | core/policy-set.json |
| P-055 | P0       | standard-release-separation     | An external StandardReleaseDecision has explicit approve or reject outcome, binds candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time, stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability, independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest, and signature. The signed decision remains immutable history; current eligibility is derived; product_acceptance_pass is false and product_public_approval is not_approved.                                                                                                                                                                                                                                       | standard_release_separation    | core/policy-set.json |

| ID    | Severity | Name                                          | Statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Validator ID                        | Owner                |
|-------|----------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------|
| P-056 | P0       | required-gates-fail-closed                    | No-degradation runs the complete required suite only through the clean installed interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion, zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor, offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped, missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass credit. | required_gates_fail_closed          | core/policy-set.json |
| P-057 | P0       | research-draft-sanitation                     | Research crosses into normative owners only through the non-persistent MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution, acceptance, or public-approval claims cannot become normative text.                                                                                                                                                                                                                                                                       | research_draft_sanitation           | core/policy-set.json |
| P-058 | P0       | doctor-result-is-live-non-authority           | DoctorResult is an exact live-only derived health report with report_authoritative=false, pass_credit=false, product_acceptance_pass=false, product_public_approval=not_approved, zero product scans, and only its Core-owned operational metrics; token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.                                                                                                                                                                                                                                                                                 | doctor_result_non_authority         | core/policy-set.json |
| P-059 | P0       | semantic-inheritance-is-structural            | The compiled projection preserves the complete SemanticModel owner: admission, five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation domain/range/rules, and structured provider forbidden boundaries.                                                                                                                                                                                                                                                                                                                                                               | semantic_inheritance                | core/policy-set.json |
| P-060 | P0       | authority-owner-is-structural                 | AuthorityModel is the single owner of the capability catalogue, delegation depth, and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not substitute hardcoded authority facts.                                                                                                                                                                                                                                                                                                                                                                                        | authority_owner                     | core/policy-set.json |
| P-061 | P0       | gate-run-definition-is-independent            | An immutable GateRunDefinition independently owns the expected evidence class, product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests. A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.                                                                                                                                                                                                                                                                                                                                | gate_run_definition_integrity       | core/policy-set.json |
| P-062 | P0       | authority-runtime-policy-is-derived-live-only | AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty rules and delegation ceiling, and is never persisted or imported as authority.                                                                                                                                                                                                                                                                                                                                                     | authority_runtime_policy_projection | core/policy-set.json |
| P-063 | P0       | semantic-export-is-export-only                | SemanticExport is a strict bounded export-only semantic-state record binding the exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or release approval.                                                                                                                                                                                                                                                                                                      | semantic_export_integrity           | core/policy-set.json |
| P-064 | P0       | operation-metrics-is-live-observations        | OperationMetrics is a strict live-only non-authoritative observation for one command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.                                                                                                                                                                                                                                                                                                                                   | operation_metrics_non_authority     | core/policy-set.json |

| ID    | Severity | Name                                    | Statement                                                                                                                                                                                                                                                                                                                      | Validator ID                  | Owner                |
|-------|----------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|
| P-065 | P0       | workcard-projection-is-v2-live-only     | RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal selected-Task materialization and exposes no continuation or cursor fields.                                                               | workcard_projection_v2        | core/policy-set.json |
| P-066 | P0       | event-store-policy-is-derived-live-only | EventStorePolicy is rebuilt from the installed AuthorityModel before normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no defaults or nullable critical fields.            | event_store_policy_projection | core/policy-set.json |
| P-067 | P0       | depends-on-graph-is-acyclic             | Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation domain and range validity alone cannot admit a cyclic dependency graph.                                                                                                                                                                        | dependency_graph_acyclicity   | core/policy-set.json |
| P-068 | P0       | ready-frontier-is-current-derived-state | ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic and truncation is explicit. | ready_frontier_projection     | core/policy-set.json |

Owner: core/policy-set.json | Validator: validator\_registry\_version=1

## Executable validator index

P-001 | no\_false\_success | core/policy-set.json  
 P-002 | activation\_binding | core/policy-set.json  
 P-003 | grant\_validity | core/policy-set.json  
 P-004 | lease\_fence | core/policy-set.json  
 P-005 | candidate\_evidence\_binding | core/policy-set.json  
 P-006 | atomic\_event\_commit | core/policy-set.json  
 P-007 | projection\_derived | core/policy-set.json  
 P-008 | minimal\_graph | core/policy-set.json  
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 P-011 | explicit\_bindings | core/policy-set.json  
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 P-013 | safe\_export | core/policy-set.json  
 P-014 | closed\_ack | core/policy-set.json  
 P-015 | critical\_fields | core/policy-set.json  
 P-016 | preset\_no\_authority | core/policy-set.json  
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 P-020 | bootstrap\_authority | core/policy-set.json  
 P-021 | normalization\_collision | core/policy-set.json  
 P-022 | provider\_substitution | core/policy-set.json  
 P-023 | exact\_init\_keyset | core/policy-set.json  
 P-024 | stale\_projection | core/policy-set.json  
 P-025 | one\_command\_one\_batch | core/policy-set.json  
 P-026 | deterministic\_retrieval | core/policy-set.json

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P-029 | root\_authorization\_limit | core/policy-set.json

P-030 | config\_exact\_idempotency | core/policy-set.json

P-031 | truthful\_observability | core/policy-set.json

P-032 | depth\_aware\_retrieval | core/policy-set.json

P-033 | lease\_only\_for\_mutation | core/policy-set.json

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P-058 | doctor\_result\_non\_authority | core/policy-set.json

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P-067 | dependency\_graph\_acyclicity | core/policy-set.json

P-068 | readv\_frontier\_projection | core/policy-set.json

Owner: core/policy-set.json | Validator: policies[].validator\_id

# Acceptance predicates - 102

| #  | Predicate ID                                        | Validator / predicate ID                            | Owner                 |
|----|-----------------------------------------------------|-----------------------------------------------------|-----------------------|
| 1  | canonical-name-only                                 | canonical-name-only                                 | core/conformance.json |
| 2  | exact-six-core-artifacts                            | exact-six-core-artifacts                            | core/conformance.json |
| 3  | preset-outside-core-digest                          | preset-outside-core-digest                          | core/conformance.json |
| 4  | exact-one-selected-preset                           | exact-one-selected-preset                           | core/conformance.json |
| 5  | exact-five-init-records                             | exact-five-init-records                             | core/conformance.json |
| 6  | zero-product-scan-during-init                       | zero-product-scan-during-init                       | core/conformance.json |
| 7  | single-pass-inventory                               | single-pass-inventory                               | core/conformance.json |
| 8  | no-false-success                                    | no-false-success                                    | core/conformance.json |
| 9  | strict-critical-fields                              | strict-critical-fields                              | core/conformance.json |
| 10 | activation-digest-binding                           | activation-digest-binding                           | core/conformance.json |
| 11 | root-ceiling-without-bootstrap-grant-cycle          | root-ceiling-without-bootstrap-grant-cycle          | core/conformance.json |
| 12 | grant-scope-expiry-nonce-proof                      | grant-scope-expiry-nonce-proof                      | core/conformance.json |
| 13 | lease-generation-fence-and-close-ack                | lease-generation-fence-and-close-ack                | core/conformance.json |
| 14 | candidate-evidence-provider-binding                 | candidate-evidence-provider-binding                 | core/conformance.json |
| 15 | one-command-one-atomic-batch                        | one-command-one-atomic-batch                        | core/conformance.json |
| 16 | deterministic-replay                                | deterministic-replay                                | core/conformance.json |
| 17 | projection-disposable-and-current                   | projection-disposable-and-current                   | core/conformance.json |
| 18 | bounded-workcard-through-depth-twelve               | bounded-workcard-through-depth-twelve               | core/conformance.json |
| 19 | typed-reference-domain-range-closure                | typed-reference-domain-range-closure                | core/conformance.json |
| 20 | dependency-graph-acyclic-and-current-ready-frontier | dependency-graph-acyclic-and-current-ready-frontier | core/conformance.json |
| 21 | safe-recursive-export                               | safe-recursive-export                               | core/conformance.json |
| 22 | idempotent-state-commands                           | idempotent-state-commands                           | core/conformance.json |
| 23 | normalization-collision-rejection                   | normalization-collision-rejection                   | core/conformance.json |
| 24 | provider-substitution-invalidates-derived-state     | provider-substitution-invalidates-derived-state     | core/conformance.json |
| 25 | rollback-and-recutover-equivalence                  | rollback-and-recutover-equivalence                  | core/conformance.json |
| 26 | zero-forbidden-name-occurrences                     | zero-forbidden-name-occurrences                     | core/conformance.json |
| 27 | zero-unowned-normative-facts                        | zero-unowned-normative-facts                        | core/conformance.json |
| 28 | zero-core-provider-lock-in                          | zero-core-provider-lock-in                          | core/conformance.json |
| 29 | zero-compiled-cache-files                           | zero-compiled-cache-files                           | core/conformance.json |
| 30 | physical-100k-one-artifact-proxy-per-file           | physical-100k-one-artifact-proxy-per-file           | core/conformance.json |
| 31 | external-standard-release-decision-exact-binding    | external-standard-release-decision-exact-binding    | core/conformance.json |
| 32 | root-bootstrap-command-limited                      | root-bootstrap-command-limited                      | core/conformance.json |
| 33 | deterministic-activation-identity                   | deterministic-activation-identity                   | core/conformance.json |
| 34 | configuration-exact-idempotency                     | configuration-exact-idempotency                     | core/conformance.json |
| 35 | truthful-operation-reporting                        | truthful-operation-reporting                        | core/conformance.json |
| 36 | depth-aware-seed-budget                             | depth-aware-seed-budget                             | core/conformance.json |
| 37 | lossless-continuation-in-reference-100k-scenario    | lossless-continuation-in-reference-100k-scenario    | core/conformance.json |
| 38 | read-workcard-without-lease                         | read-workcard-without-lease                         | core/conformance.json |
| 39 | mutation-workcard-requires-current-lease            | mutation-workcard-requires-current-lease            | core/conformance.json |
| 40 | command-capability-and-scope-resolution             | command-capability-and-scope-resolution             | core/conformance.json |
| 41 | grant-issuance-chain-closure                        | grant-issuance-chain-closure                        | core/conformance.json |
| 42 | team-signatures-cryptographically-verified          | team-signatures-cryptographically-verified          | core/conformance.json |
| 43 | required-provider-preflight-before-commit           | required-provider-preflight-before-commit           | core/conformance.json |

| #  | Predicate ID                                           | Validator / predicate ID                               | Owner                 |
|----|--------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 44 | policy-owned-task-and-lease-state-machines             | policy-owned-task-and-lease-state-machines             | core/conformance.json |
| 45 | decision-kind-target-scope-binding                     | decision-kind-target-scope-binding                     | core/conformance.json |
| 46 | non-skipped-credit-result-has-evidence                 | non-skipped-credit-result-has-evidence                 | core/conformance.json |
| 47 | exact-command-primary-event-correspondence             | exact-command-primary-event-correspondence             | core/conformance.json |
| 48 | conjunctive-scope-containment                          | conjunctive-scope-containment                          | core/conformance.json |
| 49 | authorization-binds-exact-grant-claim                  | authorization-binds-exact-grant-claim                  | core/conformance.json |
| 50 | command-effect-target-in-requested-scope               | command-effect-target-in-requested-scope               | core/conformance.json |
| 51 | lease-holder-has-task-execute-grant                    | lease-holder-has-task-execute-grant                    | core/conformance.json |
| 52 | decision-grant-matches-command-authorization           | decision-grant-matches-command-authorization           | core/conformance.json |
| 53 | artifact-kind-resolves-correct-capability              | artifact-kind-resolves-correct-capability              | core/conformance.json |
| 54 | continuation-v2-complete-page-union                    | continuation-v2-complete-page-union                    | core/conformance.json |
| 55 | candidate-recipe-applied-once                          | candidate-recipe-applied-once                          | core/conformance.json |
| 56 | creditable-candidate-immutable-vcs-tree                | creditable-candidate-immutable-vcs-tree                | core/conformance.json |
| 57 | selected-provider-adapter-evidence                     | selected-provider-adapter-evidence                     | core/conformance.json |
| 58 | team-signed-cli-end-to-end                             | team-signed-cli-end-to-end                             | core/conformance.json |
| 59 | current-release-closure-revalidation                   | current-release-closure-revalidation                   | core/conformance.json |
| 60 | finding-bound-resolution-and-waiver-evidence           | finding-bound-resolution-and-waiver-evidence           | core/conformance.json |
| 61 | candidate-delta-within-task-and-grant-scope            | candidate-delta-within-task-and-grant-scope            | core/conformance.json |
| 62 | verified-head-checkpoint-delta-replay                  | verified-head-checkpoint-delta-replay                  | core/conformance.json |
| 63 | profile-default-with-core-depth-twelve                 | profile-default-with-core-depth-twelve                 | core/conformance.json |
| 64 | canonical-exact-zip-distribution                       | canonical-exact-zip-distribution                       | core/conformance.json |
| 65 | current-interpreter-and-clean-install-modes            | current-interpreter-and-clean-install-modes            | core/conformance.json |
| 66 | required-no-degradation-tests-fail-closed              | required-no-degradation-tests-fail-closed              | core/conformance.json |
| 67 | continuation-secret-subject-grant-binding              | continuation-secret-subject-grant-binding              | core/conformance.json |
| 68 | implementation-closure-bound-and-current               | implementation-closure-bound-and-current               | core/conformance.json |
| 69 | verified-inventory-result-provenance                   | verified-inventory-result-provenance                   | core/conformance.json |
| 70 | exact-six-base-command-surface                         | exact-six-base-command-surface                         | core/conformance.json |
| 71 | zero-dead-evidence-budget-fields                       | zero-dead-evidence-budget-fields                       | core/conformance.json |
| 72 | profile-bound-physical-100k-performance                | profile-bound-physical-100k-performance                | core/conformance.json |
| 73 | windows-linux-exact-zip-matrix                         | windows-linux-exact-zip-matrix                         | core/conformance.json |
| 74 | fresh-semantic-control-state-per-saturation-iteration  | fresh-semantic-control-state-per-saturation-iteration  | core/conformance.json |
| 75 | three-zero-new-saturation-iterations                   | three-zero-new-saturation-iterations                   | core/conformance.json |
| 76 | standard-distribution-separate-from-product-acceptance | standard-distribution-separate-from-product-acceptance | core/conformance.json |
| 77 | candidate-evidence-decision-acyclic-chain              | candidate-evidence-decision-acyclic-chain              | core/conformance.json |
| 78 | evidence-manifest-physical-resolution                  | evidence-manifest-physical-resolution                  | core/conformance.json |
| 79 | evidence-manifest-exact-eight-lane-matrix-resolution   | evidence-manifest-exact-eight-lane-matrix-resolution   | core/conformance.json |
| 80 | evidence-manifest-four-cp314-supplemental-records      | evidence-manifest-four-cp314-supplemental-records      | core/conformance.json |
| 81 | evidence-manifest-path-digest-size-closure             | evidence-manifest-path-digest-size-closure             | core/conformance.json |
| 82 | evidence-manifest-nested-and-supplemental-chronology   | evidence-manifest-nested-and-supplemental-chronology   | core/conformance.json |
| 83 | seven-distinct-role-producer-scopes                    | seven-distinct-role-producer-scopes                    | core/conformance.json |
| 84 | signed-standard-decision-authority-proof               | signed-standard-decision-authority-proof               | core/conformance.json |
| 85 | semver-from-candidate-binding                          | semver-from-candidate-binding                          | core/conformance.json |

| #   | Predicate ID                                       | Validator / predicate ID                           | Owner                 |
|-----|----------------------------------------------------|----------------------------------------------------|-----------------------|
| 86  | typed-human-document-verification                  | typed-human-document-verification                  | core/conformance.json |
| 87  | broad-query-bounded-seed-refinement                | broad-query-bounded-seed-refinement                | core/conformance.json |
| 88  | continuation-token-bytes-at-most-256               | continuation-token-bytes-at-most-256               | core/conformance.json |
| 89  | inventory-stream-memory-amplification-at-most-32   | inventory-stream-memory-amplification-at-most-32   | core/conformance.json |
| 90  | explicit-scale-marker-selection                    | explicit-scale-marker-selection                    | core/conformance.json |
| 91  | authenticated-role-platform-evidence-attestations  | authenticated-role-platform-evidence-attestations  | core/conformance.json |
| 92  | installed-platform-observation-closure             | installed-platform-observation-closure             | core/conformance.json |
| 93  | handle-relative-evidence-root-safety               | handle-relative-evidence-root-safety               | core/conformance.json |
| 94  | raw-scale-summary-recomputation                    | raw-scale-summary-recomputation                    | core/conformance.json |
| 95  | bounded-incremental-commit-and-compaction          | bounded-incremental-commit-and-compaction          | core/conformance.json |
| 96  | offline-installed-no-degradation                   | offline-installed-no-degradation                   | core/conformance.json |
| 97  | offline-release-online-compatibility               | offline-release-online-compatibility               | core/conformance.json |
| 98  | decision-after-evidence-with-bounded-skew          | decision-after-evidence-with-bounded-skew          | core/conformance.json |
| 99  | single-derived-platform-closure-owner              | single-derived-platform-closure-owner              | core/conformance.json |
| 100 | truthful-process-invocation-status                 | truthful-process-invocation-status                 | core/conformance.json |
| 101 | product-public-approval-separation                 | product-public-approval-separation                 | core/conformance.json |
| 102 | historical-decision-current-eligibility-separation | historical-decision-current-eligibility-separation | core/conformance.json |

Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recutover equivalence, and an authorized human project Decision. `product_acceptance_pass` remains false for promin v1 distribution evidence.

Owner: `core/conformance.json` | Validator: `required_acceptance`



# Mutation catalog - 77 families

| #  | Mutation family                        | Mutation probe ID                      | Owner                 |
|----|----------------------------------------|----------------------------------------|-----------------------|
| 1  | status-credit-contradiction            | status-credit-contradiction            | core/conformance.json |
| 2  | missing-critical-field                 | missing-critical-field                 | core/conformance.json |
| 3  | unknown-critical-field                 | unknown-critical-field                 | core/conformance.json |
| 4  | activation-drift                       | activation-drift                       | core/conformance.json |
| 5  | grant-replay                           | grant-replay                           | core/conformance.json |
| 6  | scope-escalation                       | scope-escalation                       | core/conformance.json |
| 7  | expired-grant                          | expired-grant                          | core/conformance.json |
| 8  | revoked-grant                          | revoked-grant                          | core/conformance.json |
| 9  | stale-fence                            | stale-fence                            | core/conformance.json |
| 10 | candidate-drift                        | candidate-drift                        | core/conformance.json |
| 11 | provider-substitution                  | provider-substitution                  | core/conformance.json |
| 12 | event-fork                             | event-fork                             | core/conformance.json |
| 13 | batch-command-mismatch                 | batch-command-mismatch                 | core/conformance.json |
| 14 | duplicate-idempotency-key              | duplicate-idempotency-key              | core/conformance.json |
| 15 | projection-corruption                  | projection-corruption                  | core/conformance.json |
| 16 | projection-staleness                   | projection-staleness                   | core/conformance.json |
| 17 | reference-break                        | reference-break                        | core/conformance.json |
| 18 | relation-domain-break                  | relation-domain-break                  | core/conformance.json |
| 19 | context-flood                          | context-flood                          | core/conformance.json |
| 20 | symlink-core-input                     | symlink-core-input                     | core/conformance.json |
| 21 | shadow-core-file                       | shadow-core-file                       | core/conformance.json |
| 22 | symlink-product-entry                  | symlink-product-entry                  | core/conformance.json |
| 23 | secret-export                          | secret-export                          | core/conformance.json |
| 24 | archive-expansion                      | archive-expansion                      | core/conformance.json |
| 25 | preset-authority-escalation            | preset-authority-escalation            | core/conformance.json |
| 26 | normalization-key-collision            | normalization-key-collision            | core/conformance.json |
| 27 | path-casefold-collision                | path-casefold-collision                | core/conformance.json |
| 28 | forbidden-name-reintroduction          | forbidden-name-reintroduction          | core/conformance.json |
| 29 | reinit-configuration-conflict          | reinit-configuration-conflict          | core/conformance.json |
| 30 | clock-order-violation                  | clock-order-violation                  | core/conformance.json |
| 31 | capability-confusion                   | capability-confusion                   | core/conformance.json |
| 32 | command-scope-escalation               | command-scope-escalation               | core/conformance.json |
| 33 | grant-issuer-chain-break               | grant-issuer-chain-break               | core/conformance.json |
| 34 | unverified-signature-acceptance        | unverified-signature-acceptance        | core/conformance.json |
| 35 | provider-declared-but-unavailable      | provider-declared-but-unavailable      | core/conformance.json |
| 36 | illegal-task-transition                | illegal-task-transition                | core/conformance.json |
| 37 | illegal-lease-transition               | illegal-lease-transition               | core/conformance.json |
| 38 | decision-target-confusion              | decision-target-confusion              | core/conformance.json |
| 39 | credit-without-evidence                | credit-without-evidence                | core/conformance.json |
| 40 | command-primary-event-mismatch         | command-primary-event-mismatch         | core/conformance.json |
| 41 | duplicate-event-id                     | duplicate-event-id                     | core/conformance.json |
| 42 | scope-selector-drop-broadening         | scope-selector-drop-broadening         | core/conformance.json |
| 43 | authorization-grant-claim-substitution | authorization-grant-claim-substitution | core/conformance.json |
| 44 | command-effect-outside-requested-scope | command-effect-outside-requested-scope | core/conformance.json |
| 45 | lease-holder-grant-mismatch            | lease-holder-grant-mismatch            | core/conformance.json |
| 46 | decision-grant-provenance-mismatch     | decision-grant-provenance-mismatch     | core/conformance.json |

| #  | Mutation family                       | Mutation probe ID                     | Owner                 |
|----|---------------------------------------|---------------------------------------|-----------------------|
| 47 | artifact-kind-capability-confusion    | artifact-kind-capability-confusion    | core/conformance.json |
| 48 | continuation-frontier-loss            | continuation-frontier-loss            | core/conformance.json |
| 49 | continuation-duplicate-emission       | continuation-duplicate-emission       | core/conformance.json |
| 50 | candidate-recipe-bypass               | candidate-recipe-bypass               | core/conformance.json |
| 51 | observational-candidate-pass-credit   | observational-candidate-pass-credit   | core/conformance.json |
| 52 | snapshot-provider-substitution        | snapshot-provider-substitution        | core/conformance.json |
| 53 | unsupported-provider-adapter          | unsupported-provider-adapter          | core/conformance.json |
| 54 | stale-release-closure-credit          | stale-release-closure-credit          | core/conformance.json |
| 55 | finding-evidence-target-mismatch      | finding-evidence-target-mismatch      | core/conformance.json |
| 56 | candidate-delta-scope-escape          | candidate-delta-scope-escape          | core/conformance.json |
| 57 | checkpoint-head-mismatch              | checkpoint-head-mismatch              | core/conformance.json |
| 58 | checkpoint-corruption-credit          | checkpoint-corruption-credit          | core/conformance.json |
| 59 | normal-command-full-replay            | normal-command-full-replay            | core/conformance.json |
| 60 | profile-default-depth-bypass          | profile-default-depth-bypass          | core/conformance.json |
| 61 | continuation-public-material-mismatch | continuation-public-material-mismatch | core/conformance.json |
| 62 | inventory-provenance-override         | inventory-provenance-override         | core/conformance.json |
| 63 | implementation-closure-drift          | implementation-closure-drift          | core/conformance.json |
| 64 | noncanonical-archive-layout           | noncanonical-archive-layout           | core/conformance.json |
| 65 | inventory-graph-inflation             | inventory-graph-inflation             | core/conformance.json |
| 66 | workcard-budget-bypass                | workcard-budget-bypass                | core/conformance.json |
| 67 | fabricated-evidence-digest            | fabricated-evidence-digest            | core/conformance.json |
| 68 | unsigned-standard-decision            | unsigned-standard-decision            | core/conformance.json |
| 69 | unconfigured-standard-decision-key    | unconfigured-standard-decision-key    | core/conformance.json |
| 70 | standard-decision-version-replay      | standard-decision-version-replay      | core/conformance.json |
| 71 | candidate-decision-cycle              | candidate-decision-cycle              | core/conformance.json |
| 72 | broad-query-corpus-pagination         | broad-query-corpus-pagination         | core/conformance.json |
| 73 | continuation-token-context-inflation  | continuation-token-context-inflation  | core/conformance.json |
| 74 | inventory-stream-materialization      | inventory-stream-materialization      | core/conformance.json |
| 75 | pdf-header-only-credit                | pdf-header-only-credit                | core/conformance.json |
| 76 | scale-environment-skip-credit         | scale-environment-skip-credit         | core/conformance.json |
| 77 | content-search-omission               | content-search-omission               | core/conformance.json |

Owner: core/conformance.json | Validator: mutation\_families

## State machines

### Task

| State     | Allowed next state                 |
|-----------|------------------------------------|
| BLOCKED   | READY, CANCELLED                   |
| CANCELLED | -                                  |
| COMPLETED | -                                  |
| LEASED    | RUNNING, READY, BLOCKED, CANCELLED |
| PLANNED   | READY, BLOCKED, CANCELLED          |
| READY     | LEASED, BLOCKED, CANCELLED         |
| RUNNING   | COMPLETED, BLOCKED, CANCELLED      |

Owner: core/policy-set.json | Validator: state\_machines.task + state\_machine\_transition

## Lease

| State   | Allowed next state        |
|---------|---------------------------|
| ACTIVE  | CLOSING, EXPIRED, REVOKED |
| CLOSED  | -                         |
| CLOSING | CLOSED, EXPIRED, REVOKED  |
| EXPIRED | -                         |
| REVOKED | -                         |

Owner: *core/policy-set.json* | Validator: *state\_machines.lease + state\_machine\_transition*

```
{'closed_release': {'requires_field': 'close_ack', 'slot_reusable': True, 'state': 'CLOSED'}, '
terminal_reconciliation': {'fencing_token_binding': 'exact-lease-fencing-token', 'field': '
capacity_reconciliation', 'generation_binding': 'exact-lease-generation', 'manager_capability': '
lease.manage', 'required_fields': ['reconciled_by', 'grant_id', 'grant_claim_digest', 'reconciled_at', '
generation', 'fencing_token'], 'slot_reusable_after_record': True, 'states': ['EXPIRED', 'REVOKED']}, '
unreconciled_slot_reusable': False}
```

Owner: *core/policy-set.json* | Validator: *state\_machines.capacity\_release*

## Capabilities and command authorization

| Capability          | Purpose                                                                                                                                                                                                   | Owner                     | Validator                |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|
| standard.activate   | Activate one digest-bound repository configuration.                                                                                                                                                       | core/authority-model.json | command_capability_scope |
| authority.manage    | Issue or revoke runtime Grants within a configured root ceiling.                                                                                                                                          | core/authority-model.json | command_capability_scope |
| task.plan           | Create, split, order, or block Tasks.                                                                                                                                                                     | core/authority-model.json | command_capability_scope |
| task.execute        | Execute one Task inside its WorkCard and valid Lease.                                                                                                                                                     | core/authority-model.json | command_capability_scope |
| lease.manage        | Acquire, heartbeat, close, expire, or revoke Leases.                                                                                                                                                      | core/authority-model.json | command_capability_scope |
| evidence.publish    | Publish candidate-bound evidence Artifacts.                                                                                                                                                               | core/authority-model.json | command_capability_scope |
| validation.evaluate | Evaluate gates without promotion or release authority.                                                                                                                                                    | core/authority-model.json | command_capability_scope |
| finding.record      | Record Findings.                                                                                                                                                                                          | core/authority-model.json | command_capability_scope |
| finding.resolve     | Resolve one Finding only with evidence bound to that Finding's canonical digest.                                                                                                                          | core/authority-model.json | command_capability_scope |
| finding.waive       | Waive one Finding only through an expiring exception-policy Decision, Finding-bound evidence, and separation of duties.                                                                                   | core/authority-model.json | command_capability_scope |
| candidate.promote   | Promote a validated Candidate.                                                                                                                                                                            | core/authority-model.json | command_capability_scope |
| release.decide      | Record an immutable historical release judgement bound to an exact acceptance-closure and provider-binding digest; current eligibility remains derived.                                                   | core/authority-model.json | command_capability_scope |
| export.create       | Create a sanitized external package.                                                                                                                                                                      | core/authority-model.json | command_capability_scope |
| migration.manage    | Execute cutover, rollback, and deterministic recutover.                                                                                                                                                   | core/authority-model.json | command_capability_scope |
| projection.rebuild  | Rebuild or verify non-authoritative projections.                                                                                                                                                          | core/authority-model.json | command_capability_scope |
| projection.read     | Read one bounded current projection or resume its continuation under the exact current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard distribution authority. | core/authority-model.json | command_capability_scope |

## Command to capability resolution

| Command kind     | Capability resolution                                                                                                                                                                                                                                                              | Owner                     | Validator                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|
| task.record      | task.plan                                                                                                                                                                                                                                                                          | core/authority-model.json | command_capability_scope |
| task.transition  | task.plan                                                                                                                                                                                                                                                                          | core/authority-model.json | command_capability_scope |
| candidate.record | task.execute                                                                                                                                                                                                                                                                       | core/authority-model.json | command_capability_scope |
| artifact.record  | { "capability_by_value": { "diff": "task.execute", "evidence": "evidence.publish", "log": "task.execute", "product": "task.execute", "report": "projection.rebuild", "payload_field": "artifact_kind" }                                                                            | core/authority-model.json | command_capability_scope |
| run.record       | { "capability_by_value": { "execution": "task.execute", "validation": "validation.evaluate", "payload_field": "run_kind" }                                                                                                                                                         | core/authority-model.json | command_capability_scope |
| gate.record      | validation.evaluate                                                                                                                                                                                                                                                                | core/authority-model.json | command_capability_scope |
| finding.record   | finding.record                                                                                                                                                                                                                                                                     | core/authority-model.json | command_capability_scope |
| decision.record  | { "capability_by_value": { "accept": "validation.evaluate", "promote": "candidate.promote", "reject": "validation.evaluate", "release": "release.decide", "resolve": "finding.resolve", "revoke": "authority.manage", "waive": "finding.waive", "payload_field": "decision_kind" } | core/authority-model.json | command_capability_scope |
| grant.issue      | authority.manage                                                                                                                                                                                                                                                                   | core/authority-model.json | command_capability_scope |
| grant.revoke     | authority.manage                                                                                                                                                                                                                                                                   | core/authority-model.json | command_capability_scope |
| lease.record     | lease.manage                                                                                                                                                                                                                                                                       | core/authority-model.json | command_capability_scope |

## Command effect scope

| Command kind     | Mode                                         | Target rule                                                                                                                                                            | Owner                     | Validator                |
|------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------|
| task.record      | payload-id                                   | { "id_field": "task_id", "kind": "task" }                                                                                                                              | core/authority-model.json | command_capability_scope |
| task.transition  | payload-id                                   | { "id_field": "task_id", "kind": "task" }                                                                                                                              | core/authority-model.json | command_capability_scope |
| candidate.record | payload-id                                   | { "id_field": "candidate_id", "kind": "candidate" }                                                                                                                    | core/authority-model.json | command_capability_scope |
| artifact.record  | payload-id                                   | { "id_field": "artifact_id", "kind": "artifact" }                                                                                                                      | core/authority-model.json | command_capability_scope |
| run.record       | payload-multi-id                             | { "targets": [ { "id_field": "task_id", "kind": "task" }, { "id_field": "candidate_digest", "kind": "candidate" } ] }                                                  | core/authority-model.json | command_capability_scope |
| gate.record      | payload-id                                   | { "id_field": "candidate_digest", "kind": "candidate" }                                                                                                                | core/authority-model.json | command_capability_scope |
| finding.record   | payload-id                                   | { "id_field": "finding_id", "kind": "finding" }                                                                                                                        | core/authority-model.json | command_capability_scope |
| decision.record  | payload-dynamic-id-or-referenced-grant-scope | { "grant_target_type": "Grant", "id_field": "target_id", "kind_field": "target_type", "kind_map": { "Candidate": "candidate", "Finding": "finding", "Task": "task" } } | core/authority-model.json | command_capability_scope |
| grant.issue      | payload-scope                                | { "scope_field": "scope" }                                                                                                                                             | core/authority-model.json | command_capability_scope |
| grant.revoke     | referenced-grant-scope                       | { "grant_id_field": "grant_id" }                                                                                                                                       | core/authority-model.json | command_capability_scope |
| lease.record     | payload-id                                   | { "id_field": "task_id", "kind": "task" }                                                                                                                              | core/authority-model.json | command_capability_scope |

## Lease-bound mutation claim

A lease-bound command is admissible only when authorization proves the exact command capability, holder\_authorization independently proves task.execute, both exact Grant claims bind the same subject, Activation and conjunctive effect scope, the referenced Lease is current and ACTIVE at issued\_at, and task, generation, fence, candidate, context, and WorkCard digests match exactly.

| Claim component                                                                                                    | Canonical value                                                                                                                                                                   | Owner                         | Validator                                 |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------|
| command_authorization_source                                                                                       | CommandRequest.authorization.grant_id + grant_claim_digest                                                                                                                        | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| grant_separation_rule                                                                                              | The command authorization Grant and holder Grant are resolved and validated independently; they may be the same record only when the required command capability is task.execute. | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| holder_grant_capability                                                                                            | task.execute                                                                                                                                                                      | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| holder_grant_field                                                                                                 | holder_authorization.grant_id                                                                                                                                                     | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| lease_bound_command_kinds                                                                                          | ["artifact.record", "run.record", "gate.record", "finding.record"]                                                                                                                | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| lease_bound_task_transition_states                                                                                 | ["LEASED", "RUNNING", "COMPLETED"]                                                                                                                                                | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| lease_required_state                                                                                               | ACTIVE                                                                                                                                                                            | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| required_command_fields                                                                                            | ["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]                                                | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| required_workcard_fields                                                                                           | ["holder_grant_id", "lease_id", "lease_generation", "fencing_token", "context_digest"]                                                                                            | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| stale_or_unresolved                                                                                                | reject                                                                                                                                                                            | core/authority-model.js<br>on | lease_fence +<br>command_capability_scope |
| Binding equality                                                                                                   |                                                                                                                                                                                   | Owner                         | Validator                                 |
| CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id                                               |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.subject_id == Lease.holder_subject_id                                                               |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id                  |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id                                                     |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation                                   |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token                                      |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.context_digest == WorkCard.context_digest                                                           |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| CommandRequest.workcard_digest == sha256(canonical WorkCard)                                                       |                                                                                                                                                                                   | core/authority-model.json     | lease_fence +<br>command_capability_scope |
| <pre>{   "binding_equalities": [     "CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id",</pre> |                                                                                                                                                                                   |                               |                                           |

```

"CommandRequest.subject_id == Lease.holder_subject_id",
"CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id",
"CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id",
"CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation",
"CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token",
"CommandRequest.context_digest == WorkCard.context_digest",
"CommandRequest.workcard_digest == sha256(canonical WorkCard)"
],
"claim_rule": "A lease-bound command is admissible only when authorization proves the exact command
capability, holder_authorization independently proves task.execute, both exact Grant claims bind the same
subject, Activation and conjunctive effect scope, the referenced Lease is current and ACTIVE at issued_at,
and task, generation, fence, candidate, context, and WorkCard digests match exactly.",
"command_authorization_source": "CommandRequest.authorization.grant_id + grant_claim_digest",
"grant_separation_rule": "The command authorization Grant and holder Grant are resolved and validated
independently; they may be the same record only when the required command capability is task.execute.",
"holder_grant_capability": "task.execute",
"holder_grant_field": "holder_authorization.grant_id",
"lease_bound_command_kinds": [
  "artifact.record",
  "run.record",
  "gate.record",
  "finding.record"
],
"lease_bound_task_transition_states": [
  "LEASED",
  "RUNNING",
  "COMPLETED"
],
"lease_required_state": "ACTIVE",

"required_command_fields": [
  "workcard_task_id",
  "holder_authorization",
  "lease_id",
  "lease_generation",
  "fencing_token",
  "workcard_digest",
  "context_digest"
],
"required_workcard_fields": [
  "holder_grant_id",
  "lease_id",
  "lease_generation",
  "fencing_token",
  "context_digest"
],
"stale_or_unresolved": "reject"
}

```

Owner: core/authority-model.json; core/policy-set.json | Validator: command\_mutation\_claim\_rule + lease\_fence + command\_capability\_scope

## Replay-complete Evidence and CAS

| Evidence field                | Rule                                                                                                                        | Schema           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------|
| Artifact.required             | ["record_type", "artifact_id", "artifact_kind", "digest", "media_type", "size_bytes", "retention_class", "created_at"]      | Artifact         |
| EvidenceBinding.required      | ["activation_digest", "implementation_closure_digest", "candidate_digest", "policy_digest", "tool_digest", "input_digests"] | EvidenceBinding  |
| outcome                       | ["pass", "fail", "blocked", "skipped", "error"]                                                                             | Artifact         |
| evidence_class                | ["product-execution", "validator", "harness-generated", "migration"]                                                        | Artifact         |
| product_credit_eligible       | true only for fresh resolved passing product-execution evidence                                                             | Artifact.allOf   |
| GateResult.evidence_artifacts | exact artifact_id + artifact_record_digest for pass, fail, or blocked                                                       | GateResult.oneOf |

Owner: core/contracts.schema.json; core/semantic-model.json; core/policy-set.json | Validator: \$defs/Artifact + \$defs/EvidenceBinding + \$defs/GateResult + candidate\_evidence\_binding + credit\_requires\_evidence

## Separation of duties

| ID      | Rule                                       | Constraint                                                                                                                                                                                                  | Owner                     | Validator |
|---------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------|
| SOD-001 | lease-is-not-acceptance                    | { "forbid_lease_derived_capabilities": [ "validation.evaluate", "finding.resolve", "finding.waive", "candidate.promote", "release.decide"], "scope_kind": "lease" }                                         | core/authority-model.json | SOD-001   |
| SOD-002 | executor-not-release-decider               | { "forbid_same_subject_for_same_candidate": [ "task.execute", "release.decide"], "scope_kind": "candidate" }                                                                                                | core/authority-model.json | SOD-002   |
| SOD-003 | validator-not-self-promoter                | { "forbid_same_subject_for_same_candidate": [ "validation.evaluate", "candidate.promote"], "scope_kind": "candidate" }                                                                                      | core/authority-model.json | SOD-003   |
| SOD-004 | finder-not-sole-waiver                     | { "forbid_same_subject_for_same_finding": [ "finding.record", "finding.waive"], "scope_kind": "finding" }                                                                                                   | core/authority-model.json | SOD-004   |
| SOD-005 | evidence-producer-not-distribution-decider | { "forbid_same_key_capabilities": [ "evidence.produce", "standard.distribute"], "forbid_same_subject_for_same_candidate": [ "evidence.produce", "standard.distribute"], "scope_kind": "candidate-and-key" } | core/authority-model.json | SOD-005   |

Owner: core/authority-model.json | Validator: capabilities + command\_capability\_rules + command\_effect\_scope\_rules + separation\_of\_duties

## Trust, bootstrap, and role presets

authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim\_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json.

| Trust mode  | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Owner                     | Validator      |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|
| local-owner | { "boundary": "single-machine owner-controlled root authority", "signer_required": false }                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | core/authority-model.json | grant_validity |
| team-signed | { "boundary": "multiple actors or remote transport", "minimum_threshold": 1, "provider_capability": "signature", "provider_resolution": "Resolve the exact technologies.json signature binding for init and every ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly non-reconstructable and valid only while bound to the same project, Activation, provider, and identity; cross-process CLI workflows require a reconstructable configured adapter.", "signer_required": true, "unsupported_binding": "reject-before-workflow" } | core/authority-model.json | grant_validity |

  

| Role preset | Capabilities                                                                                                                                                                                          | Authority status  | Owner                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
| auditor     | projection.read, finding.record, finding.resolve, export.create                                                                                                                                       | non-authoritative | core/authority-model.json |
| root-owner  | standard.activate, authority.manage, task.plan, lease.manage, finding.resolve, finding.waive, candidate.promote, release.decide, export.create, migration.manage, projection.rebuild, projection.read | non-authoritative | core/authority-model.json |
| validator   | projection.read, validation.evaluate, evidence.publish, finding.record                                                                                                                                | non-authoritative | core/authority-model.json |
| worker      | projection.read, task.execute, evidence.publish                                                                                                                                                       | non-authoritative | core/authority-model.json |

Owner: core/authority-model.json | Validator: bootstrap\_rule + trust\_modes + role\_presets\_non\_authoritative

# Semantic vocabulary

| Entity               | Purpose                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                             | Owner                                                         | Validator          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------|
| Task                 | Represent one schedulable unit with one acceptance predicate and bounded mutation scope.                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Candidate            | Identify either one provider-proven immutable product snapshot or one explicitly non-creditable observation, independently from control state and bound to the exact candidate recipe.                                                                                               |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Artifact             | Identify immutable product, evidence, log, report, or diff bytes with provenance and retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the canonical Candidate-delta change manifest without creating another persistent entity. |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Run                  | Record one execution or validation attempt against exact candidate, policy, tool, and input digests.                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Finding              | Record an observed defect or policy violation with explicit effective-blocking semantics.                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Decision             | Record a project-operational acceptance, rejection, resolution, waiver, promotion, or release judgement. It is distinct from the external StandardReleaseDecision for the standard distribution.                                                                                     |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| GateResult           | Record one gate outcome against the exact immutable Task-owned GateRunDefinition, current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved evidence never receives pass credit.                                                          |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Grant                | Prove one subject capability within one activation, time interval, nonce, and scope.                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Lease                | Coordinate temporary mutation ownership through generation, fencing, expiry, heartbeat, and close acknowledgement.                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                             | core/semantic-model.json                                      | admission.entity   |
| Relation             | Source                                                                                                                                                                                                                                                                               | Target                                                                                                                                                                                                                                                                                                                                                      | Rules                                                         | Validator          |
| DEPENDS_ON           | Task                                                                                                                                                                                                                                                                                 | Task                                                                                                                                                                                                                                                                                                                                                        | acyclic-for-ready-frontier, accepted-current-outcome-required | admission.relation |
| READS                | Task, Run                                                                                                                                                                                                                                                                            | Candidate, Artifact                                                                                                                                                                                                                                                                                                                                         | read-does-not-imply-authority                                 | admission.relation |
| PRODUCES             | Task, Run                                                                                                                                                                                                                                                                            | Candidate, Artifact, Finding, Decision                                                                                                                                                                                                                                                                                                                      | content-or-record-identity-required                           | admission.relation |
| VALIDATES            | Run                                                                                                                                                                                                                                                                                  | Candidate, Artifact, Task                                                                                                                                                                                                                                                                                                                                   | candidate-policy-tool-input-binding-required                  | admission.relation |
| CITES                | Finding, Decision                                                                                                                                                                                                                                                                    | Artifact, Run, Finding                                                                                                                                                                                                                                                                                                                                      | selector-and-content-resolution-required                      | admission.relation |
| BLOCKS               | Finding                                                                                                                                                                                                                                                                              | Task, Candidate, Decision                                                                                                                                                                                                                                                                                                                                   | effective-only-while-open-and-blocking                        | admission.relation |
| Value object         |                                                                                                                                                                                                                                                                                      | Meaning                                                                                                                                                                                                                                                                                                                                                     | Owner                                                         | Validator          |
| CandidateDelta       |                                                                                                                                                                                                                                                                                      | A diff Artifact payload binding base and new Candidate digests, WorkCard digest, and the complete normalized changed-path set.                                                                                                                                                                                                                              | core/semantic-model.json                                      | design_rules       |
| Command              |                                                                                                                                                                                                                                                                                      | A state-changing request validated before commit.                                                                                                                                                                                                                                                                                                           | core/semantic-model.json                                      | design_rules       |
| Comparator           |                                                                                                                                                                                                                                                                                      | A pure ordering or equality rule.                                                                                                                                                                                                                                                                                                                           | core/semantic-model.json                                      | design_rules       |
| ContinuationToken    |                                                                                                                                                                                                                                                                                      | An authenticated bounded runtime value signed with the random per-Activation secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; it never grants authority by itself. | core/semantic-model.json                                      | design_rules       |
| CurrentReleaseStatus |                                                                                                                                                                                                                                                                                      | A derived evaluation of an immutable historical release Decision against the current closure, blockers, evidence, providers, and Activation.                                                                                                                                                                                                                | core/semantic-model.json                                      | design_rules       |



| Value object               | Meaning                                                                                                                                                                                                                            | Owner                    | Validator    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------|
| ImplementationClosure      | A canonical capability-specific binding of the promin runtime, Python interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate closure digest.                                    | core/semantic-model.json | design_rules |
| Operation                  | One completed semantic step with explicit input, output, failure, and evidence semantics.                                                                                                                                          | core/semantic-model.json | design_rules |
| PlanProposal               | A non-authoritative bounded proposal that derives candidate Tasks and relations from project intent, repository facts, backlog, and references; it cannot mutate operational state or grant authority.                             | core/semantic-model.json | design_rules |
| Policy                     | A machine-addressable invariant owned by policy-set.json.                                                                                                                                                                          | core/semantic-model.json | design_rules |
| Predicate                  | A pure decision rule.                                                                                                                                                                                                              | core/semantic-model.json | design_rules |
| ProviderInvocationEvidence | The exact capability, provider, invocation kind, identity kind and digest, and adapter ID actually used by one operation.                                                                                                          | core/semantic-model.json | design_rules |
| Query                      | A read-only request that cannot mutate authoritative state.                                                                                                                                                                        | core/semantic-model.json | design_rules |
| StandardReleaseDecision    | An external human-reviewable distribution record bound to the exact promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and evidence manifest. It never grants product acceptance. | core/semantic-model.json | design_rules |
| StateCheckpoint            | A disposable self-digested state or journal index bound to an exact authoritative HEAD; it cannot grant authority or acceptance.                                                                                                   | core/semantic-model.json | design_rules |
| WorkCard                   | A bounded generated context package for one Task that separately binds the projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance.                                                | core/semantic-model.json | design_rules |

| Derived projection     | Authority                     | Owner                    | Validator                |
|------------------------|-------------------------------|--------------------------|--------------------------|
| InventoryEntry         | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| SearchDocument         | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| InventoryProjectionRow | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| ProjectionLimits       | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| RetrievalPage          | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| ReadyFrontier          | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| WorkCardProjection     | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| WorkCard               | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| NextResult             | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| ContinueResult         | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| NavigationView         | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| StateCheckpoint        | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| CurrentReleaseStatus   | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| DoctorResult           | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| OperationMetrics       | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |
| Report                 | derived and non-authoritative | core/semantic-model.json | derived_projection_kinds |

Owner: core/semantic-model.json | Validator: admission + design\_rules + persistent\_entities + relations + value\_objects + derived\_projection\_kinds

# Technology boundaries

| Capability           | Task                                                                                                                                                                                                                              | Forbidden boundary                                                                                                                       | Owner                    | Validator               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------|
| control-runtime      | Parse explicit configuration, canonicalize JSON, and run local CLI orchestration.                                                                                                                                                 | {"effect": "reject", "forbids": ["authority-invention", "policy-invention", "provider-binding-invention"], "normative_authority": false} | core/semantic-model.json | technology_capabilities |
| shape-validation     | Validate the single Draft 2020-12 contract bundle with enforced formats and local references.                                                                                                                                     | {"effect": "reject", "forbids": ["authority-decision", "cross-record-semantic-decision"], "normative_authority": false}                  | core/semantic-model.json | technology_capabilities |
| content-identity     | Compute content digests and content-addressed object identities.                                                                                                                                                                  | {"effect": "reject", "forbids": ["digest-as-action-authority"], "normative_authority": false}                                            | core/semantic-model.json | technology_capabilities |
| local-serialization  | Provide one local-writer lock, atomic replace, file sync, directory sync, and recovery primitives.                                                                                                                                | {"effect": "reject", "forbids": ["distributed-consensus-claim"], "normative_authority": false}                                           | core/semantic-model.json | technology_capabilities |
| query-projection     | Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier, materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for the six supported CLI commands. | {"effect": "reject", "forbids": ["projection-as-normative-authority"], "normative_authority": false}                                     | core/semantic-model.json | technology_capabilities |
| filesystem-inventory | Apply the exact candidate recipe in one product pass, use no-follow reads, and publish either a provider-proven immutable VCS tree stream or an explicit non-creditable observation.                                              | {"effect": "reject", "forbids": ["implicit-initialization-scan"], "normative_authority": false}                                          | core/semantic-model.json | technology_capabilities |
| export-scan          | Perform bounded recursive export inspection, allowlisting, secret classification, and symlink rejection.                                                                                                                          | {"effect": "reject", "forbids": ["package-as-authorization-boundary"], "normative_authority": false}                                     | core/semantic-model.json | technology_capabilities |
| signature            | Dispatch the exact configured signature adapter to sign or verify activation, Grant, and Decision digests for team trust mode, and evidence the invoked adapter identity.                                                         | {"effect": "reject", "forbids": ["core-signer-implementation-lock-in"], "normative_authority": false}                                    | core/semantic-model.json | technology_capabilities |
| build-dependency     | Import project-specific build and dependency facts as derived projections.                                                                                                                                                        | {"effect": "reject", "forbids": ["semantic-model-redefinition"], "normative_authority": false}                                           | core/semantic-model.json | technology_capabilities |

Owner: core/semantic-model.json | Validator: technology\_capabilities

## Conformance budgets and scale

The physical lane uses 100,000 raw files and exactly one derived Artifact proxy per file. Raw inventory creates no Task or Relation semantics. An explicit authorized harness corpus brings the projection to at least 198,999 Core-valid typed Relations. At least 600 actual runtime queries mix exact identities, content-only and high-cardinality terms, broad bounded refinement, misses, hostile identity text, and forced continuation across depths 1-12. This bounded physical proof is not a full 1000 by 1000 execution and must not be replaced by one.

| Structural budget              | Value | Owner                 | Validator                      |
|--------------------------------|-------|-----------------------|--------------------------------|
| authority_bearing_init_records | 5     | core/conformance.json | authority_bearing_init_records |
| base_user_commands_max         | 11    | core/conformance.json | base_user_commands_max         |
| core_normative_artifacts       | 6     | core/conformance.json | core_normative_artifacts       |
| persistent_entity_kinds_max    | 9     | core/conformance.json | persistent_entity_kinds_max    |
| relation_kinds_max             | 6     | core/conformance.json | relation_kinds_max             |

| Structural budget                | Value                                                                                                                                                                                                                                                                                                                                                                                                                                              | Owner                 | Validator                             |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------|
| resolved_plan_bytes_max          | 8192                                                                                                                                                                                                                                                                                                                                                                                                                                               | core/conformance.json | resolved_plan_bytes_max               |
| selected_presets                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                  | core/conformance.json | selected_presets                      |
| resolved_plan_source_samples_max | 64                                                                                                                                                                                                                                                                                                                                                                                                                                                 | core/conformance.json | resolved_plan_source_samples_max      |
| WorkCard ceiling                 | Value                                                                                                                                                                                                                                                                                                                                                                                                                                              | Owner                 | Validator                             |
| max_bytes                        | 16384                                                                                                                                                                                                                                                                                                                                                                                                                                              | core/conformance.json | bounded-workcard-through-depth-twelve |
| max_entities                     | 32                                                                                                                                                                                                                                                                                                                                                                                                                                                 | core/conformance.json | bounded-workcard-through-depth-twelve |
| max_fanout_per_entity            | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                  | core/conformance.json | bounded-workcard-through-depth-twelve |
| max_relations                    | 48                                                                                                                                                                                                                                                                                                                                                                                                                                                 | core/conformance.json | bounded-workcard-through-depth-twelve |
| top_k                            | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                 | core/conformance.json | bounded-workcard-through-depth-twelve |
| Scale contract                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                         | Owner                 | Validator                             |
| event_commit                     | {"complexity": "O(batch)", "head_lookup": "O(1)"}                                                                                                                                                                                                                                                                                                                                                                                                  | core/conformance.json | event_commit                          |
| init                             | {"complexity": "O(core+plan)", "project_tree_passes": 0}                                                                                                                                                                                                                                                                                                                                                                                           | core/conformance.json | init                                  |
| inventory                        | {"complexity": "O(files+bytes)", "creditable_consistency_modes": ["immutable-vcs-tree"], "derived_artifact_proxies_per_raw_file": 1, "manifest_verified_jsonl": true, "memory_amplification_max": 32, "observational_mode_pass_credit": false, "project_tree_passes_max": 1, "search_text_bytes_max": 4096, "synthetic_relations_per_raw_file": 0, "synthetic_tasks_per_raw_file": 0}                                                              | core/conformance.json | inventory                             |
| normal_one_shot_replay           | {"checkpoint_authoritative": false, "complexity": "O(delta-since-verified-checkpoint)", "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"}                                                                                                                                                                                                                                                                            | core/conformance.json | normal_one_shot_replay                |
| projection                       | {"public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only", "rebuild_product_tree_passes": 0, "reserved_provenance_rejected": true}                                                                                                                                                                                                                                                                                 | core/conformance.json | projection                            |
| replay                           | {"complexity": "O(events)"}                                                                                                                                                                                                                                                                                                                                                                                                                        | core/conformance.json | replay                                |
| saturation                       | {"fresh_control_state_per_iteration": true, "init_product_tree_scans": 0, "semantic_state_reused": false}                                                                                                                                                                                                                                                                                                                                          | core/conformance.json | saturation                            |
| workcard                         | {"broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus pagination", "complexity": "O(top_k+capped_typed_closure)", "continuation_state_bytes_max": 16384, "continuation_token_bytes_max": 256, "corpus_independent_output_bound": true, "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*requested_depth))))", "selected_closure_union_completeness": 1} | core/conformance.json | workcard                              |
| Reference benchmark              | Value                                                                                                                                                                                                                                                                                                                                                                                                                                              | Owner                 | Validator                             |
| core_valid_relation_count        | 198999                                                                                                                                                                                                                                                                                                                                                                                                                                             | core/conformance.json | reference-benchmark                   |
| depths_tested                    | [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]                                                                                                                                                                                                                                                                                                                                                                                                            | core/conformance.json | reference-benchmark                   |
| file_count                       | 100000                                                                                                                                                                                                                                                                                                                                                                                                                                             | core/conformance.json | reference-benchmark                   |
| forced_continuation              | True                                                                                                                                                                                                                                                                                                                                                                                                                                               | core/conformance.json | reference-benchmark                   |
| inventory_passes                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                  | core/conformance.json | reference-benchmark                   |

| Reference benchmark                                                                                                           | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Owner                 | Validator           |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|
| performance_profiles                                                                                                          | { "portable-local-v1": { "commit_bytes_per_changed_record_max": 32768, "commit_p95_ms_max": 500, "commit_p99_ms_max": 1000, "compatible_platforms": [ "linux", "windows" ], "database_bytes_max": 402653184, "p50_ms_max": 100, "p95_ms_max": 150, "p99_ms_max": 500, "peak_rss_bytes_max": 805306368, "projection_amplification_max": 20, "requires_same_runner_no_degradation": true, "runtime_checkpoint_count_max": 24, "semantic_inflation_max": 3.25, "semantic_ingestion_seconds_max": 600 } } | core/conformance.json | reference-benchmark |
| physical_relation_corpus                                                                                                      | { "artifact_target_coverage": 1, "expected_counts_source": "fixture-manifest", "generation": "explicit-authorized-command-events", "relation_count": 198971, "relation_kind": "READS", "separate_from_raw_inventory": true, "task_count": 1567, "total_core_valid_relations": 198999 }                                                                                                                                                                                                                | core/conformance.json | reference-benchmark |
| query_count                                                                                                                   | 600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | core/conformance.json | reference-benchmark |
| raw_file_proxy_ratio                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | core/conformance.json | reference-benchmark |
| rebuild_product_passes                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | core/conformance.json | reference-benchmark |
| reference_targets_are_environment_specific                                                                                    | True                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | core/conformance.json | reference-benchmark |
| semantic_corpus                                                                                                               | { "depths": [ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 ], "expected_counts_source": "fixture-manifest", "relation_count": 28, "relation_kind": "DEPENDS_ON", "separate_from_raw_inventory": true, "task_count": 32 }                                                                                                                                                                                                                                                                                     | core/conformance.json | reference-benchmark |
| silent_truncations_max                                                                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | core/conformance.json | reference-benchmark |
| synthetic_task_ratio                                                                                                          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | core/conformance.json | reference-benchmark |
| Owner: core/conformance.json   Validator: structural_budgets + workcard_hard_ceiling + scale_contracts + reference_benchmarks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                     |

## Preset and profiles

Select bounded semantic planning, retrieval, and parallelism budgets without granting action authority.

| Profile  | Budgets                                                                                                                                                                                            | Owner                          | Validator |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------|
| baseline | { "default_dependency_depth": 2, "max_context_bytes": 8192, "max_entities": 16, "max_fanout_per_entity": 4, "max_parallel_tasks": 1, "max_relations": 24, "model_tier": "weak-local", "top_k": 8 } | presets/semantic-standard.json | profiles  |
| balanced | { "default_dependency_depth": 4, "max_context_bytes": 12288, "max_entities": 24, "max_fanout_per_entity": 6, "max_parallel_tasks": 3, "max_relations": 36, "model_tier": "capable", "top_k": 10 }  | presets/semantic-standard.json | profiles  |

| Profile                                                                                     | Budgets                                                                                                                                                                                                                                                                                                                   | Owner                          | Validator                        |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|
| extended                                                                                    | {"default_dependency_depth": 6, "max_context_bytes": 16384, "max_entities": 32, "max_fanout_per_entity": 8, "max_parallel_tasks": 6, "max_relations": 48, "model_tier": "strong", "top_k": 12}                                                                                                                            | presets/semantic-standard.json | profiles                         |
| Preset field                                                                                | Value                                                                                                                                                                                                                                                                                                                     | Owner                          | Validator                        |
| base_user_commands                                                                          | ["init", "doctor", "status", "next", "validate", "static-admission", "continue", "audit", "refresh", "context", "skills"]                                                                                                                                                                                                 | presets/semantic-standard.json | base_user_commands               |
| required_provider_capabilities                                                              | ["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection"]                                                                                                                                                                                                                    | presets/semantic-standard.json | required_provider_capabilities   |
| forbidden_implicit_behavior                                                                 | ["full-scan-on-init", "provider-discovery-as-authority", "implicit-action-grant", "unbounded-context-expansion", "search-ranking-as-acceptance-proof"]                                                                                                                                                                    | presets/semantic-standard.json | forbidden_implicit_behavior      |
| semantic_programming                                                                        | {"default_visibility": "private", "object_promotion_criteria": ["independent-invariant", "independent-policy", "replaceability", "measurable-algorithm", "failure-boundary", "reuse", "domain-significance"], "pipeline": ["validate-input", "prepare-change", "execute-operation", "record-result", "publish-evidence"]} | presets/semantic-standard.json | semantic_programming             |
| additional_provider_capabilities                                                            | ["filesystem-inventory", "export-scan", "signature", "build-dependency"]                                                                                                                                                                                                                                                  | presets/semantic-standard.json | additional_provider_capabilities |
| Owner: presets/semantic-standard.json   Validator: Preset schema + preset field identifiers |                                                                                                                                                                                                                                                                                                                           |                                |                                  |

## Top-level schema ingress

| #                                                                    | Accepted root definition   | Owner                      | Validator                  |
|----------------------------------------------------------------------|----------------------------|----------------------------|----------------------------|
| 1                                                                    | #!/\$defs/StandardManifest | core/contracts.schema.json | #!/\$defs/StandardManifest |
| 2                                                                    | #!/\$defs/SemanticModel    | core/contracts.schema.json | #!/\$defs/SemanticModel    |
| 3                                                                    | #!/\$defs/AuthorityModel   | core/contracts.schema.json | #!/\$defs/AuthorityModel   |
| 4                                                                    | #!/\$defs/PolicySet        | core/contracts.schema.json | #!/\$defs/PolicySet        |
| 5                                                                    | #!/\$defs/Conformance      | core/contracts.schema.json | #!/\$defs/Conformance      |
| 6                                                                    | #!/\$defs/Preset           | core/contracts.schema.json | #!/\$defs/Preset           |
| 7                                                                    | #!/\$defs/ProjectInit      | core/contracts.schema.json | #!/\$defs/ProjectInit      |
| 8                                                                    | #!/\$defs/StandardsInit    | core/contracts.schema.json | #!/\$defs/StandardsInit    |
| 9                                                                    | #!/\$defs/TechnologiesInit | core/contracts.schema.json | #!/\$defs/TechnologiesInit |
| 10                                                                   | #!/\$defs/AuthorityInit    | core/contracts.schema.json | #!/\$defs/AuthorityInit    |
| 11                                                                   | #!/\$defs/Activation       | core/contracts.schema.json | #!/\$defs/Activation       |
| 12                                                                   | #!/\$defs/Grant            | core/contracts.schema.json | #!/\$defs/Grant            |
| 13                                                                   | #!/\$defs/Lease            | core/contracts.schema.json | #!/\$defs/Lease            |
| 14                                                                   | #!/\$defs/Task             | core/contracts.schema.json | #!/\$defs/Task             |
| 15                                                                   | #!/\$defs/Candidate        | core/contracts.schema.json | #!/\$defs/Candidate        |
| 16                                                                   | #!/\$defs/Artifact         | core/contracts.schema.json | #!/\$defs/Artifact         |
| 17                                                                   | #!/\$defs/Run              | core/contracts.schema.json | #!/\$defs/Run              |
| 18                                                                   | #!/\$defs/Finding          | core/contracts.schema.json | #!/\$defs/Finding          |
| 19                                                                   | #!/\$defs/Decision         | core/contracts.schema.json | #!/\$defs/Decision         |
| 20                                                                   | #!/\$defs/GateResult       | core/contracts.schema.json | #!/\$defs/GateResult       |
| 21                                                                   | #!/\$defs/Relation         | core/contracts.schema.json | #!/\$defs/Relation         |
| 22                                                                   | #!/\$defs/WorkCard         | core/contracts.schema.json | #!/\$defs/WorkCard         |
| 23                                                                   | #!/\$defs/CommandRequest   | core/contracts.schema.json | #!/\$defs/CommandRequest   |
| 24                                                                   | #!/\$defs/CommandResult    | core/contracts.schema.json | #!/\$defs/CommandResult    |
| 25                                                                   | #!/\$defs/EventBatch       | core/contracts.schema.json | #!/\$defs/EventBatch       |
| Owner: core/contracts.schema.json   Validator: oneOf + Draft 2020-12 |                            |                            |                            |